

KYOS Analytical Platform

User Manual

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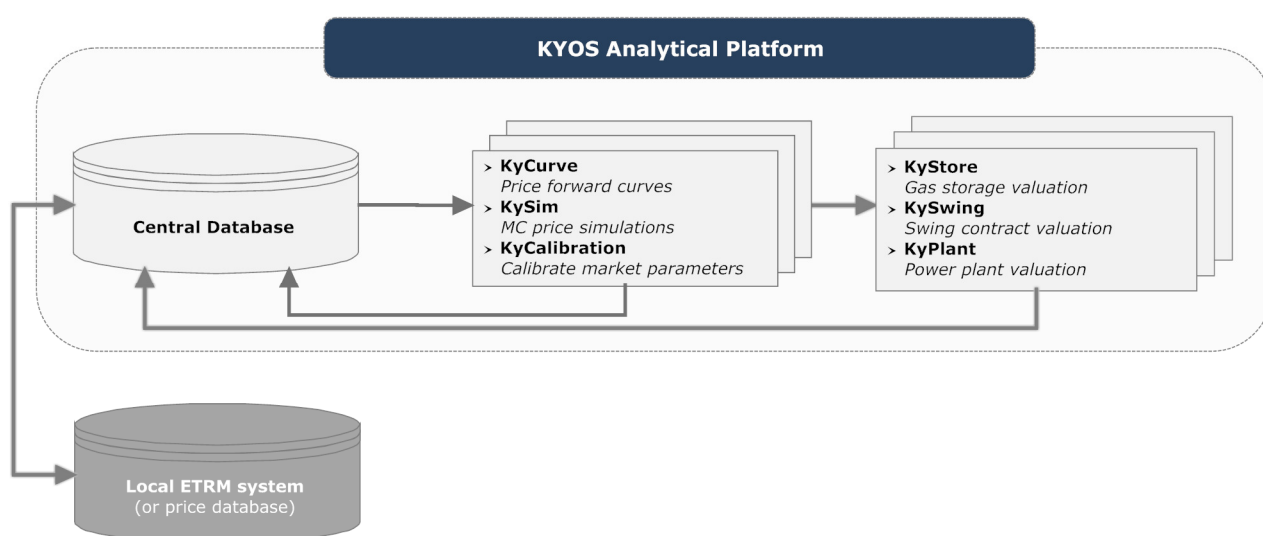
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1 Introduction

1.1 The Platform

The KYOS Analytical Platform is the central platform to run KYOS analytical software. It can be used to manage a commodity portfolio, to assess commodity risks, to value a gas storage, flexible contract or power plant, to forecast and to simulate market prices. The settings and results of various calculations are stored in a central database. Results can be displayed in the system, copied to Excel, or exported to other systems.



1.2 Conventions used in this document

→	Menu item
□	button
▼	dropdown list
<i>italics</i>	input field
Blue background	link
Kyos logo (top left)	click to go to the table of contents
▪▪▪ Kyos Energy Consulting BV ▪▪▪ (bottom left)	click to go to Kyos website
In the document index	click on a page number to go to the subject

2 Setting up the system

2.1 Commodity Settings

2.1.1 Commodity

Settings → Reference settings → Commodities

Many settings are combined in the definition of a commodity. Besides the name, the following information is required:

- Commodity group.
- Region.
- Trading definition. This input is required to calculate prices on bridge days and holidays in power markets. A centralized holiday schedule is implemented per country.
- Daylight saving. For power markets, the appropriate daylight saving definition is required for the construction of hourly price forward curves. When choosing CET (Central European Time), daylight saving is applied between hour 2 and 3 on the last Sunday in March and in October. For WET, this is 1 hour earlier. For EET, this is 1 hour later.
- Currency and Unit. For example, NCG gas is traded in EUR/MWh. Also indicate if the trading currency is actually cents. For example, NBP gas is traded in pence/therm, which is GBP cents / therm.
- Energy content in MJ. For example, 1 MWh of energy equals 3600 MJ. This is relevant for converting results into the same energy unit, if required. When the unit is not energy then this field may be left empty.
- LHV energy content in MJ. This is only shown for the commodity group natural gas and used to convert LHV energy content to HHV energy content in KyPlant and KyPowerFundamentals (KyPF)
- Carbon content per (energy) unit in kg/GJ. This is only required for commodities which are potential fuels for electricity production. The input is used in KyPlant and KyPowerFundamentals (KyPF).

Create new commodity

Name	
Group	Power v
Region	Germany v
Trading Definition	Calendar v
Daylight savings	CET v
Currency	EUR v
Is Cent	☐
Unit	MWh v
Energy	
Carbon content (kg/GJ)	

When a commodity is created, the relevant market settings can be defined and stored. With market settings the user can define:

- What products are traded on the market (how many months ahead, quarters ahead, etc.)
- How large the bid/ask spread is. Half the bid-ask spread is used as trading costs in KyStore and KySwing.

The tradable products and bid/ask spreads are only applied in calculations of KyStore and KySwing. Note that an empty field means that the product is not tradable.

2.1.2 Commodity Group

Settings → Reference settings → Commodity groups

Each commodity belongs to a commodity group. A commodity group contains (i) a name and (ii) a commodity type. Depending on the commodity type, a commodity group is treated differently in the Platform. For example:

- If the commodity type is power, then it may be a commodity for a power plant, and it can have an hourly price forward curve (or half-hourly or quarter-hourly).
- If the commodity type is natural gas, it may be used in a gas storage. The forward curve will have daily granularity.
- If commodity type is coal, carbon, currency or default, any forward curve will have at most daily granularity (not smaller).
- The commodity types coal, gas and carbon are relevant for power plant valuations and price simulations.

ID	Name	Type	
1	Oil	Default	Edit ▼
2	Natural Gas	Gas	Edit ▼
3	Power	Power	Edit ▼
4	IceCoal	Coal	Edit ▼
5	Emissions	Carbon	Edit ▼
6	FX-rate	Currency	Edit ▼
12	Fuel oil	Default	Edit ▼
15	Metals	Default	Edit ▼
17	Steam	Default	Edit ▼
18	Dredging	Default	Edit ▼
21	Soft	Default	Edit ▼

2.1.3 Calendars

Settings → Reference settings → Calendars

For creating hourly power price forward curves, a holiday and bridge day calendar is an input. The Platform includes default holiday calendars for most Western European markets. The set of standard calendars can be extended to other continents and regions on demand. In addition, the user is free to define his own holiday calendars.

There are two types of days which need to be stored:

- Public holidays (which are treated as Sundays in KyCurve)

- Bridge days, which are days around public holidays with typically lower prices (in power markets at least). An example is the period around Christmas, or Fridays following a public holiday on Thursday. A bridge is treated as a Saturday KyCurve.

Edit calendar

[Back](#)

Name

1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012
2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035
2036	2037																					

Holiday 2013

Bridge day 2013

2013-01-01	2013-01-02	+	Clear current year
2013-03-29	2013-01-03		
2013-04-01	2013-01-04		
2013-05-01	2013-05-10		
2013-05-09	2013-10-04		
2013-05-20	2013-12-23		
2013-10-03	2013-12-24		
2013-12-25	2013-12-27		
2013-12-26	2013-12-30		
yyyy-mm-dd	2013-12-31		

The user can upload holiday calendars with an Excel sheet. Alternatively, the user can make a copy of an existing holiday profile, and adjust specific days.

2.1.4 Contract Type

Settings → Reference settings → Contract types

A contract type contains a name and the definition if it is a spot product.

Spot products are treated differently in the creation of forward curves. Therefore, the user has to define:

- If the product is a spot contract type
- The granularity of the spot contract type. Possibilities are quarter-hour, half-hour, hour and day

Edit contract type

Name	Spot (Quarter-hourly)
Is spot?	☒
Delivery period	☐ Half-hour ☐ Hour ☐ Day ☒ Quarter-hour

2.1.5 Currency

Settings → Reference settings → Currency

There is no restriction on the number of currencies, but only one currency can be the system base currency.

All calculations are performed in the base currency. All other currencies are assumed to be quoted in the base currency. If a commodity is quoted in another currency than the base currency, it is converted into the base currency. In the definition of the currency, the user has to provide a forward curve and a settlement curve for conversion to the base currency.

Currency

[Create new currency](#)

4 results found.

ID	Name	Name of cents	Default	Inverse currency	Name	Settlement name	
1	USD	cent	No	No	EUR USD ECB	USD Spot m/m USD	Edit
2	EUR	cent	Yes	No	None	None	Edit
3	GBP	pence	No	No	EUR GBP ECB	GBP Spot m/m GBP	Edit

The Platform offers the possibility to store currencies with inverse values. This is useful for instance when (i) USD is defined as base currency and (ii) EUR/USD rates are imported such that x.xx USD equals 1 EUR (e.g. 1.20 USD = 1 EUR). This is the default convention for EUR/USD exchange rates in the market. In this case the system will read 1.25, but store it as $1/1.25 = 0.80$ EUR = 1 USD. This routine ensures all prices remain defined in the base currency, but the user can choose how to define the foreign exchange rates.

2.1.6 Delivery Period

Settings → Reference settings → Delivery periods

The delivery type and delivery period are system settings. Both tables are in the system, because the user has to apply these definitions when price data are imported in the system.

Delivery period

ID	Name
15	Balance of week
4	Bom
8	Calyear
3	Day
9	Gasyear
1	Half-hour
2	Hour
5	Month
10	Other
6	Quarter
14	Quarter-hour
7	Season
13	WDNW
11	Week
12	Weekend

2.1.7 Delivery Type

Settings → Reference settings → Delivery periods

The delivery type offers mutiple possible choices as follows:

ID	Name
1	Baseload
2	Peakload
3	Offpeak
4	Half Block 1a
5	Half Block 1b
6	Half Block 2a
7	Half Block 2b
8	Half Block 3a
9	Half Block 3b
10	Half Block 4a
11	Half Block 4b
12	Half Block 5a
13	Half Block 5b
14	Half Block 6a
15	Half Block 6b
16	Block 1
17	Block 2
18	Block 3
19	Block 4
20	Block 5
21	Block 6
22	Block 1 + 2
23	Block 2 + 3
24	Block 3 + 4
25	Block 4 + 5
26	Block 5 + 6
27	Block 1 + 2 + 3 + 4
28	Block 2 + 3 + 4 + 5
29	Block 3 + 4 + 5 + 6

i.

This peak, offpeak and blocks are mostly only relevant for power markets. All non-power price curves should generally be defined as baseload. This is further explained in the chapter about price sources (power baseload, peakload and offpeak prices should be defined as separate price sources).

2.1.8 Exchange, Region and Units

Settings → Reference settings → Exchanges/Regions

Exchanges, regions and units are text fields which are useful to identify and (group) commodities, prices and properties. Anything can be filled in.

2.1.9 Carbon floor

Settings → Reference settings → Carbon floor ☐ Create new carbon floor

These carbon floor settings are used in KySim. To have a non-failing KySim profile with carbon floor adjustment for CO₂ prices, a user has to enter at least one “Nominal Carbon Floor level” (CPF), as well as “Realized Carbon Price Support rate” (CPS) for historical data, where it applies.

In KySim, if CPS for a particular fiscal year is required but not found in these settings, KySim will try to calculate it from the data. In such case, a KySim profile must have enough data for this calculation. Otherwise it will fail. For any missing CPS (past or future fiscal year), the CPF will play an important part in calculations. Hence it is advised to enter reasonable and full information here as well, as it will affect the quality of CO₂ price simulations.

Create new carbon floor

[Back](#)

Name

Fiscal year	Nominal Carbon Floor level (GBP)	Realized Carbon Price Support rate (GBP)
Apr 2013 - Mar 2014	16	4.94
Apr 2014 - Mar 2015	18	9.55
Apr 2015 - Mar 2016	20	18.08
Apr 2016 - Mar 2017	22	18
Apr 2017 - Mar 2018	24	18
Apr 2018 - Mar 2019	26	18
Apr 2019 - Mar 2020	28	0
Apr 2020 - Mar 2021	30	0

How KySim is using these settings and what implications this adjustment has for power plant valuation in KyPlant is described in the chapter on KySim of this documentation, as well as in an article by Jurenas, V. & De Jong, C. (2016) “*The UK carbon floor and power plant hedging*” in Energy Risk, Vol 9 No 2 (May 2016).

2.2 Contract Settings

2.2.1 Books, Documentation, Country and Traders

Settings → Reference settings → Books/Traders/Documentation

These are simple lists with text input fields. When defining an asset or a contract, the user may select from the applicable list.

2.2.2 Business Unit and Delivery point

Settings → Reference settings → Delivery points

Settings → Reference settings → Business units

Business units can be defined as a text field and are generally used in the Platform to report results for a specific business unit.

Within the Platform, a delivery point is a location where certain volumes of a commodity are being consumed, produced or stored. A delivery point belongs to a legal entity and is situated in a country.

In Time Series -> Volumes and Actuals: the user specifies on which delivery point the volume is delivered and to which business unit(s).

In the Volume definition of a contract, the user specifies to which combination(s) of delivery point and business unit the contract volume is delivered.

2.2.3 Cost types

Settings → Reference settings → Cost types

Various costs can be part of a contract. They can be defined as a general cost type, and treated differently.

- Calculation method: Costs can be fixed, variable or based on a custom definition. Costs can be added as total costs over the contract period or calendar year, they can be added as a variable add-on to the contract price, or the customer can specify its own cost definition.
- Buy or sell: is currently not used in the system.
- Commodity: the cost definition can be applied only to a specific commodity or to all commodities.

Cost types

Name	
Calculation method	Fixed [Contract period] ▼
Buy / Sell	All ▼
Commodity	All ▼

2.2.4 Counterparties

Settings → Reference settings → Counterparties

Counterparties are relevant for contracts and (sometimes) assets. The user may connect specific trading documentation to a counterpart, and define that a certain counterpart belongs to a specific legal entity.

2.2.5 Legal entity

Settings → Reference settings → Legal entities

A legal entity is a necessary input for a contract or asset. The legal entity means the entity that buys/sells the contract, or the entity that owns the asset. Only legal entities of the own company have to be defined here. Legal entities of other companies have to be defined as Counterparties.

2.2.6 Hedging

Settings → Reference settings → Hedging strategies

A hedging strategy defines which percentage of the exposure (open position) should be hedged. This is relevant for (risk) Reports. In the profile of Time series → Volume actuals and forecasts the user defines which hedging strategy applies to the volumes of a commodity on a certain delivery point / business unit.

For example, the hedging strategy for a specific delivery point and maturity (e.g. next year) has a target hedging percentage of 80%. Suppose, in addition that at a certain date the user provides a volume forecast for this delivery point, indicating a hedging percentage of 50%. The system will then apply a hedging target of $50\% \times 80\%$ to the forecasted volume.

Note that when a volume profile is defined (Time series → Volume actuals and forecasts) the hedging strategy is an obligatory field. Therefore, it is advisable to have a default hedging strategy defined with e.g. all target percentages set to 100%.

2.3 Users and Roles

2.3.1 Roles

Settings → Roles

An administrator has the right to create roles and users. Users can have one or more roles. Each role is a combination of rights. For example, a role can have the right to view Contracts, but without the right to delete or edit anything in Contracts.

Analytics						
@Risk	☒ All	☐ View	☐ Edit	☐ Delete	☐ Run	
KyCalibration	☒ All	☐ View	☐ Edit	☐ Delete	☐ Run	
KyDispatch	☒ All	☐ View	☐ Edit	☐ Delete	☐ Run	☐ Export
KyPlant	☒ All	☐ View	☐ Edit	☐ Delete	☐ Run	☐ Export
KySim	☒ All	☐ View	☐ Edit	☐ Delete	☐ Run	
KyStore	☐ All	☐ View	☐ Edit	☐ Delete	☐ Run	☐ Export
KySwing	☐ All	☒ View	☒ Edit	☒ Delete	☒ Run	☐ Export
Assets & Contracts						
Contracts	☒ All	☐ View	☐ Edit	☐ Delete	☐ Confirm	

There are different rights depending on the function. For example, the right to Run is only applicable to models in Analytics, and the right to Confirm only to Contracts. The right Edit means the user can create and edit new elements in this part of the Platform.

The admin user can create new administrators. An administrator has the right to create new roles and users.

2.3.2 Users

Settings → Users

An administrator can create new users. Each user can get one or more roles. Based on the roles, the user has the right to perform a function.

Special treatment is given to legal entities. When a user has no restrictions, he can see all contracts and all assets. When a legal entity restriction is added to a user, the user can only see assets and contracts which are stored under the same legal entity.

Please note a user can be entitled to see contracts and assets for more than one legal entity.

2.4 System settings

Settings → System settings

2.4.1 Job clean up

Define after how many days, job data will be removed from the system, or not at all. Job data is related to Analytical models and contain inputs, outputs and a Matlab executable (of the analytical model). The default settings (1, 30, 7 and 14 respectively) are generally fine.

2.4.2 Log settings

Decide where the log files are stored, and how long to keep them. The cron log contains information about job scheduling and job execution. The pricedata import log has information about the process of importing prices. Remove after 30 days is recommended. (both are useful for debugging)

2.4.3 Email settings

You can add a sender mail address and name as well as a server name to distinguish for instance production server from the test server.

2.4.4 Import files cleanup

Decide when to remove imported files. In general remove after 30 days would be fine.

2.4.5 Simulation/Calibration disk space limits

Define how much disk space can be used for simulation and calibration data of KySim. If this limit is exceeded, then old job results are removed automatically.

System settings

Job cleanup

- ☒ After days, remove Matlab executable
- ☒ After days, remove Matlab executable, output files
- ☒ After days remove obsolete jobs
- ☒ After days remove error jobs

Log settings

- Store cron log in file .log
- ☐ After days remove cron log
- Store price data import log in file .log
- ☐ After days remove price data import log

E-mail settings

- Sender e-mail address
- Sender e-mail name
- E-mail subject prefix ☐ When prefix is empty, do not add default prefix of servername

Import files cleanup

- ☒ After days remove successfully imported price data files
- ☒ After days remove failed price data files
- ☒ After days remove successfully imported forecast data files
- ☒ After days remove failed forecast data files
- ☒ After days remove successfully imported KyInventory input data files
- ☒ After days remove successfully imported KyCavern input data files
- ☒ After days remove successfully imported storage definitions files
- ☒ After days remove failed storage definitions files

Simulations/Calibrations disk space limits

- Keep GB of Simulation data
- Keep GB of Calibration data

3 Price data

A lot of functions in the Platform rely on historical and actual market price data. Price sources are grouped into 'Price sources'. A price source generally contains a set of prices for a specific commodity, exchange (or broker) and publication source. Furthermore, separate price sources have to be defined for spot and forward data, and for baseload/peakload/offpeak forward prices (applies to power only).

3.1 Create a Price Source

Price data → Price sources □ Create price source

To create a price source click the green 'Create price source' button

A price source is a combination of:

- Commodity
- Contract type (e.g. Forward / Spot)
- Delivery Type (mostly Baseload)
- Exchange

To save a price source, a unique name should be defined. It is advised to make clear what data are in the price source.

KYOS developed direct connections to import data from EEX, Montel (paid subscriptions) and ECB (free). In addition, data can be automatically imported from company internal files. Both the general and company internal automatic uploads require data files to be placed in a specific directory to which the KYOS Analytical Platform has access. Based on the 'Path to files' and 'Start of file name' the Platform knows that the files need to be processed for price data import. File formats and further details are described further on.

New files may be placed in the applicable directory e.g. once a day. The field 'Last imported' shows when data of this price source have been uploaded the last time. It contains a text string which clarifies date and time of the most recent import.

Create price source

Name	
Commodity	German Power
Contract type	Forward
Delivery type	Baseload
Exchange	EEX
Path to files	
Start of filename	
Last imported	

3.2 Upload price data

3.2.1 Upload xls, xlsx or csv files

Price data → Price sources □ Upload

To upload, click the grey 'Upload' button.

When a price source is created, historical price data can be imported using xls, xlsx or csv price files. The format of the file is fixed, and an example file can be downloaded at Price data -> Upload

The input file needs the following:

- *download_source_id*
this should correspond with the id of the Price Source
- *trading_date_year*
a number: 2015
- *trading_date_month*
a number: 1-12
- *trading_date_day*
a number: 1-31
- *start_delivery_date_year*
a number: 2015
- *start_delivery_date_month*
a number: 1-12
- *start_delivery_date_day*
a number: 1-31
- *price*
maximum of 5 decimals and '.' separator for decimals
- *start_delivery_hour*
integer number, so 5 is treated as 5 AM. Possible values are 0-23
- *start_delivery_minute*
optional; possible values 0, 15, 30, 45
- *delivery_period_id*
see the table in **Settings → Reference Settings → Delivery periods**. For example, 5 stands for a month delivery, 8 for calendar year delivery

Note that in previous versions there were also *end_delivery_year*, *end_delivery_month*, *end_delivery_day*¹ and *product_code*².

For reasons of backward compatibility they can still be added to the xml/csv/xls(x) file, they will not produce errors but the system will ignore them.

¹ The end-delivery values are currently calculated by the system

² Ignored by the system

When prices are already available in the database, they will not be stored for a second time. When a different price for the same period and the same trading date is uploaded, the old value is overwritten by the new value. Incorrect data points can be corrected using this method.

3.2.2 Upload XML Files

As an alternative to uploading an Excel/csv file, an XML file can be used. The same information is required.

Under <root> are the nodes <product>. Each *product* contains a price (e.g. EEX power baseload, February 2013) and a specific trading date (e.g. 1 January 2013). The children of *product* are the same as the columns for the Excel upload, with the contents being the fields.

The XML file can be uploaded under Price data → Price sources ■ Upload , identical to the place where Excel files can be uploaded. An example XML file is shown below:

```
<root>
```

```
  <product>
```

```
    < download_source_id>1</download_source_id>
    < trading_date_year>2013</trading_date_year>
    < trading_date_month>1</trading_date_month>
    < trading_date_day>1</trading_date_day>
    < start_delivery_year>2013</start_delivery_year>
    < start_delivery_month>2</start_delivery_month>
    < start_delivery_day>1</start_delivery_day>

    < price>55.78</price>
    < start_delivery_hour>0</start_delivery_hour >
    < delivery_period_id>5</delivery_period_id>
```

```
  </product>
```

```
  <product>
```

```
    < download_source_id>2</ download_source_id>
    < trading_date_year>2013</trading_date_year>
    < trading_date_month>1</trading_date_month>
    < trading_date_day>2</trading_date_day>
    < start_delivery_year>2013</start_delivery_year>
    < start_delivery_month>3</start_delivery_month>
    < start_delivery_day>1</start_delivery_day>
    < price>54.33</price>
    < start_delivery_hour>0</start_delivery_hour >
    < delivery_period_id>5</delivery_period_id>
```

```
</product>
```

</root>

Note that end_delivery_year, end_delivery_month, end_delivery_day and product_code are ignored by the system. For reasons of backward compatibility they can still be added to the xml file, they will not produce errors.

3.3 Automatic upload of XML files

When a price source is defined (see 3.1), there are two fields which are optional to fill in, and one field is read only:

- Path to files
- Start of filename
- Last imported (read-only)

If the first two fields are filled in, the Analytical Platform automatically checks every 5 minutes if new data are available. When new data are available, they are imported in the Platform.

Name	DE Power FW BL EEX Montel
Commodity	German Power
Contract type	Forward
Delivery type	Baseload
Exchange	EEX
Path to files	data/montel
Start of filename	EEX.German Power.forward.bas
Last imported	20131112120428

Filename convention for price data files.

[Start of filename] [Underscore] [Extra description] [Underscore] [Timestamp] [Dot] [Extension]

Example filename: DEPower_EEX_20140922141233.xml

- The time stamp should be yyyymmddhhmmss. The Platform expects exactly 14 digits
- The Platform checks if there is a price file
- The Platform checks the date and time of the file
- The Platform looks at the 14 numbers after the filename (yyymmddhhmmss)
- When the file is processed, the Platform stores the timestamp (yyymmddhhmmss). This is visible in the field 'Last imported'
- The Platform checks if there is a new file with a time stamp later than the file which is already stored. If yes, the routine starts again.

For processing purposes, it is recommended to have one XML file for each commodity. The reasons are:

- Smaller file sizes lead to quicker processing
- The process is set up per download source id. If there is 1 file with price data for all commodities, the routine is:
 - Open the file for commodity 1 (download source 1)
 - Store the data for commodity 1 (download source 1)
 - Open the file for commodity 2 (download source 2)
 - Store the data for commodity 2 (download source 2)
 - Etc.

Therefore, 1 large XML file with all data leads to a long processing time.

3.4 View and download price data

Price data → Price sources ☐ View

By pressing view on a price source, it is possible to view the underlying price data:

- The tab 'Products by trading date' shows all prices of a single trading date. Various sub-selections are possible per product.
- The tab 'Rolling contracts' shows a graph of the price development over time of rolling contracts (e.g. 1 month ahead, 7 quarters ahead, 2 years ahead, etc. see picture below)
- The tab 'Fixed delivery' shows a graph of the price development over time of a contract for a fixed delivery period (e.g. the 2017 calendar contract)

WTI ICE Montel

[Back to price sources](#)

Brent crude oil - ICE (USD / Barrel)

Products by trading date **Rolling contracts** Fixed delivery

[Reset selection](#)

[Export to Excel](#)

[Filter](#)

Delivery period

Month

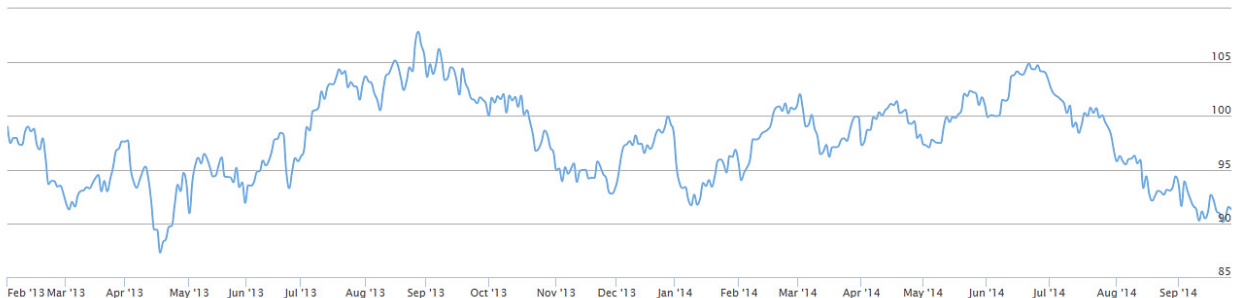
Maturity

4

Brent crude oil - Month + 4

Zoom 1m 3m 6m YTD 1y **All**

From Feb 1, 2013 To Sep 25, 2014



4 Time series

In the Time Series menu the user can define (almost) all time-series which are not price data. This currently includes forecasted and actual volumes of consumption, production (used in Reports), demand, heat, interconnect capacity (used in KyPowerFundamentals), costs (used in IntraCurve), weather data, gas storage volumes (used in KyStore), traded positions (used in KyStore), and inventory volumes (used in Reports).

4.1 Volume actuals & forecasts

4.1.1 List of profiles

Time series → Volume actuals & forecasts

In this part of the Platform the user can define expected production or consumption data (e.g. of natural gas) at a specific delivery point for a certain business unit. The volumes affect what volumes are assigned to contracts and what is shown in Reports. When the user imports volumes, then positive numbers are treated by the Platform as consumption data, while negative numbers are treated as production data.

A profile contains the settings for the volumes, for example to which commodity they apply. The underlying data are not only forecasted or actual (realized) volumes, but may also include a budget (target) price and a target hedge percentage. These data can be uploaded several times for the same profile, each time at another date, the forecast date. The last time volumes are uploaded, is typically the date that the realized volumes are known; these volumes should then be labelled as actuals.

Volume actuals & forecasts Environmental

Volume actuals & forecasts

Upload volumes ▼

18 results found.

☐	ID	Country	Group	Commodities	FX and unit	Granularity	Nr Delivery Points
☐	23	Bayern Test	Natural Gas	Gaspool	EUR / MWh	Day	1
☐	24	Bayern Test	Natural Gas	NCG	EUR / MWh	Day	1
☐	25	Bayern Test	Natural Gas	TTF	EUR / MWh	Day	1
☐	8	Belgium	IceCoal	API2 Coal ARA	USD / MT	Month	3
☐	20	Belgium	Natural Gas	NCG	EUR / MWh	Month	1

4.1.2 Create new volume profile

Time series → Volume actuals & forecasts ☐ Create new volume profile

A volume profile contains a couple of settings about the volumes on one or more delivery points / business unit combinations (DP-BU). When creating a volume profile it is important to work from top to bottom, because the form changes with the selections you choose. For example, you can only select a Commodity when you have already chosen a Commodity group. There can be only one volume profile for a specific DP-BU. Only the 'free' DP-BU combinations are shown in the screen when you define a new profile.

Country / Commodity group / Commodity: after you selected these, the list of relevant DP-BU combinations is shown on the right-hand side of the screen.

Currency / Unit: this must match with the commodity

Hedge strategy: this is a required input field. It is defined in Settings → Reference settings. If you do not actually need a hedge strategy, just choose one at random.

Cumulative costs: this is an optional field. It is defined in Settings → Reference settings and can be applied to contracts for reporting extra costs, potentially over multiple contracts.

Granularity: this is a required field and defines whether the volumes should be stored with quarter-hourly, hourly, daily or monthly granularity.

Imbalance long / Imbalance short: this is optional. It determines against which (spot imbalance) prices any imbalance volumes should be settled with the network operator.

Country	Germany	Delivery points
Commodity group	Natural Gas	☐ Barby / Starches & Sweeteners Europe
Commodities	CEGH Gaspool German Gas (m3) Natural Gas Formula	☐ Frankfurt Cargill / Cargill Malt
Currency	EUR	☐ Frankfurt Cargill / Cargill Refined Oils Europe
Amounts are in cents	☐	☐ Frankfurt Cargill / Cargill Texturizing Solutions
Unit (Energy)	Select...	☐ Frankfurt Gas Union / Gas Union BU
Hedge strategy	Select...	☐ Gelsenkirchen / Chemicals
Cumulative costs	Entry Transport Test Power REB 2012 Broker	☐ Gelsenkirchen / Polymers
Granularity	Select...	☐ Hamburg / Cargill Malt
		☐ Hamburg / Cargill Texturizing Solutions
		☐ Hannover / Chemicals
		☐ Hannover / Fertilizers

Temperature profile: this is an optional input and only used for natural gas in the KyRisk model. Natural gas consumption is generally influenced by temperature: a low temperature leads to a higher natural gas consumption for heating purposes. In the @Risk model the natural gas consumption can be jointly simulated with temperature. The relevant temperature which affects the gas consumption is defined by this input.

4.1.3 Upload volumes

Time series → Volume actuals & forecasts ☐ Upload

File format specification:

- The uploaded file can be in CSV, XLS or XLSX format. It must include the column names on the first row: upload_day, upload_month, upload_year, country, delivery_point, businessunit, commodity, forecast_granularity, forecast_year, forecast_month, forecast_day, forecast_hour, forecast_minute, volume, price, hedge percentage, type
- commodity: a single commodity per row
- forecast_granularity: supported granularities are Day, Month, Hour, Quarter-hour

- volume: positive: consumption, negative: production
- type: the forecast type - initial, updates, actual

upload_day	upload_month	upload_year	country	delivery_point	businessunit	commodity	forecast_granularity	forecast_year	forecast_month	forecast_day	forecast_hour	forecast_minute	volume	price	hedge%	type
29	10	2010	The Netherlands	Test delivery point	BU 4	Dutch power	Day	2010	10	29	24	0/15/30/45	15376	31.8	48%	initial
29	10	2010	The Netherlands	Test delivery point	BU 4	Dutch power	Day	2010	10	29	4	0/15/30/45	15450	27.4	19%	initial
29	10	2010	The Netherlands	Test delivery point	BU 5	Dutch power	Day	2010	10	29	11	0/15/30/45	13265	37.3	43%	initial
29	10	2010	The Netherlands	Test delivery point	BU 4	Dutch power	Day	2010	10	29	24	0/15/30/45	17115	39.6	24%	initial

4.1.4 View Volume actuals & forecasts

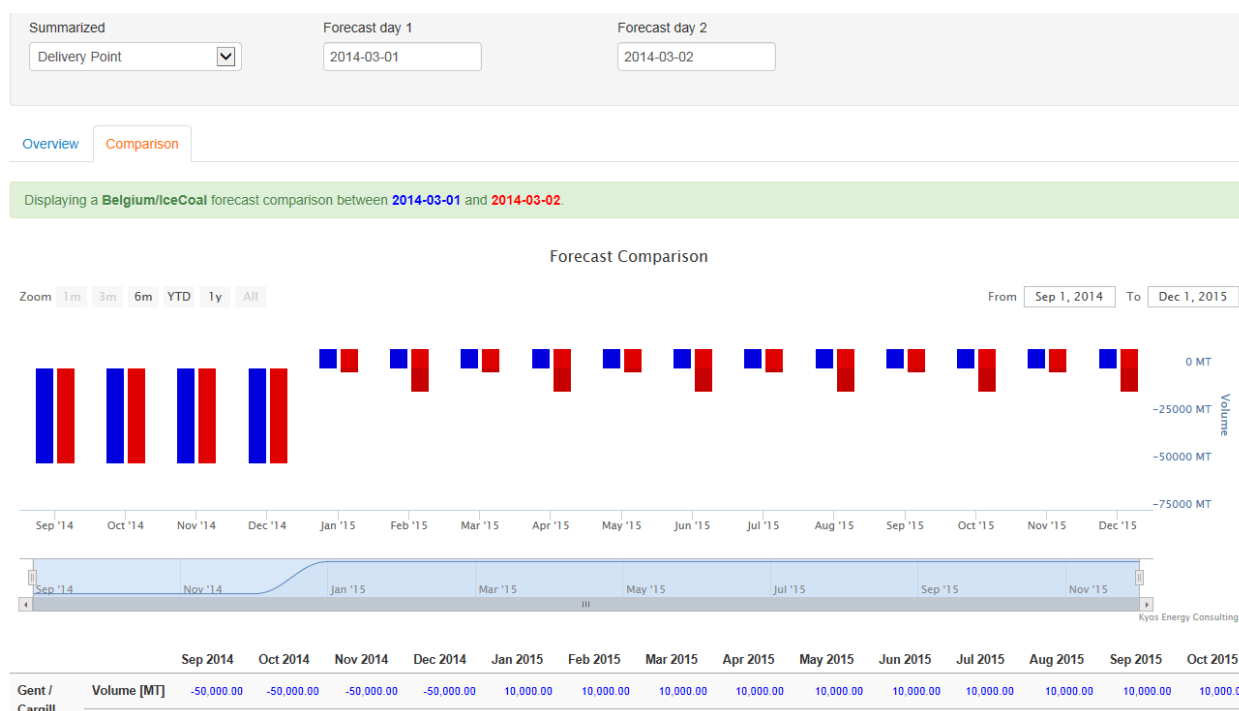
Time series → Volume actuals & forecasts ☐ View

When a user has uploaded volume data, the data can be displayed, both graphically and in a table. When the underlying data have monthly granularity, the display granularity is Monthly by default; if the data have smaller granularity (e.g. daily), there can be additional granularities to choose from.

In the Overview tab, the user sees the volumes, the budget prices and the hedge percentages. The table shows the information per delivery point / business unit and the totals. The graph also shows a volume weighted average budget price.



In the second tab (Comparison) the user can compare forecasts of different dates, either per Delivery point or the total. This second tab is only shown when there are at least two volume forecasts.



4.2 Production, Demand & Weather

In this part of the Platform the user can define and upload production, demand, and weather(temperature), costs, heat, traded positions and interconnect capacity time series. At the moment, the applications are:

- KyStore: where the Traded Positions type of time-series can be used. This time series contains the daily volumes to be delivered based on a portfolio of traded marker products. The user should do the mapping from tradable products to daily positions.
- @Risk model: where temperatures, gas demand (load) and gas spot prices may be correlated.
- KyForecast model: where the time series with seasonal patterns can be forecasted.
- KyPF model: where the forecasted load times series, heat and interconnect capacity are used as input.
- KyIntraCurve: where costs are used as input

The production, demand, or weather time series can be historical data, a forecast or both. For example, the current day is always updated with an actual temperature, while the long-term forecast stays the same. Or the first few days in the future are daily updated as well based on a short-term weather forecast.

4.2.1 Creating a time series profile

Time Series → Production, Demand & Weather ☐ Create time series

Profile name

Enter a unique name

Type

Select a type: currently choices are 'Temperature', 'Production', and 'Demand', Costs, Heat, Traded Positions, Interconnect Capacity

Linked price source (only for Type 'Temperature')

Select a price source which contains gas spot prices. Ideally, the temperature and gas spot prices have an economic relationship and therefore a potential (negative) correlation.

Granularity

Select hour or day

Create timeseries profile

Profile name	Temperature Kiel
Type	Temperature ▼
Linked price source	DE Gaspool Gas Spot EEX M ▼
Granularity	Day ▼

4.2.2 Upload time series data

Manual Upload

Time Series → Production, Demand & Weather ▾ Upload time series

The uploaded file can be in XLS, XLSX, CSV or XML format. In CSV various separators are allowed: , (comma) ; (semicolon) | (vertical bar). It doesn't have to include the column names on the first row. If column names are not included in the first row the file must have the following order of columns: 1. year, 2. month, 3. day, 4. hour, 5. value.

Example of CSV Format:

```
year,month,day,hour,value
```

```
2015,3,11,0,45.3
```

```
2015,3,11,1,45.2
```

```
2015,3,11,2,45.4
```

```
2015,3,11,3,45.5
```

Automatic Upload

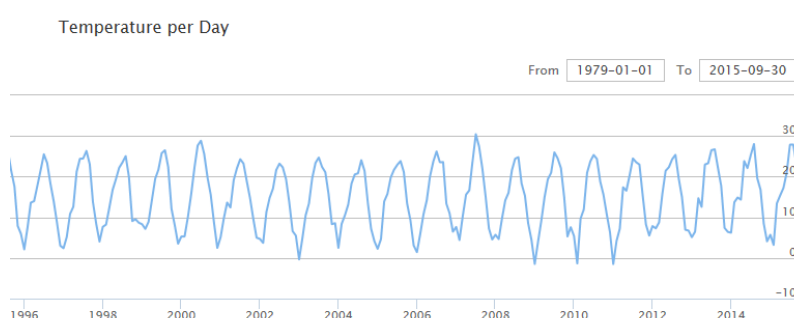
Time Series → Production, Demand & Weather ▾ Time series import settings

The file, described in the previous section, can be uploaded automatically into the platform's database. This is done by placing the file in a specific directory on the platform's ftp server. The user should define the path to the file in the "Location of files" field, and the start of the file name in "File name starts with". Note that if the automatic upload is used, there should be always a first column called "profile_id" in the uploaded file. It contains the id of the time-series for which values are uploaded.

4.2.3 View time series profiles

Time Series → Production, Demand & Weather □ View

Once temperature data have been uploaded, they can be viewed graphically.



4.3 Gas storage volumes

4.3.1 Upload scheduled volumes

Time Series → Gas storage volumes □ Upload volumes



When a gas storage is defined (Assets & Contracts → Gas storage), the Platform automatically creates a corresponding profile for Gas storage volumes. Initially, all volumes between the start date and the end date are set to zero. The user can upload storage volumes for today and historical days, but also store a forecast for the future. The gas storage volume profile is automatically deleted when the corresponding gas storage is deleted.

The stored volumes are used when storage calculations are scheduled. For example, if a storage calculation is performed on 12 January 2014 and the storage volume is then 20,000 MWh, that is taken to be the start volume (unless overwritten by the user).

The file format is:

```
storage_id;year;month;day;volume
```

```
66;2006;9;9;10000
```

```
66;2012;9;10;20000
```

```
66;2012;9;11;30000
```

```
66;2012;9;12;40000
```

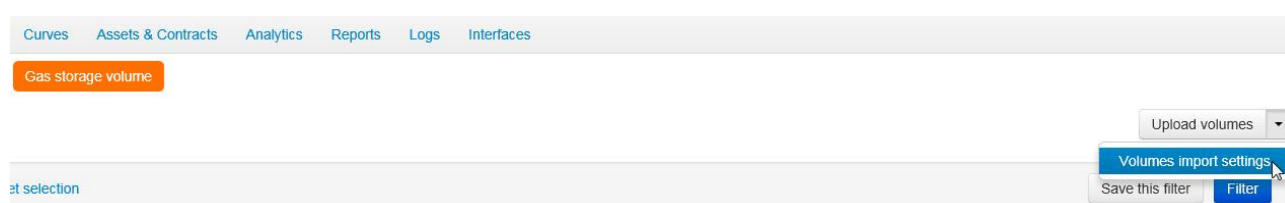
```
66;2012;9;13;50000
```

```
87;2012;9;14;60000
```

```
87;2012;9;15;70000
```

4.3.2 Automatic upload of scheduled volumes

Time Series → Gas storage volumes □ Volumes import settings ▾



The upload process can be automated: you can put csv files with the volumes in a directory. The files need to start with an identifying start of filename (see picture): the system will import the volumes automatically when there is a new data file.

Contracts Gas storage Power plants Inventory

Settings for automated import

If you want to import files automatically, fill in the fields below. The import

Location of files

File name starts with

The complete filename could be *berlin_storage.csv*

The file format is:

```
storage_id;date;volume
```

```
59;2014-08-23;56000
```

```
59;2014-08-24;57000
```

```
59;2014-08-23;58000
```

4.4 Inventory volume

Time Series → Inventory volume

Scheduled volumes for inventory can be uploaded. Volumes are at the start of the day, the default is zero and old volumes (for the same day) are overwritten.

Format specification:

- CSV separator: “;” → semicolon
- Field 1: Inventory id → a number
- Field 2: year → yyyy
- Field 3: month → mm or m (for the first nine months the leading zero is optional).
- Field 4: day → dd or d (for the first nine days of the month the leading zero is optional).
- Field 5: volume → the amount stored in the inventory a number (usually positive but can be negative as well)

An example:

```
2,2014,10,04,1000
2,2014,11,20,2050
7,2014,05,15,1500
7,2014,11,23,1900
7,2014,12,29,4000
```

5 Curves

The KYOS Analytical Platform uses forward curves in various functions. Forward curves contain prices for a specific commodity for delivery in the future. Typically, forward curves are based on market prices and then shaped into the desired granularity (hourly, daily, and monthly). The resulting curve contains a single price for every month, day or hour in the future.

The Platform can upload existing forward curves or apply the KyCurve forward curve building software to shape market prices in correctly shaped forward curves. Because there are two routes for uploading existing forward curves, the user has three approaches to choose from. If the user has the Platform without KyCurve then he can import forward curves into the system. The first two approaches are focused on doing this. If the user has the Platform with KyCurve, she/he can also use the third approach.

1. Upload a complete price forward curve into an existing price source. Then define this price source as input to a specific forward curve profile. This functionality is explained in the next section.
2. Upload a complete price forward curve directly into a forward curve profile. This functionality is explained in the section thereafter.
3. Use KyCurve to shape market prices into a consistent forward curve, shaped into hourly, daily and/or monthly granularity.

The upload via Price sources is an already longer existing functionality in the Platform. The direct upload is newer, very similar to the first approach, though more simple to use. Another advantage of this direct upload approach is that it is possible to upload an *hourly* forward curve, which is not possible with the first approach.

In the next section we first describe how curves can be uploaded. Then the application of KyCurve is explained. In the last section we switch to a quite different function, which is the definition of settlement curves; this is relevant for contracts whose prices depend on some historical ('settled') market price.

5.1 Price forward curves – upload user curves via a Price source

The general idea is that the curves are first imported as market prices (menu item 'Price data') and then copied by the Platform into the forward curve tables. This works for daily and monthly forward curves. It does not work for hourly (power) forward curves, because the amount of data is then often too large.

Step 1: Define a new price source for the forwards (or use an existing one)

Step 2: For one or more trading dates, import the forward curve into the price source. This is described in the previous chapter.

Step 3: In the menu 'Curves' → 'Forward curves' press the button 'Create profile'. Choose the right commodity and select the price source defined in step 1. Save the settings of this forward curve profile.

Step 4: When you have forward data for a specific trading date, you can run the forward curve profile for that date. In addition, if you have selected the automated or scheduled inputs in the profile settings, new curves will be constructed automatically as soon as new forward price data are imported into the underlying price source.

5.2 Price forward curves – upload user curves directly

The general idea is that the curves are placed into an input 'csv' file in a specific directory on the platform's server. Once there is a new file, the system makes a check if the forward curve within the input file is consistent, and imports it into the platform. This works for daily and hourly forward curves. In order to use this functionality, the following steps should be observed:

Step 1: Define a new price source for the forwards (or use an existing one). In the menu 'Curves' → 'Forward curves' press the button 'Create profile'. Choose 'Load complete forward curve'.

Step 2: In the 'Import Settings' tab, the start of the filename and the location where the file(s) should be placed is defined. For example, if the start of the filename is 'Gaspool_', the user could place each day a different file into that directory, e.g. 'Gaspool_20141014.csv', 'Gaspool_20141015.csv', etc. Once there are new files in the selected directory the platform automatically runs KyCurve with them.

The header line of the 'csv' file should always contain specific columns. The remaining rows contain the specific daily/hourly data. An example:

```
download_source_id;trading_date_year;trading_date_month;trading_date_day;start_delivery_year;start_delivery_month;start_delivery_day;start_delivery_hour;start_delivery_minute;delivery_period_id;price
```

```
2;2015;2;9;2018;1;1;0;0;8;88.24;
```

2	2015	2	19	2018	1	1	0	0	8	88.24
---	------	---	----	------	---	---	---	---	---	-------

Step 3: Determine if the platform should not check for new files on specific holidays. There is also an extrapolation choice which allows the KyCurve model to extrapolate some missing values at the end of the curve. For example, if the uploaded curve is till September 15, 2016. The model can extrapolate some values till the end of September.

5.3 Price forward curves – KyCurve

In this section, we describe the use of KyCurve in the Platform. With this model arbitrage free price forward curves can be created. For power, these curves are created up to (less than) hourly granularity. For all other commodities, curves are created up to daily granularity.

5.3.1 Create forward curve profile

Curves → Forward curves ☐ Create profile

The inputs for a forward curve profile can be stored under a unique profile name. The user can choose which forward and spot data profiles, he/she would like to use as inputs in the curve building model.

The following settings are required:

Definition of day types (for power and gas)

The user can indicate which days should obtain the same shaping. In the example below, the daily shape of Tuesday, Wednesday and Thursday is the same for all three days. In the same example, Monday and Friday are treated individually as transition days between weekend and the week and will have a different daily shape compared to other weekdays and between themselves.

If the “Calculate Day weights automatically” is set to “No”, then a weekend-to-working day spread can be defined manually. The model will use this percentage to make the weekend prices cheaper on average with this percentage.

The user can choose which calendar to use. Please note that public holidays are treated as Sunday. Bridge days are treated as Saturday.

Day Weight calculation

Calculate Day Weights automatically ☒ Yes ☐ No

Day Types

Monday	1
Tuesday	1
Wednesday	1
Thursday	1
Friday	1
Saturday & Bridge day	2
Sunday & Holiday	2

Choose the holiday calendar

Calendar 

Settings spot markets

Cap spot below

Cap spot above

Bom includes settled ☒ Yes ☐ No

Let model optimize weighting of historical spot months ☒ Yes ☐ No

Shaping and smoothing

Monthly shaping ☒ History ☐ User inputs

If possible use current forward prices to create monthly weights? ☒ Yes ☐ No

Use separate monthly shaping for base and peak loads ☐ Yes ☒ No

Smooth BOM and M1 ☒ Yes ☐ No

Smooth rest of the curve ☒ Yes ☐ No

Extrapolation

Extrapolate forward curve by %

Schedule

Schedule profile ☐ Yes ☒ No

Automate calculation directly when new inputs are available ☐ Yes ☒ No

Default month weight

January	1
February	1
March	1
April	1
May	1
June	1
July	1
August	1
September	1
October	1
November	1
December	1

Cap spot below/above (for power and gas)

When calculating the hourly shapes for power curves, a regression is performed on historical price data. The incorporation of negative prices or outliers in this regression can be avoided. For example, if you set 'Cap historical spot prices below' to 10, all historical spot prices below 10 will be set to 10.

Please note that capped prices are not excluded from the regression analysis.

BOM includes settled (Yes/No)

The balance of the month price can include the impact of prices which already settled earlier that month. If the BOM price includes settled prices during the month, choose yes; else, choose no. An example f where BOM includes settled prices are the power prices on EEX .

Monthly shaping (History/User inputs)

KyCurve calculates monthly weights based on historical forward quotes. The automatic procedure is executed, if 'History' is chosen. If the user prefers to provide his/her own monthly weights, he/she should choose 'User inputs', and fill in the default monthly weights values.

If possible use current forward prices to create monthly weights? (Yes/No)

The calculated monthly weights (see above) can be overruled by the shape in the current forward curve. If you want the shape in the forward curve to be leading, choose 'Yes'. Else, choose 'No'.

Smoothing (Yes/No)

The user can decide whether to smooth the short end of the forward curve, by choosing 'Yes' or 'No' in the 'Smooth BOM and M1' field. The rest of the curve can be smoothed, as well. If smoothing is applied,

the monthly curve is smoothed, so that there is a smooth transition from the last days in a month to the first days in the next month.

Extrapolate

In case that the user wants the produced forward curve to go beyond the last date, spanned by the provided current forward quotes, then an extrapolation percentage should be chosen. The price of each extrapolated month M will equal the price of month M-12 plus a fixed percentage.

Schedule profile

A profile can be scheduled on a user defined day. This means that every day a forward curve is constructed at this time with the most recent data. In case intra-day curves are necessary, multiple runs of the curve can be scheduled throw-out the day.

Automate calculation directly when new inputs are available

The curve will run once per trading date. This option is used to create forward curves based only on the settlement market data.

Forward curves

[Back to overview](#)

Select commodity
Price sources
Forward curve settings
Time intervals
Select contracts

Historical window for calibration				Calculated date
Start of forward	25	months before trading date	<input type="range"/>	2011-10-12
Start of spot	49	months before trading date	<input type="range"/>	2009-10-12

Curve length			
End of forward curve	1	years after trading date	<input type="range"/>

Back
Next

Start of forward

The calibration of the monthly shape is done based on historical forward data. The length of the history needs to be defined. The minimum is 13 months.

Start of spot

The calibration of the hourly profiles for power markets is done based on historical spot data. The length of the history needs to be defined. The minimum is 25 months.

End of curve

The end date is set at 31 December of the selected year.

Products and maturities to include

Day	☐ Do not include	☒ Include up to maturity	3
Bom	☒ Do not include	☐ Include up to maturity	10000
Month	☐ Do not include	☒ Include up to maturity	5
Quarter	☐ Do not include	☒ Include up to maturity	4
Season	☐ Do not include	☒ Include up to maturity	2
Calyear	☐ Do not include	☒ Include up to maturity	3
Gasyear	☒ Do not include	☐ Include up to maturity	10000
Other	☒ Do not include	☐ Include up to maturity	10000
Week	☒ Do not include	☐ Include up to maturity	10000
Weekend	☒ Do not include	☐ Include up to maturity	10000
WDNW	☒ Do not include	☐ Include up to maturity	10000
Balance of week	☒ Do not include	☐ Include up to maturity	10000

Select contracts

The user can limit the number of a given type of contract to be used in KyCurve in the tab 'Select contracts'. For example, the maximum number of month ahead products can be set to 5, and any monthly products further than 5 months ahead will not be used in the curve building process, e.g. because these contracts are not liquid enough. The option "Do not include" will drop any products if this type when building the forward curve.

5.3.2 Run KyCurve

Curves → Forward curves ▾ Run

In the list of forward curves click on the caret of the desired curve profile and select run.

An arbitrage free forward curve without gaps can be calculated, using a KyCurve profile:

- Automatically, using the scheduling possibility to run it at a pre-defined time. By default, the most recent market data are used to construct the detailed forward curve.
- Individually:
 - The trading day can be selected
 - The market prices can be adjusted
 - A batch of curves can be created, starting at the selected trading date, ending at the last trading date where historical price data are found in the database

Run Forward curve profile ?

Run Curve for profile Dutch Power

Select Trading day:

☐ Create jobs from selected date until today

Show trading date values ?

Start of forward	13	months before trading date		2011-11-21
Start of spot	25	months before trading date		2010-12-17
End of curve	3	years after trading date		2015-12-31

[Run profile](#)

5.3.3 View a list of all forward curve profiles

Curves → Forward curves

Forward curve profiles

[Run all profiles](#) [Compare curves](#) [Create profile](#)

[Show all filters](#) [Load filter](#) [Reset selection](#) [Save this filter](#) [Filter](#)

Commodity ×

- All
- Belgian power
- Czech Power
- Dutch power

18 results found.

20 per page

☐	ID	Name	Commodity	First trading date	Last trading date	Schedule	Automated	Curves	Curve type	View
☐	6	Belgian Power	Belgian power	2014-08-21	2015-10-27	Yes	No	299	Shaped	View
☐	10	Czech Power	Czech Power	2014-08-14	2015-08-13	No	No	238	Shaped	
☐	5	Dutch Power	Dutch power	2014-08-21	2015-10-30	No	Yes	299	Shaped	
☐	8	French Power	French power	2014-08-21	2015-10-27	No	No	291	Shaped	
☐	2	Gaspool EEX	Gaspool	2014-08-21	2015-10-27	No	No	276	Shaped	
☐	11	Gaspool EEX feed	Gaspool	2014-08-21	2015-10-27	No	No	287	Shaped	View
☐	7	German Power	German Power	2014-08-21	2015-10-27	No	No	292	Shaped	View

Run
Edit
Duplicate
Delete

5.3.4 Show all results of a specific forward profile

Curves → Forward curves □ Results

[Settings](#)
[Price data](#)
[Time Series](#)
[Curves](#)
[Assets & Contracts](#)
[Analytics](#)
[Reports](#)
[Logs](#)
[Interfaces](#)

[Forward curves](#)
[Settlement curves](#)

Curves for profile German Power EEX Montel

50 results found. Total items: 672.

« 1 2 3 4 5 6 7 8 9 »

☐	Job ID	Status	Executed	Trading date	Curve start date
☐	146124	Finished	2014-09-26 12:20:01	2014-09-25	2014-09-26
☐	146047	Finished	2014-09-25 14:40:41	2014-09-24	2014-09-25
☐	145899	Finished	2014-09-23 22:41:31	2014-09-23	2014-09-24
☐	145827	Finished	2014-09-22 22:41:12	2014-09-22	2014-09-23
☐	145700	Finished	2014-09-19 22:41:31	2014-09-19	2014-09-20
☐	145605	Finished	2014-09-18 22:41:31	2014-09-18	2014-09-19
☐	145549	Finished	2014-09-17 22:41:12	2014-09-17	2014-09-18

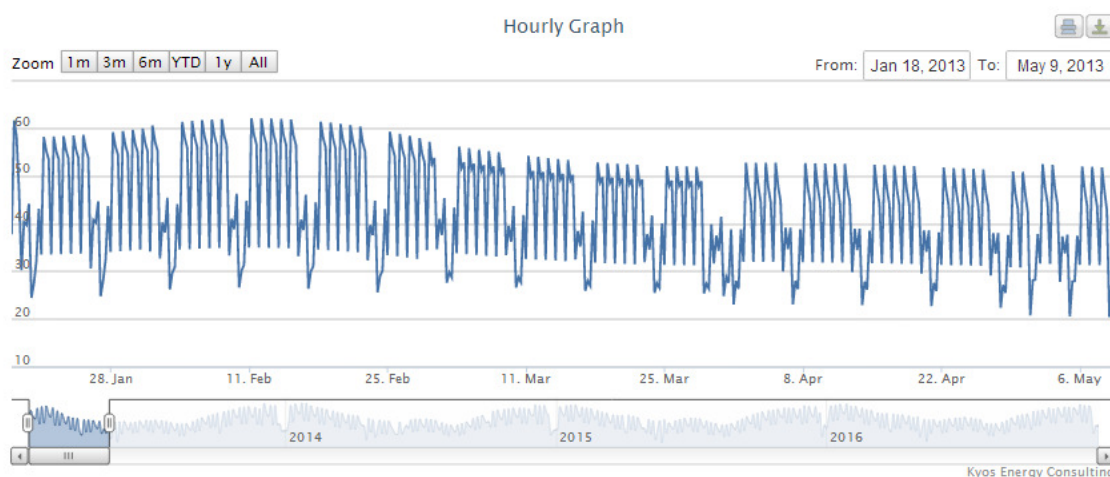
5.3.5 Show a specific result of a specific forward profile

Curves → Forward curves □ Results □ View

Graph view of German Power (German Power)

[Download results](#)

[Back to overview](#)



Curves can be displayed in the Platform and subsequently be downloaded. Each forward curve profile is presented in a list. The list displays the trading date of the first executed calculation, the trading date of the last executed calculation, and the number of calculations ('jobs') that are stored in the database.

By pressing 'View', all individual calculations for the selected curve profile are presented. The user can choose to show the constructed curve on the screen, to download all inputs and outputs (in a combination of XML, csv and txt files), or to delete the curve job.

During the construction process, the following curves are automatically produced:

- For power markets, hourly, daily and monthly curves. Monthly curves are separated in base, peak and off-peak prices
- For gas markets, daily and monthly curves. For valuation purposes, a combination is stored as valuation curve:
 - Daily granularity for spot month
 - Monthly granularity for the remainder of the curve
 - The valuation curve can be used for valuation of a gas storage, where detailed information is required at the start of the curve
- For all other commodity markets, a daily and monthly curve is constructed

5.3.6 Curve shifting

[Prices](#)
[Monthly shaping](#)
[Hourly shaping](#)
[Holidays](#)

Year	Average base	New average base	Average peak	New average peak
2015	33.923	33.923	42.6086	33.923
2016	32.9524	32.9524	41.957	32.9524
2017	31.9503	31.9503	40.95	31.9503

When viewing a forward curve for a given trading date, the user can shift the yearly prices by pressing the 'Shift results' button. As a result the daily gas, and the hourly DST power curves are re-scaled to match the new average prices.

The shifted curve can be downloaded in an Excel file by pressing the 'Download shifted curve' button. For gas, the first column contains dates, while the second prices. For power, the first column contains dates, the second hours, and the third prices.

The 'Store shifted curve on server' saves the same shifted curve in a csv format in a specific directory on the platform's server. This location can be defined based on the needs of a given client.

5.3.7 Download price curves

The user can download curves from a given forward curve profile by following the steps below:

Curves → Forward curves □ View ▾ Download

The downloaded curves are stored in an Excel file, whose name is the following:

'Forward curve <curve profile name> <trading yyyy-mm-dd>.xlsx'

The Excel file contains curves with different granularities in different sheets:

- Gas forward curve profiles produce an Excel file with 3 different sheets: 'daily', 'monthly', 'valuation'. The first column on each sheet contains dates, while the second prices.
- Power forward curve profiles produce an Excel file with 3 different sheets: 'daily', 'monthly baseload', 'monthly peakload', 'monthly offpeak'. The first column on each sheet contains dates, while the second prices. There is one additional sheet – 'hourly'. Its first column is date, the second is hour, and the last price.

5.3.8 Compare forward curves

Curves → Forward curves ☐ compare new curves ☐ Add curve or spread

In the forward curves submenu is a button 'Compare curves'. This leads to a function which allows you to:

- View and compare different forward curves, also of different trading dates
- View and compare 'spreads', which are a weighted average of different forward curves

The user can define which spreads or curves (or both) should be displayed, and select at most three different trading dates. If one of the selected curves is missing data for the selected trading date, a message will appear No data

5.4 profiles

5.4.1 Create a settlement profile

Curves → Settlement curves ☐ Create profile

The following fields have to be defined:

Select commodity

- *Settlement profile name*
Choose a unique name.
- *Select commodity*
- *Type: spot or forward data*
Settlement curves can be created for:
 - spot data (e.g. Platt's oil prices) or
 - forward data, such as [1,0,1] forward data

Settlement commodity type

- *Select historical price data*

Based on the selected commodity, a choice has to be made between all available price sources for the relevant commodity

- *Calculate settlement prices*

Settlement curves can be calculated by the system, or alternatively they are imported automatically from the price database and no calculation is done

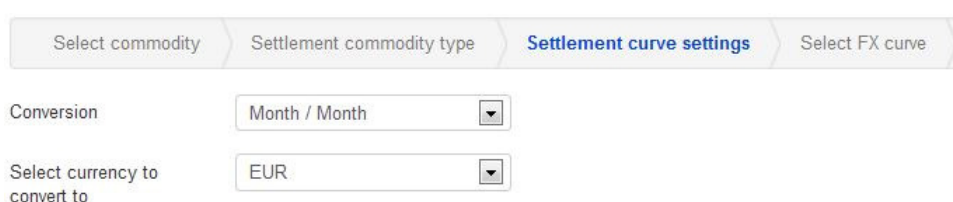
Settlement curve settings

- *Conversion* (into another currency)

A settlement curve is standard calculated in the currency of the commodity. However, it can be recalculated in another currency. There are 2 possibilities:

- Month/Month: The commodity and currency prices are averaged over the month, and thereafter translated in the selected currency.
- Day/Day: The commodity and the currency prices are translated in the selected currency on daily basis, and thereafter averaged over the valuation period

- *Select currency to convert to*



The screenshot shows a navigation bar with four tabs: "Select commodity", "Settlement commodity type", "Settlement curve settings" (which is active and highlighted in blue), and "Select FX curve". Below the navigation bar, there are two dropdown menus. The first is labeled "Conversion" and has "Month / Month" selected. The second is labeled "Select currency to convert to" and has "EUR" selected.

Select FX curve

If conversion is necessary, select an FX curve

5.4.2 Calculation of settlement prices

When a settlement curve is created, with the setting that prices should be calculated, the system automatically produces settlement prices for the last 2 years. This is necessary, because historical settlement prices are often required for calculation of oil indexed contracts.

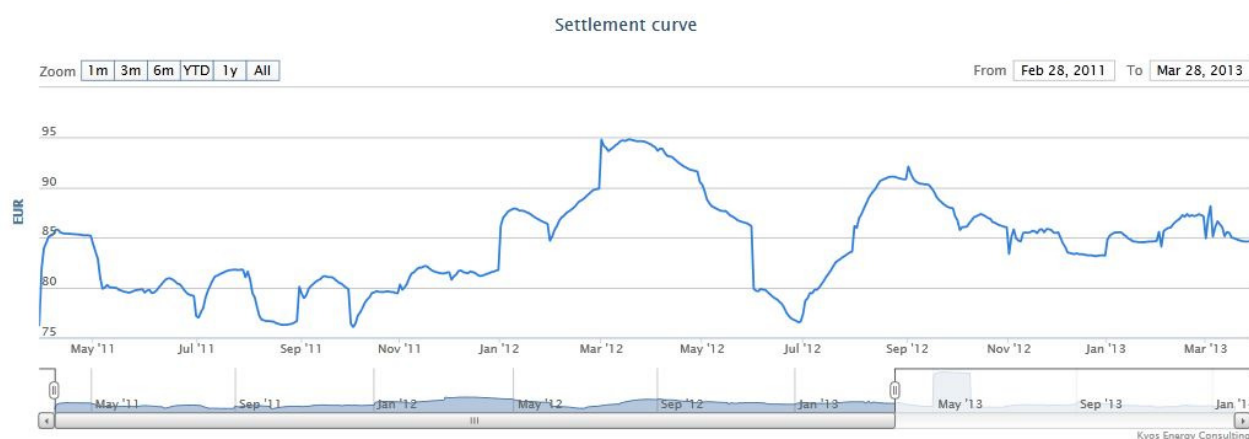
When a settlement curve is defined, it is automatically recalculated after a new price import. A routine ensures that settlement prices are recalculated for all days in the current and last month.

5.4.3 View and download settlement prices

Curves → Settlement curves □ View

Settlement prices are stored per trading day. They can be visualized on the screen, and exported to Excel.

Graph view of settlement curve Spot Oil M/M EUR

[Export to Excel](#)
[Back to the overview](#)

[Show / hide settlement prices](#)

Last trading date	Settlement price	FX rate	Commodity
2014-01-22	78.9291	1.3566	108.5200
2014-01-21	78.8474	1.3526	106.8400
2014-01-17	78.8358	1.3584	107.5500

5.5 Time Series Forecasts – KyForecast

In this section, we describe the use of KyForecast in the platform. The KyForecast model is primarily intended for use with data that have evident seasonal patterns. Furthermore, to produce a proper forecast the model must have at least one full year of input data. Time series data can have daily or hourly granularity.

Before using KyForecast, time series data must be imported at Time Series → Production, Demand & Weather ☐ Create time series.

5.5.1 Create time series forecast profile

Curves → Time Series Forecasts ☐ Create profile

The following settings are required in the *Curve type* tab:

Profile name

The inputs for a time series forecast profile can be stored under a unique profile name.

Time series history

The user must choose, for which Production, Demand & Weather data profiles he/she would like to build a forecast.

Historical window for estimation

The user can choose to use less data than is currently available in the selected Production, Demand & Weather data profile. If by accident user chooses more than is currently available, KyForecast uses all available data.

Length of forecast horizon

The smallest possible value here is 1, indicating one year of forecasts, which actually means that forecasts will be made for one full year after the current trading date will be forecasted as well as for the remaining days of the current year. Moreover, if the selected Production, Demand & Weather data profile has old data, the KyForecast model will fill the gap between the ending date in actual data and current trading date. For example, if historical data end at 2013-12-31, and we run the profile on 2015-01-01 with the *Length of forecast horizon* = 1, we will obtain a forecast for three full years: 2014-2016.

Schedule this profile to run everyday

A time series forecast profile can be scheduled to run on user selected weekdays at a user defined time. This means that every selected weekday a forecasting procedure is repeated at the specific time with the most recent data.

Automate calculation directly when new inputs are available

A time series forecast profile can be scheduled to run whenever the selected Production, Demand & Weather data profile is updated with new data.

The *Curve shaping* tab has a lot of settings similar to KyCurve:

Day types

There can be at most 7 different weekday types (*DayTypes*) possible. However, user may define less than 7 types, e.g. by grouping Tuesday, Wednesday, and Thursday into the same type.

Month types

Every month can have its own weight, so at most there can be 12 different types of months (*MonthTypes*). On the other hand, user may define some months to be of the same type, e.g. June and July can be the same type, or December, January, and February can be the same type.

Choose the holiday calendar

The user can choose which national or user-defined calendar to use. Please note, that public holidays are treated as Sunday. Bridge days are treated as Saturday. The user cannot change that.

Annual shape of the forecast

Two main options available are *Monthly averages* and *Smooth annual sine function*. The first option will result in forecast, which looks like a step function. Conditional monthly averages (one number per month type) are the basis for this forecast; daily and hourly weights are applied on top of these monthly averages. The *Smooth annual sine function* would result in much smoother looking forecast. The model

Forward curves Settlement curves **Time Series Forecasts**

Define time series forecast profile

Curve type

Curve shaping

Profile name

Time series history

testtimeseries

Historical window for estimation

12

months

Length of forecast horizon

1

years after trading date

2016-12-31

Schedule

Schedule this curve profile to run everyday

☐ Yes
 ☒ No

Automate calculation directly when new inputs are available

☐ Yes
 ☒ No

optimizes 3 parameters for either one or two oscillations per year and picks the best fit. Monthly, daily, and hourly shaping is then applied on top of this smooth functional form.

Include mean-reversion in forecast

This is only available when *Smooth annual sine function* is selected. It results in the fourth parameter, which controls reversion to mean / auto-regressive property of the trigonometric function.

Define time series forecast profile

Curve type

Curve shaping

Day types

Monday	1
Tuesday	2
Wednesday	2
Thursday	2
Friday	3
Saturday & Bridge day	4
Sunday & Holiday	5

Month types

January	1
February	1
March	3
April	4
May	5
June	6
July	6
August	8
September	9
October	10
November	11
December	1

Choose the holiday calendar

Calendar

Please select

Annual shape of the forecast:

☐ Monthly averages (using the average per month type)
☒ Smooth annual sine function

Include mean-reversion in forecast:

☐ Yes
☒ No

Capacity factor

The user has a control over yearly capacities for the historical and future data. For example, if it is known that next year the capacity utilization of the power plant will be lower due to technical maintenance or higher due to new installed capacity, this information can be incorporated into the forecast. Furthermore, if historical capacities are defined as being not equal to 100, the KyForecast makes the required adjustments to incorporate this information into monthly shaping. The user can enter up to 30 yearly capacity factors (defined in percentages).

Capacity factor

2011	100	%	2012	100	%	2013	1
2017	100	%	2018	100	%	2019	1
2023	100	%	2024	100	%	2025	1

5.5.2 Run KyForecast

Curves → Time Series Forecasts ▾ Run

In the list of profile click on the caret of the time series forecast profile and select *Run*.

In the next screen, the user can select trading date to use as the date of reference. It could be that the newest data is unreliable or includes some very significant outliers; hence, user may want to discard it and choose a day before the actual ending date of the data used in the profile.

5.5.3 View a list of all time series forecasts profiles

Curves → Forward curves

Time series forecast profiles Create profile

[Load filter](#) [Reset selection](#) Save this filter Filter

KyForecast ID's

9 results found.

20 per page ▾

ID	Name	Time series	Schedule	Automated	
15	Forecasted JP Load till 2021	JP Load2013	No	No	View
14	JP Load Forecasted20132020	JP Load2013	No	No	View
10	JP Load Forecasted20132022	JP Load2013	No	No	View
11	KyPF Japan Hydro	KyPF Japan Hydro2013	No	No	View
12	KyPF Japan Solar	KyPF Japan solar2013	No	No	View
13	KyPF Japan wind	KyPF Japan wind2013	No	No	View
7	TEST Vytas 2	TEST Load Vytas	No	No	View
8	TEST Vytas 3	TEST Load Vytas	No	No	View
3	xxxx	TEST Load CJ	No	No	View

5.5.4 Show all results of a specific time series forecasts profile

Curves → Time Series Forecasts ▢ View

Curves for profile Forecasted JP Load till 2021 Back to overview

3 results found.

20 per page ▾

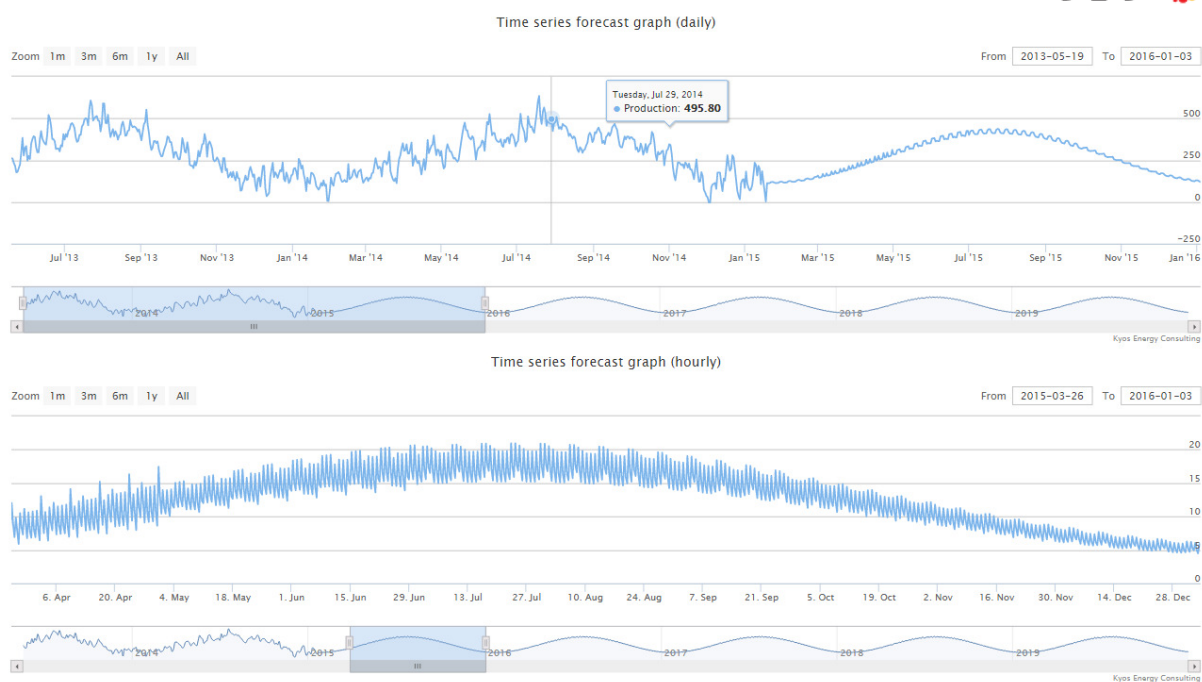
Job ID	Status	Executed	Trading date	User	
43377	Finished	2015-11-03 13:06:00	2015-11-03	administrator	View
43369	Finished	2015-11-03 12:01:22	2015-08-05	administrator	View
41540	Finished	2015-10-16 16:08:00	2015-10-16	administrator	View

20 per page ▾

The list displays the date of reference for the forecast ('Trading date') of the executed calculation, the date and time of the execution, and the number of calculation ('Job ID') that are stored in the database.

5.5.5 Show a specific result of a specific time series forecasts profile

Curves → Time Series Forecasts ▢ View ▢ View



Historical data plus forecasts can be displayed in the Platform and subsequently are available for download.

In case when hourly data is supplied as an input, the output contains hourly forecast as well as daily forecast, which is obtained by aggregating the hourly forecast. In case when daily data is used, the forecast is calculated with daily granularity only.

6 Assets & Contracts

6.1 Contracts

Assets & Contracts → Contracts

Contract information

Detailed information about contracts can be stored in the database. There are several input menus, each defining a certain part of the contract.

6.1.1 General

The General menu includes commodity, contract period, counterparty, etc.

General	Volume	Pricing	Take-or-Pay	Finance	Price Fixing	Documentation
Main Contract Info						
Commodity *	NCG					
Curve for MtM calculation	NCG					
Buy/Sell	Buy					
Volume	0 MWh					
Currency	EUR					
Start date	2016-04-01					
End date	2017-03-31					
Volume flexibility	Flexible volume					
Volumes are nominated	☐ No ☒ Yes – per day ☐ Yes – per month					
Custom option						
Realized (Y/N)	Select...					
Administrative						
Approval required	☒					
Confirmed	☒					
Confirmation number	Test Contract 1					
Contract date	2016-01-01					
Legal						
Contract for legal entity *	kyos					
Counterparty *	Trading House A					
Trading documentation	(no documentation available)					
Financial						
Book	Select...					
Physical/Financial	Physical					
☐ Own use (no proprietary trading)						
No delivery point						
☐ The contract volume will be sold (bought) to (from) other internal or external companies						
Trader						
Trader	Select...					
Counterparty trader						
Counterparty trader	Select...					
Documentation						
Upload documentation						

Curve for MtM Calculation

Determines against which forward curve the contract is valued. This is relevant for:

- KySwing : all contracts
- Reports : fixed-price contracts (not indexed), which are Marked-to-Market against the selected market curve.

Start/End Date

This defines the delivery period of the contract.

Volume flexibility

This determines whether the volume is flexible or fixed.

- Fixed block contracts are all those contracts with a certain volume. The volume may vary over time though. For example, a 10MW power forward is a fixed block contract, but a contract which delivers a different (fixed) volume per hour is also a fixed block contract.
- Flexible volume contracts have volumes which are not fully known yet, though there will typically be upper and lower bounds, which the user can define in the 'Take-or-Pay' part of the contract menu. If the volume is flexible, all calculations are based on the latest forecasts and actual consumption/production (see menu Time series), if available.

Important remarks:

- Only contracts with a flexible volume can be valued with KySwing.
- There can only be one flexible contract on a delivery point. The reason is that (e.g. for cashflow reports and invoices, see Reports) the system needs to assign volumes of a delivery point to one or more contracts. If there is more than one flexible volume contract, it is not clear which part of the volume was delivered in a contract.

Nominated Volumes

The "Volumes are nominated" choice is available only for contracts with flexible volume. There are 3 options:

- "No" – the flexible volume contract is treated by the system as described in the previous section.
- "Yes – per day" – The contract is treated as a fixed volume contract by the system. The user is able to define daily nomination volumes during the delivery period of this contract.
- "Yes – per month" – The contract is treated as a fixed volume contract by the system. The user is able to define monthly nomination volumes during the delivery period of this contract.

The nomination volumes are defined through:

Assets & Contracts → Contract nominations

Customer Label

This choice allows the user to have an arbitrary identification label per contract. The customer label can be defined through:

Settings → Reference Settings → Customer field

Approval Required

Once the "Approval Required" box is checked, an e-mail is sent to the user's supervisor. The supervisor can review the contract and uncheck the box. While the check is there:

- The contract is not included in the platform's reporting process.
- A KySwing valuation profile cannot be created for this contract.

Confirmed

If contracts are confirmed, then they cannot be changed by other users, unless the contract is unconfirmed again. A confirmation number can be assigned to the contract.

Contract Date

This is the date from which the contract is taken into account into the platform's reporting functionality.

Physical/Financial

Determined if the chosen contract is with a physical delivery, or it is settled on the financial market. If a swap contract is defined, financial should be chosen in this field.

Internal distribution

Internal distribution is used for the internal buying/selling of contracted volumes. An example is that the internal Legal Entity buys from an external counterparty as a first step (regular contract). By choosing "Internal distribution" the user cannot enter delivery points. Secondly, the user can (or should) add a second transaction for the internal resale. The volumes are bought by the formal Legal Entity from the counterparty (select the internal Legal Entity). Now the user is forced to store the relevant delivery points.

Documentation

The user is able to upload multiple pdf files.

6.1.2 Volume

The contract volume needs to be redistributed over the months, days or hours in the valuation period. Standard calculation rules for monthly, daily or hourly weighted distribution are available. Furthermore, specific volume profiles can be uploaded from a file.

Volume definition

Volume defined per

per day

Please select...

Fixed delivery

per hour

per day

per month

Manual upload

per hour

per day

per month

Distribution to delivery point

Add delivery point to contract

Create new delivery point

Total volume

Total volume

0

MWh / day x 365 days =

0

MWh

The contract volume can be distributed over multiple delivery points. The delivery points are selected based on the legal entity, which can be managed in Settings.

For physical contracts it is required that delivery points are selected. Please note that the sum of the volumes for the delivery points has to be equal to the total contract volume.

In the case when the volume has some non-standard distribution over the delivery period, the user can upload manually daily or hourly volumes. This is done by choosing “Day (user defined by upload)” / “Hour (user defined by upload)”. The possible files for upload are csv, xml, xls(x). Sample upload files can be downloaded from the platform.

6.1.3 Pricing

Commodity price
Additional costs
Additional costs nominations

Pricing model: indexed

Terms: Not applicable

Indexed

$$\text{Price} = \text{weight1} * [\text{commodity1} - \text{constant1}] + \text{weight2} * [\text{commodity2} - \text{constant2}] + \dots + \text{fixed1} [FX1] + \dots$$

Forward indexation

index product	maturity	forward curve	settlement price	weight	constant	commodity	average	lag	
month	1	TTF	NL TTF Gas FW BL ENDEX Montel	1	0	1	0		Delete

Add a forward term

Spot indexation

No spot index type term defined.

Add a spot term

Price formula

The contract can be priced based on various pricing possibilities:

- A fixed price
- An indexed price
- A financial swap
- A price formula

A physical delivery contract can have a fixed price or an indexed price. A financial contract can be either financial swap or use a price formula. When an indexed price, swap or formula is selected, also a settlement curve profile is necessary to calculate the average of the historical fixings.

The indexation terms can be forward indexation / spot indexation / oil indexation terms. A constant term can be defined as well.

Pricing > Additional costs

Additional costs can be defined (e.g. freight, tax or connection costs) as well. These costs are included in Reports, but not in a KySwing calculation. Additional costs can be from a different counterparty than the contract counterparty. This will result in separate items in the invoice Report.

6.1.4 Take-or-Pay

Contract with a flexible volume often have one or more bands around the contract volume which define the minimum and maximum takeoff. The bandwidth can apply to a period of a month, quarter, year, the total contract period or a combination of these periods. These contracts are known as contracts with Take-or-Pay (ToP) clauses.

- Daily quantity: define volumes for daily minimum and maximum takeoff. These volumes may vary over time.

Daily quantity
ToP volumes
Nomination rights

Daily contract quantity

date	2016-04-01	min	0	max	1000	MWh	Delete
date	2016-05-01	min	500	max	900	MWh	Delete

Add DCQ

- Take or pay volumes: define minimum and maximum percentages of the contract volume for different contract periods:
 - The total contract period constraint is defined as a percentage of the total contract volume.
 - There can be up to three period constraints defined as absolute volumes. Possible periods are month, quarter, season, gas year, calendar year.

Daily quantity
ToP volumes
Nomination rights

Total ToP

min %

max %

ToP policies

Quarter
Select...
Select...

start	min (MWh)	max (MWh)
2016-04-01	90000	120000
2016-07-01	90000	120000
2016-10-01	100000	130000
2017-01-01	100000	130000

- Nomination rights: define which party has the right to exercise the volume flexibility. In most cases this is the buyer of the volumes. This is also the default setting. In some contract though, the seller may nominate and this can be filled in. This input is relevant for KySwing.

[Daily quantity](#)
[ToP volumes](#)
[Nomination rights](#)

Nomination rights

Party which has the right to nominate contract volumes

Nomination Lead Time days

kyos can make nominations for the NCG offtake volumes

6.1.5 Finance

In Finance the user can define payment conditions, so when and with what frequency invoices have to be paid. This affects invoices in the Reports.

Furthermore, there is a field which can be used to define a contract as a hedge for another contract. As a consequence, the hedge contract(s) will be reported in combination with the other contract.

Hedge Accounting

This contract is linked to physical contract:

Entity	ID	Counterparty	Period	Buy/sell	Commodity	Volume	Currency
Example Entity BV	89	Shell	2015-01-01 — 2015-12-31	buy	TTF	43 800	EUR

[Select another contract](#)
[Unlink contract](#)

6.1.6 Price Fixing

Contracts with an indexation or formula are so-called floating-price contracts. The price risk may be hedged with other contracts, but it is also common that the hedge is executed with the physical counterparty as a price fixing (price click). A price fixing (or 'click') converts the floating price into a fixed price. This fixed price may apply to only part of the contract period and/or part of the contract volume. A contract owner may click multiple times. The user first defines a delivery period to which the click applies, fills in the click price and then what volume is fixed. The possibilities include:

- Fixed volume
- Percentage of the contract volume

- Percentage of the realized volume (in case there is flexibility in the offtake). In this case the volume to which the click applies is only known after the actual volumes of the contract have been provided; when making calculations for future time periods, the system therefore assumes that the click applies to the contract volume.

Note that these fields are not used for a KySwing profile. This may be extended in a future release of the software.

General
Volume
Pricing
Take-or-Pay
Finance
Price Fixing
Documentation

Edit this information
Back to overview
Save contract

Basic Contract Information

legal entity	Pure Example	buy/sell	buy	price fixing	yes
counterparty	Dow Chemicals	commodity	Chemical Formula	id/confirm	904 / unconfirmed
contract period	2014-04-01 — 2015-03-31	volume & unit	36000 MT	currency	USD
book	Procurement	physical/financial	physical	contract type	formula

Price fixing

date of fixing	start date	end date	amount		price		
2014-04-02	April 2014	March 2015	18000	MT	70.25	USD	Edit
2014-04-09	April 2014	March 2015	25	percent of contract	75.95	USD	Edit

Add a new price click

Assets & Contracts → Contracts ▢ Prices & Volumes

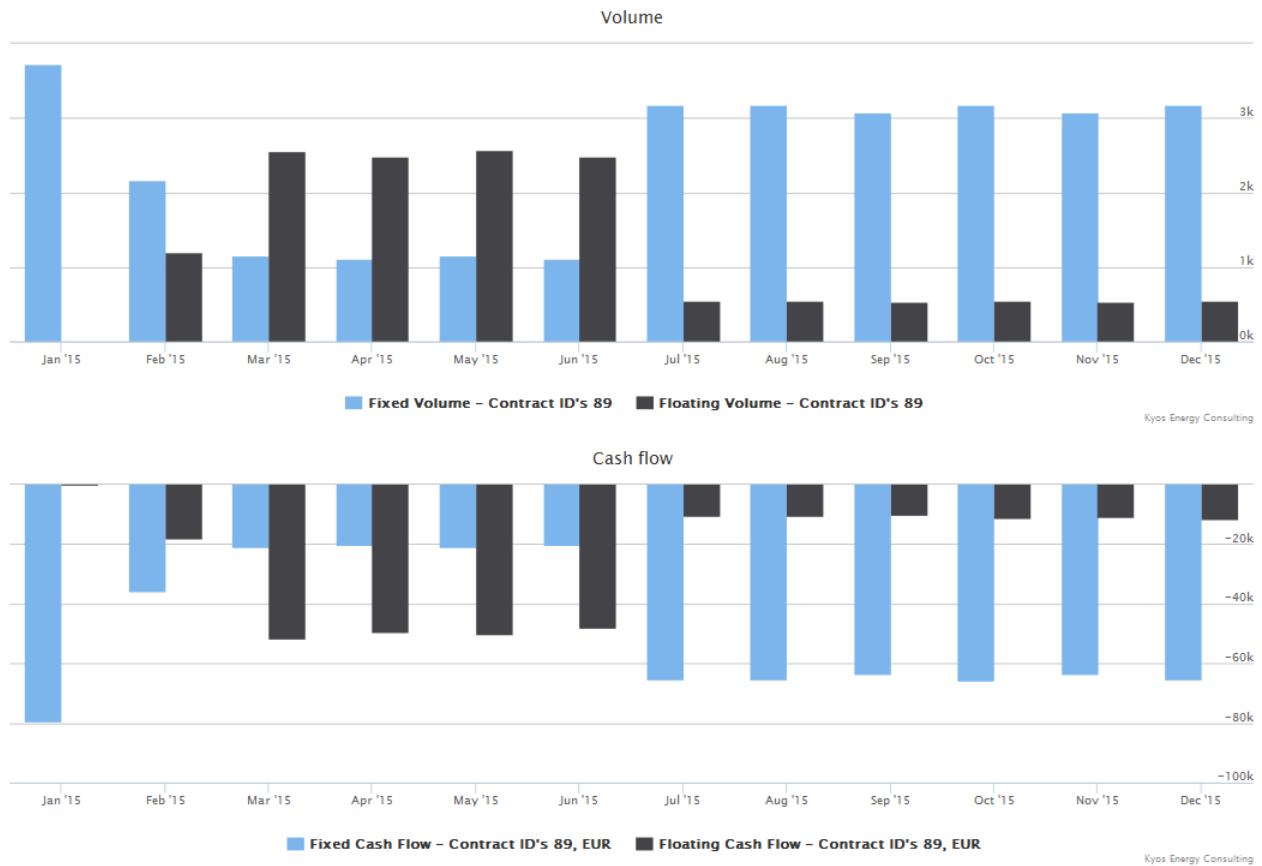
In the overview of contracts, the user can click on 'prices and Volumes' from the dropdown menu. This leads to a report which shows the volumes and cash flows (expected or realized) of the contract. Both are broken down into 'fixed' and 'floating'. The selection is done per trading date.

Trading date 2015-01-14

Export to Excel

Back to contracts

Volumes, Prices and Cash Flow for contract with ID 89



Period	Total Volume	Fixed Volume	Fixed Price	Fixed Cash Flow	Floating Volume	Market Price	Floating Cash Flow	Currency
Tot	43800	29266.605	20.046	-586688.6484	14533.395	19.431	-282401.6322	EUR
Jan 2015	3720	3720	21.352	-79430.6676	0	0	0	EUR
Feb 2015	3360	2163.406	16.543	-35789.9807	1196.594	14.981	-17926.3794	EUR
Mar 2015	3715	1151.65	18.226	-20989.75	2563.35	20.198	-51774.5433	EUR
Apr 2015	3600	1116	18.226	-20340	2484	19.909	-49453.956	EUR
May 2015	3720	1153.2	18.226	-21018	2566.8	19.609	-50332.3812	EUR
Jun 2015	3600	1116	18.226	-20340	2484	19.359	-48087.756	EUR
Jul 2015	3720	3174.481	20.629	-65486.1911	545.519	19.217	-10483.2307	EUR
Aug 2015	3720	3174.481	20.629	-65486.1911	545.519	19.367	-10565.0585	EUR
Sep 2015	3600	3072.079	20.629	-63373.7333	527.921	19.408	-10245.8949	EUR
Oct 2015	3725	3178.748	20.629	-65574.2102	546.252	20.356	-11119.5019	EUR
Nov 2015	3600	3072.079	20.629	-63373.7333	527.921	20.753	-10955.9489	EUR
Dec 2015	3720	3174.481	20.629	-65486.1911	545.519	21.002	-11456.9814	EUR

6.2 Contract Nominations

If a flexible contract uses daily or monthly nominations, the user can define the nominated volumes through:

Assets & Contracts → Contract Nominations

A list with all relevant contracts is displayed. The user can select the contracts, whose nominated volumes should be adapted, and press the “Update nominations” button. A screen containing the delivery period of the selected contracts is displayed. The user can adjust the nominations for the desired dates.

6.2.1 Uploading nominations

Assets & Contracts → Contract Nominations □ Upload Nominations

The nominations per single or multiple contracts can be uploaded through a file (csv, xls, xlsx). The columns in the uploaded files should be the following:

contract_id	year	month	day	volume	hour
1	2010	10	29	17293	19
1	2010	10	29	15693	5
1	2010	10	29	15020	2
1	2010	10	29	10457	3

Assets & Contracts → Contract Nominations □ Import Settings

The file, described in the previous section, can be uploaded automatically into the platform’s database. This is done by placing the file in a specific directory on the platform’s ftp server. The user should define the path to the file in the “Location of files” field, and the start of the file name in “File name starts with”.

6.3 Gas storage

Assets & Contracts → Gas storage

To define a gas storage:

Assets & Contracts → Gas storage □ Create new storage

The following sections describe the content of the different tabs when defining a new storage.

6.3.1 General

Storage name

The inputs for a gas storage can be stored under a unique profile name.

Storage market

Define the gas market, including the corresponding forward curve (from KyCurve).

Edit gas storage: Fast Storage

General Rates and costs Volume restrictions Injection & Withdrawal

Main storage info

Storage name

Storage physical location

Location coordinates found.



Gas market *

Market forward curve

Storage type ☒ Physical ☐ Financial

Delivery point

Buy or sell ☒ Buy ☐ Sell

Gas storage unit

Energy MJ

Start date

Start volume MWh

End date

End volume MWh

Legal

Legal entity *

Business unit

Counterparty *

Documentation

Counterparty trader

Administration

Status ☐ Confirmed ☒ Not confirmed ☐ Confirmed ☒ Not confirmed

Confirmation number

Contract date

Trader

Financial

Book

Energy Content Storage

The storage can be defined in another energy unit than the market (e.g. storage is defined in m3, while market trades in MWh). The energy content is required to convert the storage values (working gas volume, injection rates, etc.) to the market gas unit.

Dates and volumes

The start and end date have to be defined, together with the corresponding volumes.

Legal information

Information about the legal entity, who is buying/selling the storage contract, including counterparty details, can be stored.

Administration

Contract date, confirmation number and contract status (confirmed, unconfirmed) can be stored. Furthermore, the contract can be assigned to a book.

6.3.2 Rates and costs

Storage costs

The user can provide 8 different cost components, divided in normal costs and extra (e.g. interruptible/optional) costs. The logic is as follows:

- 'Firm' (first four inputs): these apply to all injections and withdrawals which are 'firmly' available.

- 'Extra' (last four inputs): these apply to all injections and withdrawals which are either 'optional' or 'interruptible'.
- 'Fixed': these are costs expressed as a fixed money amount per volume injected or withdrawn. The fixed injection and withdrawal costs must be expressed in the same unit as the gas price.
- 'Variable': these are costs expressed as a percentage of the gas price.
- The fixed firm costs are allowed to change from a certain date onwards.
- In the case, when there is potentially some additional injection and/or withdrawal capacity on top of the firm injections and withdrawals, additional inputs are required. First, the table 'Injection & Withdrawal – default' can be filled in. This extra capacity may be seen as 'interruptible' or 'optional' or both because it may have the following characteristics:
 - It may have higher costs to use. In that case the 'interruptible' cost table will contain higher values than the 'firm' cost table. This would make it an option to use.
 - It may not always be available. The user defines the probability it is available on any given day.
- The last table defines how likely it is that EXTRA rates are available and how much of it is available.

General
Rates and costs
Volume restrictions
Injection & Withdrawal

Storage costs

Valid from	Injection cost - fixed, firm	Withdrawal cost - fixed, firm	Injection cost - variable, firm	Withdrawal cost - variable, firm
2014-01-01	0.48 EUR / MWh	0 EUR / MWh	0 % of gas price	0 % of gas price

Fill in below fields when there are time varying storage costs

Valid from	Injection cost alternate	Withdrawal cost alternate	
2014-03-01	0.52 EUR / MWh		Reset

Check this box if the storage has optional or interruptible (= extra) rates ☒

Extra rates

Valid from	Injection cost - fixed, extra	Withdrawal cost - fixed, extra	Injection cost - variable, extra	Withdrawal cost - variable, extra
2014-01-01	0 EUR / MWh	0 EUR / MWh	0 % of gas price	0 % of gas price

Availability for interruptible/optional injection and withdrawal rates

Capacity	Availability extra injection rate	Availability extra withdrawal rate
100 %	100 %	100 %

There is a 100% chance the extra withdrawal capacity is available up to 100% of the defined interruptible injection rates. There is a 100% chance the extra withdrawal capacity is available up to 100% of the defined interruptible injection rates.

+ Add row

6.3.3 Volume restrictions

Volume restrictions

This tab defines the capacity (working volume) of the storage.

- If the working volume is constant over time, then only the first line is needed. The storage volume may vary between the minimum and maximum value over the complete valuation period

- If there are limitations in storage volume, they can be entered. For instance, the user can choose to have a minimum of 80% of the storage on 1 December (when the winter starts). Please remember that the day thereafter the storage volume should be put back at zero, else the storage must remain filled for 80% all days thereafter.
- If the user chooses 'yes' for linear interpolation, the volume restrictions are linearly interpolated between the different dates.
- Please make sure that the volume restrictions match with all other details of the storage or otherwise the model might fail to give a solution.

General
Rates and costs
Volume restrictions
Injection & Withdrawal

Volume restrictions

Start date	Min (MWh)	Max (MWh)
2014-01-01	0	50000

+ Add row

Linear interpolation
☐ Yes
☒ No

6.3.4 Injection & Withdrawal

Injection and withdrawal rates

- If the injection and withdrawal rate do not depend on time, inventory level, and are fully 'firm', then the user needs to provide only two inputs: (i) the daily injection rate and (ii) the daily withdrawal rate.
- To allow the injection and withdrawal rates to vary with the inventory level, some additional rows should be added. The sample inputs in the table should be interpreted as:
 - from an inventory of 0 the injection rate is 400, withdrawal rate 300
 - from an inventory of 25000 the injection rate is 400, withdrawal rate 1200
 - from an inventory of 50000 the injection rate is 240, withdrawal rate 1200
- If the user chooses 'yes' for linear interpolation, the withdrawal rate increases linearly from 300 to 1200 from the volume level of 0 to the level of 25000 MWh. Likewise, the injection rate would slowly go down between the volume levels of 25000 and 50000.
- In the table, the rates are chosen in absolute terms. Alternatively, the user can specify the rates as a percentage of the (overall) maximum working volume.
- Extra injection/withdrawal rates can be provided in the next two columns.
- The next two *Exceptions* give the possibility to handle e.g. maintenance periods. Note that such restrictions might make it difficult or even impossible to define a monthly or tradable (rolling) intrinsic value.

Injection/withdrawal rate defined as: ☐ A percentage of the maximum inventory level ☒ Absolute numbers

Default rates (daily)

Inventory level	Firm injection rate	Firm withdrawal rate	Extra injection rate	Extra withdrawal rate	
0	400	300	0	0	
25000	400	1200	0	0	x
50000	240	1200	0	0	x
+ Add row					

Linear interpolation ☒ Yes ☐ No

Exceptions (as % of default rates)

Start date	End date	Injection rate %	Withdrawal rate %	
2014-03-01	2014-03-31	80	70	x
+ Add row				

6.3.5 List of all storages

Assets & Contracts → Gas storage

Shows a list of defined gas storages. Each gas storage profile can be edited, duplicated, or deleted.

Settings Price data Time Series Curves **Assets & Contracts** Analytics Reports Logs Interfaces

Contracts **Gas storage** Power plants Inventory

Storage import settings [Create new storage](#)

Show all filters Load filter Reset selection [Save this filter](#) [Filter](#)

Commodity x

All
NBP
NCG
TTF

List [Map](#)

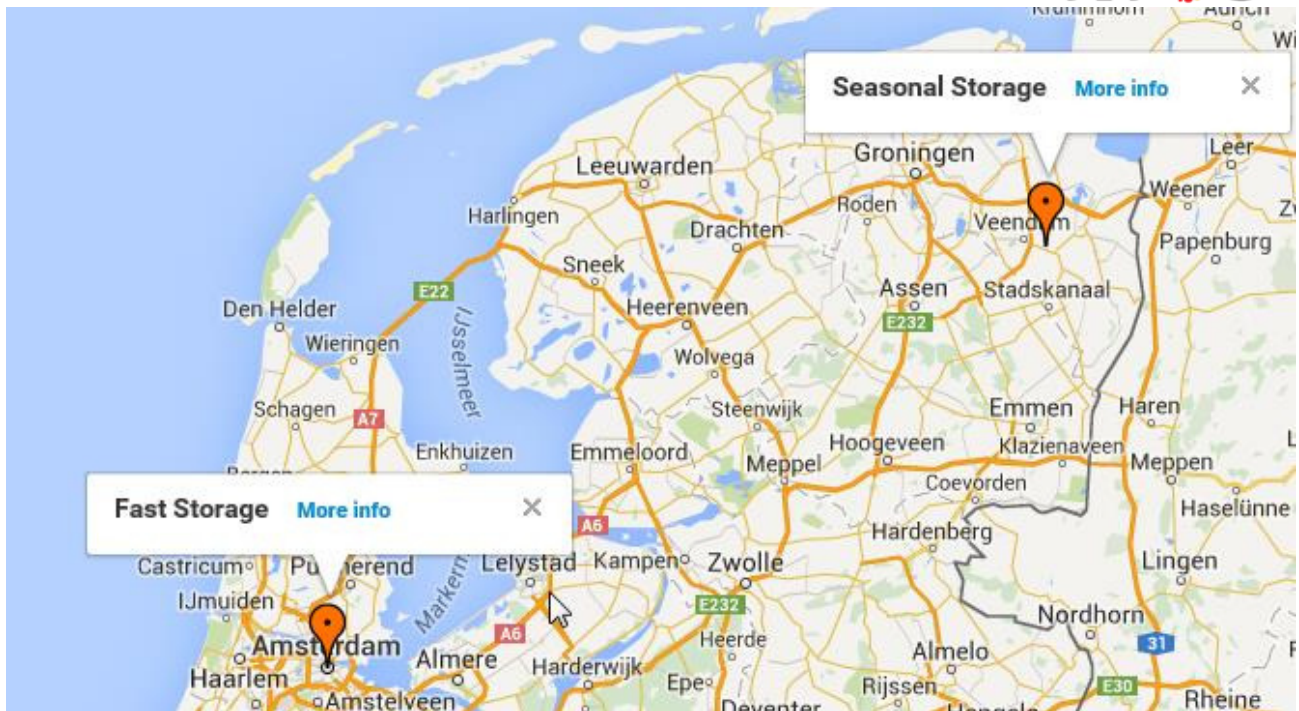
5 results found.

☐	ID	Name	Gas Market	Date created	Start date	End date	Curve profile	
☐	156	test	NCG	2014-03-27	2015-04-01	2016-04-01	NCG EEX Montel	Edit
☐	154	Fast Storage	NCG	2014-03-10	2020-04-01	2021-04-01	NCG ICE Montel	Edit
☐	173	New Volume	TTF	2014-10-08	2015-04-01	2016-05-01	TTF ENDEX Montel	Edit

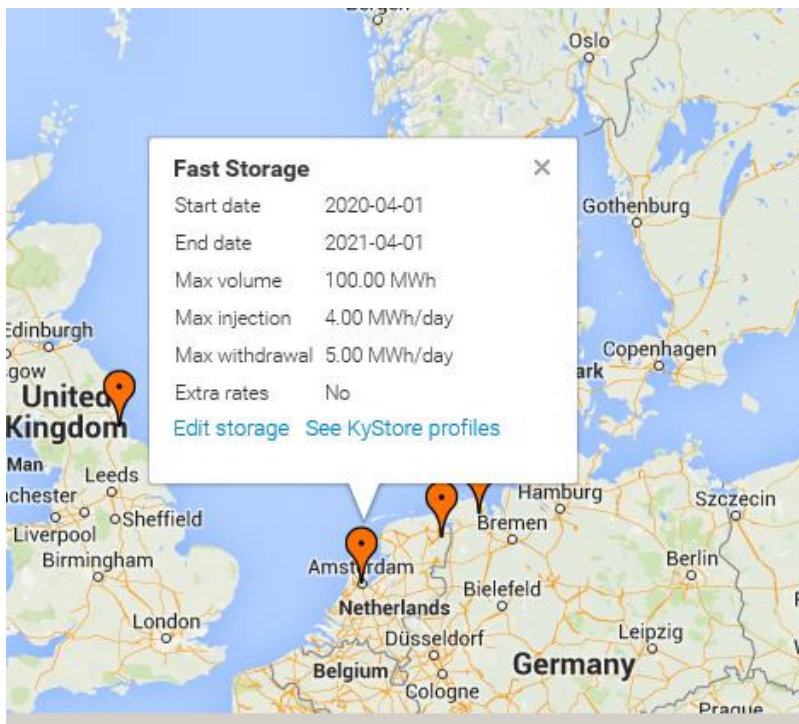
6.3.6 Map

Assets & Contracts → Gas storage [Map](#)

Shows a geographical map with the physical location of each gas storage.



The map can be used to navigate to the desired storage/storage profiles, or to see the specifications of the storage.



6.3.7 Upload storage

Assets & Contracts → Gas storage ☐ Storage import settings

[Settings](#)
[Price data](#)
[Time Series](#)
[Curves](#)
[Assets & Contracts](#)
[Analytics](#)
[Reports](#)
[Logs](#)
[Interfaces](#)

[Contracts](#)
[Gas storage](#)
[Power plants](#)
[Inventory](#)

Gas storage

Storage import settings

Create new storage

[Show all filters](#)
[Load filter](#)
[Reset selection](#)

Save this filter

Filter

By specifying a directory and the start of a filename, the user can upload a gas storage specification automatically.

[Settings](#)
[Price data](#)
[Time Series](#)
[Curves](#)
[Assets & Contracts](#)
[Analytics](#)
[Reports](#)
[Logs](#)
[Interfaces](#)

[Contracts](#)
[Gas storage](#)
[Power plants](#)
[Inventory](#)

Settings for automated import

If you want to import files automatically, fill in the fields below. The import will automatically import files in the given directory, starting with the file-pattern.

Location of files

File name starts with

If the complete filename is *Berlin_storage.xml* enter *Berlin* as start of filename.

Below is an example of the contents of a storage definition xml file. A sample xml file is also available for download in the platform.

```
<?xml version="1.0" encoding="UTF-8"?>
<KyosStorageImport>
  <Storage>
    <KeepAfterCalculationYesNo>Yes</KeepAfterCalculationYesNo>
    <KyStoreReferenceProfileID>123</KyStoreReferenceProfileID>
    <UniqueExportID>12345678</UniqueExportID>
    <TradingDate> <!-- optional -->
      <Day>1</Day>
      <Month>1</Month>
      <Year>2014</Year>
    </TradingDate>
    <ValuationDate> <!-- optional -->
      <Day>1</Day>
      <Month>1</Month>
      <Year>2014</Year>
    </ValuationDate>
    <Name>Reference storage 1</Name>
    <Location>Nieuwe Gracht, 2012 NG, Haarlem, Netherlands</Location> <!-- optional -->
    <CommodityID>123</CommodityID>
    <PhysicalFinancial>Physical</PhysicalFinancial>
    <BuySell>Buy</BuySell>
    <UnitID>12</UnitID>
    <Energy>3600</Energy>
    <ContractDate>
      <Day>1</Day>
      <Month>1</Month>
      <Year>2014</Year>
    </ContractDate>
    <StartDate>
      <Day>1</Day>
      <Month>4</Month>
      <Year>2014</Year>
    </StartDate>
    <EndDate>
      <Day>31</Day>
      <Month>12</Month>
      <Year>2015</Year>
    </EndDate>
    <StartVolume>10</StartVolume>
    <EndVolume>100</EndVolume>
    <LegalEntityID>123</LegalEntityID>
    <CounterpartyID>456</CounterpartyID>
    <ConfirmedYesNo>Yes</ConfirmedYesNo>
    <Confirmation>456-XY-789</Confirmation> <!-- optional -->
    <BookID>789</BookID> <!-- optional -->
    <Costs>
      <Firm>
        <Injection>
```



```

    <Fixed>0.25</Fixed>
    <Variable>0.02</Variable>
  </Injection>
  <Withdrawal>
    <Fixed>0.1</Fixed>
    <Variable>0.015</Variable>
  </Withdrawal>
</Firm>
<Alternates> <!-- optional -->
  <Alternate>
    <StartDate>
      <Day>18</Day>
      <Month>6</Month>
      <Year>2014</Year>
    </StartDate>
    <Injection>
      <Fixed>0.22</Fixed>
    </Injection>
    <Withdrawal>
      <Fixed>0.12</Fixed>
    </Withdrawal>
  </Alternate>
  <Alternate>
    <StartDate>
      <Day>28</Day>
      <Month>6</Month>
      <Year>2014</Year>
    </StartDate>
    <Injection>
      <Fixed>0.21</Fixed>
    </Injection>
    <Withdrawal>
      <Fixed>0.11</Fixed>
    </Withdrawal>
  </Alternate>
</Alternates>
<Extra> <!-- optional -->
  <Injection>
    <Fixed>0.20</Fixed>
    <Variable>0.01</Variable>
  </Injection>
  <Withdrawal>
    <Fixed>0.18</Fixed>
    <Variable>0.03</Variable>
  </Withdrawal>
  <Availabilities>
    <Availability>
      <Capacity>0.66</Capacity>
      <Injection>0.78</Injection>
      <Withdrawal>0.65</Withdrawal>
    </Availability>
    <Availability>
      <Capacity>0.30</Capacity>
      <Injection>0.98</Injection>
      <Withdrawal>0.85</Withdrawal>
    </Availability>
  </Availabilities>
</Extra>
</Costs>
<VolumeRestrictions>
  <VolumeRestriction>
    <StartDate>
      <Day>1</Day>
      <Month>4</Month>
      <Year>2014</Year>
    </StartDate>
    <Minimum>10</Minimum>
    <Maximum>35</Maximum>
  </VolumeRestriction>
  <VolumeRestriction> <!-- more than 1 VolumeRestriction is optional -->
    <StartDate>
      <Day>1</Day>
      <Month>9</Month>
      <Year>2014</Year>
    </StartDate>
    <Minimum>15</Minimum>
    <Maximum>50</Maximum>
  </VolumeRestriction>
  <LinearInterpolationYesNo>Yes</LinearInterpolationYesNo>
</VolumeRestrictions>
<InjectionWithDrawalRates>
  <ValuesAbsolutePercentage>Absolute</ValuesAbsolutePercentage>
  <LinearInterPolationYesNo>Yes</LinearInterPolationYesNo>
  <Rates>
    <Rate>
      <Level>0.4</Level>
      <Firm>
        <Injection>12.5</Injection>

```

```

    <Withdrawal>28</Withdrawal>
  </Firm>
  <Extra>
    <Injection>10.1</Injection>
    <Withdrawal>20.5</Withdrawal>
  </Extra>
</Rate>
<Rate> <!-- more than 1 Rate is optional -->
  <Level>0.65</Level>
  <Firm>
    <Injection>8.5</Injection>
    <Withdrawal>20</Withdrawal>
  </Firm>
  <Extra>
    <Injection>6.5</Injection>
    <Withdrawal>15</Withdrawal>
  </Extra>
</Rate>
</Rates>
<Exceptions> <!-- optional -->
  <Exception>
    <StartDate>
      <Day>1</Day>
      <Month>4</Month>
      <Year>2014</Year>
    </StartDate>
    <EndDate>
      <Day>5</Day>
      <Month>4</Month>
      <Year>2014</Year>
    </EndDate>
    <Injection>0.5</Injection>
    <Withdrawal>1.0</Withdrawal>
  </Exception>
  <Exception>
    <StartDate>
      <Day>1</Day>
      <Month>5</Month>
      <Year>2014</Year>
    </StartDate>
    <EndDate>
      <Day>9</Day>
      <Month>5</Month>
      <Year>2014</Year>
    </EndDate>
    <Injection>0.75</Injection>
    <Withdrawal>0.75</Withdrawal>
  </Exception>
</Exceptions>
</InjectionWithdrawalRates>
</Storage>
</KyosStorageImport>

```

6.3.8 Upload storage characteristics

The user can modify some storage characteristics by manually or automatically uploading xml, csv, or Excel files. This option is an alternative to defining the characteristics manually in the platform when defining a gas storage profile (described in earlier sections).

6.3.8.1 Upload volume restrictions

Assets & Contracts → Gas storage □ Upload volume restrictions

A sample csv file is presented below. Sample xls and xml files can be download from the platform.

storage_id;year;month;day;minimum;maximum

123;2012;6;1;0;1000

123;2012;7;16;100;2000

123;2015;8;1;200;2000

6.3.8.2 Upload default rates

Assets & Contracts → Gas storage □ Upload default rates

A sample csv file is presented below. Sample xls and xml files can be download from the platform.

storage_id;inventory_level;firm_injection_rate;firm_withdrawal_rate;extra_injection_rate;extra_withdrawal_rate

123;0;3000;5000;0;0

123;20;2500;4000;0;0

123;40;2000;3000;0;0

6.3.8.3 Upload exception rates

Assets & Contracts → Gas storage □ Upload exception rates

A sample csv file is presented below. Sample xls and xml files can be download from the platform.

storage_id;start_year;start_month;start_day;end_year;end_month;end_day;injection_rate;withdrawal_rate

123;2015;10;1;2015;10;15;10;15

123;2015;12;16;2015;12;31;25;5

123;2015;1;1;2015;1;10;10;20

6.3.8.4 Storage import settings

Assets & Contracts → Gas storage □ Storage import settings

The files which modify the storage characteristics can be imported automatically in the platform. The path to the files on the server should be defined. The start of the file name of the different type of characteristics should be filled in. When this information is filled in, the platform will check regularly, if there are new files in the defined directory. If they are, they will be automatically imported.

6.4 Power plants (txt upload)

Assets & Contracts → Power plants txt

There are two ways in the KYOS Platform to define the characteristics of a power plant:

1. In a pre-formatted Excel file, which is exported to txt and then uploaded into the Platform. Only a few input fields are defined in the Platform interface.
2. With all input fields defined in the Platform interface.

Depending on the user only one of the two approaches for plant definition may be available. In this section we explain the first approach, in the next section the second approach.

The general power plant data, using the txt-based approach, consists of:

Name

The inputs for a power plant can be stored under a unique profile name.

Plant physical location (Optional)

The address of the physical plant can be specified and is used to show the plant in Google Maps.

Plant settings (as TXT file)

The txt file containing the plant settings has to be uploaded.

Legal entity

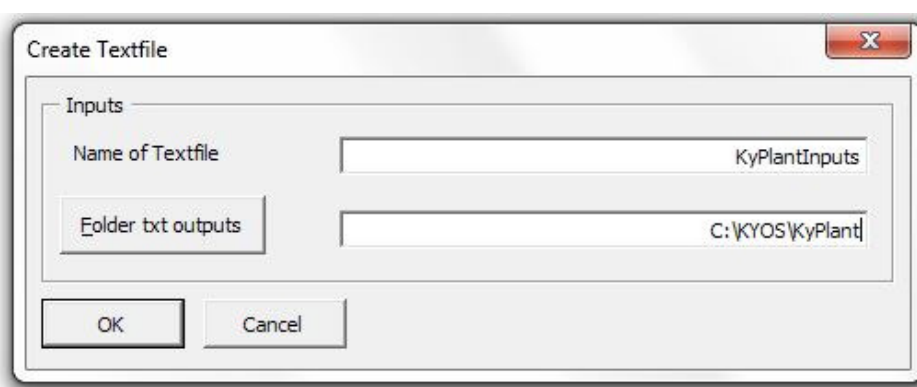
The name of the legal entity

Use this profile for

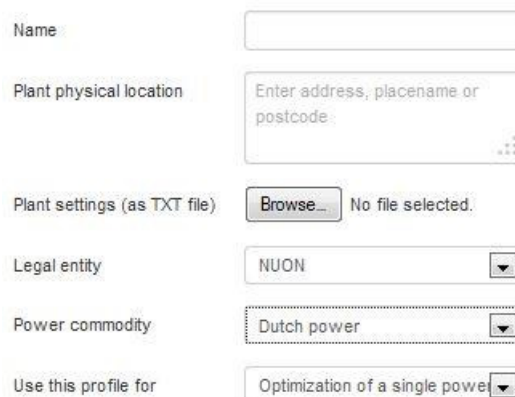
Choose between optimization of a single power station and optimization of a portfolio of power stations

The more detailed power plant settings are defined in the file “KyPlant Input SETTINGS.xls”. This input file is used for the Analytical platform as well as in the stand-alone models. To create a txt file that is uploaded to the Power Plant profile, go to sheet “Instructions”, press button “Create txt file from inputs”.

Create txt file from inputs



Create power plant profile



6.4.1 General Settings

Plant characteristics

Fuel

Fuel type of power plant (gas or coal)

Min and Max Capacity

Minimum and maximum stable generation in MWe

Minimum runtime

Minimum production time for which a power plant is ON

Min Offtime

Minimum period in which the power plant does not produce power

Use ramp speeds

Indicate if ramp speeds are to be considered in the valuation

Ramp-Up speed

Speed to ramp up between minimum stable generation to maximum capacity.

Ramp-down speed

Speed to ramp down between maximum capacity to minimum stable generation.

Additional load point

It may be allowed to reduce output to a fixed set. Indicate if such provisions are applicable or not (yes/no)

Production at additional load point

Output of the additional load point

Efficiency at additional load point

Efficiency associated with the additional load point

Minimum runtime of additional load

Minimum runtime restrictions applicable to the additional load point

Plant characteristics		
Fuel	[type]	Gas
Min Capacity	[MWe]	180.00
Max Capacity	[MWe]	435.00
Runtime		
Minimum runtime	[Hour]	1
Minimum offtime	[Hour]	1
Ramp speeds		
Use ramp speeds	[yes/no]	Yes
Ramp-Up speed	[% of max capacity / minute]	3.00%
Ramp-Down speed	[% of max capacity / minute]	3.00%
Additional load point		
Include additional load point	[yes / no]	No
Production at additional load point [MWe]		100
Efficiency at additional load point [%]		30.00%
Minimum runtime of additional load [Hour]		1

Efficiency

Efficiency curve

Indicate type of curve information to be considered in the optimization. Here one may indicate to use the full curve based on a regression, or only (a subset of) the indicated set points. (Complete curve, Min/Max Capacity Only, Max Capacity Only).

Efficiency definition

Defines if the efficiency is applied to LHV or HHV level of gas.

Min and max efficiency

Power plant efficiency at minimum stable generation and Power plant efficiency at maximum capacity.

Mid capacity and mid efficiency

Intermediate set point of plant output between min and max capacity and power plant efficiency at this mid capacity.

Seasonal variation

Both efficiency and output of a power plant may show seasonal variations as it not only depends on the technical characteristics of the facility but also on ambient temperatures. Monthly average corrections for such variations may be accounted for through use of the entries in frame 'Seasonal Variation' on the tab 'General'. Corrections for seasonal variations in efficiency are directly applied to the parameterization of the efficiency curve, while corrections for output are applied to minimum and maximum output.

Efficiency		
Efficiency Curve	[choice]	Complete_Curve (default)
Efficiency definition	[LHV / HHV]	LHV
Min Efficiency	[%]	52.00%
Max Efficiency	[%]	59.00%
Mid Capacity	[MWe]	300.00
Mid Efficiency	[%]	57.00%
Regression on efficiency correct		
41%	180.0 MW	52.00%
69%	300.0 MW	57.00%
100%	435.0 MW	59.00%

Seasonal variation				
	Output	Result	Efficiency	Result
January	100%	435 MW	100%	59.00%
February	100%	435 MW	100%	59.00%
March	100%	435 MW	100%	59.00%
April	100%	435 MW	100%	59.00%
May	100%	435 MW	100%	59.00%
June	100%	435 MW	100%	59.00%
July	100%	435 MW	100%	59.00%
August	100%	435 MW	100%	59.00%
September	100%	435 MW	100%	59.00%
October	100%	435 MW	100%	59.00%
November	100%	435 MW	100%	59.00%
December	100%	435 MW	100%	59.00%

Additional settings

Avoided Grid Cost

The subsidy for avoided grid costs is translated as an additional income defined in EUR/MWh. Mathematically, it is additive to the power price.

Avoided grid costs		
Include avoided grid costs	[yes/no]	Yes
Avoided grid costs	[EUR/MWh]	5

Gas Booking Costs

For each individual gas day, when power plant consumes gas for 1 or more hours, i.e. when it runs, starts or is shutting down, a capacity payment has to be paid.

This payment is subtracted from the margin on the first hour of a gas day during which gas is consumed.

Gas Discount		
Discount	[yes/no]	Yes
Available volume of gas (HHV)	[MWh/year]	10,000
Taken volume in first year (HHV)	[MWh/year]	1,200
Gas price discount	[EUR/MWh]	2

Nomination lead time

Nomination Lead Time		
Nomination lead time	[yes/no]	No
Number of days	[#]	5

It may be the case, for example for contractual reasons, that a lead time t_n may be associated with nomination of production. Method that incorporates nomination lead time into the model was developed – partially in KyPlant and partially in KySim models.

6.4.2 Shut Down and Start Curves

Start-ups and shut downs have specific costs associated with them and impose a limitation in the form of start-up/shut down time and a start-up/shut down output profile. The costs associated with a start-up, the start-up costs, are largely determined by the temperature of the boiler or turbine at start-up and thus the number of hours passed since the last switch off. These costs consist of fuel (gas) costs, carbon emissions and fixed costs. Start-up and shut down costs are incorporated in the optimal dispatch schedule. These costs directly affect the decisions (not) to start, and (not) to shut down.

Shut down

Shut down may be restrictive in terms of both the shutdown cost, shut down time and the shutdown profile. In this case only one profile has been assumed to apply. The following inputs are required:

- Specification of fuel type and energy content
- Specification if a calculated curve or a user-specified curve is to be used
- If a user-specified curve is to be used, a specification of average and final

Shut down curve					
Applied	[choice]	Yes			
Fuel unit	[bt]	GJ			
Energy	[MJ]	1,000.00			
LHV/HHV	[choice]	LHV			
Shut down curve	[choice]	User_Input			
	Power Avg MWh	Power MW end	Power Avg MWh	Power MW end	Gas GJ [LHV]
	Calculated		User input		
Startlevel	0	180	180		
Hour 1	21	0	145	0	1,260
Hour 2	0	0			
Hour 3	0	0			

production and fuel consumption during shut down for each time unit of the shut down profile

Start Curves

Up to twenty different start curves may be specified within KyPlant. These curves should range from hot start curves to cold start curves. For each of these curves several inputs are required:

- Definition of each start type in terms of the number of hours after shut down
- The number of Equivalent Operating Hours (EOH) for each start. Note that the EOH may determine at what dates maintenance takes place.
- The fixed costs per start. These costs may vary per year.
- A specification of average and final power production as well as consumption of up to 3 start fuels during start up for each time unit of the start-up profile

Start curves														
Used number of start curves		[#]	2											
Curve [#]	Used [#]	From Hour [#]	To Hour [#]	EOH [#]				Hour 1	Hour 2					
1	TRUE	1	7	1	Power	Avg MWh		90						
					Power	MW end		180						
					Start Commodity 1 (TTF)	MWh		200.0						
					Start Commodity 2 (API2)	MT		0						
					Start Commodity 3 (API2)	MT		0						
					CO2	ton		18.09						
					Fixed Cost	year		2014	2015					
					Fixed Cost	[EUR]		5,000	5,000					
					2	TRUE	8	>8	3	Power	Avg MWh		45	135
										Power	MW end		90	180
Start Commodity 1 (TTF)	MWh		100.0	300.0										
Start Commodity 2 (API2)	MT		0	0										
Start Commodity 3 (API2)	MT		0	0										
CO2	ton		9.04	27.13										
Fixed Cost	year		2014	2015										
Fixed Cost	[EUR]		5,000	5,000										

6.4.3 Technical settings

Outages

KyPlant can simulate plant outages. Forced outages are user defined inputs, which can vary per calendar year. The input is a percentage of the total number of hours in a year. These are entered on a year-by-year basis in the frame Forced Outages in the tab Technical. Note that since the number of hours per year differs, so does the number of outage hours. Outages are split into scheduled and unscheduled outages. Scheduled outages should not be confused with maintenance, as these outages are foreseen on a short notice only (e.g. there is small problem, and the plant operator asks if he can shut down the power station in the weekend or night to repair). This short notice period typically allows replacement of positions which may avoid imbalance costs and a large increase in operating expenses (included the Equivalent Operating Hours, EOH). An unscheduled shutdown on the other hand, has a large negative impact on the technical installation, and therefore EOH will need to be attributed to a trip. Also, alternative sourcing depends highly on the market liquidity so that imbalance costs may be imposed.

Required inputs of an outage are:

- Yearly outage time as percentage of yearly number of hours
- Distribution outage time between scheduled and unscheduled
- Mean time to repair
- EOH

Plant trips				
Include trips	[yes/no]		No	
	Distribute	Mean time to repair (hours)	EOH per outage	Starts Equivalent
Trip type 1	60.0%	36	10	5
Trip type 2	40.0%	50	20	10

Trip probability					
Year	Trips % of all hours	Trip type 1		Trip type 2	
		Hrs / year	Nr / year	Hrs / year	Nr / year
2013	5.0%	263	8	176	4
2014	5.0%	263	8	176	4
2015	5.0%	263	8	176	4

Maintenance

During maintenance, the capacity of the power plant is set at zero. Maintenance scheduling in KyPlant is implemented in two variants, a fixed schedule and a dynamic scheduling procedure. Such a dynamic schedule is based on equivalent operating hours (EOH).

A fixed maintenance schedule ensures that the power plant is not producing in a pre-determined period.

An alternative to a fixed maintenance schedule is to plan maintenance based on equivalent operating hours. This means that in scenarios/periods with high spreads and more production, more maintenance can be expected. In scenarios with lower spreads, less maintenance will take place. Input for this approach is an EOH-based maintenance schedule. This schedule determines when maintenance takes place, and for how long (A, B or C inspection). Such a dynamic schedule may lead to maintenance in unfavourable periods, i.e. periods in which it is profitable to produce. Therefore, KyPlant incorporates a provision that allows maintenance to be shifted from a high margin period to a low margin period. For example, if maintenance should take place in January, it is possible to transfer the start of the maintenance period to March.

6.4.4 Heat

KyPlant can take into account heat obligations. Two modes of operation may be selected for heat production:

- heat production is taken as the driver for plant operation, while the boilers are used for back-up purposes
- optimal (least-cost) dispatch of the joint boiler-plant system

The specification of the installation's PQ diagram can be specified. In addition, heat transport limitations may be imposed to be specified in this frame as well. Such system limitations impose a maximum on the heat production.

For each of the boilers specify the maximum heat capacity and LHV efficiency.

Heat supply	
☐ No heat	
☒ Heat (optimize between plant and boiler)	
Selection	2

Heat Power Plant					
	Power [MWe]	Heat [MWth]	Gas [MWh]	Eff Electric [%]	Eff Total [%]
Plant characteristics					
Max Load combined with max Heat	390	115	737	59.0%	68.5%
Min Load combined with max Heat	205	75	346	52.0%	80.9%
System characteristics					
Max Heat (system limitation)	[MWth]	115			
Heat Income for Producing Heat by Power Plant					
Heat Income	[EUR/MWth]	7.50			

Boilers				
Include gas costs of boilers?	[yes/no]	Yes		
		Boiler 1	Boiler 2	Boiler 3
Use Boiler	[yes/no]	Yes	No	Yes
Max heat capacity	[MWth]	20	50	1,000
Efficiency Boiler [LHV]	[%]	90%	95%	85%

6.5 Power plants (using Platform interface)

Assets & Contracts → Power plants

There are two ways in the KYOS Platform to define the characteristics of a power plant:

1. In a pre-formatted Excel file, which is exported to txt and then uploaded into the Platform. Only a few input fields are defined in the Platform interface.
2. With all input fields defined in the Platform interface.

Depending on the user only one of the two approaches for plant definition may be available. In this section we explain the second approach, in the previous section the first approach.

General characteristics

Name

The inputs for a power plant can be stored under a unique profile name.

Plant physical location (Optional)

The address of the physical plant can be specified and is used to show the plant in Google Maps.

Legal entity

The name of the legal entity.

Start date / end date

If the plant is used in KyPlant, then the valuation can only be in between the start and end date.

Granularity

This defines the level of granularity for defining the power plant characteristics. It will determine whether KyPlant is based on an hourly or less than hourly granularity. At the moment, only hourly and half-hourly are supported, not quarter-hourly.

Country / Region

Optional inputs. Not actively used.

Commodity inputs

General characteristics	Commodity inputs	Technical characteristics	Seasonality
Power commodity	German Power		
Fuel type	Gas	Fuel commodity multiplier	1
Fuel commodity	NCG		
CO2 commodity	EUA Carbon futures		
Start commodity 1 (optional)	Please select...		

Fuel commodity multiplier

This input is not used in KyPlant. Best to keep the default value of 1.

Technical characteristics

Production capacities (MWe)

Minimum stable level (SEL) MWe

Maximum generation level (MEL) MWe

Midpoint generation level MWe

Efficiency of the plant

For gas this is LHV.

Efficiency curve

Minimum stable level (SEL) %

Maximum generation level (MEL) %

Midpoint generation level %

Other technical plant parameters

Minimum ontime Hour

Minimum offtime Hour

Use ramp speeds ☒ Yes ☐ No

Ramp-Up speed % of max capacity / minute

Ramp-Down speed % of max capacity / minute

Variable operational costs EUR/MWh

Variable operational costs EUR/OH

Fixed operational costs EUR/kW/Year

Fixed start costs EUR

Availability %

Must Run ☐ Yes ☒ No

Is CHP ☐ Yes ☒ No

Minimum and maximum production capacity

This determines the electrical output of the plant at minimum stable generation and at maximum generation level.

Efficiency curve

Indicate type of curve information to be considered in the optimization. Here one may indicate to use the full curve based on a quadratic regression, or only (a subset of) the indicated set points. (Complete curve, Min/Max Capacity Only, Max Capacity Only). Note that all efficiencies are in LHV.

Min and max efficiency

Power plant efficiency at minimum stable generation and Power plant efficiency at maximum capacity.

Mid capacity and mid efficiency

Intermediate point of plant output between min and max capacity and power plant efficiency at this mid capacity. This is used to fit a quadratic curve through the minimum, maximum and mid load efficiencies / capacity points.

Minimum ontime

Minimum production time for which a power plant has to be ON, i.e. producing power. This is measured in the granularity of the power station, e.g. hourly or half-hourly.

Minimum offtime

Minimum time between two production phases.

Use ramp speeds

Indicate if ramp speeds are to be considered in the valuation (as a post-process).

Ramp-Up speed

Speed to ramp up between minimum stable generation to maximum capacity.

Ramp-down speed

Speed to ramp down between maximum capacity to minimum stable generation.

Variable operational costs

Can be used for all costs which vary with the production output (in MWh) or production hours (in hours).

Fixed operational costs

Can be used for all costs which are fixed over time. These costs do not affect the optimal dispatch, are only deducted at the end of a KyPlant valuation.

Fixed start costs

This version of plant definition assumes there is just one start type, i.e. without distinguishing between warm, hot and cold starts. Furthermore, a start is simply characterized by a fixed cost per start.

Must Run / Is CHP

Inputs are not yet actively used, but already defined for later use.

Seasonality

Both efficiency and output of a power plant may show seasonal variations, especially due to variations in ambient temperatures. Monthly average corrections for such variations can be defined by the user. Corrections for seasonal variations in efficiency are directly applied to the parameterization of the efficiency curve, while corrections for output are applied to minimum and maximum output.

6.6 Inventory

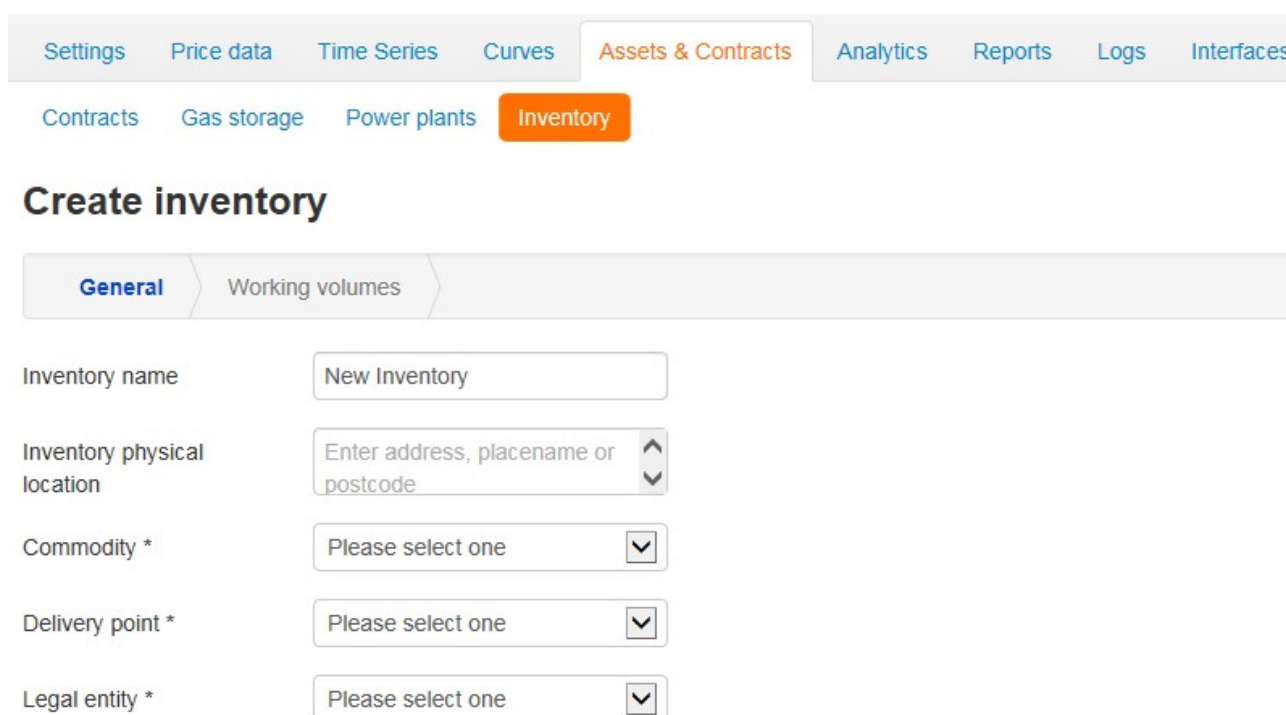
The Inventory refers to a gas storage facility that needs to be defined by the user prior to using the KyInventory model. The Inventory definition involves the specification of a location, commodity, delivery point and legal entity. Further, the volume capacities of the various tranches in the gas

6.6.1 Create KyInventory profile – General Settings

This subsection provides a practical demonstration to create a new Inventory. The profile can be created on the KYOS analytical platform at:

Assets & Contracts → Inventory □ Create profile

Refer to the Figure below for a snapshot from the KYOS analytical platform and a description of the various entries.



Inventory Name

Enter a unique profile name.

Inventory Physical Location

Enter an address, place name or a postal code.

Commodity

Select the commodity from the drop down menu list.

Delivery Point

Select the delivery point from the drop down menu list. Delivery points can be created on the platform at:

Settings → Reference Settings → Delivery Points ☐ Create Delivery Point

Legal Entity

Select the legal entity from the drop down menu list. Legal entity can be created on the platform at:

Settings → Reference Settings → Legal Entities ☐ Create New Legal Entity

6.6.2 KyInventory profile – Tranche Volumes

The user has to mention the minimum and maximum volume capacities of the various tranches in the storage facility. This is part of the Inventory creation process. Refer to the Figure below for a snapshot from the KYOS analytical platform.

[Contracts](#)
[Gas storage](#)
[Power plants](#)
[Inventory](#)

Edit Inventory : Aneesh
[Back to overview](#)

[General](#)
[Working volumes](#)

Working volumes

Start date	Tranche 1 (min)	Tranche 1 (max)	Tranche 2 (min opt)	Tranche 2 (max opt)	Tranche 3 (min opt)	Tranche 3 (max opt)
2015-01-01	0	200	0	10000	0	10000

[+ Add row](#)

[Back](#)
[Save](#)

The user needs to specify a date and enter the tranche capacities for that particular inventory. If, at a future date, the user wishes to modify the tranche capacities, that is possible by adding a new row of volumes by selecting 'Add Row'.

The three tranches refer to the so-called PAD, Back-Up and As-Available tranches and the filled volume capacities will be made use in the KyInventory model.

7 KyCalibration

Analytics → KyCalibration

7.1 Calibration of statistical parameters

The KyCalibration model has been developed to support the KyStore (gas storage valuation) and KySwing (swing contract valuation) models. Based on historical price data, KyCalibration calibrates market parameters, which are later used to generate the internal Monte Carlo price simulations within the KyStore and KySwing models. The assumption when generating the internal KyStore/KySwing simulations is that gas forward prices are driven by a 3-factor process. The commodity prices and FX rate are driven by 1 factor following a Geometric Brownian Motion (GBM) process. The gas spot prices are mean-reverting to the gas forward price. In addition, the returns of all commodities can be correlated, and the prices cointegrated. Some characteristics of the simulation engine in KySwing and KyStore:

- The price model is defined for either:
 - Two gas markets, where the second gas market is simulated on forward basis only
 - Or one gas market plus up to 4 commodities and possibly 1 FX rate
- It is assumed that the spot volatility also affects the nearby contracts and hence also the month-ahead contract.
- All residual returns are assumed to come from a normal distribution. No GARCH, regime-switches or jumps are included.

Using the KyCalibration model, the user can estimate the parameters that drive:

- The spot gas price
- The forward gas prices, forward commodity prices (e.g. oil), and 1 FX rate.

More specifically, KyCalibration estimates:

- Volatilities
- Spot mean-reversion rate
- Correlations
- Cointegration parameters

7.2 Create calibration profile

Analytics → KyCalibration □ Create profile

TabCommodity selection

Commodity selection
Underlying commodities
Cointegration

Create calibration profile
Back to overview

Name

Gas commodity

☒ Calibration by KyCalibration
☐ Enter manual calibration values

Choose the historical period to calculate

Volatilities and correlations:
 months

Apply Exponential Weighting
☐

Mean reversion and cointegration:
 months

Apply Exponential Weighting
☐

Price source forward

Price source spot

Number of gas markets included

By default, choose 1 gas market. Only choose 2 gas markets if this profile is for pricing contracts with KySwing AND the gas market for valuation is different from the gas market in the gas-indexed contract price.

☒ 1 gas market
☐ 2 gas markets

Next

Name

The inputs for a KyCalibration estimation can be stored under a unique profile name.

Gas commodity

Select the primary gas commodity from the drop down menu list. For a storage valuation, this is the gas market on which the storage operates. For a swing contract, this is the contract gas.

Calibration by KyCalibration OR Enter manual calibration values

The default option is using the KyCalibration model. The second choice gives the option for the user to create a profile with manually edited calibration values. Such a profile does not use historical data to estimate market parameters. Instead, the user defines them according to his/her views on the market.

Choose the historical period to calculate

Based on historical price data, volatilities and correlations and, mean reversion and cointegration are calculated. The user has to provide the time window over which estimations are performed. The default time horizon for volatilities is set to 6 months whereas for mean reversion is 24 months. It is advisable to use a longer time horizon to calculate cointegration and mean reversion. The reason is that results might be unstable, when using a short time horizon. Furthermore, the user should take into account the

length of the delivery period of the storage / swing contract when deciding how long the historical window should be.

Apply Exponential Weighting

If the option is not selected, then all parameters (volatilities, correlations, etc.) are calculated, using the standard equally weighted method. Under this method, all returns in the historical period are equally weighted. In this case, a return observed a year ago has the same weight as a return observed yesterday, when calculating, for example, the spot volatility.

If the option is selected, then the model uses the Exponentially Weighted Moving Average (EWMA) method to calculate the market parameters. The user has to specify the decay factor λ for volatilities and correlations and for the mean-reversion and cointegration. In this case, a historical return closer to the current moment will have a higher weight. Thus, the historical returns will be multiplied with the following weight, when calculating volatilities:

$$(1 - \lambda)\lambda^{t-1}$$

Where t denotes how many days in the past the return is, e.g. $t = 1$ means the most recently observed return.

The EWMA methodology is commonly used to calculate volatilities and correlations. Compared to the equally weighted method, it reflects faster changes in the market. This happens because the most recent returns have higher weights. Another feature is that there might be unexplained jumps in the volatility calculated by equally weighted method. This will happen when there was a relatively large return at the beginning of the historical period (e.g. a year ago) which is dropped as the historical window moved forward. This behavior is mitigated if the EWMA methodology is used.

Price source forward

Select the forward price source (for the chosen commodity) from the drop down menu list.

Price source spot

Select the spot price source (for the chosen commodity) from the drop down menu list.

Number of gas markets included

1 or 2 gas markets can be chosen. 2 gas markets should be chosen, only if the profile is for pricing a swing contracts with KySwing, and the contract gas is different than the gas, used in the indexation.

Tab Underlying commodities

Commodity selection
Underlying commodities
Cointegration

Create calibration profile

Back to overview

Select 0 to 4 commodities

Select commodity

Select commodity

Select commodity

Select commodity

Select a currency (optional)

Select currency

Back

Next

In this tab the user can choose up to four commodities, for which historical data is used in the calibration. This is relevant when pricing a swing contract and the indexation terms include non-gas commodities (oil, gas-oil, etc.).

Tab Cointegration

Commodity selection
Underlying commodities
Cointegration

Create calibration profile

Back to overview

Cointegration constant

Include cointegration constant ☒ Yes ☐ No

Define cointegration relations

	Gasoil	API2 Coal ARA
NCG	☒	☐
Gasoil		☐
API2 Coal ARA	☐	

Back

Next

The user can define if cointegration parameters need to be estimated. In the example, the user has chosen to cointegrate the gas market (NCG) with Gasoil. Rules related to cointegration:

- A commodity can never be cointegrated with itself
- There can be a maximum of 1 cointegration relation per commodity
- There has to be at least 1 commodity which is not cointegrated with another commodity

7.3 Calibration results

Analytics → KyCalibration ☐ View

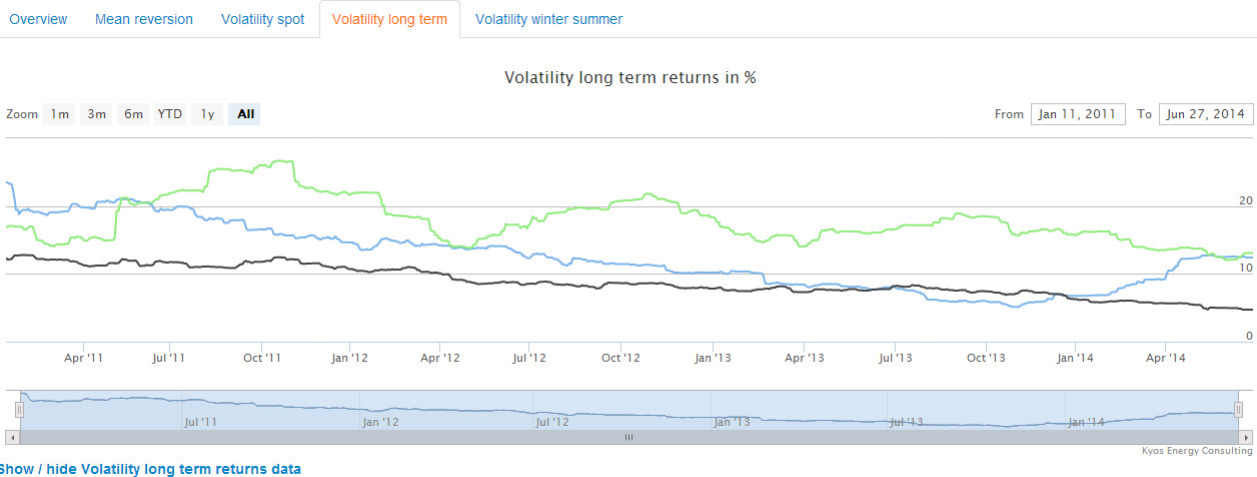
The calibration results can be presented by clicking on view in the main calibration menu.

[Overview](#)
[Mean reversion](#)
[Volatility spot](#)
[Volatility long term](#)
[Volatility winter summer](#)
[Monthly volatility](#)

In the overview menu, volatilities, correlations and cointegration for the most recent trading date are provided.

The evolution of the estimated mean-reversion, spot volatility, long-term volatility and winter-summervolatility of the primary gas market are presented in the other tabs.

The long-term volatilities are also calculated and presented for all other commodities included in the calibration.



Overriding market parameters

Overview
Mean reversion
Volatility spot
Volatility long term
Volatility winter summer
Monthly volatility

Display calibration result for date:

☒ Use manual override values instead of calculated volatilities

Gas Market Volatilities

	Calculated value	Override
Mean reversion	32.386 %	32.386 %
Volatility spot	58.346 %	58.346 %
Volatility long term	27.63 %	27.63 %
Volatility winter summer	10.042 %	10.042 %

Other longterm volatilities

	Calculated value	Override
Volatility long term TTF	28.137 %	28.137 %

Correlation matrix

	NCG	Override	TTF	Override
NCG	100.00 %	100.00 %	99.56 %	99.56 %

When viewing the calibrated market parameters on a given trading date, the user has the option to manually override some or all of the estimated parameters. First, the “Use manual override values instead of calculated volatilities” box should be checked. Then, the user can adapt the desired parameters. After pressing the “Save override settings” button, the adapted parameters will be saved, and used in the KyStore and KySwing models connected to this KyCalibration profile.

8 KyStore - Gas storage valuation

Analytics → KyStore

8.1 Gas storage valuation, optimization and hedging

KyStore is developed to value gas storage facilities and gas storage contracts. It identifies storage value, optimal trading decisions, exposures and hedging strategies. It is based on Monte Carlo simulation and uses mixed-integer linear programming, dynamic programming and least-squares Monte Carlo techniques. It quickly calculates intrinsic, rolling intrinsic and full option value in forward and spot markets. The storage methodology has been developed in-house, and refined on the basis of customer feedbacks. The core methodology has been published in the Journal of Derivatives (Boogert and De Jong, spring 2008).

KyStore accounts for typical storage constraints:

- Time-dependent and inventory-dependent injection and withdrawal rates;
- Time-dependent minimum and maximum inventory levels;
- Incorporation of operational and trading costs, including bid-ask spreads;
- Optional and/or interruptible injection and withdrawal;
- Maintenance.

KyStore is based on Monte Carlo simulation techniques. Important characteristics are:

- Mean-reverting multi-factor model with long-term, short-term and seasonal dynamics;
- Least Squares Monte Carlo methodology to find optimal trading and operating decisions;
- Prices matched to user-provided forward curve with up to daily precision;
- Volatility term structure and other simulation inputs can be derived from historical data;
- The user has the choice to use external simulations, coming from KySim.

KyStore calculates storage value based on different trading strategies:

- Intrinsic: maximum value that may be locked in on forward and spot of today;
- Rolling Intrinsic: nearby forward positions are adjusted over time to enhance intrinsic value;
- Full Option Value: storage decisions are optimized based on spot market dynamics.

The model supports day-to-day trading decisions:

- Advise on optimal trading strategies;
- Calculate delta positions;
- Provide distribution of cash-flows and inventory;
- Track individual simulation results over time of prices, volumes, injections and withdrawals.

8.2 Defining a KyStore valuation profile

Analytics → KyStore ☐ Create profile

A KyStore valuation profile is the combination of a gas storage and some valuation settings. To define a gas storage valuation profile, you can either

- Go to Analytics → KyStore. Press the green button “Create profile” and choose a storage from the list of storages provided.
- In Assets & Contracts → Gas storage, select a specific storage, click on the drop-down list, and select “Create valuation profile”. The screen to define a new gas profile appears.

ID	Name	Gas Market	Date created	Start date	End date	Curve profile	
103	Gaspool EEX Storage TEST	Gaspool EEX Storage TEST	2015-11-05	2015-09-01	2016-11-24	Gaspool EEX Storage TEST	Edit

20 per page

Delete selection Save selection as filter

- Create valuation profile
- Duplicate
- Upload volume restrictions
- Upload default rates
- Upload exception rates
- Delete

8.2.1 Type of valuation

Create KyStore profile

[Back to overview](#)

Type of valuation

Rolling intrinsic

Calculations

Automation

Basic information Gaspool EEX Storage TEST

Entity	Example Europe LE	Buy/sell	buy	Start volume	0
Counterparty	EON	Commodity	Gaspool	End volume	0
Contract period	01-09-2015 — 24-11-2016	Physical/financial	physical	Gas storage unit	MWh

Profile name

Select the market parameters ☒ User defined

Market curve

Market Parameters (Gas)

Which forward curve? ☒ Daily
☐ Monthly
☐ Mixed (BOM = daily, rest = monthly)

Spot volatility (factor 1) %

Mean reversion rate %

Discount rate %

Volatility long-term returns (factor 2) %

Type of simulations ☒ Internal (generated by KyStore)
☐ External (generated earlier by KySim)

Volatility winter-summer returns (factor 3) %

Traded positions

At the top, the user sees a short description of the gas storage specification. After that, some general KyStore valuation options should be chosen:

Which forward curve

This choice determines which forward curve should be sent as an input to the KyStore model. The KyCurve curve building model produces 3 different forward curves for gas commodities:

- Monthly curve = a forward curve with a different price for each month. However, each day within a particular month has the same price.
- Daily curve = a forward curve with a potentially different price for each future day. The average daily price in a month equals the monthly price in the previous curve type. For example, the daily curve is smoothed, so prices end of March are lower than the beginning of March. The daily curve may also contain a (small) weekend-workday spread.
- Mixed curve = a curve which has a monthly granularity from the front-month onwards, but daily until then. For example, on 16 February, there is a daily price from 17 Feb to 28 Feb, then a price for March, April and so on.

The choice of a forward curve can affect the calculated storage value, following the different trading strategies. For example, there is a storage which delivers in a single month (September). If the daily forward curve implies a lower price at the beginning of September, than at the end of September, it might be profitable to inject at the beginning of the month and to withdraw at the end, following the intrinsic strategy (all weeks can be traded). However, if the monthly curve is used, all days within September have the same price, so no intrinsic trade would be suggested by the model.

Discount Rate

The discount rate is applied in all calculations to discount back future cash-flows to the current (valuation) date. For example, if you receive cash-flows of 1000 GBP (from gas withdrawals) 1 year from now, and the discount rate is 5%, this is included in the valuation as $1000 * \exp(-5\%) = 951$ GBP. Note that the discount rate is interpreted as a continuous rate, which leads to the use of the exponential function.

Type of simulations

The user can choose whether KyStore should generate simulations internally or use external simulations from KySim.

- If *external* is chosen, the user can select which KySim profile's simulations to use. The simulations are directly used, so the user does not have to define any market parameters. The characteristics of the simulations can be defined in the KySim profile instead. Please refer to the documentation of KySim for more details.
- Through using the *external* option, the user can use his/her own simulations. This is done through creating a KySim profile, which uses user uploaded simulations. Please refer to the section on KySim, explaining how this can be done.
- If *internal* is chosen, the user should choose what market parameters to use. Prices are simulated following a 3 factor model. Inputs for the three factors are spot volatility (factor 1), long-term volatility (factor 2) and winter-summer volatility (factor 3). Furthermore, an input is required for the mean reversion rate. The mean reversion rate determines how quickly spot

prices revert back to the forward price level. It is often around 10% per day for natural gas markets (usually between 5 and 30%). Roughly speaking, this means that the expected spot price will move 10% closer to the (front-month) forward price each day. For example, if the spot price today is 40 and the forward price 30, the expected spot price for tomorrow is 39, excluding any stochastic component from the simulation process.

Inputs for market parameters can be provided:

- Manually, as hard coded numbers. In this case, the user determines directly the parameter values. The KyCalibration section in the platform documentation explains the meaning of the four market parameters.
- Automatically, using a calibration profile (KyCalibration). When a calibration profile is selected, the market parameters (volatility, mean reversion rate, etc.) on the specific valuation date are taken from this profile. In a calibration profile the parameters are estimated based on historical market price data (though they may be fixed, using the overwrite function). Please refer to the documentation of KyCalibration for more details.
- Both of the above options result in a single number for the spot volatility. However, the user has the flexibility to apply a different spot volatility per each month. This can be applied through defining monthly weights, which are used to rescale the spot volatility. The weights don't have to sum to 100 %: for each month in the price simulations, the weights are multiplied with the spot volatility, which has either been set by the user or calibrated on historical data. If the weights do not sum to 100%, then the model automatically rescales them to an average of 100%. For example, suppose that the user defines a weight of 140% to the first 3 months in the year (Jan-Mar), and 100% to the others, and then the actual weights in the calculations are 130% for Jan-Mar and 90% for Apr-Dec. If the annual spot volatility is 50%, the model uses $130\% \times 50\% = 65\%$ for Jan-Mar, and $90\% \times 50\% = 45\%$ for Apr-Dec.

Monthly weights applied to the spot volatility

Use numbers larger than 0%. The model will automatically adjust the weights, so that the average is 100%. The resulting weights are then multiplied with the spot volatility (user input or calibrated).

Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
100 %	100 %	100 %	100 %	100 %	100 %	100 %	100 %	100 %	100 %	100 %	100 %

[Next](#)

Traded positions

In case that some products have already been traded in order to hedge the gas storage, their volumes can be inputted through creating a *Traded positions* time-series:

Time Series → Production, Demand & Weather ☐ Create time series

Once the profile is created, the daily gas volumes can be uploaded. The volumes of all traded products should be mapped by the user to daily gas volumes and uploaded to a time-series. These volumes should be updated each time a product is traded on the market.

For example, the model has estimated that there should be an intrinsic hedge of going short 6720 MWh in February'2017. Without any traded positions input, the model will show in its outputs the 6720MWh hedge. However, assume that one trading lot of 5MW has already been sold in the past. In this case, the user should upload in the traded positions time-series, the traded volume each day in February'17. This is $24h * 5 MW = -120 MWh$. The total is -3360MWh over the whole month. Using this traded position, the model will output as an intrinsic hedge in February'17 only -3360MWh, i.e. $(-6720 MWh - (-3360MWh))$. The trading costs when estimating the intrinsic/rolling intrinsic values will also be lower in this case. Avoiding this trading costs will result in a little intrinsic/rolling intrinsic values, given everything else is the same.

If there are already some traded positions time-series for the given gas storage, then the platform will automatically show them in a drop-down menu. Please refer to the section on *Traded positions* in the platform documentation for more information.

8.2.2 Rolling Intrinsic

Create KyStore profile

[Back](#)

Type of valuation **Rolling intrinsic** Calculations Automation

Basic information Gaspool EEX Storage TEST

Entity	Example Europe LE	Buy/sell	buy	Start volume	0
Counterparty	EON	Commodity	Gaspool	End volume	0
Contract period	01-09-2015 — 24-11-2016	Physical/financial	physical	Gas storage unit	MWh

Calculate rolling intrinsic ☒ Yes ☐ No

Hedge type for spot strategy ☐ Static monthly delta hedge at valuation date (default) ☒ Dynamic monthly delta hedge ☐ Dynamic intrinsic hedge (longer calculation time)

Which rolling intrinsic ☒ Tradable (as defined in the market settings of the commodity) ☐ Monthly (each month is a tradable contract) ☐ Daily (each day is a tradable contract)

Re-hedging frequency for rolling intrinsic ☒ Monthly ☐ Weekly ☐ Daily

Minimum re-hedge profit per re-hedge

A *rolling intrinsic calculation* can be performed, where a rolling forward hedging strategy is applied. When this setting is chosen, the user should select how many forward simulations to be used; what products should be used as a hedge (tradable, monthly, daily); how often re-hedging takes place (monthly, weekly). The rolling intrinsic trading strategy is explained [over here](#).

Minimum re-hedge profit per re-hedge

The minimum re-hedge profit is e.g. 0.20 EUR/MWh. It ensures that a set of trades (a re-hedge) is only executed when it generates at least this net profit, after deducting trading costs (determined by the bid-

ask spread). For example, if it is optimal to buy 2000 MWh and sell 2000 MWh with a net margin of only 0.18 EUR/MWh, then the trade will not be executed.

Hedges for the spot strategy

The spot trading strategy is automatically calculated by the model. The choice of a hedge defines an extra calculation on top of the spot trading strategy. The results of this calculation are displayed in the second column ('Spot & Hedge') of the Values tab when you view results. A screenshot is included in the section, discussing the KyStore outputs. The possible hedge types are a static delta hedge, a dynamic delta hedge, and a dynamic intrinsic hedge. In the first case, a single delta hedge is applied once at the valuation date. In the second case, a monthly re-hedging to the spot strategy is done, using delta hedging. It is assumed that only monthly products can be traded during the delivery horizon. In the third case, the hedging is based on the rolling intrinsic strategy. If a static delta hedge is chosen, the average value of the spot strategy is not affected (or only marginally), only the distribution of outcomes, which will typically become narrower (less risk). If a dynamic delta hedges is chosen the trading costs associated with changing the hedge are usually higher. A more detailed explanation of the spot strategy and its hedging [is over here](#).

Note that when the user selects the static or dynamic monthly delta hedge, independent from that hedge the model may calculate the rolling intrinsic value as a stand-alone trading strategy. That calculation is independent from the spot trading strategy. In that case, the user can select all re-hedging frequencies, including weekly and daily.

8.2.3 Calculations

Type of valuation
Rolling intrinsic
Calculations
Automation

Calculate actual delta, or volume based delta positions?

☒ Actual delta, value neutral
☐ Volume neutral delta

Calculate gamma sensitivities

☐ Yes
☐ Yes, with cross gammas
☒ No

Seed type for simulations

☒ Automatic (random number)
☐ Manual

Number of spot simulations

Number of forward simulations

Create extra txt files with detailed information for

simulations

Calculate actual delta, or volume based delta positions?

The user can choose which kind of hedges should be calculated, based on the spot strategy. The aim of these hedges is to decrease the risk associated with trading on the spot market. The two different types of hedges (actual and volume-based) are discussed in section [rolling intrinsic strategy](#). The link should be to the section Spot Trading Strategy instead.

Calculate gamma sensitivities

Gammas or cross-gammas may be calculated and saved in a separate txt-file in the job folder (not displayed in the user screen). Gammas measure how fast deltas change after a change in the gas price. Deltas represent the change in storage value, resulting from a change in the gas price.

Random seed

A random seed can be provided, setting the seed of the price simulations. Choosing the “Manual” option will result in the same price simulations. You can fill in any number. As long as your input is not changed, the simulation engine will generate the same random numbers. This option can be used to allow different KyStore profiles to run over the same set of price simulations. This could be useful when comparing the same storage’s value, after small changes in the valuation settings. If “Automatic (random number)” is selected, then the model would randomly draw a random seed number.

Number of simulations

The number of spot and forward simulations can be chosen separately. Spot and forward price simulations are used to infer the (rolling) intrinsic strategy. The estimation of the spot trading strategy is based only on the spot price simulations. The spot simulations should be at least as many as the forward simulations, because the Least Squares Monte Carlo method, used to estimate the spot trading strategy, usually requires more simulations.

Create extra files with detailed information for XX simulations

Not everything can be shown in the KyStore output screens in the platform. Thus, the model can output additional txt files with information from the valuation run (simulated spot/forward prices, decisions per simulation path, etc.). The user selects how many simulations should be saved. The txt files are located in the job folder, and can be downloaded. More information about the additional outputs can be found in section “Additional Text Files Output” (include a link to section).

8.2.4 Scheduling & Automation

Type of valuation
Rolling intrinsic
Calculations
Automation

Automatic runs

Schedule calculation daily at a specific time ☐ Yes ☒ No

Automate calculation directly when new inputs are available ☐ Yes ☒ No

Start date and start volume for the valuation

How many days after the trading date should the valuation start? days

Change start volume by

Automatically store expected storage volume in the storage volume table? ☐ Yes ☒ No

Schedule calculation daily at a specific time

The user can schedule to run the profile every day at a certain time. The model uses the latest available gas forward curve as an input. The scheduling can be done multiple times per day and in different week-days.

Automate calculation directly when new inputs are available

KyStore runs automatically when a new forward curve for the profile is available. The default valuation date is equal to the trading date plus one day (the valuation date is the first day when a trading decision should be made). The start storage volume is taken from the gas storage volume time series. If “Yes” is selected, one more choice should be done.

Run automatically when there is a new forward curve:

- *For a new trading date:* The model is going to run only when a forward curve for a new trading date becomes available. This option should be used if the user is interested in running a single valuation per day based on the forward market’s settlement data.
- *Also for the latest trading date:* The model is going to run each time a forward curve is created. This is going to happen even if some forward curves for the same trading date already exist. The option should be used if the user is interested in running multiple valuations per single trading date. These multiple forward curves can for example be based on the intra-day forward market’s data.

Start date and start volume for the valuation

The user can modify the default automation valuation date and start volume by adjusting these fields.

Automatically store spot decision in storage volume table

This option can be useful if the user follows the spot trading strategy suggested by the KyStore model. Once the storage is in delivery, estimating the spot trading strategy results in a trading signal (inject/withdraw/do nothing). The day-ahead decision can be stored automatically in the storage volume table. Please note that per storage, there can be only one KyStore profile which stores the results in the volume table.

8.3 Running KyStore

Analytics → KyStore ☐ Run

A user can choose to run a particular KyStore profile manually. On the run screen, the user can select:

- *trading date:* The forward curve created from this trading day’s market data is used as an input to KyStore.
- *valuation date:* This is the first day when a trading decision should be taken.
- *Inventory level:* This is the volume within the gas storage before the first decision is taken on the valuation date. This volume is taken by default from the gas storage volume time series, and it can be modified by the user.

- **Run backtest:** If the delivery period of the storage is in the past, the user can run a backtest. Several options can be selected for the backtest:
 - *Hedge Contracts:* What contracts should be used for the intrinsic and delta hedges? These can be either monthly or tradable products.
 - *Frequency of forward hedging to reduce the spot trading exposure:* How often should delta hedging be performed – daily/weekly/monthly?
 - *How much of the spot hedge will be unwinded before the delivery month?*
 - If this number is 100%, then the whole delta hedge product will be closed on the market at the end of the month before the product's delivery starts. For example, if there has been a delta hedge being long March'15, then at the end of February'15, the model will go short March'15 with the same volume as the delta hedge.
 - If this number is 0%, then the product will settle against the spot market. For example, the March'15 product will settle against the spot prices during March'15. This option is the one usually used for backtests.
 - Any number between 0% and 100% is possible. The same logic as above applies.
- **Bid-ask costs:** The user can modify the bid-ask costs of the tradable products through the “Bid-ask spreads” tab. The lot size and the way in which the traded product delivers can also be defined here.

The initial values in this tab are coming from the market setting for the specific gas commodity. The user can overwrite the existing gas market settings by choosing *Save inputs as default* in the *Bid-Ask spreads* tab.

8.4 Outputs KyStore

Analytics → KyStore ☐ View ☐ View

Go to Analytics → KyStore and click the “View” button next to the profile. In the list with all available runs, the valuation per particular run can be seen by pressing the “View” button again.

Various outputs are stored per calculation:

- The storage's value based on different valuation techniques (including distributions)
- Trading decision support
- Volume distributions
- Hedge values:
 - Monthly
 - Tradable
 - Daily

View results for TEST Dynamic Intrinsic Hedging

[Download PDF](#)
[Export to Excel](#)
[Back to profiles](#)

Storage id	86	Market	NCG	Start date	2016-01-01	End date	2017-01-01
Job id	43896	Job created at	2015-11-06 00:43:30	Valuation date	2015-11-06	Job created by	Automation

[Values](#)
[Volumes](#)
[Monthly hedges](#)
[Tradable hedges](#)
[Daily hedges](#)

Intrinsic

	Daily (EUR)	Monthly (EUR)	Tradable (EUR)
Value	2 463	1 865	779.18

Total value based on Monte Carlo price simulations

Strategy	Spot (EUR)	Spot & Hedge (EUR)	Rolling intrinsic (EUR)
Hedge Product	No	Months	Days
Rehedge	No	Monthly dynamic intrinsic	Monthly
Avg	3 816	3 811	2 559
Std Deviation	1 324	521.22	63.13
5 % low	1 426	3 057	2 479
10 % low	2 109	3 182	2 494
25 % low	2 969	3 518	2 509
Median	3 771	3 810	2 546
75 % high	4 626	4 107	2 588
90 % high	5 589	4 373	2 632
95 % high	5 852	4 584	2 680

8.4.1 Intrinsic valuation

Intrinsic valuation is the optimization of the storage given a single forward curve. This value is calculated based on three types of forward curves:

1. A 'theoretical' curve with daily granularity. This means:
 - a. Every day has potentially its own price. For example, the user could provide a curve such that early March is more expensive than end of March, or introduce weekday effects.
 - b. Every day a different decision can be taken: e.g. early March you withdraw, while end of March you do nothing.
 - c. The spot bid-ask spread is used for the calculations.
2. A 'theoretical' curve with monthly granularity. This means:
 - a. From the user-provided forward curve the model derives an average price per month. This forms the basis for the valuation.
 - b. Every month a different decision can be taken. However, during the month, the same injection/withdrawal rate applies.
 - c. The M1 bid-ask spread is used for the calculations. If this is not provided, the BOM bid-ask is used.
3. A 'realistic' curve based on tradable contracts defined by the user as part of the Market Setting. This is an intrinsic value that could really be locked in. This means
 - a. From the user-provided curve, the model calculates the average price for each tradable product.

- b. Every tradable product can be bought or sold. Because tradable products could be overlapping, the gas storage flow can vary during the contract period. For example, when you buy 30 MWh in April-12 and 91 MWh in Q2-12, then the injection in April is 2 MWh/day and 1 MWh in May and June.
- c. As trading costs, the model takes the bid-ask spreads defined by the user, per tradable product.
- d. The available tradable products do not have to span the whole storage delivery period. For example, a storage can end delivery on April 1, 2018, but the traded products could cover the period till December 31, 2017. In this case, a tradable intrinsic value will be calculated. The last injection/withdrawal in this case will be at the latest in December'2017. The required end volume will also be reached on December 31, 2017.

Finding the intrinsic values comes down to finding the actions that lead to the highest cumulative cash-flow while fulfilling all storage constraints. If there is no feasible solution, the model will report a value of 0. The intrinsic valuation is carried out by Mixed-Integer Linear Programming in case the storage has no volume dependent injection and withdrawal rates otherwise dynamic programming is used.

8.4.2 Spot trading strategy

Depending on the user selection, KyStore calculates the value of up to two different dynamic trading strategies using Monte Carlo simulations. The first is the spot trading approach (Strategy 'Spot'), which assumes you only trade in the spot market, every calendar day. The optimal spot trading strategy is determined by applying the Least-Squares Monte Carlo approach (LSMC) and avoids perfect foresight. This leads to daily injections and withdrawals (buy and sell in the market) that differ per simulated spot price path. Each simulation will therefore have its own value (discounted cumulative cash flow). The distribution of these values is summarized by the reported standard deviation and quartiles in the first two columns of the Contract Value table.

- Column 1: without any forward hedge: based on pure spot trading. Would be very risky and therefore typically wide distribution.
- Column 2: with a forward hedge put in place at the date of valuation. This is either the static or dynamic delta hedge, or a (dynamic) rolling intrinsic hedge. The choice is made in the valuation profile settings:
 - Static monthly delta hedge at valuation date: it is assumed that the trader executes the delta hedge volumes at the valuation date in the forward market (note: static means that the hedges are not adjusted over time). These are the volumes displayed in the results under 'Monthly hedges'. The combination of spot trading revenues and P&L on the hedges leads to a lower risk than achieved with only spot trading.

If the "volume based delta" setting is selected when defining the KyStore valuation profile, then the delta hedge over a specific day is derived from the expected injection / withdrawal volume on this day. The expected volume is calculated by averaging over the actions per simulation, suggested by the spot strategy. The monthly hedge is just a sum over the daily hedges.

If the “value based delta” setting is selected, the methodology is similar. The only difference is that the expectation is calculated by weighting the actions’ volumes by the simulated spot prices when calculating the average. KYOS recommends using this delta for hedging purposes.

- Dynamic monthly delta hedge: it is assumed that the trader executes the delta hedge volumes at the valuation date in the forward market (note: dynamic means that the hedges are adjusted over time). Afterwards, at monthly intervals, the initial delta hedge is adjusted to match the newly calculated delta hedges at each re-hedge date. Bid-ask costs are taken into account while calculating the cash-flows of the dynamic hedge. This is a theoretical value, since a trader can take a hedging position only on the current valuation date. It is an estimation of the expected return and risk, if the trader follows the dynamic delta strategy till the expiry of the storage.
- Dynamic (rolling) intrinsic hedge: it is assumed that the trader executes the intrinsic hedge volumes at the valuation date in the forward market. Thereafter, he adjusts the hedges every month. Note that more frequent re-hedging is not supported for this combination of (option-based) spot trading and rolling intrinsic hedging, due to the rather heavy calculations. This is why the other re-hedging frequency options for rolling intrinsic (daily / weekly) are blended out.

For simplicity we assume no trading costs, i.e. zero bid-ask. If the static hedge is chosen the average value is the same as in column 1, which depends on the fact that the spot simulations on average are equal the forward prices. However, for the dynamic delta and intrinsic hedging strategies, the values are not necessarily the same as you can trade the forward contracts dynamically over the time and incur trading costs.

8.4.3 Rolling intrinsic strategy

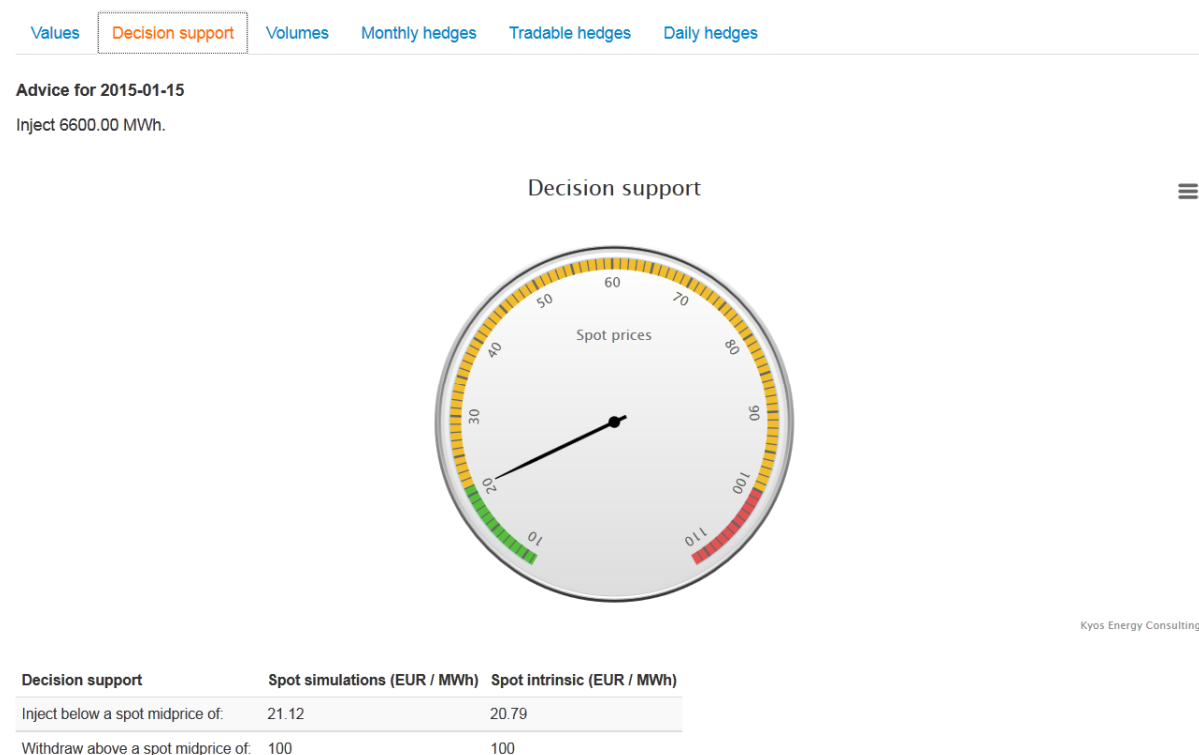
A rolling intrinsic valuation acknowledges that over time the forward curve will change, and the original intrinsic hedge may not be optimal anymore. At that moment, the existing position can be unwound, and a new optimal position can be locked in. In that way, value is slowly added to the initial intrinsic valuation. The question is how often and when re-balancing should be done. There is no standard consensus in the market. In the platform, the user can set the re-balancing frequency to daily, weekly, and monthly. At each re-hedge date, KyStore performs an optimization to check if a re-hedge generates extra income. This is subject to the condition that there are already positions in place and that the storage constraints continue to be fulfilled. When re-hedging transaction costs (bid-ask) and also a potential change in the injection and withdrawal costs are taken into account; for example, when the re-hedge leads to fewer cycles, these costs will decrease and hence add value. Note that the available products which can be traded on the market on each re-hedge date are each day, each month, or the tradable products defined by the user. This choice is made when defining the KyStore profile.

In almost all cases, the rolling intrinsic valuation converges faster than the spot valuation. For this reason, the user can set the number of rolling intrinsic simulations to a lower number than for the spot simulations.

8.4.4 Decision support

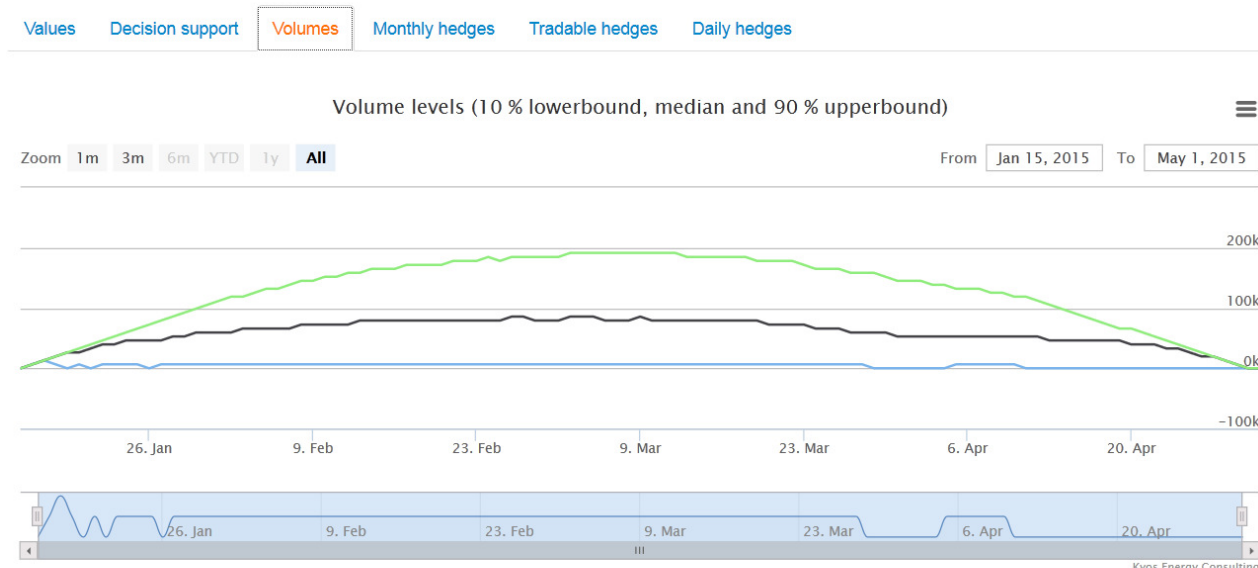
This output is only available if the storage is operational today, so if the current day is in the storage operation period. Otherwise the tab is missing.

The different storage values are each based on a certain trading strategy. For the spot value (Monte Carlo) and the daily intrinsic value (static curve) this includes an action on the current day. On the current day, the spot price is known. The storage model informs the trader or storage operator above which market price it is optimal to withdraw gas from storage, and below which market price it is optimal to inject. The difference can e.g. be equal to injection or withdrawal costs. It also provides the user the amount of gas it should inject or withdraw according to the spot strategy.



8.4.5 Volumes

The expected volume distribution is plotted in the menu Volumes. The graph shows the 10% lowest, the average and the 90% highest, all measured across all simulations of the spot trading strategy (LSMC).



[Show / hide table](#)

8.4.6 Hedges

The model produces various statistics that help the user to secure the reported values over time. The numbers in the hedge output contain the volumes that should be traded in the market according to the two strategies.

- The intrinsic hedge which captures the intrinsic value: labelled 'Intrinsic'
- The delta hedge which captures the extrinsic value: labelled 'Delta'. The delta hedge has an opposite sign to the delta sensitivity, which is the sensitivity of the storage value with respect to the storage value.

Both the intrinsic and the delta hedge are calculated with three types of granularity of the forward curve:

- Monthly: characterized by the first day of the month
- Tradable products: characterized by the first and last day of the delivery period
- Daily: a number for each day



[Show/hide table](#)

Intrinsic hedge

When the user would implement the intrinsic hedges, and take trading, storage costs and discount rate into account, he would secure the corresponding intrinsic value.

Delta hedge

When the user would implement the delta hedges, and adjust this delta hedge over time, he/she would stabilize the spot trading value as much as possible. It will narrow the distribution of outcomes and ensure that at least a large part of the extrinsic value is realized. However, there is no full guarantee that the exact spot value will be realized over the lifetime of the option, for example because the volatility in the market may change.

The delta hedge can be calculated using either the 'actual' deltas or the 'volume based' deltas. The 'volume based' deltas are the average actions in the spot trading strategy (averaged across the simulations). For example, suppose for simplicity there are 2 simulations. In simulation 1, the average volume for July 2014 is 1000 therms (injection), in simulation 2 it is 2000 therms. Then the volume-based delta hedge for July is 1500 therms, which should be bought in the forward market as a hedge.

Actual delta hedge; this is equivalent to the value-based delta hedge and reflects more accurately the change in expected storage value due to a change in the forward price. To calculate this type of delta, the actions are multiplied with realized spot price divided by the initial forward price. Using the same example, assume that the average spot price for the injections is 60 pence/therm in simulation 1 and 40 pence/therm in simulation 2. Then the actual delta hedge is weighted by these prices, i.e. $(1000 \times 60 + 2000 \times 40) / (60 + 40) = 1400$ therms.

Comparison: the actual delta hedge yields a slightly better stabilization of total revenues, at least according to the theory. For example, if the delta hedge is 1400 therms, the market forward price for July 2014 goes up by 1 pence, then you may expect a decrease of 1400 pence in the storage value. When you have a forward position of 1400 therms, the decrease in storage value is exactly compensated for by a mark-to-market profit on your hedge. However, the disadvantage of an actual delta hedge is that the volumes of the hedges will not sum up as you would intuitively expect, which they do for a volume-based hedge. For example, if the end volume of the storage has to be the same as

the begin volume, the sum of the volume based hedges will always be zero. That is why many people in practice prefer the volume based hedges.

The delta hedge is originally generated for daily forward contracts. The monthly hedges are derived by summing the daily deltas over the months. The tradable hedges may have overlapping periods. KyStore calculates the tradable hedge which is as close as possible to the daily hedge in a least squares sense.

Volume based delta: the average volumes across the spot simulations. For example, suppose for simplicity there are 2 simulations. In simulation 1 the average volume for July 2014 is 1000 therms (injection), in simulation 2 it is 2000 therms. Then the volume-based delta hedge for July is 1500 therms, which should be bought in the forward market as a hedge.

8.4.7 Backtest

If the storage has expired the user can run a backtest. There are two output tabs which contain outputs related to the backtest.

View results for TEST GasTerra SBU 14/15 - Backtest

[Download PDF](#)
[Export to Excel](#)
[Back to profiles](#)

Storage id	37	Market	TTF	Start date	2014-04-01	End date	2015-04-01
Job id	28314	Job created at	2015-11-04 14:28:08	Valuation date	2014-04-01	Job created by	Admin istrator

[Values](#)
[Decision support](#)
[Volumes](#)
[Monthly hedges](#)
[Tradable hedges](#)
[Daily hedges](#)
[Backtest Values](#)
[Backtest Spot Decisions](#)

Backtest Values

Strategy	Projected value (EUR)	Realized value (EUR)
Intrinsic daily	4 694	
Intrinsic monthly	4 619	
Intrinsic tradable	4 280	
Rolling intrinsic	4 641	4 979
Spot	5 451	4 940
Spot with static intrinsic hedge		4 940
Spot with static delta hedge		5 479
Spot with dynamic intrinsic hedge		7 215
Spot with dynamic delta hedge		6 244

The tab “Backtest Values” contains values per different types of strategy. There are two columns. The “Projected value” contains the values estimated by KyStore when the valuation date is the first day in the backtest. The “Realized value” column contains the actual values realized during the backtest period by implementing a particular trading strategy.



The tab “Backtest Spot Decisions” contains the spot prices and the daily decisions related to the spot trading strategy.

8.4.8 Additional Text Files Output

KyStore can produce more detailed output than displayed in the platform. If the user chooses to create an additional output for a particular number of simulations, then some additional text files will be created in the “Output” directory of the valuation run. The whole valuation run directory can be downloaded in a zip file, through the Jobs list in the platform.

All files are with an extension “.txt”, and a tab is used as a delimiter. All dates follow the Excel serial date convention.

Spot Trading Strategy

The spot simulations and the inventories implied by following the spot trading strategy are in files (their number of columns matches and is equal to the number of spot simulations):

PriceSimulationsSpot: Contains simulated spot prices across time (the rows) and simulations (the columns).

InventorySpot: Contains the storage level for the spot strategy across simulations (the rows) and time (the columns).

Rolling Intrinsic Strategy

The forward curve simulations can be a different number than the spot simulations (usually less). They are used to derive the rolling intrinsic strategy

PriceSimulationFwd:

- The first column contains the hedge date.
- The second column contains the forward contract's start of delivery date.
- The third column contains the forward contract's end of delivery date.
- The remaining columns (Column 4 – end) contain the simulated forward prices at the hedge date (column 1), for the contract in question. The number of columns depends on the number of forward simulations.

RollIntrTrades: same format as PriceSimulationFwd, but it contains the traded volume in a particular forward contract per particular simulation.

InventoryRoll: Contains the physical storage level per day and simulation, resulting from following the rolling intrinsic strategy. The file is across simulations (the columns) and time (the rows).

Backtest Text Files Output

When performing a backtest some additional text files are created in the "Output" folder of the valuation run. The set of files can include the following:

backtest_static_hedge: Contains the forward trades, associated with the static intrinsic and the delta hedge strategies. These are executed on the forward market at the initial valuation/projection date. The cash flows of these hedges are taken into account when reporting the realized "Spot with static intrinsic hedge" and "Spot with static delta hedge" values.

- The first column and second columns contain the forward delivery period, start and end date, respectively.
- The third column contains the spread between the average realized spot price over the given period, and the forward price for this period on the initial valuation/projection date.
- The fourth column contains the intrinsic hedge volume.
- The fifth column contains the delta hedge volume.

backtest_dynamic_intr_hedge: Contains the forward trades at all re-hedge dates, associated with implementing the “Spot with dynamic intrinsic hedge” strategy.

- The first row contains the re-hedge dates.
- The first and second rows contain the forward product delivery dates.
- The remaining rows and columns contain the hedges per specific re-hedge date and forward product.

backtest_dynamic_delta_hedge: Contains the forward trades at all re-hedge dates, associated with implementing the “Spot with dynamic delta hedge” strategy.

- The first row contains the re-hedge dates.
- The first and second rows contain the forward product delivery dates.
- The remaining rows and columns contain the hedges per specific re-hedge date and forward product.

backtest_forwardprices_monthlycontracts: Contains the monthly forward trades at all re-hedge dates, associated with implementing the dynamic hedge strategies.

- The first row contains the re-hedge dates.
- The first and second rows contain the forward product delivery dates.
- The remaining rows and columns contain the forward price per specific re-hedge date and forward product.

Note that the dynamic hedge volumes are always reported on a monthly basis, no matter if the hedge products have been daily, monthly, or tradable.

Backtest_rollingintrinsic_trades: Contains the forward trades at all the re-hedging dates, associated with the rolling intrinsic strategy.

- The first column is the trading/re-hedge date,
- The second and third columns contain the swapping period, start and end date, respectively.
- The fourth column is the actual traded volume.

Backtest_rollingintrinsic_fwdprices: Contains the forward prices at all the re-hedging dates, used in the rolling intrinsic strategy.

- The first column is trade date
- The second and third columns contain the swapping period, start and end date, respectively.
- The fourth column is the associated forward price.

8.5 Value Report

Above the overview of the KyStore profiles, there is a grey button 'Value report'. The purpose of this button is to export the historical evolution of intrinsic/spot storage value in an Excel file. The user can select in what currency the outputted values should be. A filter containing the desired KyStore profiles can be chosen. Furthermore, the start and end date of historical time series can be selected. The valuation approach behind the outputted storage value can be selected as well. Currently, there are two choices – daily intrinsic and spot strategy value.

8.6 Export Interfaces

Different outputs of the KyStore model can be exported automatically from the KYOS platform. This functionality can be used to integrate the platform with other systems. A KyStore export interface can be created by going to:

tab 'Interfaces' -> tab 'Export' -> button 'Create' -> select 'KyStore' form drop-down.

A description of the content of the different export types is available upon request.

9 KySwing – Gas contract valuation

The KySwing model is developed to value optionality in long-term gas contracts, but it can be generally applied to value other types of swing options in power or gas. It identifies swing value, optimal trading decisions, exposures and hedging strategies. It employs the Least Squares Monte Carlo (LSMC) technique to determine optimal spot decisions and the mixed-integer linear programming (MILP) technique to determine other strategies, e.g. in forward markets.

KySwing calculates swing value based on three different trading strategies. The *intrinsic strategy* finds the optimal actions based on the current forward curve, being daily, monthly or tradable. The *rolling intrinsic strategy* finds out how the intrinsic value can be enhanced by adjusting forward positions over time. The rebalancing frequency, the moments where the position is adjusted, can be set to daily, weekly or monthly. The *full option strategy* finds the optimal spot trading decisions by optimally using all available information about current and future price levels and optionality.

The model supports day-to-day trading decisions:

- Advise on optimal trading strategies;
- Calculate delta positions;
- Provide distribution of cash-flows and inventory;
- Track individual simulation results over time of prices, volumes, injections and withdrawals.

KySwing accounts for typical gas swing constraints:

- Various levels of volume restrictions;
- Fixed (monthly constant) or an indexed lagged/average strike price with up to 4 commodities;
- Penalties for violation of volume constraints;
- Carry-forward and make-up conditions.

KySwing is based on Monte Carlo simulation techniques. Important characteristics are:

- Multi-commodity simulation model to enable indexed strike prices;
- Mean-reverting multi-factor model with long-term, short-term and seasonal dynamics;
- Cointegration between commodities on long-term level;
- The user has the choice to use external simulations, coming from KySim or some other simulation module;
- Prices matched to user-provided forward curve with up to daily precision;
- Least Squares Monte Carlo methodology to find optimal trading and operating decisions.

The swing methodology has been developed in-house, and refined on the basis of customer feedback. The spot valuation approach is an extension of the gas storage methodology whose core methodology is published in the Journal of Derivatives (Boogert and De Jong, Spring 2008) and the Journal of Energy Markets (Boogert and De Jong, January 2012).

9.1 Defining a (swing) contract

[See Assets & Contracts -> Contracts.](#)

The following items are required to define a valid contract to be valued by KySwing:

- The contract needs to be defined as flexible contract (bandwidth flexibility), or it should be a contract with nominations.
- The user can choose if daily/monthly nominations should be used within the contract.
- One or more delivery points have to be defined.
- The contract has to be defined as physical.
- Daily contract quantity constraints have to be defined. Furthermore, inputs for monthly, quarterly, annual take-or-pay constraints are common.

9.2 Gas offtake from a swing contract

If the contract uses daily/monthly nominations, offtake volumes come from the contract nominations data in Assets & Contracts -> Contract nominations. The user can modify the nominations on this screen. If nominations are used, the relevant nominated volumes are used to calculate the taken volumes.

If a contract with no nominations is chosen, then expected offtakes and actual contract offtakes can be stored by uploading them in tab Time Series -> tab Volume Actuals & Forecasts -> button upload volumes. For gas contracts, the granularity can be daily or monthly. It is recommended to upload actuals on a daily basis for KySwing.

The delivery point selection is an essential part in the set-up of the swing calculation in this case. The actuals are connected to a delivery point. When a KySwing calculation is made, the actuals are automatically included as taken volumes.

It is also important to remember there can be only 1 flexible contract per delivery point. The model (i) looks at the total consumption per delivery point and (ii) looks at all contracts with fixed volume per delivery point (if any). The remaining volume between the actuals and the fixed volumes is considered as the volume delivered by the flexible contract.

It is not possible to store multiple contracts for the same delivery point and the same period. The reason is that with more delivery points, the flexible volume cannot be allocated to a contract.

9.3 KySwing valuation profile

Analytics → KySwing ☐ Create profile

A KySwing valuation profile is the combination of a swing contract and a set of valuation settings. First, go to Analytics -> KySwing and press “Create profile”. A list of all flexible, physical contracts appears. Select a contract and define a profile for this contract.

9.3.1 Type of valuation

Type of valuation
Rolling intrinsic
Calculations
Automation

Basic contract information

entity	kyos	buy/sell	buy	price fixing	no
counter party	Kyos Energy	commodity	NGC	id/confirm	324 / yes: TEST KySwing M1 17/18
contract period	01-01-2017 — 31-12-2018	volume & unit	2400000 MWh	currency	EUR
book	Procurement	physical/financial	physical	contract type	market_indexation

Profile name

TEST NCG - M1 2017/18

Market curve

NGC

Which forward curve?

☒ Daily
☐ Monthly
☐ Mixed (BOM = daily, rest = monthly)

Discount rate

0 %

Calculate spot value based on perfect foresight?

☒ Yes
☐ No

Type of simulations

☒ Internal (generated by KySwing)
☐ External (generated earlier by KySim)

Select the market parameters

☐ Use calibration
☒ User defined

Market Parameters (Gas)

Spot volatility (factor 1)

50 %

Mean reversion rate

7 %

Volatility long-term returns (factor 2)

15 %

Volatility summer-winter returns (factor 3)

14 %

Monthly weights applied to the spot volatility

Use numbers larger than 0%. The model will automatically adjust the weights, so that the average is 100%. The resulting weights are then multiplied with the spot volatility (user input or calibrated).

Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
100 %	100 %	100 %	100 %	100 %	100 %	100 %	100 %	100 %	100 %	100 %	100 %

Make-up

Include a make-up right

☒ Yes
☐ No

At the top, the user sees a short description of the contract specification. After that, some general KySwing valuation options should be chosen:

Which forward curve

This choice determines which forward curve should be sent as an input to the KySwing model. The KyCurve curve building model produces 3 different forward curves for gas commodities:

- Monthly curve = a forward curve with a different price for each month. However, each day within a particular month has the same price.
- Daily curve = a forward curve with a potentially different price for each future day. The average daily price in a month equals the monthly price in the previous curve type. For example, the daily curve is smoothed, so prices end of March are lower than the beginning of March. The daily curve may also contain a (small) weekend-workday spread.
- Mixed curve = a curve which has a monthly granularity from the front-month onwards, but daily until then. For example, on 16 February, there is a daily price from 17 Feb to 28 Feb, then a price for March, April and so on.

The choice of a forward curve can affect the calculated swing contract value, following the different trading strategies. For example, there is a swing contract which delivers in a single month (September). If the daily forward curve implies a negative expected gas/strike price spread at the beginning of September, and a positive at the end, it might be profitable to offtake only at the end of the month. However, if the monthly curve is used, the expected gas/strike price spread might be 0 during the whole of September, and no offtake during September suggested.

Discount Rate

The discount rate is applied in all calculations to discount back future cash-flows to the current (valuation) date. For example, if you receive cash-flows of 1000 GBP (from gas offtakes) 1 year from now, and the discount rate is 5%, this is included in the valuation as $1000 * \exp(-5\%) = 951$ GBP. Note that the discount rate is interpreted as a continuous rate, which leads to the use of the exponential function.

Calculate spot value based on perfect foresight

Choose if the spot value should be calculated based on perfect foresight. The perfect foresight value is an upper bound on the value generated by the spot strategy. Have in mind, that the Least Squares Monte Carlo techniques is always applied by default when deriving the spot value. The LSMC technique avoids perfect foresight. See [over here](#) for a more detailed explanation regarding this strategy.

Type of simulations

The user can choose whether KySwing should generate simulations internally or use external simulations from KySim.

- If *external* is chosen, the user can select which KySim profile's simulations to use. The simulations are directly used, so the user does not have to define any market parameters. The characteristics of the simulations can be defined in the KySim profile instead. Please refer to the documentation of KySim for more details.
- Through using the *external* option, the user can use his/her own simulations. This is done through creating a KySim profile, which uses user uploaded simulations. Please refer to the section on KySim, explaining how this can be done.
- If *internal* is chosen, the user should choose what market parameters to use. Prices are simulated following a 3 factor model. Inputs for the three factors are spot volatility (factor 1), long-term volatility (factor 2) and winter-summer volatility (factor 3). Furthermore, an input is required for the mean reversion rate. The mean reversion rate determines how quickly spot prices revert back to the forward price level. It is often around 10% per day for natural gas markets (usually between 5 and 30%). Roughly speaking, this means that the expected spot price will move 10% closer to the (front-month) forward price each day. For example, if the spot price today is 40 and the forward price 30, the expected spot price for tomorrow is 39, excluding any stochastic component from the simulation process.

Inputs for market parameters can be provided:

- Manually, as hard coded numbers. In this case, the user determines directly the parameter values. The KyCalibration section in the platform documentation explains the meaning of the four market parameters.
- Automatically, using a calibration profile (KyCalibration). When a calibration profile is selected, the market parameters (volatility, mean reversion rate, etc.) on the specific valuation date are taken from this profile. In a calibration profile the parameters are estimated based on historical market price data (though they may be fixed, using the overwrite function). Please refer to the documentation of KyCalibration for more details.

- Both of the above options result in a single number for the spot volatility. However, the user has the flexibility to apply a different spot volatility per each month. This can be applied through defining monthly weights, which are used to rescale the spot volatility. The weights don't have to sum to 100 %: for each month in the price simulations, the weights are multiplied with the spot volatility, which has either been set by the user or calibrated on historical data. If the weights do not sum to 100%, then the model automatically rescales them to an average of 100%. For example, suppose that the user defines a weight of 140% to the first 3 months in the year (Jan-Mar), and 100% to the others, and then the actual weights in the calculations are 130% for Jan-Mar and 90% for Apr-Dec. If the annual spot volatility is 50%, the model uses $130\% \times 50\% = 65\%$ for Jan-Mar, and $90\% \times 50\% = 45\%$ for Apr-Dec.

Monthly weights applied to the spot volatility

Use numbers larger than 0%. The model will automatically adjust the weights, so that the average is 100%. The resulting weights are then multiplied with the spot volatility (user input or calibrated).

Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
100 %	100 %	100 %	100 %	100 %	100 %	100 %	100 %	100 %	100 %	100 %	100 %

Make-up

If the contract has some period quantity constraints (PCQs), then the user can define a make-up clause. If “Yes” is selected in “Include a make-up right”, then a new tab “Make-up” will appear at the end. In this tab, the user can define the make-up specifications. This is discussed in a later section.

9.3.2 Model Currency

Profile name	TEST system currency	Select the market parameters	☒ Use calibration ☐ User defined
Market curve	NBP	Calibration profile	NBP + FX
Model currency	☐ Perform valuation in market currency (GBP) ☒ Perform valuation in system currency (EUR)		
FX conversion	☐ Spot (day-by-day) ☒ Previous month (average) ☐ Current month (average)		
Which forward curve?	☒ Daily ☐ Monthly ☐ Mixed (BOM = daily, rest = monthly)		
Discount rate	0 %		
Calculate spot value based on perfect foresight?	☒ Yes ☐ No		
Type of simulations	☒ Internal (generated by KySwing) ☐ External (generated earlier by KySim)		

If the gas market is in a particular currency, e.g. GBP, but the domestic market is in a different currency, e.g. EUR, the exchange rates will influence the offtake decisions. The user can select to perform the optimizations in the domestic currency (the platform's system currency). If this option is selected the market gas prices will be converted to the domestic market using the daily simulated FX rate. The strike prices will be converted according to the chosen FX averaging.

9.3.3 Market parameters (other)

This screen is filled only if the user has chosen internal simulation and user defined market parameters. Commodities different from the contract gas are simulated with a 1 factor model. Inputs are automatically provided with a calibration profile, or alternatively they are provided as hard-coded inputs:

- Long term volatility
- Correlation matrix (long term returns)
- Cointegration values (long term returns)

The software uses a one factor model for the different commodities. The one factor shifts the complete forward curve up or down, and contains no mean-reversion. The user can enter the volatilities and correlations inside the respective fields. The user has to specify a complete correlation matrix for all commodities. If the true correlation matrix (containing only the active commodities / forex) is not positive definite, the software will make the matrix positive definite.

In addition to the volatility-correlation, the user is allowed to include cointegration on the long-term basis. The level is adjusted to be in line with the current forward curve, whereas mean-reversion rate and cointegration speeds are estimated from historical data. Some remarks related to cointegration:

- A commodity can never be cointegrated with itself.
- There can be a maximum of 1 cointegration relation per commodity.
- There has to be at least 1 commodity which is not cointegrated with another commodity.

9.3.4 Rolling intrinsic

Select contract
Type of valuation
Market parameters
Rolling intrinsic
Calculations
Automation

Basic contract information

entity	Swing Trading	buy/sell	buy	price fixing	no
counter party	Kyos Energy	commodity	NCG	id/confirm	8 / no
contract period	01-05-2015 — 30-04-2016	volume & unit	12000 MWh	currency	EUR
book		physical/financial	physical	contract type	formula

Calculate contract value based on rolling intrinsic strategy? ☒ Yes ☐ No

Hedge type for spot strategy ☒ Static monthly delta hedge at valuation date (default) ☐ Rolling intrinsic hedge (longer calculation time)

Which rolling intrinsic strategy? ☒ Tradable (as defined in the market settings of the commodity) ☐ Monthly (each month is a tradable contract) ☐ Daily (each day is a tradable contract)

Re-hedging frequency for rolling intrinsic ☒ Monthly ☐ Weekly ☐ Daily

Minimum rehedging profit

A *rolling intrinsic calculation* can be performed, where a rolling forward hedging strategy is applied. When this setting is chosen, the user should select what products should be hedged (tradable, monthly, daily); how often re-hedging takes place (monthly, weekly); and the minimum-rehedge profit. The rolling intrinsic trading strategy is explained [over here](#).

Minimum re-hedge profit per re-hedge

The minimum re-hedge profit is e.g. 0.20 EUR/MWh. It ensures that a set of trades (a re-hedge) is only executed when it generates at least this net profit, after deducting trading costs (determined by the bid-ask spread). For example, if it is optimal to buy 2000 MWh and sell 2000 MWh with a net margin of only 0.18 EUR/MWh, then the trade will not be executed.

Hedges for the spot strategy

The spot trading strategy is automatically calculated by the model. The choice of a hedge defines an extra calculation on top of the spot trading strategy. The results of this calculation are displayed in the second column ('Spot & Hedge') of the Values tab when you view results. A screenshot is included in the section, discussing the KySwing outputs. Whatever the choice of this setting is, it does not affect the average value of the spot strategy (or only marginally), only the distribution of outcomes, which will typically become narrower (less risk). Hedge types are static monthly delta hedge and dynamic (rolling) intrinsic hedge. In the first case, a single hedge is applied once at the valuation date. In the second case, a re-hedging is done according to the specified frequency. Please refer to section 9.5.2 for a more detailed explanation of the spot strategy and its hedging.

Note that when the user selects the ‘static monthly hedge’, independent from that hedge the model may calculate the rolling intrinsic value as a stand-alone trading strategy. That calculation is independent from the spot trading strategy. In that case, the user can select all re-hedging frequencies, including weekly and daily.

9.3.5 Calculations

Calculate actual delta, or volume based delta positions?

The user can choose which kind of hedges should be calculated, based on the spot strategy. The aim of these hedges is to decrease the risk associated with trading on the spot market. The two different types of hedges (actual and volume-based) are discussed [over here](#).

Calculate gamma sensitivities

Gammas or cross-gammas may be calculated and saved in a separate txt-file in the job folder (not displayed in the user screen). Gammas measure how fast deltas change after a change in the gas price. Deltas represent the change in storage value, resulting from a change in the gas price.

Find strike formula resulting in zero intrinsic value?

Yes, additive – the model finds a constant such that when it is added to the strike formula, the monthly intrinsic value is zero.

Yes, multiplicative – the model finds a constant such that when it is multiplied with the strike formula, the product results in a zero monthly intrinsic value.

Random seed

A random seed can be provided, setting the seed of the price simulations. Choosing the “Manual” option will result in the same price simulations. You can fill in any number. As long as your input is not changed, the simulation engine will generate the same random numbers. This option can be used to allow different KySwing profiles to run over the same set of price simulations. This could be useful when comparing the same swing contract’s value, after small changes in the valuation settings. If “Automatic (random number)” is selected, then the model would randomly draw a random seed number.

Number of simulations

The number of spot and forward simulations can be chosen separately. Spot and forward price simulations are used to infer the (rolling) intrinsic strategy. The estimation of the spot trading strategy is based only on the spot price simulations. The spot simulations should be at least as many as the forward simulations, because the Least Squares Monte Carlo method, used to estimate the spot trading strategy, usually requires more simulations.

Create extra files with detailed information for XX simulations

Not everything can be shown in the KySwing output screens in the platform. Thus, the model can output additional txt files with information from the valuation run (simulated spot/forward prices, decisions per simulation path, etc.). The user selects how many simulations should be saved. The txt files are located in the job folder, and can be downloaded. More information about the additional outputs can be found in section “Additional Text Files Output” (include a link to section).

9.3.6 Scheduling & Automation

Schedule calculation daily at a specific time

The user can schedule to run the profile every day at a certain time. The model would use the latest available relevant gas forward curve as an input. The scheduling can be done multiple times per day and in different week-days.

Automate calculation directly when new inputs are available

The user can choose if automation should be used. The automation procedure results in an automatic run each time new forward curve inputs for the model are available. If “Yes” is selected, one more choice should be done.

Run automatically when there is a new forward curve:

- *For a new trading date:* The model is going to run only when a forward curve for a new trading date becomes available. This option should be used if the user is interested in running a single valuation per day based on the forward market’s settlement data.
- *Also for the latest trading date:* The model is going to run each time a forward curve is created. This is going to happen even if some forward curves for the same trading date already exist. The option should be used if the user is interested in running multiple valuations per single trading date. These multiple forward curves could be based on the intra-day forward market’s data.

9.3.7 Make-up

Make-up Characteristics

Quarter

Max make-up rights you may use per Quarter: MWh

Max make-up rights you may create per Quarter: MWh

Max duration of make-up gas: Quarters

Allow violation of Quarter constraints due to make-up usage: ☐ Yes ☒ No

Make-up Pricing

Pay percentage of the current Quarter price during creation of make-up: %

Pay percentage of current Quarter price when using make-up: %

In the Make-up tab, the user can define the different characteristics of the contract's make-up clause. The make-up is defined always only on the first period quantity constraint (PCQ). In the example above, the contracts has some quarterly volume constraints. The different fields are self-explanatory.

9.4 Running KySwing

Analytics → KySwing □ Run

A user can choose to run a particular KySwing profile manually. On the run screen, the user can select:

- *trading date*: The forward curve created from this trading day's market data is used as an input to KySwing.
- *valuation date*: This is the first day when an offtake can be exercised.
- *volumes already taken*: This is the volume offtake in the past. It is defined per month.
- *Run backtest*: If the delivery period of the contract is in the past, the user can run a backtest. Several options can be selected for the backtest:
 - *Hedge Contracts*: What contracts should be used for the intrinsic and delta hedges? These can be either monthly or tradable products.
 - *How much of the spot hedge will be unwinded before the delivery month?*
 - If this number is 100%, then the whole delta hedge product will be closed on the market at the end of the month before the product's delivery starts. For example, if there has been a delta hedge being long March'15, then at the end of February'15, the model will go short March'15 with the same volume as the delta hedge.
 - If this number is 0%, then the product will settle against the spot market. For example, the March'15 product will settle against the spot prices during March'15. This option is the one usually used for backtests.
 - Any number between 0% and 100% is possible. The same logic as above applies.
- *Bid-ask costs*: The user can modify the bid-ask costs of the tradable products through the "Bid-ask spreads" tab. The lot size and the way in which the traded product delivers can also be defined here.

The initial values in this tab are coming from the market setting for the specific gas commodity. The user can overwrite the existing gas market settings by choosing *Save inputs as default* in the *Bid-Ask spreads tab*.

9.5 KySwing Output

Analytics → KySwing ☐ View ☐ View

Various outputs are stored per calculation:

- The value, including distributions.
- Spot decision support (when contract is in delivery).
- Volume distributions.
- Trading support: various delta positions (Monthly, Tradable, Daily).

Values	Decision support	Volumes/Offtakes	Monthly hedges	Tradable hedges	Daily hedges
Intrinsic					
Granularity	Daily		Monthly		Tradable
Value (EUR)	367 876		367 877		338 776
Total off-take (MWh)	785 301		785 301		785 301
Total penalty (EUR)	0		0		0
Total value based on Monte Carlo price simulations					
Strategy	Spot		Spot & Hedge		Rolling intrinsic
Hedge forward products	No		Months (delta)		Tradable
Rehedge	No		Rolling intrinsic hedge (longer calculation time) hedge type for spot strategy		Monthly
Avg (EUR)	821 561		805 804		614 939
Std Deviation (EUR)	691 595		228 863		187 747
5 % low (EUR)	76 390		485 951		387 962
10 % low (EUR)	117 637		551 860		411 890
25 % low (EUR)	292 145		639 090		489 426
Median (EUR)	602 988		768 491		569 738
75 % high (EUR)	1 149 928		902 435		691 394
90% high (EUR)	1 774 925		1 120 501		854 545
95 % high (EUR)	2 283 258		1 207 080		966 655
Average total off-take (MWh)	721 242		721 242		776 244
Average total penalty (EUR)	0		0		0
Value with perfect foresight (EUR)	853 676				
Calculated strike - zero intrinsic value					
Additive factor					NaN
Multiplicative factor					NaN

9.5.1 Intrinsic values

Intrinsic valuation is the optimization of the swing contract, given a single forward curve. This value is calculated based on three types of forward curves:

- A ‘theoretical’ curve with daily granularity. This means:
 - Every day has potentially its own price. For example, the user could provide a curve such that early March is more expensive than end of March, or introduce weekday effects.

- Every day a different decision can be taken: e.g. early March you take more gas than end of March.
- The spot bid-ask spread is used for the calculations.
- A ‘theoretical’ curve with monthly granularity. This means:
 - From the user-provided forward curve the model derives an average price per month. This forms the basis for the valuation.
 - Every month a different decision can be taken. However, during the month, the off-take is constant.
 - The M1 bid-ask spread is used for the calculations. If this is not provided, the BOM bid-ask is used.
- A ‘realistic’ curve based on tradable contracts defined by the user as part of the Market Setting. This is an intrinsic value that could really be locked in. This means:
 - From the user-provided curve, the model calculates the average price for each tradable product.
 - Every tradable product can be bought or sold. Because tradable products could be overlapping, the off-take can vary during the contract period. For example, when you sell 30 MWh in April-12 and 91 MWh in Q2-12 in the gas market, then the contract off-take in April is 2 MWh/day and 1 MWh in May and June.
 - As trading costs, the model takes the bid-ask spreads defined by the user, per tradable product.
 - The same tradable products are assumed for your strike as for the gas market. A typical assumption is that trading in the gas market is leading the trading in the potentially other commodities of the strike price.

Finding the intrinsic values comes down to finding the actions that lead to the highest cumulative cash-flow while fulfilling all contract constraints. If there is no feasible solution, the model will report a value of 0. The intrinsic valuation used to be carried out Mixed Integer Linear Programming (MILP). Please note that a contract which has gas-market indexation may have an intrinsic value close to 0. This is because as of today, there is no arbitrage possibility in the forward market.

9.5.2 Spot trading strategy

Depending on the user selection, KySwing calculates the value of up to two different dynamic trading strategies using Monte Carlo simulations. The first is the spot trading approach (Strategy ‘Spot’), which assumes you only trade in the spot market, every calendar day. The optimal spot trading strategy is determined by applying the Least-Squares Monte Carlo approach (LSMC) and avoids perfect foresight. This leads to daily offtake decisions (sold in the gas market) that differ per simulated spot price path. Each simulation will therefore have its own value (discounted cumulative cash flow). The distribution of these values is summarized by the reported standard deviation and quartiles in the first two columns of the Contract Value table.

- Column 1: without any forward hedge: based on pure spot trading. Would be very risky and therefore typically shows a wide distribution.
- Column 2: with a forward hedge put in place at the date of valuation. This is either the static or the dynamic rolling intrinsic hedge defined in the profile. The choice is made in the valuation profile settings:

- Static monthly delta hedge at valuation date: it is assumed that the trader executes the delta hedge volumes at the valuation date in the forward market (note: static means that the hedges are not adjusted over time). These are the volumes displayed in the results under 'Monthly hedges'. The combination of spot trading revenues and P&L on the hedges leads to a lower risk than achieved with only spot trading.
- Rolling intrinsic hedge: it is assumed that the trader executes the intrinsic hedge volumes at the valuation date in the forward market. Thereafter, he adjusts the hedges every month. Note that more frequent rehedgeing is not supported for this combination of (option-based) spot trading and rolling intrinsic hedging, due to the rather heavy calculations. This is why the other rehedgeing frequency options for rolling intrinsic (daily / weekly) are blended out.

For simplicity we assume no trading costs, i.e. zero bid-ask. If the static hedge is chosen the average value is the same as in column 1, which depends on the fact that the spot simulations on average are equal the forward prices. However, for the rolling intrinsic hedge the values are not necessarily the same as you can trade the forward contracts dynamically over the time.

9.5.3 Rolling intrinsic value

A rolling intrinsic valuation acknowledges that over time the forward curve will change, and the original intrinsic hedge may not be optimal anymore. At that moment, the existing position can be unwound, and a new optimal position can be locked in. In that way, value is slowly added to the initial intrinsic valuation. The question is how often and when re-balancing should be done. There is no standard consensus in the market. In the platform, the user can set the re-balancing frequency to daily, weekly, and monthly. At each re-hedge date, KySwing performs a MILP optimization to check if a re-hedge generates extra income. This is subject to the condition that there are already positions in place and that the contract constraints continue to be fulfilled. When re-hedging transaction costs (bid-ask) are taken into account.

In almost all cases, the rolling intrinsic valuation converges faster than the spot valuation. For this reason, the user can set the number of rolling intrinsic simulations to a lower number than for the spot simulations.

9.5.4 Perfect foresight spot value

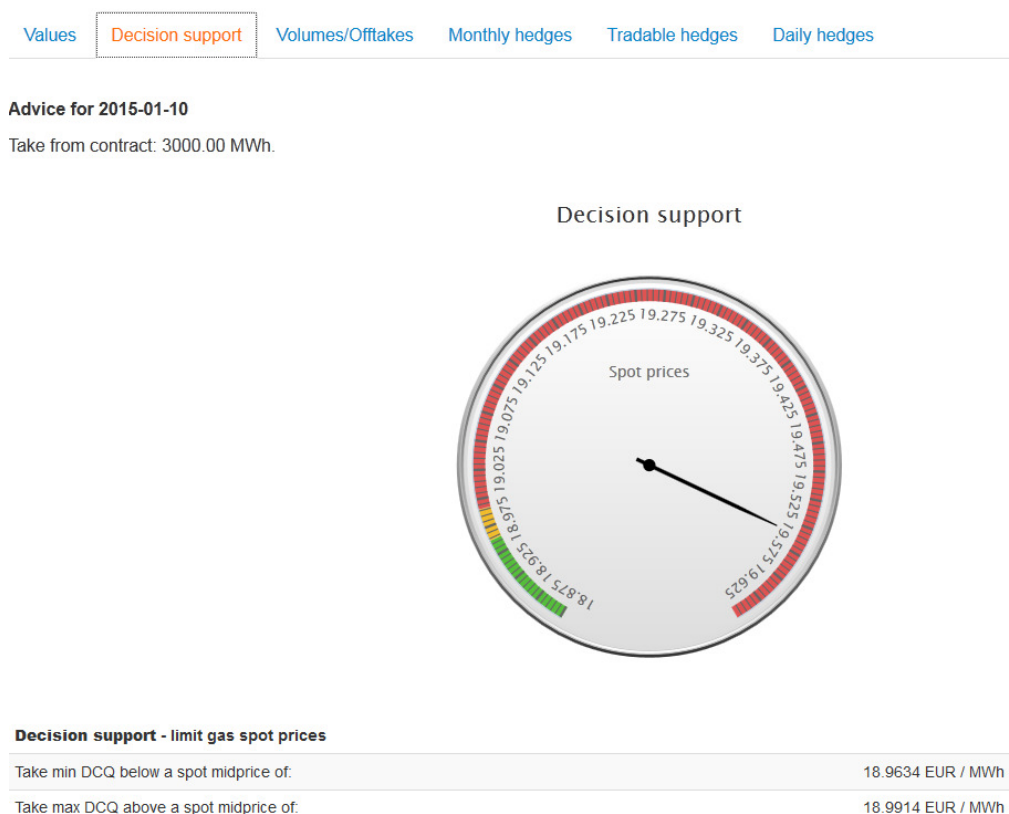
The 'perfect foresight spot value' is derived from an optimization per spot price scenario. For each scenario an optimal set of daily actions is determined (using MILP) assuming the trader knows exactly how prices will evolve in the future. Because perfect foresight is assumed, this amount constitutes an upper-bound on the contract value, and it is a number that could not be realized in practice. However, this is a useful reference amount:

- It helps to judge whether the spot trading value calculated without perfect foresight (derived from LSMC) is away from the theoretical optimum. For example, if it is much lower than the perfect foresight value, then the optimization might be very difficult.
- It allows the incorporation of all contract constraints. This is in contrast to the approximation that is necessary in some cases in the LSMC approach.

9.5.5 Decision support

This output is only available if the contract is operational today, so if the current day is in the delivery period of the contract. Otherwise the tab is missing.

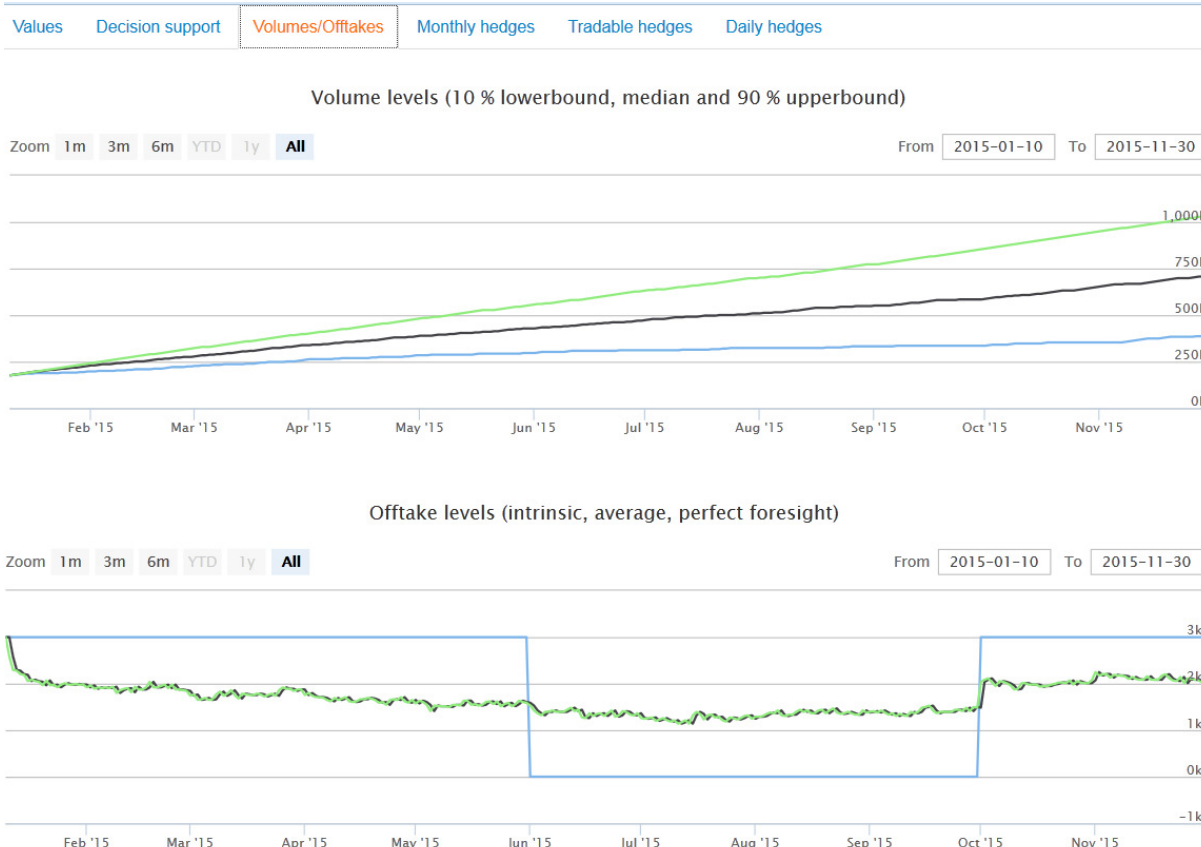
The different contract values are each based on a certain trading strategy. For the spot value (Monte Carlo) and the daily intrinsic value (static curve) this includes an action on the current day. On the current day, the spot price is known. The swing model informs the trader or portfolio manager above which market price it is optimal to take the maximum volume of gas from the contract, and below which market price it is optimal to take the minimum. It also provides the amount of gas the trader should take according to the spot strategy.



9.5.6 Volumes/Offtakes

The expected volume distribution is plotted in the 'Volume levels' graph. The Platform reports the 10% lower bound, average and 90% upper bound across all simulations, based on the spot trading strategy (LSMC).

The offtake levels per day for the different strategies are shown in the 'Offtake levels' graph.



9.5.7 Hedges

The model produces various statistics that help the user to secure the reported values over time. KySwing provides both the intrinsic and the delta hedges:

- Intrinsic hedge: the volumes that should be traded in the market to capture the intrinsic value
- Delta hedge and delta sensitivity:
 - The delta sensitivity is defined as the change in contract value divided by a small change in the underlying price.
 - Based on this number, a trader can take an opposite position in the market ('delta hedge') in order to minimize the risk and move with the market.

Both types of hedges are calculated with three types of granularity of the forward curve:

- Monthly: characterized by the 1st day of the month
- Tradable products: characterized by the 1st and last day of the delivery period
- Daily: single date

Intrinsic hedge

When the user would implement the intrinsic hedges, and take trading costs and discount rate into account, he would secure the corresponding intrinsic value.

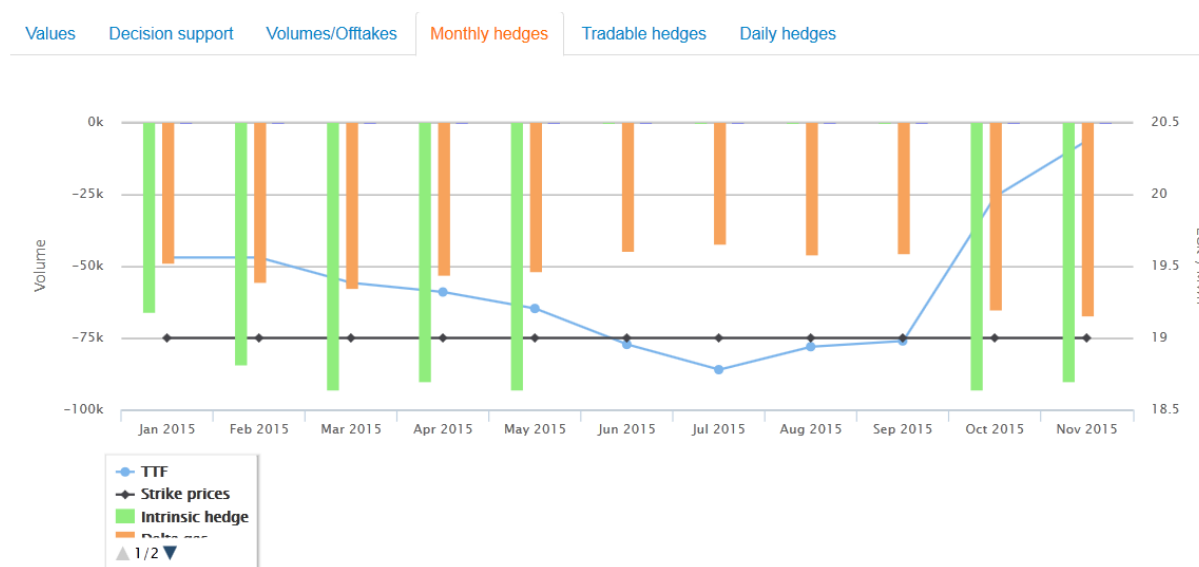
Delta hedge

When the user would implement the delta hedges, and adjust this delta hedge over time, he would stabilize the spot trading value as much as possible. It will narrow the distribution of outcomes and ensure that at least a large part of the extrinsic value is realized. However, there is no full guarantee

that the exact spot value will be generated over the lifetime of the option. For example, the volatility in the market may change.

The delta hedges are calculated for the exposure to the gas price, but also (if applicable) to the other prices underlying the strike price:

- With an oil-indexed contract, there will typically be a negative delta sensitivity to the oil prices: a higher oil price means a lower contract value
- With an oil-indexed contract involving lagging, there will be no delta sensitivity in the last few months: oil prices in this part of the curve no longer affect the strike price
- With an oil-index contract involving an FX-conversion, there is also an FX delta. In the example case, the numbers indicate how many US dollars should be bought as a hedge



Show / hide table

Begin date	TTF (EUR / MWh)	Strike price (EUR / MWh)	Intrinsic hedge (MWh)	Delta gas (MWh)	Delta FX
2015-01-10	19.56	19	-66 000	-48 644	0
2015-02-01	19.56	19	-84 000	-55 517	0
2015-03-01	19.38	19	-93 000	-57 445	0

9.5.8 Additional Text Files Output

SPOT STRATEGY

SpotPricesGas.txt: Contains simulated gas spot prices across time (the rows) and simulations (the columns). The time starts at the valuation date.

SpotPricesStrike.txt: Contains (simulated) strike prices across time (the rows) and simulations (the columns). The time starts at the valuation date.

SpotInv.txt: Contains the offtake level for the spot strategy across time (the rows) and simulations (the columns).

SpotCF.txt: Contains the cash flows for the spot strategy for different dates (the rows) and simulations (the columns).

FORWARD STRATEGY

FwdPricesGas.txt: Column 1: contains the re-hedge date.

Column 2: contains the forward contract's start of delivery date.

Column 3: contains the forward contract's end of delivery date.

The remaining columns contain the simulated forward prices at the hedge date (column 1), for the contract in question.

FwdPricesStrike.txt: The file contains strike prices. The format is similar to the file containing forward gas price.

RollIntrTrades.txt: The file contains the trades associated with the rolling intrinsic strategy. The format is similar to the file containing forward gas price.

RIInv.txt: Contains the offtakes associated with the rolling intrinsic strategy. In each row there is a separate simulation path, and in each column a separate day on the forward curve. The first date is the valuation date.

9.6 Value Report

Above the overview of the KySwing profiles, there is a grey button 'Value report'. The purpose of this button is to export the historical evolution of intrinsic/spot storage value in an Excel file. The user can select in what currency the outputted values should be. A filter containing the desired KySwing profiles can be chosen. Furthermore, the start and end date of historical time series can be selected. The valuation approach behind the outputted swing value can be selected as well. Currently, there are two choices – daily intrinsic and spot strategy value.

9.7 Export Interfaces

Different outputs of the KySwing model can be exported automatically from the KYOS platform. This functionality can be used to integrate the platform with other systems. A KySwing export interface can be created by going to:

tab 'Interfaces' -> tab 'Export' -> button 'Create' -> select 'KySwing' form drop-down.

A description of the content of the different export types is available upon request.

10 KySim – Price simulations

10.1 Price Simulations

KySim is a multi-commodity forward and spot price simulation model for commodity markets and FX rates. It uses a mix of volatility structures, correlations, co-integration, regime-switches, fundamental and statistical inputs. Therefore, it realistically matches the statistical and fundamental characteristics of commodity markets.

The model uses historical forward and spot data to calibrate parameters. This means that price sources for the commodities that are going to be simulated have to be already created. KySim utilizes the forward curves: the average of the simulated prices is equal to the forward prices. Therefore, curve profiles for the relevant commodities have to be created as well.

The results of KySim can be used in other KYOS models: KyPlant, KyStore, KyRisk and KySwing.

The KySim simulation model has two important steps, thus one profile will be used for two separate jobs: calibration and simulation. However, simulation directly depends on calibration, so by creating KySim profile we are actually defining two profiles at once:

- Calibration. The simulation model has parameters in the forward and spot model that need to be calibrated based on historical price data. This can be performed every day or only once a week or month, always using the latest history of market prices.
- Simulation. When the parameters have been estimated, the model can simulate price scenarios. In the simulation profile the user should refer to an existing calibration profile.

10.2 Defining a calibration profile

Analytics → KySim □ Create profile

10.2.1 Monte Carlo

There are two options. By selecting “Use KySim to generate Monte Carlo simulations” a user is required to define calibration and simulation settings for a standard KySim functionality. By selecting “Import your own Monte Carlo simulations” a user is required to have simulations already. In such case KySim runs in the copy-mode, it just reformats the simulations and creates the output files necessary for the proper functioning of other applications, such as KyPlant or KyStore. In the following sections user guide describes the standard KySim functionality. For the usage of copy-mode, see the section on using external simulations.

10.2.2 General Settings

Name

The inputs for a KySim profile must be stored under a unique profile name.

Model base currency

The base currency determines the currency in which all calculations are performed. All commodity prices provided in a currency different from this currency are converted into the base currency with the appropriate spot exchange rate.

Calibration settings: Time steps for correlation

The calculation of the correlation matrix is based on daily returns or returns based on more distant prices. When filling in a higher number (e.g. 5 days) it may be possible to eliminate day-to-day noise; in general this leads to somewhat higher correlations.

Calibration settings: Historical forward window

Calibration of forward prices is performed on the historical time period defined in this field. Hence, if the trading date is 15th September 2016 and the historical forward window is 24 months, then historical prices starting on the 15th September 2014 until the trading date are used to calibrate the forward model.

Calibration settings: Historical spot window

Similar setting as for forward prices: determines the historical window for calibration. It is generally advised to use at least 1 year for the spot calibration.

Calibration settings: Forward volatility model

The deepest level of volatility modelling can use either GARCH(1,1) process or assume constant volatility. It is up to a user to decide. GARCH(1,1) process estimation as well as simulation is very fast and does not result in significantly longer running time. For details where exactly this modelling occurs, see KySim technical documentation.

Monte Carlo
General settings
Commodities settings

Create KySim profile
Back to overview

General settings

Name
New KySim profile
Model base currency
EUR

Simulation settings

End of simulation window
2 years after trading date (2018-12-31)
Spot and forward number of simulations
100
Seed type for simulations
Automatic (random number)
Manual
Schedule calculation daily at a specific time
Yes
No
Automate calculation directly when new inputs are available
Yes
No
Automation type
Simulation

Calibration settings

Time steps for correlation
1 days
Historical forward window
24 months
Historical spot window
24 months
Forward volatility model
GARCH(1,1)
Constant volatility

Hourly power simulations
yes
Start year of hourly random sampling
2014
Generate spreads
No
Yes

Back
Next

Simulation settings: Simulation window

The future period for simulating forward and spot prices, measured in years. Zero years will result in simulations until the end of the current year.

Simulation settings: Spot and forward number of simulations

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Total number of generated forward and spot simulations. Up to around 1,000 simulations will generally work within reasonable time, depending also on the number of commodities and the length of the simulation window. If the combination (multiplication) of these parameters is high, calculation time will be longer and your computer may experience memory problems.

Simulation settings: Seed type for simulations

The random number seed can be fixed ("Manual"), which allows the system to repeat simulation runs with the exact same random numbers. Just fill in a number (which ever) and the random number generator of the simulation engine generates the same numbers in every run. For a truly random seed, the random seed should be set to "Automatic"; then every time you rerun the simulations with the same settings, the scenarios will be different.

Simulation settings: Schedule calculation, Schedule time and schedule type

Simulations can be run automatically, using the scheduling possibility to run it at a pre-defined time on selected weekdays. It is possible to schedule 'Simulation' only or both 'Calibration and Simulation' can be performed at the scheduled time. In case 'Simulation' is selected, the last calibration results are used to generate simulation at each scheduled trading date.

Simulation settings: Automate calculation and automation type

Very similar to schedule, but dependant on the new data. Platform will run Simulation job or Calibration and Simulation jobs whenever new data becomes available.

Simulation settings: Hourly power simulations and Start year of hourly random sampling

When it is selected to generate hourly simulations for power commodities a start year for hourly random sampling has to be chosen as well. It is suggested to have at least two years, though the choice depends on what data (hourly price differences) are assumed to be the most representative for the simulation period. Model will use the last day available in historical data as the end date of the sampling period.

Simulation settings: Generate spreads

After generating Monte Carlo simulations, KySim has the ability to generate user-defined spreads using one or more commodities. A built-in spread analyser uses simulations on a rolling basis and works in a daily granularity. Since spreads are user-defined, e.g. to get a realistic spark spread, you will have to fill in weights based on net calorific values of the fuels that are used for generating electricity and the efficiency of the power plant. A separate tab will be used for defining the spreads, if this option is selected.

10.2.3 Commodities Settings

Select commodities from a list of commodities provided in the drop-down box.

Create KySim profile

Select commodities

Commodity 1	Commodity 2	Commodity 3	Commodity 4
Dutch power - EUR	API2 Coal ARA - USD	TTF - EUR	EUA - EUR
Forward data profile (Base) NL Power FW BL ICE Montel	Forward data profile (Base) API2 Coal ICE FW Montel (20	Forward data profile (Base) NL TTF Gas FW BL ENDEX I	Forward data profile (Base) EU EUA carbon FW EEX Mor
Forward data profile (Peak) NL Power FW PL Endex Mon	Forward curve profile API2 Coal	Spot data profile NL TTF Gas Spot EEX Montel	Forward curve profile EUA Carbon
Spot data profile NL Power Spot APX Montel (I	Half life in years: creates forward mean-reversion 20	Forward curve profile TTF	Half life in years: creates forward mean-reversion 5
Forward curve profile Dutch BasePower KyPF		Spot price model Two-Regimes Normal	Carbon floor profile None
Spot price model Three-Regimes Normal		Half life in years: creates forward mean-reversion 10	
Daily caps (min-max) 0 - 200		☐ Apply monthly weighting for spot volatility	
Hourly caps (min-max) -50 - 500			
Half life in years: creates forward mean-reversion no			

☒ Apply monthly weighting for spot volatility

- Forward profile. This field refers to a previously defined price source. For power commodities both base and peak profiles have to be selected.
- Spot data profile. This field refers to a price source which contains spot data.
- Forward curve profile. Price simulations will be generated around the selected forward curve.
- For power and gas: choose one of six spot models. Recommended models are 'Two-Regimes Normal' for gas and 'Three-Regimes Normal' for power. Since all six models are calibrated, you can select the one, which has the highest likelihood as indicated in the calibration output, but before actually running simulations.
- For power: a lower and upper bound for daily and hourly spot prices.
- Half-life in years: Half-life in long-term prices allows you to make the distribution of simulated forward prices smaller for longer horizons. It will impose a mean-reversion on the simulated prices towards the forward curve. E.g. with a half-life of 5 years, and simulated level 50 above original curve, prices will tend to move to 25 above the original curve in 5 years, on average.
- For power and gas: Apply monthly weighting for spot volatility. Enables the additional settings in the next tab that allow volatility scaling for specific months in spot markets.
- For carbon: Carbon floor profile. Here user can select from profiles defined in 'Carbon floor' section under Commodity settings (see the relevant chapter). As of September 2016, carbon floor policy is applied in the UK power market only. Hence, it is advised to use carbon floor adjustment with profile that includes UK power and EUA carbon futures. In such profile, KySim will create additional simulations for CO₂ prices adjusted for the policy rates. For more details how this adjustment works and what impact does it have, please see KySim technical documentation.

In KySim, all simulated prices are generated in the base currency. In case one or more commodities are quoted in a currency different from the base currency, the commodity prices have to be converted to the base currency. In such a case, FX rates are automatically added to the profile. The spot data and forward curve profiles for the exchange rate have to be specified.

Select FX rates to be simulated (optional) platform currency: EUR | model currency: EUR

FX EUR USD

Spot data profile
FX EUR USD ECB (1999-01-04 2013)

Forward curve profile
EUR USD ECB

10.2.4 Cointegration matrix

The forward model incorporates short-term and long-term cointegration between price levels. The motivation is that time-series behaviour is not sufficiently well described on the basis of daily returns alone. In particular, spreads between contracts of different commodities, e.g. clean spark spread or the spread between two gas markets, often display mean-reverting behaviour in practice.

The power commodities can be simulated following two different methodologies. Following the first methodology, power is treated as any other commodity. The user can define arbitrary cointegration relationships between the power and other commodities. The model assumes by default that power peak and offpeak are cointegrated. To use this methodology, the user should *not* check the 'Use merit order' box.

If the 'Use merit order' box is checked, the model uses a merit order methodology when simulating power. More details about the merit order are presented in the next section.

Cointegration constant

Use a cointegration constant? ☒ Yes ☐ No

Define cointegration relations

	German Power	API2 Coal ARA	NCG	EUA Carbon futures	Use merit order
German Power		☒	☒	☒	☒
API2 Coal ARA	☐		☐	☐	
NCG	☐	☐		☐	
EUA Carbon futures	☐	☐	☐		

10.2.5 Monthly weights applied to the spot volatility

If a user selected 'Apply monthly weighting for spot volatility' for at least one gas or power commodity, the following table is displayed. Adjusting simulated volatility in the simulations is useful when some patterns are expected in the future, due to any kind of fundamental change, but are not yet present in the historical data. For definition of monthly volatility scaling parameters, use numbers larger than 0%. The model will automatically adjust the weights, so that the average is 100%. The resulting weights are then multiplied with the spot volatility (user input or calibrated).

Monthly weights applied to the spot volatility

Use numbers larger than 0%. The model will automatically adjust the weights, so that the average is 100%. The resulting weights are then multiplied with the spot volatility (user input or calibrated).

Commodity	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	%
Dutch power	100	100	100	100	100	100	100	100	100	100	100	100	100

10.2.6 Power market using merit order

When using merit order, power commodities are always assumed to be cointegrated with gas, coal and emissions. So, in order to simulate power prices, the user has to simulate gas, coal, and emissions with it.

The historical relationships between power prices and fuel/CO₂ prices are considered to be weak predictors of the future “stable” relationship for at least two reasons. First, power prices exhibit relatively low cointegration with other commodities and the stable relationship is therefore hard to estimate. Second, the stable relationship will evolve over time due to changes in the market's merit order.

The KySim model assumes that marginal production costs eventually drive power prices. The marginal production costs are calculated for 4 different **representative production technologies**. These production “plants” are specified in terms of the fuel(s) used for “production” of power, the (fixed) efficiency level of the plant and the amount of CO₂ emissions needed. The representative plants are defined with the following efficiencies

- High efficiency coal plant: 40% efficiency with assume increase of 0.2% per year
- Low efficiency coal: 34% efficiency with assume increase of 0.2% per year
- High efficiency gas: 50% efficiency with assume increase of 0.2% per year
- Low efficiency gas: 40% efficiency with assume increase of 0.2% per year

In a real merit order model you would associate every production technology with a capacity, order them by cost and find an intersection with a demand function separately for all peak and offpeak hours. In order to avoid this tremendous increase in complexity KySim uses the following approximation:

- Four **cost slots** are defined in ascending cost order, low being the cheapest, and high the most expensive. Both peak and off peak get a fixed percentage in all 4 cost slots. For more information, we refer to the KySim User Manual. The input screen is the following:

Edit KySim profile: German Power (500 sims)

Merit order German Power

Historical Weight ☒ user definition ☐ model estimation

Fuel	Commodity name	Offpeak historical weight	Peak historical weight
Carbon	ELIA Carbon futures	0.9	0.5
Coal	API2 Coal ARA	0.4	0.05
Gas	NGG	0.25	2

Plant efficiencies

Definition CCGT ☒ LHV ☐ HHV

Energy Conversion Rate
LHV/HHV

Name of plant type	Efficiencies
High eff coal	40 %
Low eff coal	34 %
High eff gas LHV	50 %
Low eff gas LHV	40 %

Production costs

Year		2014	2015	2016	2017
Mean level - OffPeak	Cost price low	50 %	50 %	50 %	50 %
	Cost price med-low	50 %	50 %	50 %	50 %
	Cost price med-high	0 %	0 %	0 %	0 %
	Cost price high	0 %	0 %	0 %	0 %
Mean level - Peak	Cost price low	0 %	0 %	0 %	0 %
	Cost price med-low	0 %	0 %	0 %	0 %
	Cost price med-high	50 %	50 %	50 %	50 %
	Cost price high	50 %	50 %	50 %	50 %

The sum over both 'Peak' and 'OffPeak' cannot exceed 100%, but can be less. These numbers are fixed for the whole simulation run. For every single simulated day the marginal production costs are sorted from lowest to highest and each production plant is associated to an explicit cost slot. Please note the difference between technology and cost slot; the first is allocated to the latter.

The 'expected' power prices are then determined as a weighted average of these sorted marginal costs, plus a constant. To the percentage-weighted marginal production costs a constant is added to make the expected peak and off peak power prices in line with the original forward curve.

If sum over 'Peak' or 'OffPeak' is less than 100%, the original forward curve will have more impact, whereas, the fuel-mix will have less: simulated power prices will be not fully mean-reverting to the fuel mix; consequently, spark spreads will have wider distributions.

10.2.7 Spread settings

You can add up to 5 spread definitions and up to 10 terms in each of them. Spreads are calculated in daily granularity. Forward products that are allowed to be selected, are defined on a rolling basis. You can define spreads between forward and spot terms.

Generate spreads

Spread 1

Clean spark spread M1

	Commodity	Product	Weight
Term 1	Dutch power (Peakload)	M01	1
Term 2	TTF	M01	-2
Term 3	EUA	M01	-0.3408

+ Add term

Spread 2

Clean spark spread Spot

	Commodity	Product	Weight
Term 1	Dutch power (Peakload)	Spot	1
Term 2	TTF	Spot	-2
Term 3	EUA	Spot	-0.3804

+ Add term

Weights must be entered by a user. For spreads involving different commodities, enter weights with respect to different definitions. To make weights reasonable, first take into account the units in which the commodity is being traded, net calorific value of the fuels, and also assumptions about the efficiency of the power plant. For power commodities you can choose from 'Baseload', 'Peakload', and 'Offpeak', however, peak load will result in spread, which is defined during the peak hours only.

10.2.8 Running KySim profile

If the user runs simulations manually (not automatically scheduled), a 'Trading date' will have to be selected. All forward curves for the selected commodities for this trading date are automatically sent to KySim. If the run is scheduled or automated, the 'Trading date' is set to be the current date. The platform will try to find curves defined in Commodities settings specifically for this trading date.

For running manually, you have three options:

- Run only KySim calibration: for days indicated by a green colour, this is the only available option, because all forward curves required to run simulation were not built on those days.
- Run only KySim simulation: only available on days indicated by a light blue colour. Additionally, you have to choose an existing calibration job, which must be from the same or earlier date than the 'Trading date' selected.
- Run both: only available on days indicated by a light blue colour. In this case, a simulation job will be run immediately after a calibration job finished running.

Trading date

Aug

2016

Su	Mo	Tu	We	Th	Fr	Sa
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30	31			

Choose calibration job

job #90566 for 2016-09-02

Run KySim calibration

Run KySim simulation

Run both

In order to take calibration results into account (e.g. which spot models best fits the data) or override calibrated values, run only calibration, adjust profile settings, and run simulation by selecting an appropriate calibration job.

10.3 Using external simulations

Kyos analytical platform allows importing third party (external) simulations. Hence, a user can analyse storages, swing contracts, or power plants based on his/her own simulations, not only those generated by KySim (internal). However, for other Kyos models to accept external simulations, they must be in the same format as internal. For this purpose, a user must create a KySim profile with so-called copy-mode. As of April 2016, platform accepts external simulations from *.csv files only.

10.3.1 Monte Carlo

While creating a KySim profile, external simulations can be set up by selecting “Import your own Monte Carlo simulations” in the first screen called Monte Carlo.

10.3.2 General Settings

Model base currency

In external simulations case, KySim assumes that all simulated commodity prices are supplied in one currency: model base currency. No internal conversions are performed. However, if your profile includes commodities in another currency than model base currency, you will need to supply simulations of spot exchange rates between those two currencies as well.

10.3.3 Commodities Settings

In this screen a user can add commodities to a profile in the same way as with a regular Kysim profile. In this case, however, there are only two (four in case of power commodity) fields for every commodity to

fill. First select a commodity from the dropdown list of all commodities registered on the platform. Then fill in the name of the CSV file that contains simulations (both spot and forward) for this (non-power) commodity. Simulation for each commodity added will have to be in a separate CSV file. For a power commodity three files must be supplied: 1. forward simulations (base); 2. forward simulations (peak); 3. hourly spot simulations. A user is expected to supply all those files in a single ZIP folder while running the profile. The following section explains the format of these files.

It is important to note, that if a commodity is registered in a currency, which is different from model currency and (or) platform currency, a user will also have to supply simulations for the exchange rate between two currencies in a separate CSV file.

There are no other settings to be defined. A button “Save” will create a profile.

10.3.4 Format of input files

First row in all files is the header row with column names. Column names are important for identifying forward / spot products. It is necessary to use the standard product names as indicated below. Symbol # indicates an integer number in the columns name. For instance, “M+#” is a name for month-ahead products. “M+1” indicates 1 month-ahead forward contract, “M+2” indicates 2 months-head forward, etc. “WK+#” is the name layout for week-ahead products, “WKND+#” is the name layout for weekend-ahead products.

Non-power commodity or currency file

Column 1: “Date” – Excel date

Column 2: “SimNumber” – index for simulations

Columns 3-N: “Spot”, “BOM”, “M+#”, “WK+#”, “WKND+#” – products, the order of columns here is not important.

Power commodity base or peak file

Column 1: “Date” – Excel date

Column 2: “SimNumber” – index for simulations

Columns 3-N: “BOM”, “M+#”, “WK+#”, “WKND+#” – products, the order of columns here is not important. KySim expects “WK+#” and “WKND+#” products in the file for base load power only

Power commodity spot file

Column 1: “Date” – Excel date

Column 2: “SimNumber” – index for simulations

Columns 3-26(50): “#” – integer number indicating hourly (half-hourly) period in a day

Note 1

Simulations are sorted in the ascending order by first two columns, first by “Date”, then by “SimNumber”. Each commodity or each CSV file can have a different number of simulations. KySim will pick the minimum number of simulations available in all files.

Note 2

KySim assumes no gaps in simulation period. Thus, external simulations must include every day between trading day + 1 and end of simulation period, including weekends and holidays. Respectively, hourly simulations must include every hour, but no daylight saving time adjustment is assumed.

10.3.5 Running a profile with external simulations

In case of KySim profile for external simulations, the “Run” screen with calendar will look a bit different than the usual KySim “Run” screen.

First, it is important to select a trading date, which is suitable for all the commodities in a profile, i.e. every CSV file you are uploading must start maximum one day later than the trading date you selecting on the calendar. Note, that simulations for various commodities can have different length. For example, let’s assume you are creating a profile with two commodities. For the first commodity, simulation dates start at 21 April 2016 and end at 31 March 2018. For the second commodity, simulation dates start at 11 April 2016 and end at 31 December 2017. Thus, the first day you can select as a trading date for such a profile is 20 April 2016. KySim will automatically detect what is the maximum possible profile length and will use it for all commodities included in a profile. In our example, the end of simulation period will be 31 December 2017.

Second, a ZIP archive must be uploaded with all the CSV files indicated in a profile. In general, the name of ZIP archive does not matter. When a user saves the profile, names of CSV files in that profile cannot change, whereas the name of ZIP archive can be different every time the profile is run. It can include a trading date. Consequently, it is possible to set up automatic uploading of the ZIP archive and running of the profile based on the ZIP filename. Contact the IT support of Kyos Energy Consulting to set up such environment.

After a job finished running, a user will have an options to download or delete it, but no option to view.

10.4 Calibration and Simulation results

Calibration and Simulation results can be seen by clicking at ‘View’ in the main menu (Analytics -> KySim).

Analytics → KySim □ View

Then a list of jobs relevant for a specific profile is presented.

10.4.1 Calibration output

Selecting ‘View’ of a particular calibration job, estimated parameters and statistics will be displayed on screen in tabs ‘Forward output’ and ‘Spot output’. The graphs of the calibration job, found in tab ‘PDF output’, provide a good check on the availability and quality of the historical spot and forward data. The list of graphs:

- Forward calibration
 - Plots of historical forward prices for year-ahead, month-ahead and winter-summer ahead. All prices are rescaled to average of 100 to be easily visually comparable.
 - Plots of returns for year-ahead, month-ahead and winter-summer spreads.
 - Plots of Availability of historical data for all commodities for month-ahead, year-ahead and for winter-summer is presented to the user.
 - Plots showing the result of the estimated cointegration.
 - Volatility terms structures of M1, M2 and Y1 products for all commodities. The volatility term structure should follow a Samuelson effect, i.e. with increasing time to maturity the should volatility decrease.
 - Annualized volatilities for all commodities as a function of maturity.
 - Estimated factor returns
- Spot calibration
 - For the spot model, historical spot and month-ahead log-prices and spreads are saved for all power and gas commodities.

Additionally calibration results include tab 'Statistics output', with relevant statistics concerning the estimation. If user have used GARCH(1,1) modelling in three factor model for forward prices, the table 'Forward model loglikelihoods before & after GARCH(1,1) estimation' contains likelihoods of constant volatility and conditional volatility modelling, as well as Likelihood ratio (LR) test results. The second table 'Spot model likelihood ratio tests and p-values for nested models' shows the comparison between various nested volatility modelling approaches and uses a simple algorithm based on LR test, likelihood, and number of parameters to suggest the best fitting model for each of the commodities.

10.4.2 Simulation output

Selecting 'View' of a particular simulation job, realized volatilities (annualized) for simulated commodities as a function of maturity will be displayed on screen. In general, they should also display Samuelson effect. Moreover, curve of realized volatility of a power commodity should be seasonal.

Additionally simulation results include the table 'Realized correlations of the rolling contracts' in the 'Rolling correlations' tab. Correlations between all commodities are displayed for two benchmark simulated time series: 1-month-ahead and 1-year-ahead. Correlations of all rolling series that are generated by KySim are available in the job files.

Tab 'Graphs' contains:

- Plots of simulated prices for the Forward model.
- Plots of the quantiles of simulated spot price distributions for the Spot model.
- Plots of user-defined spread distributions (if spread option was selected).

10.4.3 Downloading results

It is possible to download the statistics displayed on screen by selecting 'Download Excel' in the job screen. Graphs in *.pdf format for a specific job can be downloaded in the tab showing these graphs by

clicking button 'Download PDF'. To download the complete set of simulations including the graphs and statistics, go to 'Logs' and select 'Download' from the job menu.

10.4.4 Using overrides

You can override parameters calibrated in the Forward model. It might be very useful, if the calibrated parameters for some reason do not correspond with what you perceive as the normal level of volatility for a particular commodity. Overrides can be also used to investigate some more extreme price behaviour patterns. However, be aware that the realized volatility of Monte Carlo simulations is a nonlinear product of the factor volatilities. Thus, use overrides with care and check the realized volatilities in the simulation job outputs afterwards.

By viewing the calibration job profile you will see inactive override cells, containing the same values as calibrated parameters:

	Annualized volatility					
	Factor 1		Factor 2		Factor 3	
TTF	23.978%	23.978	21.272%	21.272	2.936%	2.936
API2 Coal ARA	16.162%	16.162	22.154%	22.154	0.005%	0.005

Begin with checking the box: ☐ Use manual override values instead of calculated parameters

Cells will activate:

	Annualized volatility					
	Factor 1		Factor 2		Factor 3	
TTF	23.978%	23.978	21.272%	21.272	2.936%	2.936
API2 Coal ARA	16.162%	16.162	22.154%	22.154	0.005%	0.005

Enter override values and finish by clicking:

Add override settings

You can also override mean-reversion in long- or short-term cointegration:

Mean reversion		
TTF	0%	0
API2 Coal ARA	0%	0
EUA	0%	0
Dutch power Offpeak	1.069%	1.069
Dutch power Peakload	0.450%	0.450

However, if you change values for relationships that were not estimated, they won't be used. In the example above, TTF has 0% of mean-reversion, this means it was not estimated. Consequently override will be saved but will not be used in simulation.

When you 'Add override settings', next time KySim simulation will be run with those settings. Settings are added to all calibration jobs in the current profile. If you run KySim calibration, override settings will be copied to the new job as well. To sum up, override settings are not job- but profile-dependent.

11 KyPlant and KyPlant txt

Analytics → KyPlant txt / KyPlant

The KyPlant software determines the value of (a portfolio of) power plants by quickly calculating the optimal dispatch over one or more price scenarios. The real options-based model combines actual plant behaviour with realistic price simulations. The model quickly identifies plant value, supports trading decisions and reports exposures.

KyPlant incorporates the most important physical and contractual plant constraints:

- Operational constraints: maximum capacity and minimum stable level, minimum runtimes, ramp rates, efficiency curve from minimum to maximum level
- Costs: fixed, variable operations & maintenance costs, inflation
- Start costs: fixed and variable fuel costs for cold, warm & hot starts, start buckets to increase costs with more starts
- Periodic: valuation horizon, seasonal influences on capacity and efficiency, impact degradation on efficiency and capacity
- Outages: planned maintenance schedule (fixed or based on equivalent operating hours), forced outages
- Contractual constraints: Gas Take-or-Pay contract (for CCGT), constrained load

The output of KyPlant includes:

- Intrinsic plant value and real-option value
- (Half) Hourly dispatch schedule
- Detailed statistics: P&L, cash-flow, power production (full load & part load), fuel consumption, emissions, starts, equivalent operating hours, outages, costs, etc.

11.1 Defining a power plant valuation profile

Analytics → KyPlant txt □ Create profile

General settings

Profile name

A unique name of the KyPlant profile

Start and End date

Start and end date of plant valuation

Use simulations

Yes/No. Without simulations, only an intrinsic calculation is performed.

Hedging strategy

Select the hedging strategy to be used in the KyRisk model.

Schedule and Automate

Specify if the valuation is supposed to run at a specified time or if it is automated. Automated means that the model runs automatically when new forward curves or new price simulations are available.

Settings

Obligations

General settings

Name

TEST Gas plant 234MW ⓘ

Start date

2017-01-01

End date

2017-12-31

Use simulations

☒ Yes
 ☐ No

Hedging strategy

None ▼

Schedule

Schedule profile

☐ Yes
 ☒ No

Automated

☒ Yes
 ☐ No

Simulation settings

If simulations are used for the plant valuation, the following fields have to be filled.

Select simulation profile

Based on the power commodity of the power plant, a list of simulation profiles is displayed in the drop-down window.

Number of simulations

The default number of simulations is equal to the number of simulations in the selected simulation profile. If you fill in a lower number, then not all simulations are used. If you choose a higher number, KyPlant will use only the number of simulations which are available.

Optimization method

The power plant dispatch is optimized by a) dynamic programming (DP), assuming perfect foresight about future price levels per scenario, or b) with Least Squares Monte Carlo (LSMC) method, avoiding such perfect foresight. The LSMC optimization always leads to a somewhat lower value than dynamic programming.

Commodity settings (for KyPlant txt only)

Specify the fuels and start commodities.

Initial and end conditions

If not specified otherwise, the power plant is OFF and in a cold state on the beginning of the valuation. That means that the moment the plant starts, it follows the cold start curve. On the other hand, the end state is determined by the optimization.

The KyPlant model allows to specify the start and end conditions. For the start condition, the user can specify how many hours the plant is running, how many hours the plant is off or in which start curve it is currently running. There is an optional input to specify the end condition, where the user can select if the plant ends in an OFF state or in an ON state.

Select simulations

Simulation profile	TEST German power (250 si) ▼
Number of simulations	250
Optimization method	Dynamic programming ▼

Initial condition

☒ Off	1000	Hour
☐ Start curve	1 ▼	
☐ On	1	Hour

End condition

Define end condition	☒ Yes ☐ No
☐ Off	
☒ On	

[Next](#)

Curve settings

If only intrinsic valuation is performed, the user has to select power, fuel and relevant start and boiler commodities from the list of all available forward curves.

For KyPlant txt only: select also relevant currencies. It is required to select either gas or coal forward curve. Start commodities and currencies are optional.

Curve settings

Power	German Power
Gas	NCG
CO2	EUA Carbon
Start commodity 1	NCG

Obligations

Must-run

In case forward commitments have been made with regard to power production, for example through forward sales, such commitments may be accounted for in the KyPlant model. Here one may specify if only the committed load should be run (*Run at predefined production level only*), or if additional capacity should be traded on the market (*Optimize on market, given predefined production level as minimum*).

Must-run

Must-run types	☐ No must-run obligations
	☒ Run at predefined production level only
	☐ Optimize on market, given predefined production level as minimum

Predefined production time series	Must Run GE 2016 - 2017
-----------------------------------	--

Heat demand

Heat demand type	☒ Static value
	☐ Define by time series
	200 MWth

If the must-run obligations are selected, select the predefined production level from the drop-down list of the time series.

Heat demand

KyPlant also allows for modelling of heat obligations. If CHP has been defined in the power plant profile, then in the KyPlant profile, the user has to specify heat obligations either as a static value or as a time series.

11.2 Outputs KyPlant

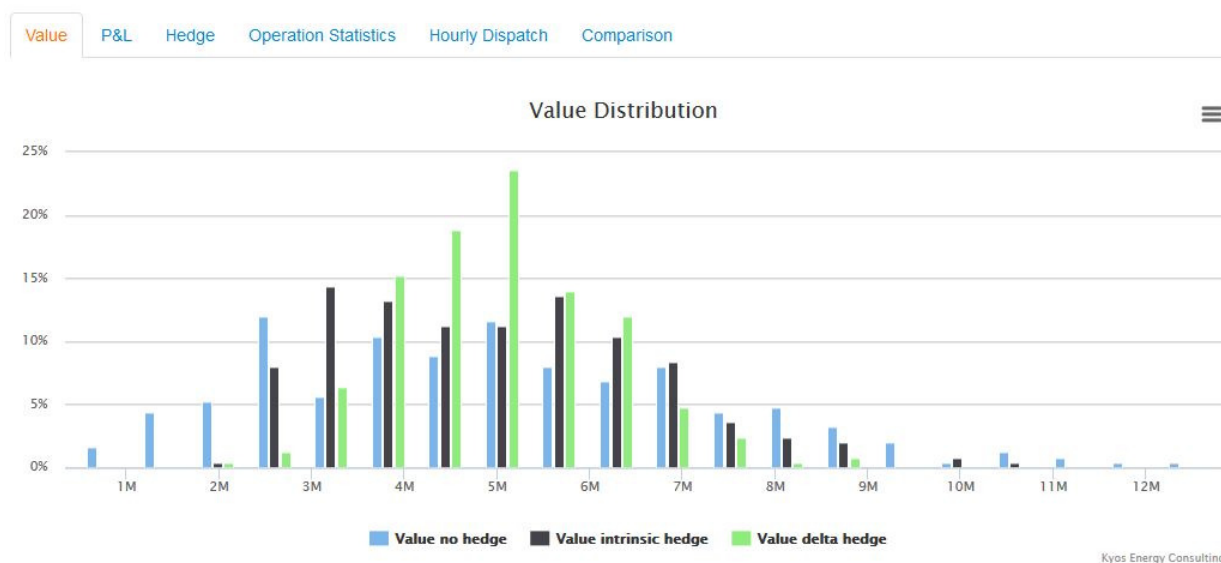
Analytics → KyPlant txt ☐ View ☐ View

Various outputs statistics are stored per calculation:

- The plant value, including distributions
- Monthly and yearly P&L for intrinsic run as well as average across all simulations
- Delta and intrinsic hedge
- Operation statistics
- Hourly dispatch for intrinsic run as well as average across all simulations
- Comparison

11.2.1 Value distribution

In the first tab, you can see the histogram showing the distribution of plant values for all simulations. Histograms showing the values including a static intrinsic or static delta hedge show the narrowing of the distribution after hedging based on hedging decisions calculated by KyPlant and presented in tab Hedge. Additionally, table containing the intrinsic, simulation and extrinsic values is presented.



Power Plant Value

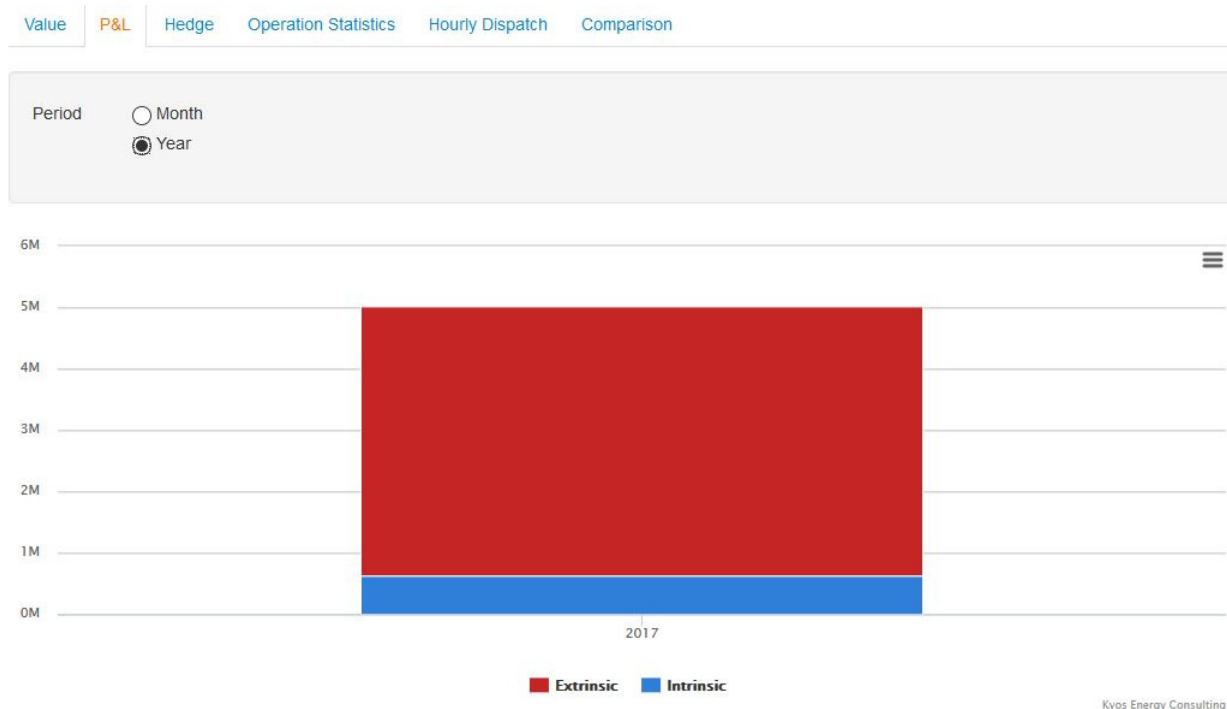
Intrinsic Value	EUR	631 101
Simulation Value	EUR	5 017 696
Extrinsic Value	EUR	4 386 595

11.2.2 P&L

In the P&L graph and table the monthly or yearly, intrinsic and simulation values are presented. Income for power generation, heat income, intraday income, income from imbalance market and avoided grid

costs is presented per chosen period in the power commodity currency. The costs consist of the fuel costs, start fuels costs, carbon emission costs, gas booking costs, variable O&M, fixed start costs, fixed O&M costs and transportation costs. The total plant value in a period is equal to income minus costs.

The Extrinsic plant value is equal to Simulation total value minus intrinsic total value.



Intrinsic		2017
Income		
Power	EUR	12 103 896
Heat	EUR	0
Intraday	EUR	0
Imbalance	EUR	0
Avoided Grid Costs	EUR	0
Costs		
Gas	EUR	9 119 354
Coal	EUR	0
Fuel starts	EUR	1 291 981
Carbon	EUR	583 259
Gas Booking	EUR	0
Variable O&M	EUR	387 120
Start Costs	EUR	91 080
Fixed O&M	EUR	0
Transport Costs	EUR	0
Total	EUR	631 101

11.2.3 Hedge

In the third tab, the recommended intrinsic and delta hedge for each month over the valuation period is presented. Delta sensitivity (is equal to minus the delta hedge) is defined as the change in plant value divided by a small change in the commodity price. Executing the delta hedge volumes in the market limits the price exposure and more certainty than the full plant value without hedge is actually realized.

11.2.4 Operation statistic

Operation statistic for monthly or yearly period and for intrinsic or simulation value are presented in this tab.

The technical characteristics include the number of peak and offpeak operation hours, the total number of operation hours. Furthermore information on load factor, number of maintenance hours, trip duration, number of starts in selected period and number of gas bookings is shown here.

Value	P&L	Delta Hedge	Operation Statistic		Comparison				
Period	☒ Month ☐ Year		Type	☒ Intrinsic ☐ Simulation					
Technical Characteristics		May 2014	Jun 2014	Jul 2014	Aug 2014	Sep 2014	Oct 2014	Nov 2014	Dec 2014
Peak OH	#	211.00	244.00	132.00	84.00	264.00	276.00	240.00	180.00
Offpeak OH	#	142.00	179.00	50.00	62.00	203.00	140.00	194.00	30.00
Total OH	#	353.00	423.00	182.00	146.00	467.00	416.00	434.00	210.00
Load factor	#	51.71%	53.53%	22.42%	17.18%	59.66%	50.21%	54.38%	24.64%
Maintenance	[hrs]	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Trip duration	[hrs]	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Starts	[#]	26.00	29.00	13.00	13.00	31.00	31.00	32.00	15.00
Gas Bookings	[#]	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Baseload show / hide									
Production	[MWh]	138,059	167,645	72,570	55,614	186,856	162,515	170,321	79,754
Heat Plant	[MWh]	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Heat Boiler	[MWh]	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Gas consumption	[MWh]	254,074	309,678	134,087	101,679	346,296	298,572	313,934	145,264
Gas Boiler	[MWh]	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Coal Consumption	[MT]	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Start fuel 1 (TTF)	[MWh]	7,400	7,600	3,200	3,800	7,200	9,200	8,600	6,000
CO2 emissions	[MT]	47,427	57,678	24,966	19,056	64,381	55,779	58,552	27,276
Peak show / hide									

Separately for baseload, peakload and offpeak the following information is summarized:

- The number of MWh hours of power production
- Heat produced by the power plant or by boiler
- Fuel consumption, start fuel consumption and CO2 emissions

11.2.5 Hourly dispatch

Detailed hourly dispatch for the whole valuation period for the intrinsic run as well as average across simulations is presented in this tab. Additionally, it is possible to export the hourly dispatch to Excel.

Graph ☒ Intrinsic ☐ Simulation

Export hourly dispatch

Hourly dispatch



11.2.6 Comparison

Comparison between two runs- the current run and one past run is presented in a set of graphs and tables. For each month, the values are compared for the following:

- Plant value
- Base and peak spreads
- Power production
- Plant availability
- And number of starts in each month

11.3 Power plants (using Platform interface)

Assets & Contracts → Power plants

There are two ways in the KYOS Platform to define the characteristics of a power plant:

- In a pre-formatted Excel file, which is exported to txt and then uploaded into the Platform. Only a few input fields are defined in the Platform interface.
- With all input fields defined in the Platform interface.

Depending on the user only one of the two approaches for plant definition may be available. In this section we explain the second approach, in the previous section the first approach.

General characteristics

Name

The inputs for a power plant can be stored under a unique profile name.

Plant physical location (Optional)

The address of the physical plant can be specified and is used to show the plant in Google Maps.

Legal entity

The name of the legal entity.

Start date / end date

If the plant is used in KyPlant, then the valuation can only be in between the start and end date.

Granularity

This defines the level of granularity for defining the power plant characteristics. It will determine whether KyPlant is based on an hourly or less than hourly granularity. At the moment, only hourly and half-hourly are supported, not quarter-hourly.

	General	Commodities	Technical	Seasonality
Name	TEST Gas plant 234MW			
Plant physical location	Enter address, placename or postcode			
Legal entity	kyos			
Start date	2000-01-01			
End date	2037-12-31			
Granularity	Hour			

Commodity inputs

General	Commodities	Technical	Seasonality	Start curves	Maintenance & degradation	Grid losses
Power commodity	German Power					
Fuel type	Gas	Fuel commodity multiplier	1			
Fuel commodity	NCG					
CO2 commodity	EUA					
Start commodity 1 (optional)	NCG					
Start commodity 2 (optional)	Please select...					
						Back Next

Fuel commodity multiplier

The fuel price is multiplied with this value. In most cases, it is the best to keep the default value of 1.

Technical characteristics

General	Commodities	Technical	Seasonality	Start curves	Maintenance & degradation	Grid losses
Production capacities (MWe)						
Minimum stable level (SEL)	60	MWe				
Maximum generation level (MEL)	234	MWe				
Midpoint generation level (optional)		MWe				
Efficiency of the plant						
For gas this is LHV.						
Efficiency curve	Complete curve					
Minimum stable level (SEL)	35	%				
Maximum generation level (MEL)	55	%				
Midpoint generation level (optional)		%				
Other technical plant parameters						
Minimum ontime	4	Hour				
Minimum offtime	4	Hour				
Use ramp speeds	☒ Yes ☐ No					
Ramp-Up speed	5	% of max capacity / minute				
Ramp-Down speed	5	% of max capacity / minute				
Variable operational costs per MWh	☒ Static value: 0	EUR/MWh	☐ Defined by time series			
Variable operational costs per Hour	☒ Static value: 240	EUR/Operating Hour	☐ Defined by time series			
Fixed operational costs	5.6	EUR/kW/Year				
Is CHP	☐ Yes ☒ No					
						Back Next

Minimum and maximum production capacity

This determines the electrical output of the plant at minimum stable generation and at maximum generation level.

Efficiency curve

Indicate type of curve information to be considered in the optimization. Here one may indicate to use the full curve based on a quadratic regression, or only (a subset of) the indicated set points. (Complete curve, Min/Max Capacity Only, Max Capacity Only). Note that all efficiencies are in LHV.

Min and max efficiency

Power plant efficiency at minimum stable generation and Power plant efficiency at maximum capacity.

Mid capacity and mid efficiency

Intermediate point of plant output between min and max capacity and power plant efficiency at this mid capacity. This is used to fit a quadratic curve through the minimum, maximum and mid load efficiencies / capacity points.

Minimum ontime

Minimum production time for which a power plant has to be ON, i.e. producing power. This is measured in the granularity of the power station, e.g. hourly or half-hourly.

Minimum offtime

Minimum time between two production phases.

Use ramp speeds

Indicate if ramp speeds are to be considered in the valuation (as a post-process).

Ramp-Up speed

Speed to ramp up between minimum stable generation to maximum capacity.

Ramp-down speed

Speed to ramp down between maximum capacity to minimum stable generation.

Variable operational costs

Can be used for all costs which vary with the production output (in MWh) or production hours (in hours).

The OM costs can be defined as a value or as a time-series. For time-varying OM costs, the user first has to upload the O&M time series into Time Series -> Production, Demand and Weather.

Fixed operational costs

Can be used for all costs which are fixed over time. These costs do not affect the optimal dispatch, are only deducted at the end of a KyPlant valuation.

Fixed start costs

This version of plant definition assumes there is just one start type, i.e. without distinguishing between warm, hot and cold starts. Furthermore, a start is simply characterized by a fixed cost per start.

Gas Booking

Gas booking is an additional cost per day applied when the plant consumes gas. For each individual gas day, when power plant consumes gas for 1 or more hours, i.e. when the plant runs, starts or is shutting down, a capacity payment has to be paid. This payment is subtracted from the margin on the first hour of the gas day during which gas is consumed.

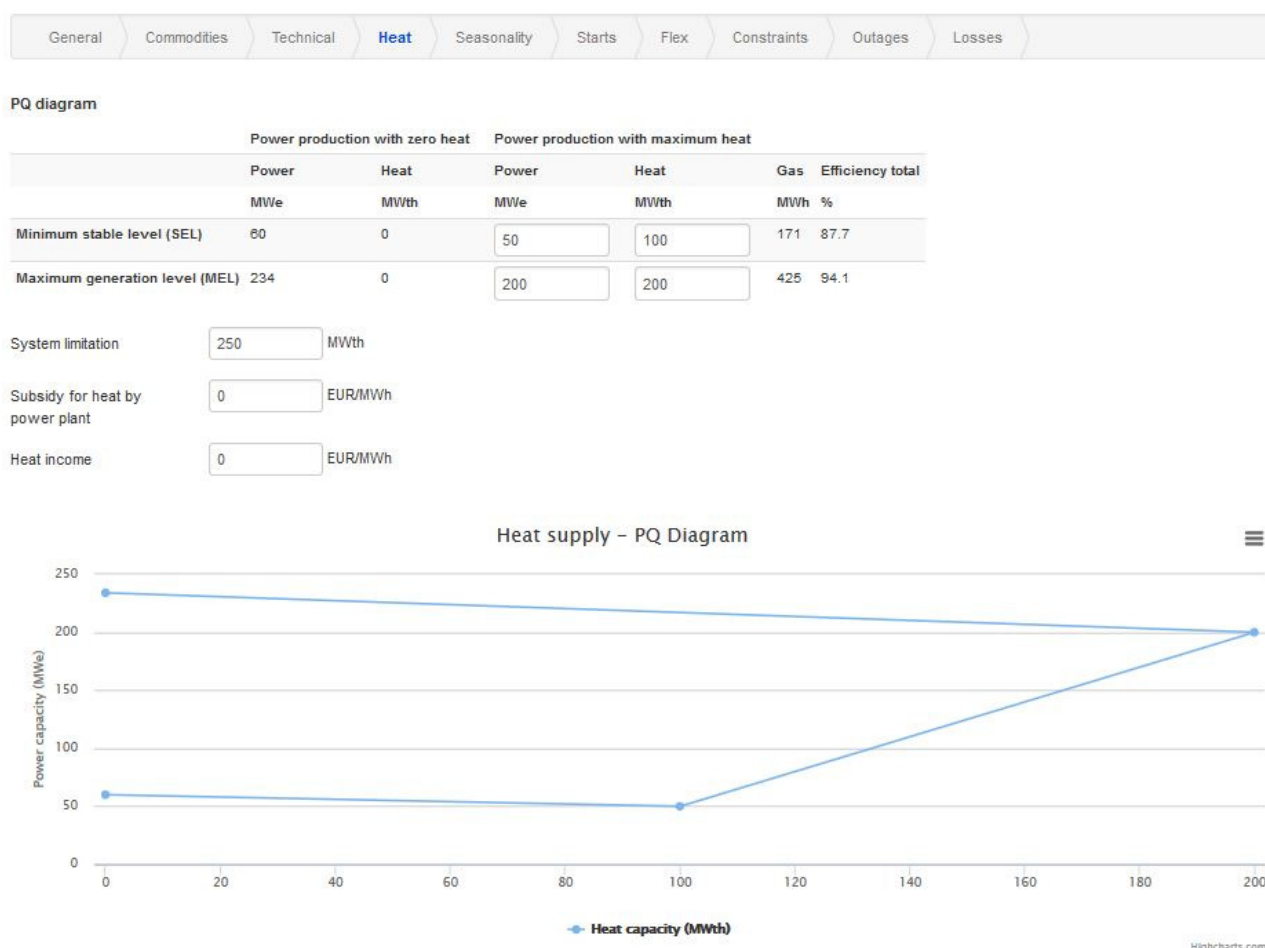
Gas booking costs can be defined as a value or as a time-series. For time-varying Gas booking costs, the user first has to upload the Gas booking time series into *Time Series -> Production, Demand and Weather*. Note that unlike the other time-varying costs which are hourly or half-hourly, the gas booking costs are always daily.

The Gas booking costs are sometimes applicable in the German market.

Is CHP

This input is not yet actively used, but already defined for later use.

Heat



The Heat table allows for specification of the power plant's PQ diagram. Minimum stable level (SEL) and maximum generation level (MEL) decrease, when heat is generated by the power plant. In the PQ diagram table, the user specifies 2 additional points - the new SEL point and the new MEL point with maximum possible heat at these production levels. In addition, heat transport limitations may be imposed. This system limitations impose a maximum on the total heat production.

In some markets, the power plant owner can receive a *subsidy for heat produced by the power plant*. This subsidy influences the plant optimization, since the subsidy makes heat production by power plant cheaper than without the subsidy.

On the other hand, *heat income* applies to both heat produced by power plant and heat produced by boilers. The heat income does not influence the optimization.

Boilers

Commodity	Max capacity	Efficiency (%)	Production costs
☒ Brent - USD	100	MWth 85	0 USD/Barrel
☒ NCG - EUR	100	MWth 90 LHV	5 EUR/MWth
☒ cost only	150	MWth	25 EUR/MWth

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As an alternative to heat produced by the power plant, heat can also be produced by boilers. There are up to 3 boilers that can be specified. For each boiler the user has to provide maximum capacity, efficiency and production costs. For gas boilers, the *efficiency* is defined as LHV efficiency. *The production costs* are added on top of the market price of the fuel used in the boiler.

User can specify any commodity as the boiler fuel. Additionally, he can select “cost only” instead of commodity. This selection means that a fixed price is applied for each MWth of heat generated.

Seasonality

Both efficiency and output of a power plant may show seasonal variations, especially due to variations in ambient temperatures. Monthly average corrections for such variations can be defined by the user. Corrections for seasonal variations in efficiency are directly applied to the parameterization of the efficiency curve, while corrections for output are applied to minimum and maximum output.

Start curves

Number of start curves

Limit number of starts ☐ Yes ☒ No Not used with LSMC

Curve number	Start hour	Curve length	Equivalent operating hours
	Hour	Hour	#
1	1	5	3
2	10	7	8

Fixed costs (EUR)

Curve number

1	960
2	1980

Fuel consumption start curve 1

	Power avg	Power end	Start 1: NCG	CO2: EUA
Hour	MWth	MW	MWth LHV	MT
1	9	18	60	13
2	35	52	75	16
3	106	160	194	40
4	180	200	370	78
5	220	240	450	93

Start curves

Number of start curves has to be specified. These curves should range from hot start curves to cold start curves. For each of these curves several inputs are required:

- Definition of each start type in terms of the number of hours after shut down

- The number of Equivalent Operating Hours (EOH) for each start. Note that the EOH may determine at what dates maintenance takes place.

- The fixed costs per start. These costs may vary per year.

- A specification of average and final power production as well as consumption of up to 3 start fuels during start up for each time unit of the start-up profile

Shut-down

Shut down profile can be defined. A specification of average and final production and fuel consumption during shut down for each time unit of the shut-down profile is required.

Shutdown curve

Use shutdown curve ☒ Yes ☐ No

Shutdown curve length Hour

Shutdown fuel consumption

	Power avg	Power end	NCG
Hour	MWh	MW	MWh LHV
1	15	0	28

Flex

Additional load point

For specific installations, provisions may have been created to avoid shut down. In such cases it may be allowable to reduce output to a fixed point between minimum stable generation and full shut down. Alternatively, for specific plant, there may be a fixed point above the maximum generation level. The Additional load point definition allows for accommodating both plant specifications. This means that another state is introduced into the state space with a fixed generation level, associated efficiency and associated minimum ontime.

General
Commodities
Technical
Seasonality
Starts
Flex
Constraints
Outages
Losses

Include additional load? ☐ No ☒ Yes, point ☐ Yes, range

Additional load point definition

Production capacities (MWe)

Maximum generation level (MEL) MWe

Efficiency of the plant

For gas this is LHV.

Maximum generation level (MEL) %

Other technical plant parameters

Minimum ontime Hour

Additional load range

This additional load reflects the states when the plant produces with 2 GTs instead of 1 GT. When compared to the additional load point, this setting allows for more flexibility in defining the extra range of production.

General
Commodities
Technical
Seasonality
Starts
Flex
Constraints
Outages
Losses

Include additional load? ☐ No
☐ Yes, point
☒ Yes, range

Additional load range definition

Production capacities (MWe)

Minimum stable level (SEL) MWe

Maximum generation level (MEL) MWe

Midpoint generation level (optional) MWe

Other technical plant parameters

Minimum ontime Hour

Minimum offtime Hour

Additional operational costs per Hour ☒ Static value:
 EUR/Operating Hour
☐ Defined by time series

Efficiency of the plant

For gas this is LHV.

Minimum stable level (SEL) %

Maximum generation level (MEL) %

Midpoint generation level (optional) %

Minimum stable level, maximum generation level when producing in the additional load with 2 GTs, as well as corresponding efficiencies have to be defined.

Minimum ontime specifies at least how many hours should the plant run in the additional load. When this minimum period is reached, the plant can continue producing at the additional load point, it can return to the production level of 1GT or the plant can completely shut-down.

Minimum offtime determines at least how many hours should the plant stay off or should the plant run with only 1 GT after leaving the production level of the additional load range.

Additional operational costs per hour

These costs are the additional operating costs per operating hour for running in the additional load.

Start curve

Curve length	Equivalent operating hours
Hour	#
2	1

Additional fixed start costs (EUR)

2016	+
980	

Total fuel consumption

	Power avg	Power end	Start 1: NCG
Hour	MWh	MW	MWh LHV
1	280	300	560
2	325	350	650

Shutdown curve

Shutdown curve length	1	Hour
-----------------------	---	------

Shutdown fuel consumption

	Power avg	Power end	NCG
Hour	MWh	MW	MWh LHV
1	100	0	200

Back Next

Start curve

One start curve can be defined, representing the ramp-up from one GT to two GTs mode. When the second GT starts, there is no flexibility allowed in the first GT, i.e. the first GT has to be at MEL. Therefore, the fuel consumption and power output during the start of the second GT are the total consumption and total output of the first and the second GT combined.

Shutdown curve

The shutdown curve represents the ramp-down trajectory for the whole plant, i.e. two GTs going OFF.

Constraints

Limit the number of starts or Limit the number of operating hours

An optimal schedule, as generated by the optimization module in KyPlant, will allow any number of start-ups and any number of operating hours per year, if this is economically justified. This can lead to a high number of starts (or operating hours), which may be justified from a financial perspective but may not be in line with common plant technical dispatch restrictions, VPP constraints or operating conditions that are not explicitly modelled.

In KyPlant a functionality has been developed that allows for limitation of the number of starts or number of operating hours. These constraints can be defined per year.

General
Commodities
Technical
Heat
Seasonality
Starts
Flex
Constraints
Outages
Losses

Constraints
Not used with LSMC

☐ No constraints
☒ Limit the number of starts
☐ Limit the number of operating hours

Year
Max number of starts

2017
150
+

Back
Next

To incorporate the limitation of the number of starts or operating hours, the plant dispatch optimization will need to be evaluated multiple times.

Maintenance and degradation

During maintenance, the capacity of the power plant is set to zero. Maintenance scheduling in KyPlant is implemented based on a fixed schedule, a dynamic scheduling procedure or a combination of both.

Static maintenance

A fixed maintenance schedule ensures that the power plant is not producing in a pre-determined period. Required inputs are the start date, duration in (half-)hours and output and efficiency degradation. It is also possible to define degradation without maintenance period. In that case, the user has to fill in 0 for the duration of maintenance.

Dynamic maintenance

Dynamic maintenance is based on equivalent operating hours and/or equivalent starts. Such a dynamic schedule may lead to maintenance in unfavourable periods, i.e. periods in which it is profitable to produce. Therefore, KyPlant incorporates a provision that allows maintenance to be shifted from a high margin period to a low margin period. For example, if maintenance should take place in January, it is possible to transfer the start of the maintenance period to March.

When defining maintenance, it is possible to define static maintenance only, dynamic maintenance only and it is also possible to combine static and dynamic maintenance schedules. Often, the maintenance in the close periods is known already, so it can be defined as static maintenance. On the other hand, the maintenance for future periods is dependent on the number of hours the plant is in operation. The number of operating hours or starts for dynamic maintenance is calculated from the moment of the last hour of the last static maintenance.

Maintenance definition type

Maintenance Plant trips

☒ Use maintenance

Planned maintenance

Degradation after maintenance

Start date	Duration Hours	Output degradation	Efficiency degradation	
2016-03-10	168	99 %	99 %	x
2017-01-01	0	98 %	98 %	x

+

Dynamic maintenance and degradation

☒ Use equivalent starts

Degradation after maintenance

Equivalent Operating Hours #	Equiv. starts (#)	Duration Hours	Output degradation	Efficiency degradation	
1000	200	336	98 %	98 %	x

+

Transfer maintenance periods

From	To
January	January
February	February
March	March
April	April
May	May
June	June
July	July
August	August
September	September
October	October
November	November
December	December

Maintenance and Additional load range

For all maintenance periods, the user has to indicate if the whole plant is in maintenance or only 1 GT is in maintenance. Note that the degradation percentages always apply to the whole plant.

Trips

Forced outages are user defined inputs, which can vary per calendar year. The input is a percentage of the total number of hours in a year. Note that since the number of hours per year differs, so does the number of outage hours.

It is possible to define 2 types of outages. Required inputs of an outage are:

- Yearly outage time as percentage of yearly number of hours
- Distribution outage time between trip type 1 and trip type 2
- Mean time to repair
- EOH per outage
- Starts equivalent per outage

Maintenance definition type

[Maintenance](#) [Plant trips](#)

☒ Use plant trips

Trip types

	Distribution	Mean time to repair	EOH per outage	Starts equivalent
		Hours	#	#
Trip type 1	60 %	36	0	0
Trip type 2	40 %	40	110	10

Trip probability

Year	Trips	Trip type 1		Trip type 2		
	% of all Hours	Hours per year	# per year	Hours per year	# per year	
2016	5	263	7	175	4	x
2017	7	367	10	245	6	x

For cases with Additional load range, it is assumed that once a trip occurs the whole plant trips.

Grid losses

In case, producers are charged for grid losses this may be accounted for explicitly in KyPlant. Such losses are to be characterized as losses to either a *grid supply point* or on a *transmission line*. 100% represents zero loss, while lower levels do represent a loss.

[General](#) [Commodities](#) [Technical](#) [Seasonality](#) [Start curves](#) [Maintenance & degradation](#) [Grid losses](#)

Apply grid losses

☒ Yes ☐ No

LLF: Power loss from Plant to Grid Supply Point

TLM: Power loss in Transmission Lines

	LLF Week	LLF Weekend	LLF Week	LLF Weekend	TLM
	Nov - Mar	Nov - Mar	Apr - Oct	Apr - Oct	
Hour 1	100%	100%	100%	100%	100%
Hour 2	100%	100%	100%	100%	100%
Hour 3	100%	100%	100%	100%	100%
Hour 4	100%	100%	100%	100%	100%
Hour 5	100%	100%	100%	100%	100%

12 KyDispatch

The KyDispatch model is a short term optimization model of a power plant. It can be used for two purposes:

- It is used as the short term model for day-ahead optimization.
- Or as a model for optimization of the power plant for the spot and the next month.

The model creates different type of outputs for these two cases.

Cost calculator model has been integrated with the short-term dispatch model. It uses the same input screen. In a separate chapter the Cost calculator model will be explained. In this chapter, we explain purely the KyDispatch model.

12.1 Create profile

Analytics → KyDispatch ☐ Create profile

General

Name

Legal entity

Efficiency %

Variable O&M

Power plant settings No file selected.

Output

Profile type

☒ Output to PN

☐ Output to Align

☐ Output to Cost Calculator

PN settings

Create PN until days after start

Minimum ramp MW

Inputs

Input directory

Power

Filename

Commodity

Warn if older than minutes

Gas

Filename

Commodity

Warn if older than minutes

Carbon

Filename

Commodity

Warn if older than minutes

Availability

Filename

Warn if older than minutes

Initial conditions

Filename

Warn if older than minutes

The user selects (i) what plant settings have to be used, (ii) what output format is required and (iii) which prices have to be applied. There is no limitation in the number of KyDispatch profiles that can be created.

General

Name

The inputs for a KyDispatch can be stored under a unique profile name.

Legal entity

Select a legal entity from the drop-down list.

Efficiency and variable O&M

Plant efficiency and variable O&M are used to check the price inputs. These values are not used in the calculation!

Power plant settings

Upload a txt file. The txt file can be generated from an Excel input file.

Output

Output to

Select from three options: a) Output to PN, b) Output to Align or c) Output to Cost Calculator. Depending on the user choice, additional inputs are required.

PN settings

For Output to PN, specify how many days are outputted and what is the Minimum Ramp used in the outputs.

Align settings

For Output to Align, specify the KyPlant profile to which this KyDispatch profile is connected. Additionally, the state of the power plant at the end of a user specified month is used in the KyPlant model as the starting point for the long term optimization. Specify the end of the month, for which the state has to be passed to KyPlant.

Cost Calculator settings

Cost calculator mode explained in a separate chapter

Inputs

Input directory

Enter a directory where the input files are stored

Input files: Power, Gas, Carbon, Availability, Initial conditions

KyDispatch expects price files to be in GMT, i.e. always containing 48 values. The availability and initial conditions are in local time (BST in the summer, GMT in the winter) and contain 46/48/50 values.

Define the file name, commodity type and the time when the file is considered old for each of these five files.

12.2 Run KyDispatch

Before running the KyDispatch model, the user can check the inputs. When the user presses 'Check inputs', he enters a new screen.

KyDispatch profiles Create profile

Show all filters Load filter Reset selection Save this filter Filter

Legal Entity x

All
kyos

2 results found.

20 per page

ID	Name	Entity	Output to	Jobs	
3	CORB	kyos	PN	8	View
4	CORB (Aligne)	kyos	Aligne	4	Check inputs Edit Delete

20 per page

Delete selection Save selection as filter

The user enters a screen, where he can check if the input files are up-to-date. On the background, the system looks for the most price and availability files. The time stamps are presented on the screen. When all time stamps are less than few minutes old (user defines the time in the KyDispatch profile), they are assumed to be correct. When 1 or more files have older time stamps, they are marked red.

Start optimization

The optimization can start at next EFA midnight (23:00 BST) or at the Gate closure.

Run KyDispatch profile 3 (CORB) Back to overview View Jobs

Price files Graph

Settings

Time stamp

Power prices 2014-12-22 13:43:48

Gas prices 2014-12-22 13:43:48

Carbon prices 2014-12-22 13:43:48

Availability 2014-12-22 13:43:49

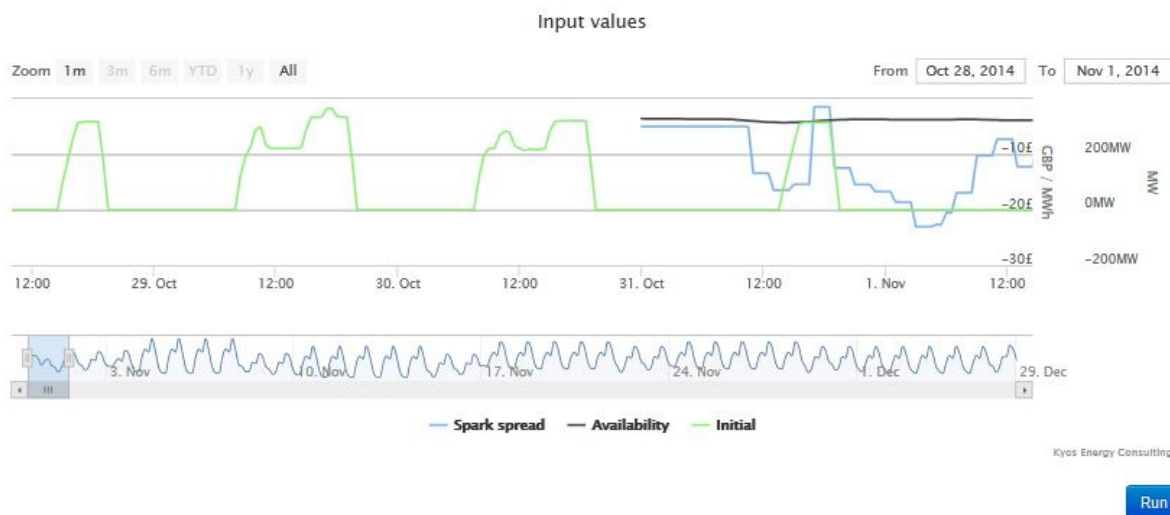
Initial 2014-12-22 13:43:49

Start optimization Gate Closure

Run

In the next tab, Graph, the price inputs together with the availability data and initial conditions are plotted.

Run KyDispatch profile 3 (CORB)

[Back to overview](#)
[View Jobs](#)
[Price files](#)
[Graph](#)


When the user presses 'Run', the most recent files are copied from the input directories. They are applied in the calculation.

12.3 KyDispatch output

Analytics → KyDispatch ☐ View

In the input file user chooses between two kinds of outputs: a) Output to PN and b) Output to Align.

12.3.1 Results

For both selections, the user can see tab Results. In this tab, the calculated dispatch together with the spark spread is shown in one graph. In another graph, margins and costs are shown.

View results for CORB

[Back to overview](#)

KyDispatch ID 3

Job id 9241

Job created at 2014-12-23 11:19:02

Job created by Sasa Bundalova

[Export to Excel](#)

Results

Plant dispatch

Committed load



12.3.2 Plant dispatch

On the second tab, the calculated plant dispatch is shown – both graphically as well as numerically in a table below the graph.

The user can edit the PN by creating or overwriting new elbow points. The user can add as many changes to the PN file as he desires. The changes can be added manually by filling in the start time, end time and the volumes. Alternatively, if the user clicks on a line in the PN file, the start and the end dates are automatically filled in and the user has to fill in only the volume field.

When the user presses 'Save', the changes are stored in the database. Furthermore, the PN is adjusted accordingly. The PN will always consist of half-hourly steps, potentially with elbow points in the middle.

View results for CORB

[Back to overview](#)

KyDispatch ID 3

Job id 9241

Job created at 2014-12-23 11:19:02

Job created by Sasa Bundalova

[Export to Excel](#)

Results

Plant dispatch

Committed load

[Export PN to Siemens](#)

Plant dispatch



Start date and time

End date and time

Volume

Kyos Energy Consulting

+ Add row

[Save](#)

You can click on a not highlighted row in the table below to edit the PN value.

Start date and time	Start volume	End date and time	End volume
2014-10-31 11:00	0	2014-10-31 11:30	0
2014-10-31 11:30	0	2014-10-31 12:00	0
2014-10-31 12:00	0	2014-10-31 12:30	0

Overwrite Elbow Point

Case 1 – Overwrite an existing elbow point with a new value

Suppose there is an elbow point at 7:23 with a value of 100, which should be overwritten with 125. The inputs are:

- Start time = 7:23
- End time = 7:30
- Volume = 125

From 7:30 onwards, the old values remain valid.

Case 2 – Define an elbow point at an earlier time

Suppose there is an elbow point at 7:23 with a value of 100. The elbow point should be moved to 7:21.

The inputs are:

- Start time = 7:21
- End time = 7:30
- Volume = 100

From 7:30 onwards, the old values remain valid.

Case 3 – Define an elbow point at a later time

Suppose that at 7:00 there is a production of 0 and an elbow point at 7:23 with a value of 100. The elbow point should be replaced to 7:25. The inputs are:

- Start time = 7:00
- End time = 7:25
- Volume = 0

And

- Start time = 7:25
- End time = 7:30
- Volume = 100

The first step ensures the original elbow point is removed. The second step is necessary to add a new elbow point at 7:25. From 7:30 onwards, all existing values remain valid.

Add or Remove Elbow Point

Case 4 – Add an elbow point

Suppose there is no elbow point, and you would like to add an elbow point at 7:23 with a value of 100.

The inputs are:

- Start time = 7:23
- End time = 7:30
- Volume = 100

Case 5 – Remove an elbow point

Suppose that at 7:00 there is a production of 0, and an elbow point at 7:23 with a value of 100. The elbow point should be removed. The inputs are:

- Start time = 7:00
- End time = 7:30
- Volume = 0

Define power output for several consecutive hours

Case 6 – Define power output for several consecutive hours, not ending on an elbow point

Suppose you want to define the plant runs from 9 August 2 AM until 13 August 17:00 PM at 250 MW.

The inputs are:

- Start time = 9 August, 02:00
- End time = 13 August, 17:30
- Volume = 250

Please note the end time is set at 17:30. This is because the PN is adjusted until the end time. From 17:30 onwards, the existing PN is applied.

Case 7 – Define power output for several consecutive hours, ending on an elbow point

Suppose you want to define the plant runs from 9 August 2 AM until 13 August 17:15 PM at 250 MW.

The inputs are:

- Start time = 9 August, 02:00
- End time = 13 August, 17:15
- Volume = 250

12.3.3 Committed load

A traded position can be entered as committed load. KyDispatch ensures that the committed load is produced during the optimization. There are 2 possibilities:

- Only the committed load is produced. The plant will be shut down as soon as the committed load ends.
- The committed load is produced as a minimum load

Committed load is constrained by technical plant conditions. When the maximum allowed capacity of the power station is lower than the committed load, the committed load simply cannot be produced and the technical characteristics determine the output.

A similar situation occurs when the plant is running, and the committed load is lower than the minimum stable generation level. The plant may ramp down to Minimum Level, but cannot go below this level.

With committed load, the power station should start even if this is economically not optimal, or continue to produce when this is suboptimal.

Optimize on market, given committed load as minimum

When the committed load is entered as minimal production, the model has to optimize between running the committed load and running the optimal load. When the optimal load should be lower than the committed load, then the optimal load is ignored and overruled with the committed load.

Committed load only

When the committed load is entered as the only production, the model will only produce the committed load irrespective of prices. The commercial optimization is ignored completely

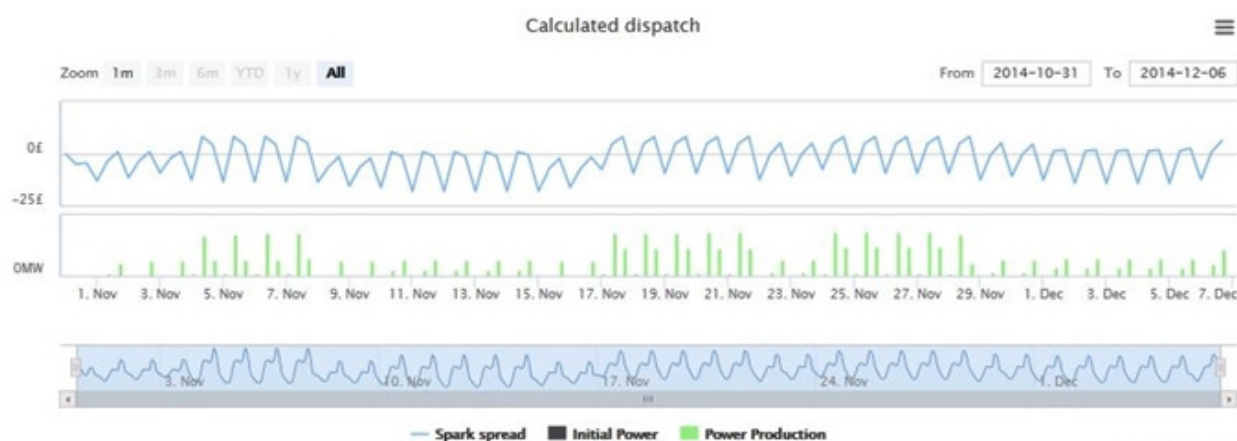
View results for CORB

[Back to overview](#)

KyDispatch ID 3		
Job id 9241	Job created at 2014-12-23 11:19:02	Job created by Sasa Bundalova

[Export to Excel](#)

Results Plant dispatch **Committed load**



Define committed load

Committed load type

Start date and time End date and time Tradable volume

[+ Add row](#)

[Rerun](#)

When committed load type is defined, the optimization is rerun. When the user presses 'Rerun', he can see the original Run screen, with the list of commodity files and the time when they were last time updated. Additionally, the committed load settings are displayed. When the user presses 'Run', the optimization with committed load is executed.

Run KyDispatch profile 3 (CORB)

[Back to overview](#)
[View Jobs](#)
[Price files](#)
[Graph](#)

Settings

Time stamp

Power prices	2014-12-22 13:43:48
Gas prices	2014-12-22 13:43:48
Carbon prices	2014-12-22 13:43:48
Availability	2014-12-22 13:43:49
Initial	2014-12-22 13:43:49

Start optimization ▼

Committed loads

Type: Committed load only

Start date and time	End date and time	Tradable volume
2014-10-03 01:00	2014-10-10 01:00	300

[Run](#)

13 Cost calculator

Cost calculator model has been integrated with the short-term dispatch model. It uses the same input screen. In this chapter we explain how to create and run a cost calculator profile.

Cost calculator model has been integrated with the short-term dispatch model. The costs calculator can be used for 2 purposes:

1. Price bids and offers submitted to the Balancing Mechanism
2. Show production costs and SRMC for forward blocks, which can be used for intraday markets.

13.1 Create profile

Analytics → KyDispatch □ Create profile

The user selects (i) what plant settings have to be used, (ii) what output format is required and (iii) which prices have to be applied.

General

Name

The inputs for a KyDispatch can be stored under a unique profile name.

Legal entity

Select a legal entity from the drop-down list.

Efficiency and variable O&M

Plant efficiency and variable O&M are used to check the price inputs. These values are not used in the calculation!

Power plant settings

Upload a Cost Calculator txt file. The txt file can be generated from an Excel input file.

Output

Output to

Select from three options: a) Output to PN, b) Output to Aligner or c) Output to Cost Calculator. Depending on the user choice, additional inputs are required. For Cost calculator choose the 3rd option.

Cost Calculator settings

Cost Calculator settings

Create output until	5	days after GC
fixed start costs	2500	GBP
APC@MEL+ adder	5	% of the APC@SEL
MNZN@MEL adder	3	% of the MNZN@SEL
Super red	10	GBP/MWh

Bid offer baseline data

Pair 2	350	GBP/MWh
--------	----------------------------------	---------

Only one of the following pairs (pair -1, pair -2) can be used. When one is edited the other will become empty.

Pair -1	☒	-350	GBP/MWh
Pair -2	☐		GBP/MWh

Acceptable Bid offer levels

Lowest offer level	30	GBP/MWh
Highest offer level	100	GBP/MWh
Lowest bid level	20	GBP/MWh
Highest bid level	80	GBP/MWh

Create output until... days after GC

Specify how many days are outputted

Fixed start costs

Specify the fixed start costs

APC@MEL+ adder & MNZN@MEL adder

The model infers the average costs at MEL (APC@MEL+) as $APC_MEL+ = APC_SEL * (1 + APCAdder)$, and MNZN costs at MEL as $MNZN_MEL = MNZN_SEL * (1 + MNZNAdder)$

Super red

This input is not used at the moment.

Bid offer baseline data

The baseline data are not used in the calculation, they are used only in the output presentation. Define the baseline data at Pair 2 and Pair -1 or Pair -2

Acceptable Bid offer levels

Define the lowest and highest acceptable offer and bid levels.

Inputs

Input directory

Enter a directory where the input files are stored

Input files: Gas, Carbon, Availability, Initial conditions

Cost Calculator expects price files to be in GMT, i.e. always containing 48 values. The availability and initial conditions are in local time (BST in the summer, GMT in the winter) and contain 46/48/50 values.

Define the file name, commodity type and the time when the file is considered old for each of these five files.

13.2 Cost calculator output

[Analytics](#) → [KyDispatch](#) ☐ [View](#)

13.2.1 Results

Various costs are calculated for each point in time:

- MNZT@SEL, MNZT@MEL: marginal costs at SEL and MEL
- APC@SEL, APC@MEL: average production costs (in GBP/MWh) at SEL and MEL
- Inc SEL_MEL: marginal costs (in GBP/MWh) for increasing the production from SEL to MEL
- Bid down: marginal costs (in GBP/MWh) for decreasing the production from planned level to SEL
- Bid off early: for each (half-)hour when the plant is producing, costs of early turn-off are calculated

[Results](#) [Bid/Offer](#)

Summary - Results at Gate Closure: 28-May-2015 10:30:00

MNZT@SEL (£)	69.82
APC@SEL (£)	58.04
APC@MEL (£)	54.52
SEL to MEL (£)	48.58
Bid down (£)	58.04
Bid off early (£)	-

Date	Block	PN (MW)	Periods to next on	MNZT SEL (£)	MNZT MEL (£)	APC SEL (£)	APC MEL (£)	APC MEL+ (£)	Inc SEL-MEL (£)	Bid down (£)	Bid off early (£)
28-May-2015 10:30:00	3B	0	54	69.82	71.91	58.04	54.52	60.94	48.58	58.04	-
28-May-2015 11:00:00	4A	0	53	69.82	71.91	58.04	54.52	60.94	48.58	58.04	-
28-May-2015 11:30:00	4A	0	52	69.82	71.91	58.04	54.52	60.94	48.58	58.04	-
28-May-2015 12:00:00	4A	0	51	69.82	71.91	58.04	54.52	60.94	48.58	58.04	-
28-May-2015 12:30:00	4A	0	50	69.82	71.91	58.04	54.52	60.94	48.58	58.04	-
28-May-2015 13:00:00	4B	0	49	69.82	71.91	58.04	54.52	60.94	48.58	58.04	-

13.2.2 Bid / Offer

On the Bid/Offer screen the baseline bids and offers are shown for each half-hour and pair. Initially, the calculated values for each half-hour are presented. The user can apply template (one or multiple) on top of these calculated values.

Apply template

Select Template

Baseload 23:00 - 23:00

Price or Premium

price

Amount

0

 £

Select time period

All

Apply to:

☐ All
☐ All

☐ Pair 2
☐ Bids

☐ Pair 1
☐ Offers

☐ Pair -1

Apply Template

Select template

From a drop-down of user-defined templates select one.

Select time period

All/Within Day/Day Ahead

Price or premium

The user can choose premium or price to be added to the template.

Apply to

Select pairs and bid/offer combinations to which the template is to be applied

The applied templates can be tracked and also can be undone.

History

	Template	Hours	Price or Premium	Amount	Select time period	Apply to:	Affected bids/offers
1	Baseload	23:00 - 23:00	premium	5.00%	Within Day	pair: all Bids	81

Undo Last Template

Bid/Offers

Click on a highlighted cell to see which templates are applied to it. The selected bid/offer cell and the applied templates will be marked with a black arrow. ✕

Date	Block	Pair	MW	Offer (£)	Bid (£)
28-May-2015 10:30:00	3B	5	0	-	-
		4	0	-	-
		3	0	-	-
		2	350	72.00	55.10
		1	220	70.00	55.10
		-1	-350	58.50	55.10
		-2	0	-	-
		-3	0	-	-
		-4	0	-	-
		-5	0	-	-

14 KyRisk

The main purpose of the KyRisk module is to provide insight in the distribution of future cash-flows and profit margin (earnings), i.e. to compute the Cashflow-at-Risk and Earnings-at-Risk. In addition, it computes distributions of the contract volumes and suggest trades on the market to follow a particular hedging strategy. The KyRisk model can analyse different energy contracts (including gas swing contracts), gas storages, and power plants. Depending on the chosen portfolio, the KySwing, KyStore, and/or the KyPlant model can be used in addition. The KyRisk module uses forward and spot price simulations for all relevant commodities. These simulations are generated by the KySim model within the KYOS Analytical Platform,

The output screens of the KyRisk module in the platform show:

1. Earnings@Risk: The portfolio's Earnings@Risk measure (at 99% or 95%) is presented on top. At the bottom, the "Percentiles" table is visible. It contains some percentiles of the earnings distributions per particular commodity, energy contract or real option (swing contract, gas storage, or a power plant). The user can click on any row in this table. After the click, the graphs above will change accordingly. These graphs contain further information regarding the earnings distribution for the given commodity, contract, etc.
2. Cash-flow@Risk: The portfolio's Cash-flow@Risk measure (at 99% or 95%) is presented on top. At the bottom, the "Percentiles" table is visible. It contains some percentiles of the cash-flow distributions per particular commodity, energy contract or real option (swing contract, gas storage, or a power plant). The user can click on any row in this table. After the click, the graphs above will change accordingly. These graphs contain further information regarding the cash-flow distribution for the given commodity, contract, etc.
3. Hedges@Risk: Optimal hedge trades in order to close the intrinsic/delta exposure of the portfolio are reported. The exposure before and after the hedge are graphed. A histogram shows the portfolio earnings before and after the different hedging strategies (static/dynamic).
4. Volumes@Risk: At the bottom, the "Percentiles" table is visible. It contains some percentiles of the volume distributions per particular commodity, energy contract or real option (swing contract, gas storage, or a power plant). The user can click on any row in this table. After the click, the graphs above will change accordingly. These graphs contain further information regarding the volume distribution for the given commodity, contract, etc.

14.1 Create KyRisk profile

This subsection provides a practical demonstration on how to create and run an KyRisk profile. The profile can be created on the KYOS Analytical Platform at:

Analytics → @Risk □ Create profile

Refer to the Figure below for a snapshot from the KYOS platform and a description of the various entries.

Edit Risk Profile

Name	TTF Gas Plant + hedge contract	
Contract filter	TTF Hedge Contract	▼
KyPlant filter	Test @Risk: Gas Plant LSMC	▼
KyStore filter	None	▼
KySwing filter	None	▼
KySim profile	TEST NL Power, TTF, Coal, C	
Confidence level	95	▼ %
EaR limit	2000000	empty or integer greater than 0
Forecast end month	Dec	▼ 2018 ▼
Load simulation	☐ Yes ☒ No	
Should the correlation between load and temperature changes be estimated historically?		
Historical	☒	
User defined	☐	

Name

Enter a unique profile name.

Contract filter

All available contract filters will be shown in the drop-down. Only one filter can be selected. Thus, a filter containing all contracts which should be a part of the portfolio analysis should be created first. This can be done on the KYOS Analytical Platform:

Assets & Contracts → Contracts → Save selection as filter

KyPlant filter

All available KyPlant filters will be shown in the drop-down. Only one filter can be selected. Thus, a filter containing all KyPlant power plant valuation profiles which should be a part of the portfolio analysis should be created first.

KyStore filter

All available KyStore filters will be shown in the drop-down. Only one filter can be selected. Thus, a contract filter containing all KyStore gas storage valuation profiles which should be a part of the portfolio analysis should be created first.

KySwing filter

All available KySwing filters will be shown in the drop-down. Only one filter can be selected. Thus, a filter containing all KySwing swing contract valuation profiles which should be a part of the portfolio analysis should be created first.

KySim profile

Select the appropriate KySim profile from the list. It must simulate all the commodities which are in the chosen portfolio. The KyPlant/KyStore/KySwing valuation profiles should use the selected KySim profile as well. The reason is that, for example, each KyPlant valuation uses a set of price simulations in order to derive the optimal plant strategies. Thus, it is important to choose the same KySim profile which has been used in the plant valuation for the KyRisk analysis. If that is not the case, these will be dropped from the calculations.

Confidence level

Either 99% or 95% can be selected. This is the confidence level associated with the reported Earnings@Risk and Cash-flow@Risk values. These values can be seen in the summary tables at the top of the KyRisk output screens.

EaR Limit

This is the Earnings@Risk limit. It should be an integer bigger than 0. If the Earnings@Risk (95% or 99%) of the total portfolio or of the portfolio on the target hedge strategy (if such has been selected) is above this limit, then the value of the EaR is shown on a red background in the list with KyRisk profiles:

[Settings](#)
[Price data](#)
[Time Series](#)
[Curves](#)
[Assets & Contracts](#)
[Analytics](#)
[Reports](#)
[Logs](#)
[Interfaces](#)

[KyCalibration](#)
[KyStore](#)
[KySwing](#)
[KyPlant](#)
[KyPlant txt](#)
[KyDispatch](#)
[KySim](#)
[@Risk](#)
[KyInventory](#)
[KyCavern](#)
[KyVaR](#)
[KyPF](#)
[KyWhat-if](#)

Risk profiles

Create profile

Load filter

Reset selection

Save this filter

Filter

ID's

94

1 result found.

200 per page

ID	Name	EaR	EaR limit	Portfolio selection	Confidence level	KySim profile	Calibration period	Jobs
94	TTF Gas Plant + hedge contract	2 001 344	2 000 000	Contract: TTF Hedge Contract	95 %	TEST NL Power, TTF, Coal, Carbon	0	2

Forecast end month

Select the forecast end month and year. This defines the risk horizon. The analysis will be performed up to this point.

Load Simulation Settings

Load Simulation

This setting is relevant if load risk should be taken into account for the flexible gas contracts within the portfolio. Specify whether the loads should be simulated or not by choosing between the 'Yes' and 'No' options.

Should the correlation between load and temperature changes be estimated historically?

The user can choose whether to use historical load/temperature time series for calibrating the parameters of the temperature/load simulation model. When the 'Historical' field is chosen, then the model will use the historical data. If the 'User defined' field is chosen, then the user should fill in the correlation coefficient between load and temperature.

14.2 Gas Contract's Load Risk

For physical natural gas contracts with a flexible volume, the KyRisk module can create internally gas load simulations. In that manner, load risk can be taken into account. In order to create the simulations, the KyRisk model can use historical loads, historical temperatures and historical gas spot prices. The relationship between load, price and temperature can be established through an internal calibration phase. As a next step, joint scenarios of temperature, load and gas prices are generated and subsequently used in the model. In the KyRisk's output screens, the distributions of future volumes (load) and their effect on future cash-flows and earnings are presented.

In order to simulate loads in the KyRisk module, the user has to take the following steps:

1. Volume and delivery point information:

If the volume of one particular contract needs to be simulated, the user has to define a forecast for the gas volumes of this contract. The forecast is defined per each delivery point individually. It contains historically realized volumes and forecasted volumes for this particular delivery point. There can be only one contract with flexible volume per forecast. If the gas volumes of multiple contracts have to be simulated, each contract should have its own delivery point with associated volume history and forecast. Forecasts can be created on the KYOS Analytical Platform at:

Time Series → Volume actuals & forecasts □ Create new volume profile

When defining a forecast, the commodity and delivery points must be selected. The temperature profile which is associated with the forecast should be selected as well.

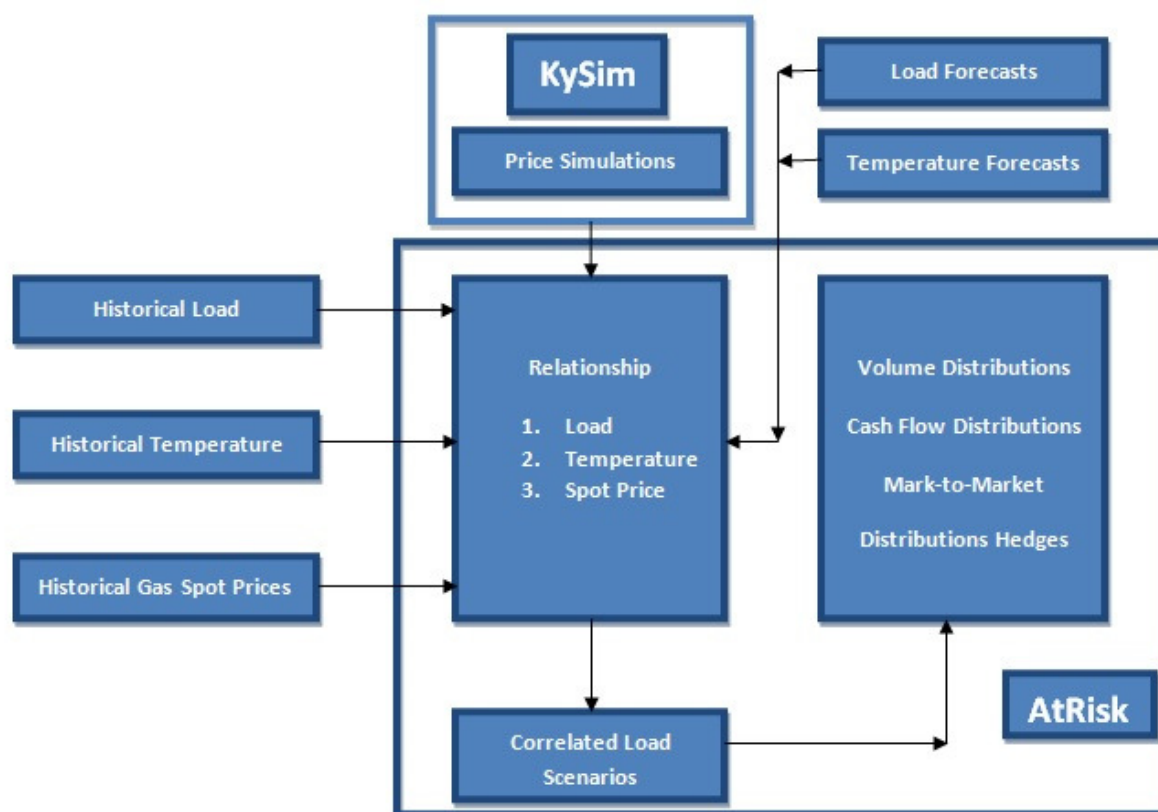
A dataset of historical loads and future (expected) load values must be uploaded for every delivery point. This can be done after defining the forecast, by pressing the 'Upload volumes' button in the 'Volumes actuals & forecasts' screen.

2. Temperature Profile: The creation of a temperature profile is necessary for load simulation. Each temperature profile should be connected to a particular gas spot price source. Data containing historically realized temperature values and the future expected values must be uploaded. This can be done on the KYOS Analytical Platform at:

Time Series → Production,Demand & Weather □ Create Time Series

In the profile definition screen, 'Temperature' should be selected in the 'Type' field.

The figure below shows a flow chart of the relevant inputs and-outputs, and summarises the functionalities of KyRisk related to gas load and price risk:



14.3 Contract Assumptions

This section specifies some of the main model assumptions regarding contracts.

1. For physical contracts with fixed volumes and all financial contracts, the contractually defined volumes will be used for computing cash-flows and earnings.
2. For physical contracts with flexible volumes, the model can simulate load. The following criteria apply:
 - a. The model verifies the existence of a defined forecast and an associated temperature profile.
 - b. In case the forecast exists and the historical load series and forecasts are available, a simulated load distribution will be generated and shall be used for computing cash-flows and earnings. This is usually the case for all gas commodities.
3. When a forecast cannot be associated with a temperature profile, the load simulation will not be done and the volumes defined in the forecast will be used for computing the cash-flows and earnings. Thus, there will be no load risk taken into account in the calculations. This situation also arises when the deliverable commodity is a not a gas and hence will not have a temperature profile.

4. If the user has selected the option of not simulating the load, then the contract volumes or the forecasts volumes (in case a forecast is defined) are assumed for future loads. There is no load risk in this case.
5. When creating a physical contract, information on Take-or-Pay is can be supplied by the user. However, the KyRisk model does not incorporate this information directly from the contract definition. Swing contracts should be evaluated first by the KySwing model. Then they can be added to the KyRisk profile definition.
6. The KyRisk model takes into account price clicks in physical contracts that essentially fix the contract price for a defined volume of the commodity during a defined period.
7. When calculating the portfolio hedge, the intrinsic and delta contract exposure equals the contract volumes. In case there is a load simulation for a particular flexible contract, the expected load is used. In effect, this is the forecasted volume.

14.4 View profiles and results

This section explains the outputs after an KyRisk profile has been created, and a job for a particular trading date has been executed.

14.4.1 List all profiles

A list of all the existing KyRisk profiles can be viewed at

Analytics → @Risk

Any profile that has already been created can be edited or deleted or run in its current form. All these choices appear in the dropdown menu next to each profile:

Risk profiles

[Create profile](#)

[Load filter](#)
[Reset selection](#)
[Save this filter](#)
[Filter](#)

ID's

1 result found.

200 per page

ID	Name	EaR	EaR limit	Portfolio selection	Confidence level	KySim profile	Calibration period	Jobs
94	TTF Gas Plant + hedge contract	2 001 344	2 000 000	Contract: TTF Hedge Contract KyPlant: Test @Risk: Gas Plant LSMC	95 %	TEST NL Power, TTF, Coal, Carbon	0	2

[Delete selection](#)
[Save selection as filter](#)

Results

- Run
- Edit
- Duplicate
- Delete

200 per page

14.4.2 Show all results of a specific profile

After a profile run has been completed, the results can viewed at

[Analytics](#) → [KyRisk](#) □ [Results](#)

The user will be able to view only a single set of results per trading date. All the previous runs for that trading date will automatically become obsolete.

14.4.3 Display results for a specific trading date

The user can view the results on a particular trading date and the chosen profile at

[Analytics](#) → [KyRisk](#) □ [Results](#) □ [View](#)

The results include pictorial representations and statistical results in tabular form with regard to distributions of volume, cash-flow and earnings. Additionally, portfolio hedge trades are presented.

The outputs are separated in several tabs containing the earnings, cash-flows, volumes, and hedges information.

[Earnings @Risk](#)
[Cash-flow @Risk](#)
[Hedges @Risk](#)
[Volumes @Risk](#)

Once in a particular tab, some graphs and tables are displayed:

- On the very top, there can be a summary table. It contains Earnings@Risk or Cash-flow@Risk (95% or 99%) over the whole portfolio.
- At the bottom, the “Percentiles” table is visible. It contains some percentiles of the cash-flow distributions per particular commodity, energy contract or real option (swing contract, gas storage, or a power plant. The user can click on any row in this table. After the click, the graphs

above will change accordingly. These graphs contain further information regarding the cash-flow distribution for the given commodity, contract, etc:

Percentiles

Commodity	Currency	Avg	5%	10%	90%	95%	99%
Total	EUR	6 072 124	4 070 779	4 545 299	7 631 762	8 699 306	9 708 354
TTF	EUR	1 641	959.33	1 070	2 202	2 509	2 818
Dutch power	EUR	6 070 482	4 069 270	4 543 349	7 630 067	8 697 536	9 706 343
EUA	EUR	0	0	0	0	0	0

Contract	Commodity	Currency	Avg	5%	10%	90%	95%	99%
Total		EUR	1 641	959.33	1 070	2 202	2 509	2 818
855 TTF Daily Nomination	TTF	EUR	1 641	959.33	1 070	2 202	2 509	2 818

KyPlant	Commodity	Currency	Avg	5%	10%	90%	95%	99%
Total		EUR	6 070 482	4 069 270	4 543 349	7 630 067	8 697 536	9 706 343
7 TEST Gas Plant LSMC	Dutch power	EUR	6 070 482	4 069 270	4 543 349	7 630 067	8 697 536	9 706 343

14.5 Contract Portfolio Sample Results

Several test cases were considered and an @Risk profile was run for them. The profiles included:

1. Fixed/flexible, physical/financial contracts, one underlying commodity, single delivery point.
2. Fixed/flexible, physical/financial contracts, multiple underlying commodities, single temperature profile and single delivery point.
3. Fixed/flexible, physical/financial contracts, multiple underlying commodities, single temperature profile and multiple delivery points.
4. Fixed/flexible, physical/financial contracts, multiple underlying commodities, multiple temperature profiles and multiple delivery points.

14.5.1 Portfolio Selection

For illustration purposes, the following cases are going to be discussed:

Contract	Start Date	End Date	Commodity	Volume	Currency	Pricing	Type
17	1-Jan-15	31-Dec-15	NCG	1200	EUR	Indexed	Physical
16	1-Jan-15	31-Dec-15	TTF	-900	EUR	Fixed	Physical
15	1-Jan-15	31-Dec-15	NCG	600	EUR	Formula	Physical

Two BUY contracts on NCG and one SELL contract on TTF were selected as a part of the portfolio. Negative volumes correspond to 'SELL' contracts, and positive volumes correspond to 'BUY' contracts. All of the contracts are physical: Contract 16 has fixed price; Contract 15 has a formula based price which is spot indexed to TTF with 6-0-3 indexation and weight of 0.75; Contract 17 is forward indexed to TTF with one month lag and weight of 0.6. Kindly note that the 6-0-3 type of indexations are usually

done on oil commodities and not on gas commodities as assumed in contract 15. We, however, include this example for illustration purposes only, in order to highlight the functionalities of the KyRisk model.

There were no defined budgets and hence the contractual volumes were used to compute cash-flows and earnings. The loads were not simulated in this case.

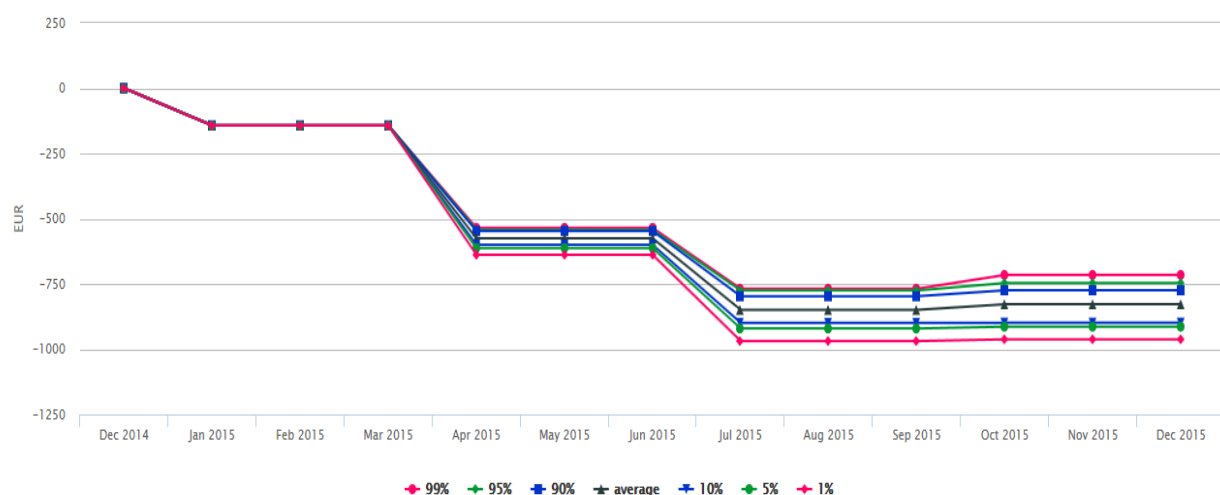
14.5.2 Volume Distribution

Since, no load simulation was done; the monthly volume will stay constant and is simply an equally weighted average of the total volume. Other choices to divide the contractual volumes over the months could be: depending on the number of hours or days in a month or user specified.

14.5.3 Cash-flow/Earnings Distributions

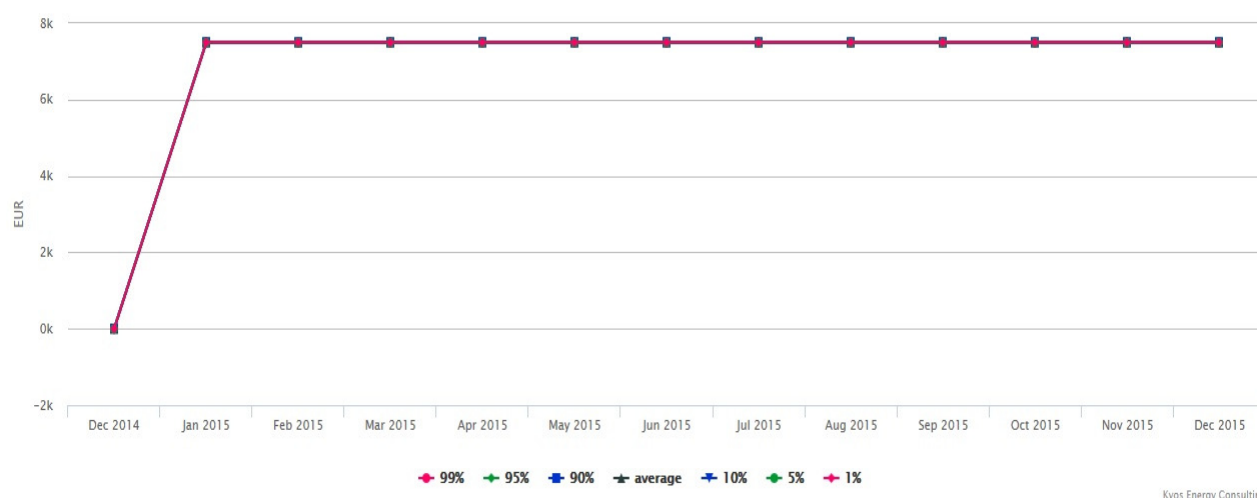
The following snapshots show the Cash-flow and Earnings distributions for each individual contract as displayed on the platform interface.

Monthly Cashflow Distribution



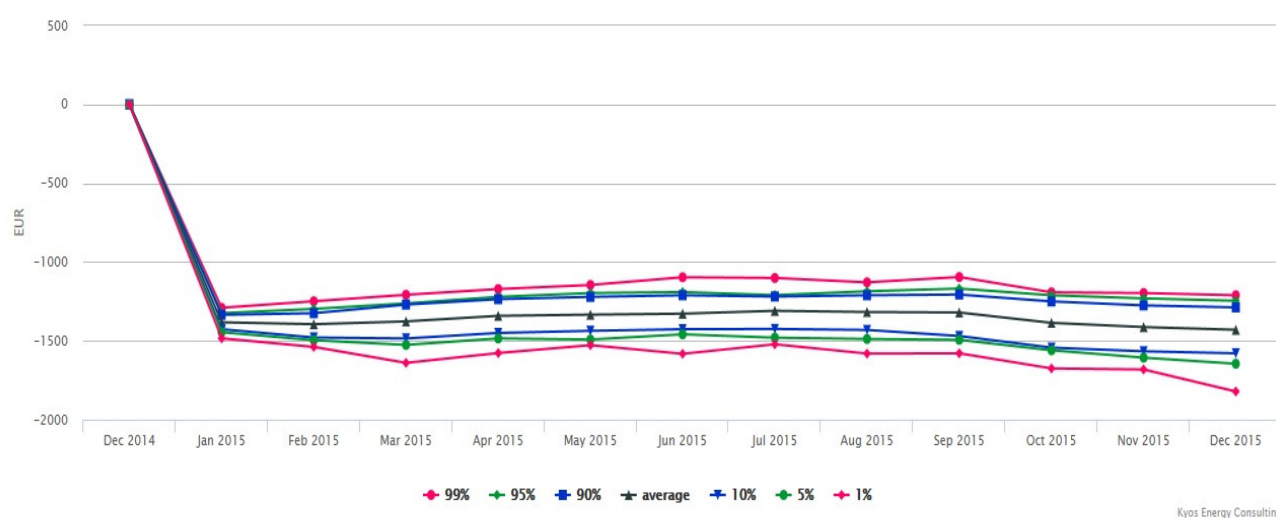
Contract 15

Monthly Cashflow Distribution



Contract 16

Monthly Cashflow Distribution



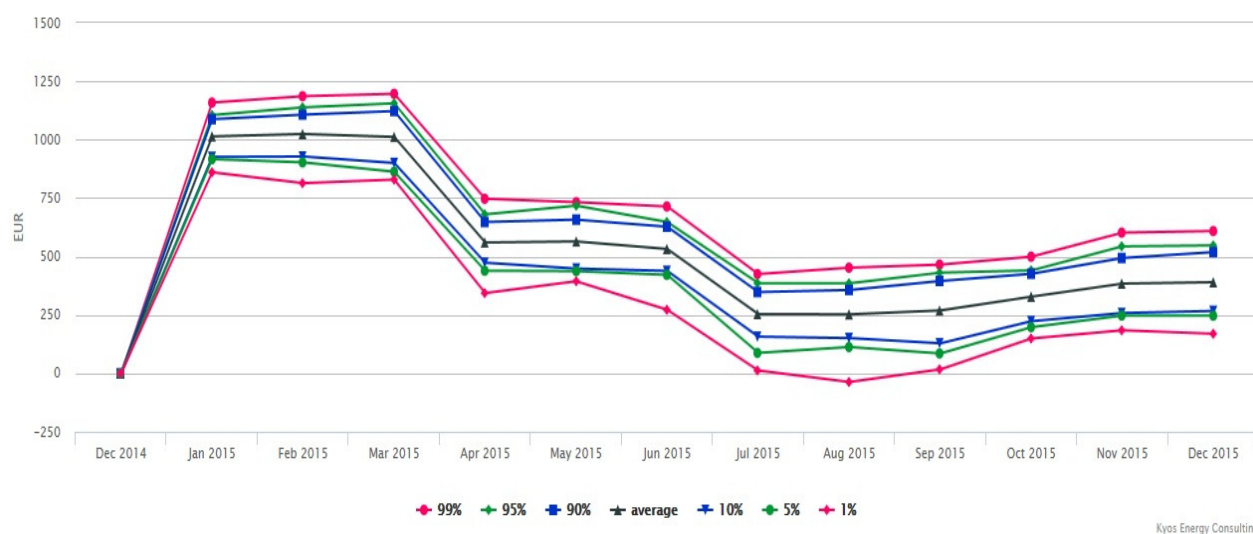
Contract 17

A straightforward observation is that the cash flows for contract 16 stay fixed due to it being a fixed price and fixed volume contract. The cash flow distribution for contracts 15 and 17 remain entirely negative because they are BUY contracts.

Furthermore, it can be observed that the shape of the cash-flow distribution for contract 15 follows a stepwise function with a step size of 3 months. This happens due to the type of indexation. In particular, with a 6-0-3 indexation, the price for the future three months is determined from the average of the spot price index over the past 6 months. The price distribution over those future three months remains the same in the absence of any volume uncertainty. This is evidently observed in the plots.

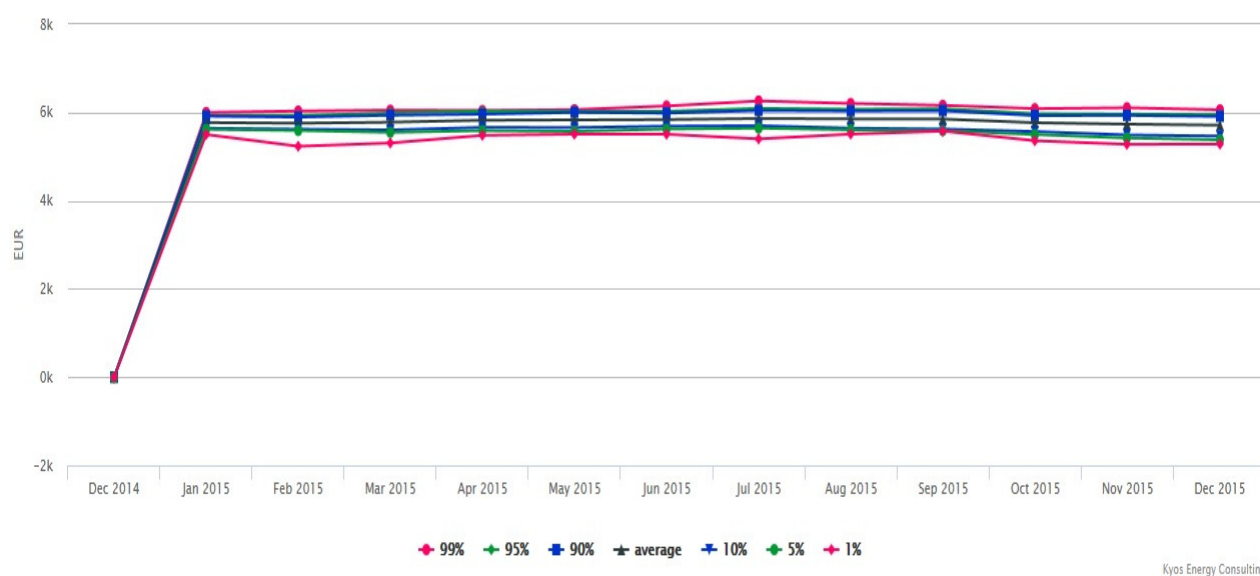
The next three plots show the Earnings (i.e. Mark-to-Market) distribution for the three contracts.

Monthly Earnings Distribution



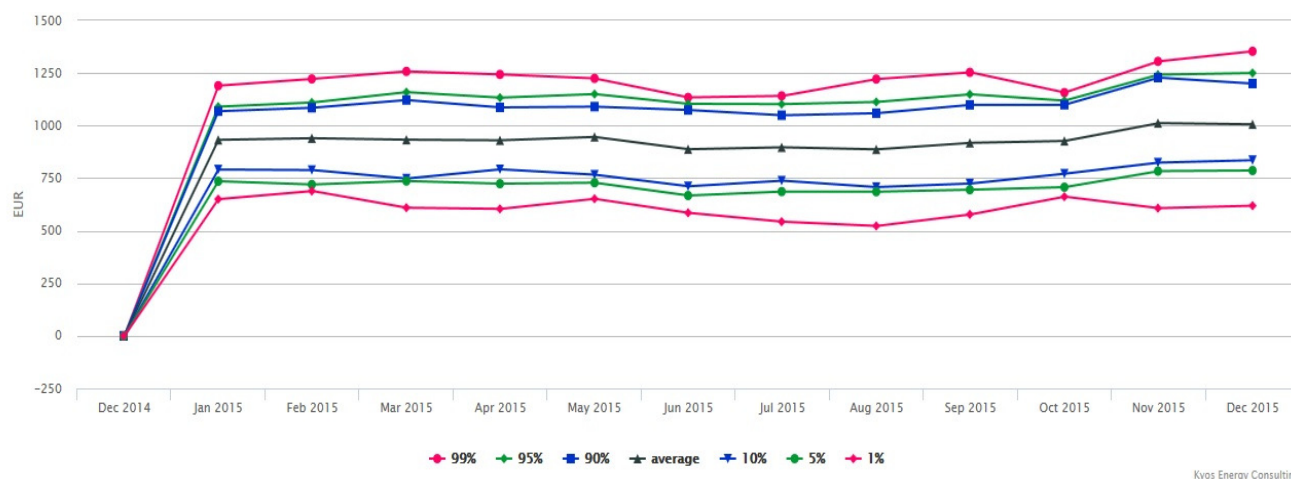
Contract 15

Monthly Earnings Distribution



Contract 16

Monthly Earnings Distribution



Contract 17

As expected, the earnings distribution for contract 16 is quite narrow throughout because of zero price and volume uncertainty but we do have a TTF spot price uncertainty. The Earnings distribution for contracts 15 and 17 reflect both, spot price and contract price uncertainty and, hence, there are wider distributions.

14.6 Example of Power Plant Results

The sample @Risk profile contains only one Dutch gas power plant. The Dutch power plant uses TTF gas as a fuel, and EUA carbon certificates are also required. All prices are converted to the platform currency, if necessary. In the sample case, this is euros. One KyPlant valuation of the power plant has been run using settlement price data on the trading date of 22/09/2016. The gas plant is operational over the complete year of 2017. An KyRisk profile containing only a selected KyPlant filter has been created, and a job executed for the same trading date.

14.6.1 Earnings@Risk

At the top of the tab, the Earnings@Risk measure (95% or 99%) is presented. For power plants, earnings are defined as the margin between the power price and all related costs (fuel costs, operation costs, trading costs, etc.) at the time of power production. The power production decisions follow the optimal exercise strategy, which is calculated by the KyPlant model.

Risk Profile: AtRisk Analysis Gas Plant 2017 2016-09-22

Earnings @Risk Cash-flow @Risk Hedges @Risk Volumes @Risk		
Earnings @Risk Summary		
	Currency	95% @Risk
Total	EUR	5 723 668

Note that this is the Earnings@Risk over the complete portfolio without any hedging strategy applied. In this case, the complete earnings over the delivery horizon of the power plant are used.

The 'Percentiles' table is located at the bottom of the screen. The user can see the earnings distribution percentiles information per individual commodity, per individual power plant, over the whole portfolio, and over all power plants.

Percentiles

Commodity	Currency	Avg	5%	10%	90%	95%	99%
Total	EUR	14 640 582	8 916 914	9 890 919	20 117 325	22 539 252	26 356 124
TTF	EUR	0	0	0	0	0	0
Dutch power	EUR	14 640 582	8 916 914	9 890 919	20 117 325	22 539 252	26 356 124
EUA	EUR	0	0	0	0	0	0

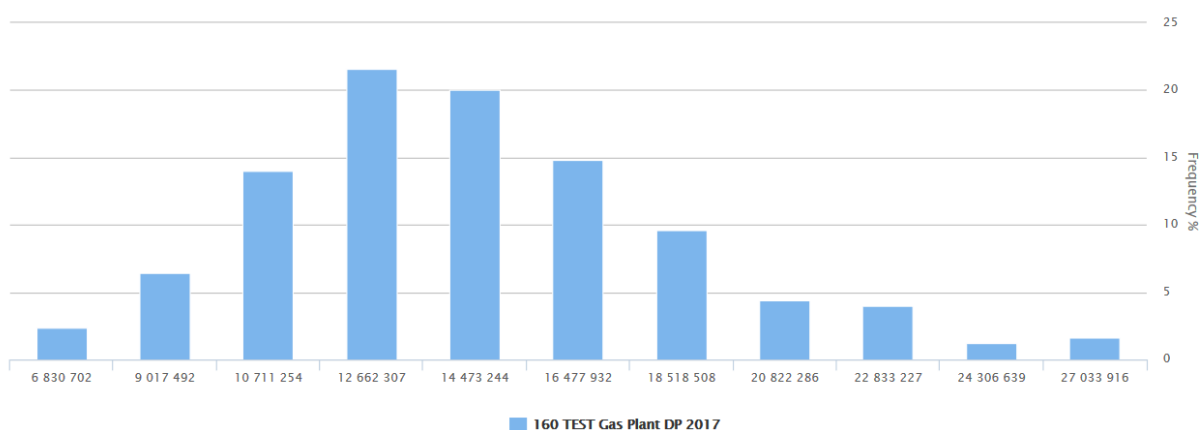
KyPlant	Commodity	Currency	Avg	5%	10%	90%	95%	99%
Total		EUR	14 640 582	8 916 914	9 890 919	20 117 325	22 539 252	26 356 124
160 TEST Gas Plant DP 2017	Dutch power	EUR	14 640 582	8 916 914	9 890 919	20 117 325	22 539 252	26 356 124

Note that there are earnings only for the Dutch power in the 'Commodity' columns. The reason is that the plant's earnings are assigned to the plant's power commodity. This is always the case.

Assume that the user clicks on the power plant row in the 'Percentiles' table above. The first graph which is displayed for this power plant is the histogram of the cumulative earnings over the complete power plant operation period. This histogram shows the probability of observing total earnings in a particular range. The earnings are a result of following the optimal power plant dispatch strategy which is derived by the KyPlant model. Note that earnings are based on power, fuel and CO2 market prices, and all operational costs are taken into account. The numbers, displayed in the histogram, can be viewed in a table by pressing the "Show/hide histogram graph table" underneath.

Kyplant: 160 TEST Gas Plant DP 2017 - Dutch power

Histogram graph

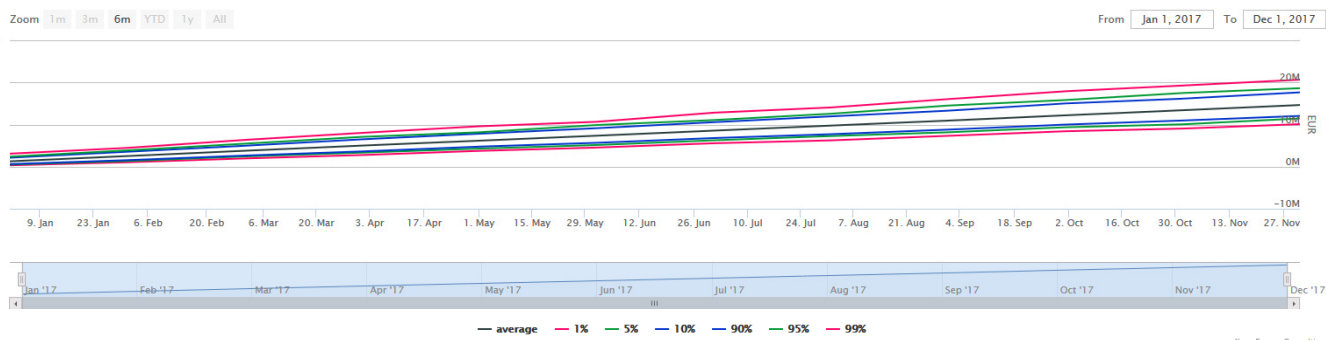


Show/hide histogram graph table

Currency	Lower bound	Average earnings	Upper bound	Frequency %
EUR	5 356 458	6 830 702	7 417 308	2.40
EUR	7 417 308	9 017 492	9 478 157	6.40
EUR	9 478 157	10 711 254	11 539 007	14
EUR	11 539 007	12 662 307	13 599 856	21.60

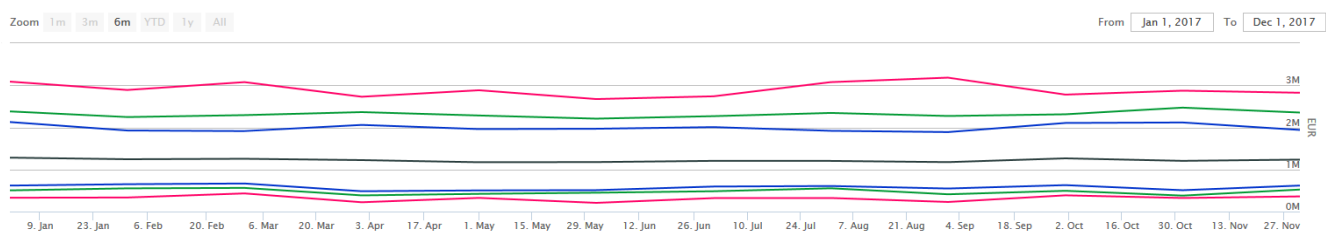
The following two graphs display the percentiles of the earnings distribution over each month of power plant operation. The “Monthly Earnings Distribution” contains the distribution over each month individually. The “Cumulative Monthly Earnings Distribution” contains the cumulative distribution. This means the cumulative earnings over the whole period from the valuation date of the power plant up to the end of the particular month.

Cumulative Monthly Earnings Distribution



[Show/hide cumulative monthly earnings distribution table](#)

Monthly Earnings Distribution



The show/hide table under each of these graphs can be used to display a table with the numerical values, on which the graphs are based.

14.6.2 Cash-flow@Risk

At the top of the tab, the Cash-flow@Risk measure (95% or 99%) is presented. For power plants, cash-flows are defined as the volume bought or sold at the market price at the time of power production. No additional trading and operational costs are taken into account. The power production decisions follow the optimal exercise strategy, which is calculated by the KyPlant model.

[Earnings @Risk](#)

Cash-flow @Risk

[Hedges @Risk](#)

[Volumes @Risk](#)

Cash-flow @Risk Summary

Total

Currency

EUR

95% @Risk

5 690 530

At the bottom of the screen, the user can choose to see more detailed information per individual commodity, per individual power plant commodity, and over the whole portfolio. There are positive cash-flows per power base/peak/offpeak, since power is sold. The cash-flows per TTF and EUA are negative, since they have to be bought in order to produce power.

Percentiles

Commodity	Currency	Avg	5%	10%	90%	95%	99%
Total	EUR	15 074 130	9 383 600	10 326 370	20 527 325	22 930 252	26 714 124
TTF	EUR	-49 797 570	-63 081 923	-60 826 341	-38 804 755	-37 034 204	-32 651 232
Dutch power	EUR	67 599 636	48 719 731	52 170 566	84 292 345	89 659 190	102 762 130
Dutch power (peakload)	EUR	42 110 496	32 181 875	33 870 403	49 610 966	52 324 679	61 793 917
Dutch power (offpeak)	EUR	25 489 140	15 881 754	17 101 889	35 119 498	38 909 496	46 439 948
EUA	EUR	-2 727 937	-4 379 549	-3 881 424	-1 787 752	-1 588 015	-1 308 141

KyPlant	Commodity	Currency	Avg	5%	10%	90%	95%	99%
160 TEST Gas Plant DP 2017	TTF	EUR	-49 797 570	-63 081 923	-60 826 341	-38 804 755	-37 034 204	-32 651 232
160 TEST Gas Plant DP 2017	Dutch power	EUR	67 599 636	48 719 731	52 170 566	84 292 345	89 659 190	102 762 130
160 TEST Gas Plant DP 2017	Dutch power (peakload)	EUR	42 110 496	32 181 875	33 870 403	49 610 966	52 324 679	61 793 917
160 TEST Gas Plant DP 2017	Dutch power (offpeak)	EUR	25 489 140	15 881 754	17 101 889	35 119 498	38 909 496	46 439 948
160 TEST Gas Plant DP 2017	EUA	EUR	-2 727 937	-4 379 549	-3 881 424	-1 787 752	-1 588 015	-1 308 141

The other graphs are similar to the graphs in the Earnings@Risk section.

14.6.3 Volumes@Risk

The volume distributions presented in this section are based on the power plant optimal dispatch, calculated by the KyPlant model. The graphs and tables are similar to the Earnings@Risk and Cash-flow@Risk tabs. The only difference is that distributions cannot be presented per portfolio or plant, but rather only per individual commodity. The reason is that volumes in different units cannot be added up.

14.6.4 Hedges@Risk

At the top of the tab, the Earnings@Risk summary table, given that a hedge strategy is applied, is presented. The Earnings@Risk are either at 95% or 99%.

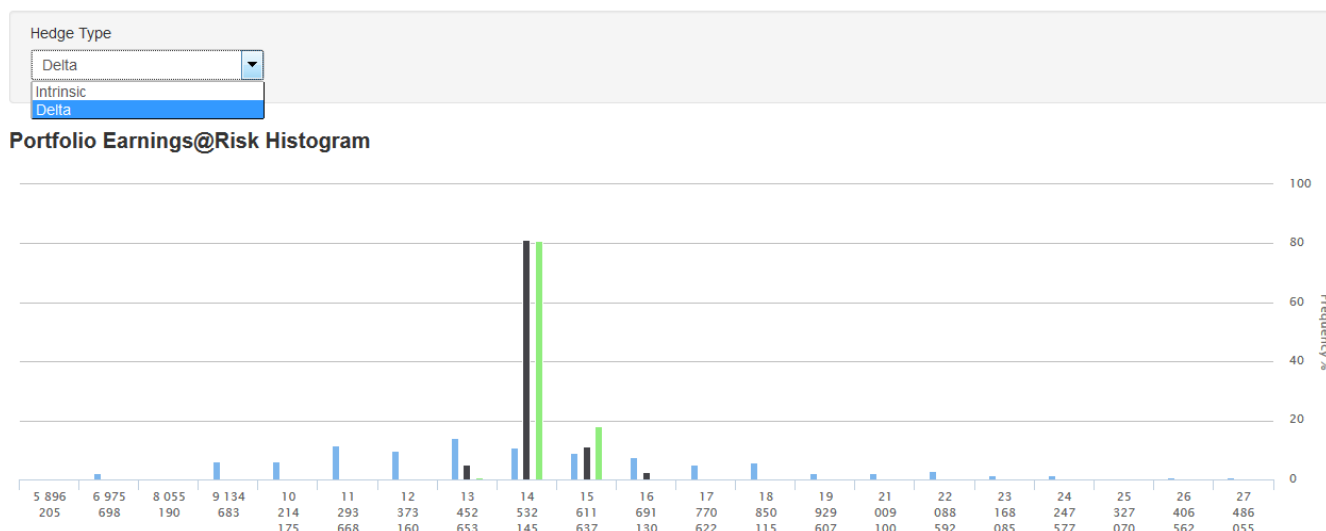
[Earnings @Risk](#)
[Cash-flow @Risk](#)
[Hedges @Risk](#)
[Volumes @Risk](#)

Earnings @Risk Summary	Currency	95% @Risk
EaR	EUR	5 723 668
EaR after optimal delta static hedge	EUR	653 544
EaR after optimal delta dynamic hedge	EUR	611 373

- **EaR:** this measure is based on the complete existing portfolio. The existing portfolio consists of all contracts and plants, which have been selected in the KyRisk profile definition. Thus, the reported EaR number is the same as the one reported in the Earnings@Risk tab. For each plant, the plant's earnings at the optimal spot strategy are used. The plant earnings at the optimal strategy are calculated by KyPlant. In this sample case, the portfolio consists only of a single plant.
- **EaR after optimal delta static hedge:** this number is based on the sum of the earnings from the existing portfolio and the earnings resulting from the static delta trade, suggested by the KyRisk model.
 - The static hedge trade is locked in on the forward market at the trading date, and it is settled against the spot market.
 - The forward market trade takes into account the existing contract positions. For example, if the static delta exposure is 1000 therms of NBP, and there is already a position for 500 therms, then only 500 therms will be traded.
 - Note that the earnings from the static hedge include also transaction costs and bid-ask costs.
- **EaR after optimal delta dynamic hedge:** this number is based on the sum of the earnings from the existing portfolio and the earnings resulting from the dynamic delta hedges, suggested by the KyPlant model.
 - The static hedge is locked in on the forward market at the trading date.
 - At the end of each month, a re-hedge is done to take into account changes in the power dispatch strategy and the market expectations.
 - All trading and bid-ask costs, associated with the dynamic hedges are included in hedge earnings' calculation.

A user should choose which plant hedging strategy should be examined. There are always two options:

Intrinsic/Delta.

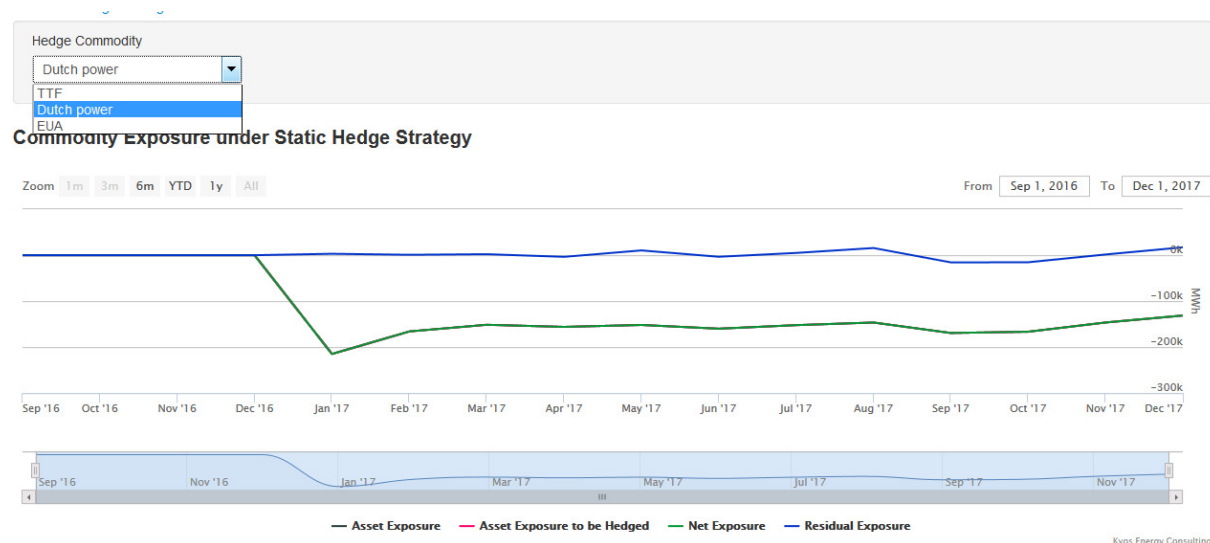


The complete portfolio's total earnings histogram is presented first. The "Before Hedge" bars depict the frequency with which the plant's earnings fall in a particular range. No static or dynamic financial hedge is done. The *EaR* measure is derived from this distribution.

The "After optimal delta static hedge" bars present the case when a single hedge at the valuation date is executed. The hedge is based on the intrinsic or delta exposure volumes, which are calculated by the KyPlant model for the particular plant. In case that there are more than one plant in the portfolio, the intrinsic/delta exposures will be added up over all plants. The addition is per hour for the power commodities and per day for the fuel and CO2 commodities. All hedge costs are taken into account when calculating the earnings. The earnings are based on the sum of the earnings resulting from the power plant dispatch and the earnings from the static hedge.

The "After optimal dynamic hedge" bars are based on the earnings when a dynamic hedging strategy is followed. The hedge on the valuation date is the same as under the static hedging strategy. However, at later point of time, a re-hedge is done, if necessary. The re-hedge is based on the intrinsic/delta exposures calculated by the KyPlant model at the later re-hedge dates. The hedges on future re-hedge dates cannot be executed at the valuation date. Thus, this value presents an approximation. Currently, the dynamic intrinsic hedging is not calculated.

The next choice which the user should make is to determine for which commodity, more information about the static hedge should be displayed.



The first presented graph underneath is the commodity exposure under the static hedge strategy. Monthly asset exposures are presented. Note that all hedge calculations are done on an hourly or daily basis. The graph just adds up the volumes within a specific month.

Asset Exposure: This is the complete intrinsic/delta exposure for the particular commodity. In this case it is negative starting from January'2017 because the power plant starts producing in this period. The plant operator should sell this amount of power on the market to hedge the power plant at 100%.

Asset Exposure to be Hedged: This is the exposure which has to be hedged based on a particular hedging strategy. The hedging strategy is defined per individual power plant. It is chosen in the KyPlant power

plant valuation profile. For example, hedge only 60% of the production in particular month. In the graph above, the choice is that 100% of the exposure should be hedged.

$$\text{Asset Exposure to be Hedged} = \text{Asset Exposure} * \text{Hedging \%}$$

Net Exposure: If some futures/forward contracts have already been traded in order to hedge the plant, this will be taken into account:

$$\text{Net Exposure} = \text{Asset Exposure to be Hedged} - \text{Existing Market Position}$$

Residual Exposure: This is the exposure which is remaining after the suggested hedges have been executed:

$$\text{Residual Exposure} = \text{Net Exposure} - \text{Traded Hedge Volume}$$

The last table presents the optimal market trades in order to cover the Net Exposure. The trades with the lowest trading costs which result in the lowest residual exposure are calculated.

Optimal Trades

Date	Product	Delivery Type	Lots	Volume	Unit	Mid Price	Trade Prices	Currency	Bid-Ask Costs	Trading Costs	Currency
2016-09-22	Jan 2017	Baseload	-23	-85 560	MWh	36.82	36.82	EUR/MWh	427.80	0	EUR
	Feb 2017	Baseload	-10	-33 600	MWh	34.69	34.69	EUR/MWh	168	0	EUR
	Q2 2017	Baseload	-3	-32 760	MWh	30.70	30.70	EUR/MWh	163.80	0	EUR
	Summer - 17	Baseload	-3	-65 880	MWh	30.41	30.40	EUR/MWh	329.40	0	EUR
	Cal - 2017	Baseload	-25	-1 095 000	MWh	31.81	31.80	EUR/MWh	5 475	0	EUR
	Jan 2017	Peakload	14	18 480	MWh	42.84	42.85	EUR/MWh	92.40	0	EUR
	Feb 2017	Peakload	3	3 600	MWh	42.46	42.47	EUR/MWh	18	0	EUR
	Q2 2017	Peakload	8	31 200	MWh	36.58	36.59	EUR/MWh	156	0	EUR
	Summer - 17	Peakload	2	15 600	MWh	37.10	37.10	EUR/MWh	78	0	EUR
	Cal - 2017	Peakload	-44	-686 400	MWh	38.95	38.94	EUR/MWh	3 432	0	EUR

Dynamic Delta Hedge Strategy Outputs

There are some additional 'csv' files which contain information about the dynamic delta hedging strategy. These files can be downloaded as a part of the job outputs.

It can be argued that the dynamic hedge strategy's earnings can be decomposed into 3 components. The earnings from the existing portfolio (contracts + plants), the earnings from the static hedge at the valuation date, and the earnings from re-hedging at later hedge dates (dynamic hedges). The purpose of these additional output files is to show the effect of the dynamic hedges only:

The columns in the following three output files are:

1. 'Hedge date', 2. 'Commodity id', 3. 'Delivery type id',
4. 'Delivery year', 5. 'Delivery month', 6. 'Expectation',
7. – end. 'Value in Simulation Scenario 1, Value in Simulation Scenario 2, ...'

- The hedge position per commodity and re-hedge date is contained in the file '**DynamicDelta_HedgePosition_Month.csv**'. Note, that these are volume positions per calendar month.

The positions reported on the valuation date (the first date) take into account the existing contracts and the static hedge trade.

- The dynamic hedge trades are contained in the file '**DynamicDelta_HedgeTrades_Month.csv**'. Sold volumes are with a minus, and bought volumes with a plus.

There are no trades reported on the valuation date. The reason is that on this date the static hedge trades take place. The static trades are reported on the output screens in the platform.

Note that on each re-hedge date (after the valuation date) the model decides on the optimal trades based on all available tradable products. The possible set of tradable products is restricted to base and peak, and to months and all products delivering within full months (quarters, years, etc.). In this output file, all trades are mapped to months. For example if Q1 was traded, the volume will be split accordingly into Jan, Feb, March.

- The simulated monthly forward prices are in the file '**DynamicDelta_ForwardPrice_Month.csv**'.
- The transaction costs and bid-ask costs, associated with hedging are contained in '**DynamicDelta_TransactionCosts_Month.csv**' and '**DynamicDelta_BidAskCosts_Month.csv**'. The columns in the file are

1. 'Hedge date', 2. 'Commodity id', 3. 'Delivery type id',
4. 'Expected cost', 5. 'Cost per simulation path'

There are no costs reported on the valuation date. The reason is that on this date the static hedge trades take place. The static trades and associated costs are reported on the output screens in the platform.

14.6.5 Hedge Contracts

The sample case discussed in this section is focused on a single power plant. There are no contracts in the portfolio. However, if there is already an existing futures/forward position in a particular commodity, it can be taken into account when calculating the net commodity exposure (see previous section). In the particular example, power, TTF and EUA contracts can be created in the platform. A contract filter containing the contracts containing the existing position can be created. This filter can be selected when defining the KyRisk valuation profile. In this way, the KyRisk model will take into account both the plant commodity exposures and the existing commodity market positions.

15 Reports

The system offers a flexible format to create reports to present e.g. commodity positions, cash flows, Mark-to-Market, average price per unit of measurement, hedging overviews and invoice calculations.

The set up is divided in different stages of tasks and reporting. In Risk Management the user can create multiple reports concerning contracted positions. In Risk Physical the system combines budget data from the Time Series with data from the contracted volumes and the hedging strategy to create position reports. In invoices the user is able to create overviews of invoice amounts using realized volumes. The user is also able to calculate expected invoices using forward curves and expected consumption/production data.

Before being able to create reports the standard procedure is that a user uploads historical and actual price data to create relevant curves. The user has to store relevant budget data (see chapter Time Series) such as forecasted and actual consumption/production figures. When the user has also stored contracts and the hedging strategy the required reports can be produced. The calculations have to be executed first in the “Run” area.

15.1 Run

Reports → Run

The user is able to schedule the calculation of a report at a specific hour and minute. The user is also able to re run reports e.g. after a correction of historical price data. The user can also select “Multiple jobs” meaning that multiple reports are produced using data of multiple trading days.

Reporting settings

[Run calculations](#)
[Define schedule](#)

Next run

The next automatic run will be at **06:00**.

Job type ☒ Single job ☐ Multiple jobs

Select Trading day

[Run now](#)

Previous runs

Select a previous run date to re-run that date's calculation.

October 2014

Su	Mo	Tu	We	Th	Fr	Sa
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

[Run again](#)

In “Define schedule” the user is able to store the hour/minute that the calculation for the reports have to be created. The logic is that users can schedule to upload price data first, then create forward curves and after both processes are ready the calculation of positions, cash flows and MtM is executed.

Run Risk management Risk physical Invoices

Reporting settings

Run calculations Define schedule

Scheduled run times

Reports will automatically be created at the following times:

Hour	Minute	
06	00	Delete
Add a time		Save run times

After execution of the calculation job the user can create several reports in Risk Management.

15.2 Risk management

Reports → Risk management

This section shows reports of commodity positions including the underlying financial exposures, cash flows and Mark-to-Market values. The information for the Risk Management report is taken from the contracts. Once the data for the report has been selected, the user can switch between the different reports by clicking on the tabs. The outcome of each report can be exported to Excel.

Positions Positions + underlying Cash flow Average price Cash flow + costs Average price + costs Mark to market

Positions	: Positions per unit
Positions + underlying	: Positions including the underlying exposure per unit
Cash flow	: Cash flow in system currency or per contracted currency
Average price	: Average price per unit
Cash flow + costs	: Separation of cash flow in commodity related cashflow and cost components
Average price + costs	: Average price per unit, divided in commodity and costs
Mark-to-market	: Mark-to-market of physical and/or financial contracts

The following three cases will be used as example:

- Contract #99 Buy natural gas based upon an oil based formula (Brent & GasOil)
- Contract #100 Buy natural gas based upon a gas index (TTF)
- Contract #125 Buy natural gas against a fixed price

In all three cases the same cost structure is stored in the system. One cost component is EUR 1 per MWh (unit of measurement). The second cost component is stored as a fixed cost of EUR 175,200 per year.

Additional costs

cost	type	distribution	value (EUR)	counterparty	settlement	
TAX	Variable [Price add-on]	Distributed relative to volume	1	GdF Suez	monthly	Delete
Flex Fee	Fixed [Calendar year]	Distributed relative to volume	175200	GdF Suez	monthly	Delete

Example case A (Contract#99):

Client buys the commodity natural gas with physical delivery in 2015. The total contract volume is 87,600 MWh, monthly distributed and weighted by the hours per month. The pricing formula is based upon Gasoil and Brent. The gasoil weight is 0.027 with a 6.0.3 calculation method and the weight of Brent is 0.13 with a 1.0.1 calculation method.

[Commodity price](#)
[Additional costs](#)
[Additional costs nominations](#)

Pricing model: formula

Formula price

Price = $weight1 * [commodity1 - constant1] + weight2 * [commodity2 - constant2] + \dots + fixed1 [FX1] + \dots$

Variable terms

commodity profile	settlement profile	weight	constant	commodity			direct conv.	FX			
				average	lag	period		average	lag	period	
E-Brent	Brent	0.13	0	1	0	1	yes				Delete
0.1 % GO Barges	0.1 % GO Barges	0.027	0	6	0	3	yes				Delete

Example case B (Contract#100):

Client buys the commodity natural gas with physical delivery in 2015. The total contract volume is 87,600 MWh, monthly distributed and weighted by the hours per month. The agreed pricing method is based upon an indexed price. The contracted index is the TTF index with a weighting of 1 times the TTF index.

[Commodity price](#)
[Additional costs](#)
[Additional costs nominations](#)

Pricing model: indexed price

Indexed price

Price = $weight * commodity + constant$

commodity profile	settlement profile	weight	constant (EUR)	
TTF	TTF	1	0	Delete

Example case C (Contract#125):

Client buys the commodity natural gas with physical delivery in 2015. The total contract volume is 87,600 MWh, monthly distributed and weighted by the hours per month. The client has agreed a fixed price of EUR 19.92 per MWh.

Commodity price Additional costs Additional costs nominations

Pricing model: fixed price

Fixed price

- ☒ constant price per month
- ☐ manually enter price per month

price 19.92 EUR per MWh TTF

Creating a risk management report involves the following steps:

Selection

- The user selects the contracts that have to be included in the Risk management report.
- The user can select a single contract or a selection of contracts using several filter settings.

In this example the user has selected contract IDs 99, 100 and 125:

Risk management Risk physical Invoices Run

Selection Load report filter Load other filters Hide unused filters Reset selection Save this filter

Contract ID's 99, 100, 125	Commodity All 0.1 % GO Barges FOB Rdarr 0.1 AAA 3.5 % Fuel Oil Barges FOB R	Country All Australia Austria Belgium	Business Unit All -Coatings, Adhesives, Specia -Polycarbonates -Polyurethanes
Delivery point All Antwerp Baytown Belford Roxo	Legal Entity All Bayer MaterialScience Ltd BMS Belgium NV BMS China	Counterparty All Barclays Int BASF BNP	Book All Energy Example I Example II
Buy/Sell All	Physical/Financial All	Contract Type All	Volume flexibility All
Base/Peak/Offpeak All			

The next step is to define the presentation of the data; this function is performed in the “Display options”.

Display options

- Trading date: determines which forward price curves are used.

- Report period: determines which delivery period of the contracts is shown in the reports.
- Reporting properties: are used to define the aggregation order of the selected data.
- Report in original currency: when this checkbox is ticked, the cash flows and MtM are presented in the original currency instead of the base currency of the system.

Display options [Reset selection](#)

Trading date

Report period
from
to

Reporting properties

☐ Report in original currency

If the user selects “Cost type” in the Reporting properties then the cashflows of the costs can be viewed separately from the cashflows of the contract prices.

The settings of a report may be stored under a user-defined name, so a report can easily be recreated another time. This is called a report filter. A filter may only be visible to the user or to all users (default).

Save current filter setting

Provide a name to remember this filter setting

Or select an existing filtersetting to overwrite

Dutch Power Only
Commodity position per Counterparty
Financial hedges per Counterparty
Commodity position BU "Commercial"
Cashflows financial swaps QIII and QIV 2014
Commodity Position Benelux

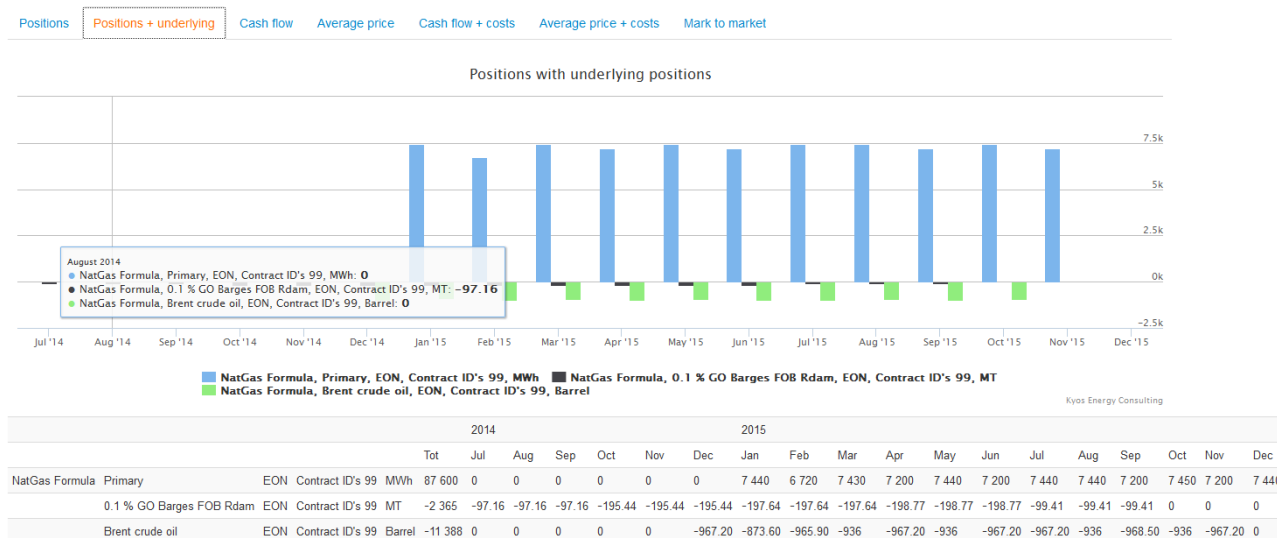
Make this private?
☐ Yes (only you can see it)

In the following example the user created a position report of contract ID 99, 100 and 125 with the following order in the display options : Commodity and Counterparty. The positions are presented as monthly volumes. Because the commodity is purchased (a buy contract), the positions are positive.

All contracted volumes of the selected contracts (99,100 and 125) have the same volume. This is the reason for presenting this position example in a single display.



In the next tab “Positions + underlying” the user can view the physical position plus the underlying financial exposures. In this example the user has selected contract 99. The blue bar represents the bought volumes of the natural gas in MWh whilst the underlying financial positions are negative, because there is a negative exposure to the underlying commodities: the contract becomes more costly when the Brent and Gasoil prices go up. Note that Brent is quoted in bbl and gasoil in MT.



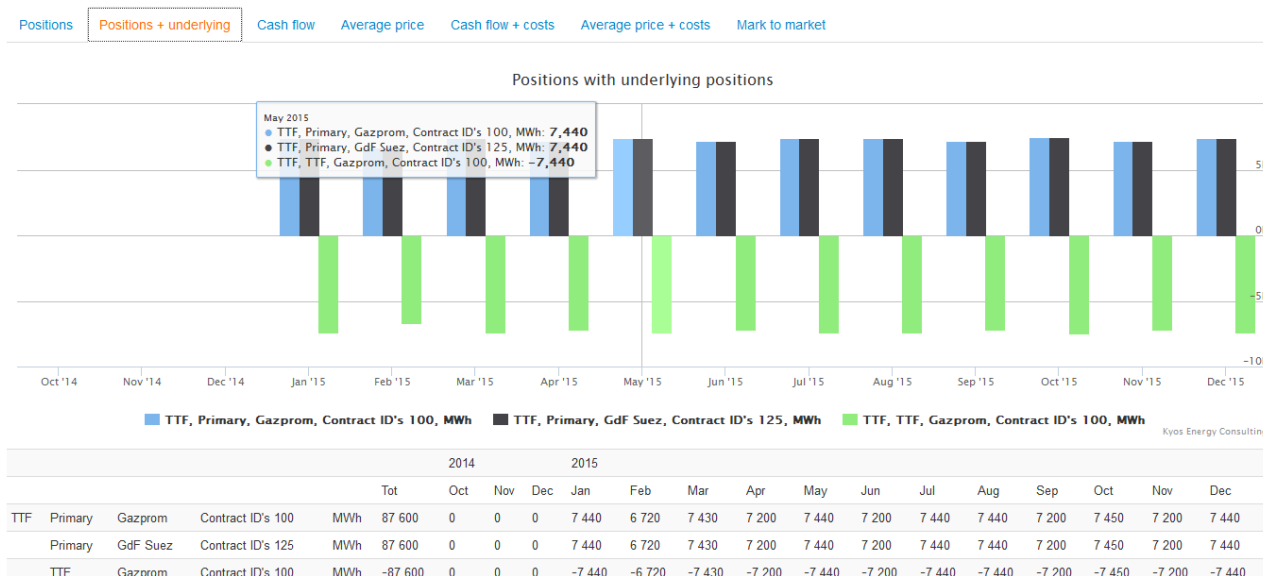
The underlying financial Brent exposure (volume) is calculated by multiplying the weight 0.13 with the natural gas volume relevant for the pricing period. The 1.0.1 formula means that the natural gas volume in the first contract month (7,440 MWh) is multiplied with 0.13 to calculate the financial exposure of Brent the month prior to the first contract month. This leads to a one month lag in exposures.

The underlying financial Gasoil exposure (volume) has a 6.0.3 averaging. The underlying financial exposure for physical delivery in January–March 2015 is calculated by multiplying the weight 0.027 with the NatGas volume of the first 3 months of 2015 (21,590 MWh) then divided by 6. The outcome (-97.16) represents the monthly financial exposures of Gasoil expressed in MT prior to the start of the 3 monthly period. This routine is performed for every 3 monthly period during the contract. Because certain months (Oct-14 to Jun-15) are included in the pricing period for 2 of the quarters in the delivery period (Jan-15 to Dec-15), the underlying exposure is twice as high in QII and in QIII versus QI and QIV.

				2014							2015													
				Tot	Jul	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
NatGas Formula	Primary	EON	Physical	Contract ID's 99	MWh	87 600	0	0	0	0	0	7 440	6 720	7 430	7 200	7 440	7 200	7 440	7 200	7 450	7 200	7 440		
	0.1 % GO Barges FOB Rdam	EON	Financial	Contract ID's 99	MT	-2 365	-97.16	-97.16	-97.16	-195.44	-195.44	-195.44	-197.64	-197.64	-197.64	-198.77	-198.77	-198.77	-99.41	-99.41	0	0	0	
	Brent crude oil	EON	Financial	Contract ID's 99	Barrel	-11 388	0	0	0	0	0	-967.20	-873.60	-965.90	-936	-967.20	-936	-967.20	-967.20	-936	-968.50	-936	-967.20	0

For contract 100 and 125 the underlying financial exposure is different. The bought volumes are the same (blue and black bar) but contract 125 has no underlying financial exposure as this is a fixed price contract. Contract 100 is stored as an indexed contract using the TTF index. In this example the financial exposure has the same weight (1) as the physical exposure.

Physical price clicks lower the remaining financial exposure.



The cashflow is presented in the default system currency but the user is able to present the cashflow in the original contract currency using the applicable tick box.

Display options [Reset selection](#)

Trading date

2014-11-15

Report period

from

Jul

2014

to

Dec

2015

Reporting properties

Commodity

Counterparty

Cost type

(None)

☒ Report in original currency

[Export to Excel](#)

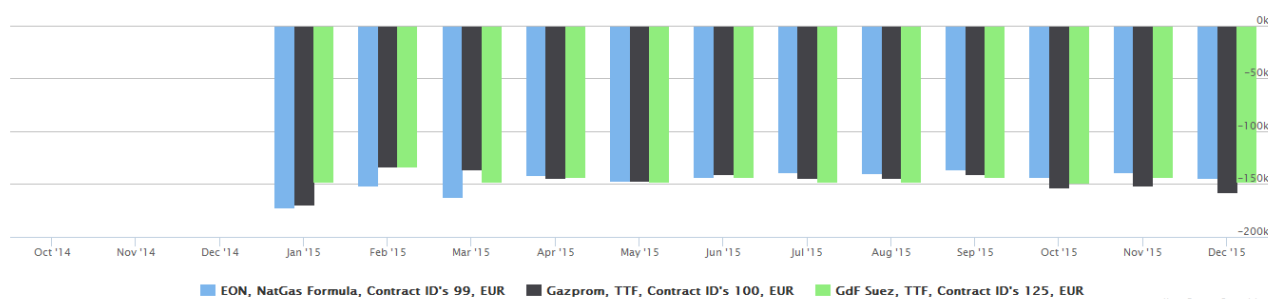
[Create report](#)

The next tab shows the expected cashflows of the contract 99, 100 and 125. The expected cash-flows are derived from the stored price structures (formula, indexation, fixed). The cashflow report presents the commodity cashflows of the contract and the calculated cash-flows are based on a mix of:

- settlement prices of TTF, Brent and Gasoil in historical pricing periods (contract 99,100)
- forward prices of TTF, Brent and Gasoil for future pricing periods (contract 99,100)
- the fixed price of 19.91 multiplied with the volumes (contract 125)

[Positions](#) [Positions + underlying](#) [Cash flow](#) [Average price](#) [Cash flow + costs](#) [Average price + costs](#) [Mark to market](#)

Cash flow



		2014				2015											
		Tot	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
EON	NatGas Formula	Contract ID's 99	EUR	-1 756 906	0	0	0	-171 910	-151 017	-162 269	-141 335	-146 893	-143 032	-138 783	-139 638	-135 920	-143 040
Gazprom	TTF	Contract ID's 100	EUR	-1 762 210	0	0	0	-169 312	-133 479	-135 736	-144 540	-146 888	-141 005	-144 090	-144 544	-140 494	-153 239
GdF Suez	TTF	Contract ID's 125	EUR	-1 744 992	0	0	0	-148 205	-133 862	-148 006	-143 424	-148 205	-143 424	-148 205	-143 424	-148 404	-143 424
Total			EUR	-5 264 108	0	0	0	-489 427	-418 358	-446 011	-429 299	-441 986	-427 461	-431 078	-432 387	-419 838	-444 683

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Note that the presented cashflows are without cost components.

The user can include the contract costs as well (additional to the contract price). The total cashflows can be presented separated into a commodity (contract) cashflow and a cashflow of the additional costs. The costs can be viewed separately using the Display option "Cost type".

Display options [Reset selection](#)

Trading date

2015-02-03

Report period

from

Oct

2014

to

Dec

2015

Reporting properties

Counterparty

Commodity

Contract ID's

Cost type

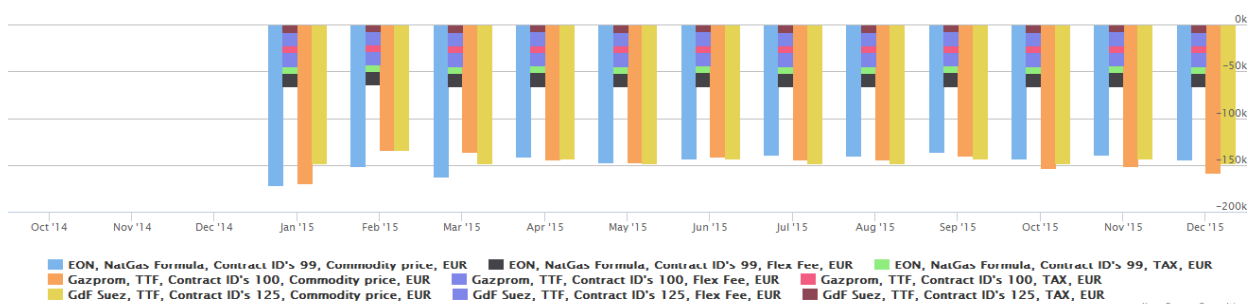
☐ Report in original currency

[Export to Excel](#)

[Create report](#)

[Positions](#) [Positions + underlying](#) [Cash flow](#) [Average price](#) [Cash flow + costs](#) [Average price + costs](#) [Mark to market](#)

Cash flow incl. additional costs



				2014				2015												
				Tot	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
EON	NatGas Formula	Contract ID's 99	Commodity price	EUR	-1 756 906	0	0	0	-171 910	-151 017	-162 269	-141 335	-146 893	-143 032	-138 783	-139 638	-135 920	-143 404	-138 894	-144 174
			Flex Fee	EUR	-175 200	0	0	0	-14 600	-14 600	-14 600	-14 600	-14 600	-14 600	-14 600	-14 600	-14 600	-14 600	-14 600	-14 600
			TAX	EUR	-87 600	0	0	0	-7 440	-6 720	-7 430	-7 200	-7 440	-7 200	-7 440	-7 440	-7 200	-7 450	-7 200	-7 440
Gazprom	TTF	Contract ID's 100	Commodity price	EUR	-1 762 210	0	0	0	-169 312	-133 479	-135 736	-144 540	-146 888	-141 005	-144 090	-144 544	-140 494	-153 239	-150 991	-157 892
			Flex Fee	EUR	-175 200	0	0	0	-14 600	-14 600	-14 600	-14 600	-14 600	-14 600	-14 600	-14 600	-14 600	-14 600	-14 600	-14 600
			TAX	EUR	-87 600	0	0	0	-7 440	-6 720	-7 430	-7 200	-7 440	-7 200	-7 440	-7 440	-7 200	-7 450	-7 200	-7 440
GdF Suez	TTF	Contract ID's 125	Commodity price	EUR	-1 744 992	0	0	0	-148 205	-133 862	-148 006	-143 424	-148 205	-143 424	-148 205	-148 205	-143 424	-148 404	-143 424	-148 205
			Flex Fee	EUR	-175 200	0	0	0	-14 600	-14 600	-14 600	-14 600	-14 600	-14 600	-14 600	-14 600	-14 600	-14 600	-14 600	-14 600
			TAX	EUR	-87 600	0	0	0	-7 440	-6 720	-7 430	-7 200	-7 440	-7 200	-7 440	-7 440	-7 200	-7 450	-7 200	-7 440
Total				EUR	-6 052 508	0	0	0	-555 547	-482 318	-512 101	-494 699	-508 106	-492 861	-497 198	-498 507	-485 238	-510 833	-498 709	-516 390

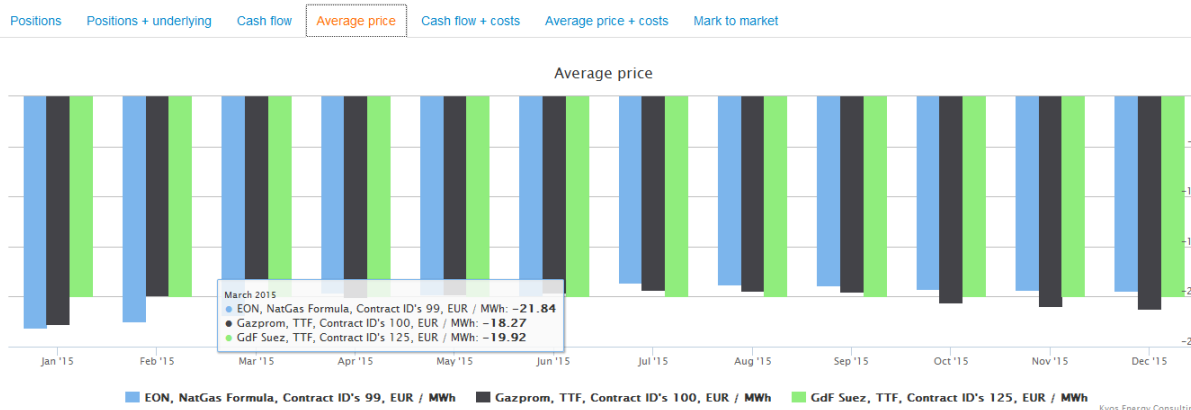
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Note that the costs for each of the contract is the same:

- One cost component (TAX) is calculated by multiplying the contract volume with EUR 1 per MWh.
- The second cost component (Flex Fee) is stored as a fixed cost of EUR 175,200 per year.

This functionality can be used in a tender phase to compare different providers.

In the tab "Average price" the user is able to view the average commodity costs per unit. In the following example the user selected contract 99, 100 and 125 to view the "Average price" per counterparty and per counterparty, which is displayed here in EUR/MWh.



2015				Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
EON	NatGas Formula	Contract ID's 99	EUR / MWh	-23.11	-22.47	-21.84	-19.63	-19.74	-19.87	-18.65	-18.77	-18.88	-19.20	-19.29	-19.38
Gazprom	TTF	Contract ID's 100	EUR / MWh	-22.76	-19.86	-18.27	-20.08	-19.74	-19.58	-19.37	-19.43	-19.51	-20.57	-20.97	-21.22
GdF Suez	TTF	Contract ID's 125	EUR / MWh	-19.92	-19.92	-19.92	-19.92	-19.92	-19.92	-19.92	-19.92	-19.92	-19.92	-19.92	-19.92

In the tab “Average price + costs” the user is able to view the average costs per unit including the cost components. The user has to select ‘cost type’ in the Display options.

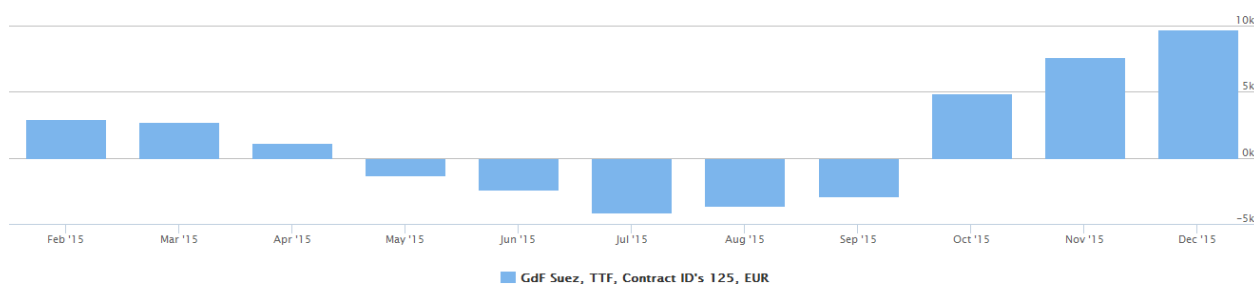


2015				Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
EON	NatGas Formula	Contract ID's 99	Commodity price	EUR / MWh	-23.11	-22.47	-21.84	-19.63	-19.74	-19.87	-18.65	-18.77	-18.88	-19.20	-19.29	-19.38
			Flex Fee	EUR / MWh	-1.96	-2.17	-1.97	-2.03	-1.96	-2.03	-1.96	-1.96	-2.03	-1.96	-2.03	-1.96
			TAX	EUR / MWh	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1
Gazprom	TTF	Contract ID's 100	Commodity price	EUR / MWh	-22.76	-19.86	-18.27	-20.08	-19.74	-19.58	-19.37	-19.43	-19.51	-20.57	-20.97	-21.22
			Flex Fee	EUR / MWh	-1.96	-2.17	-1.97	-2.03	-1.96	-2.03	-1.96	-1.96	-2.03	-1.96	-2.03	-1.96
			TAX	EUR / MWh	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1
GdF Suez	TTF	Contract ID's 125	Commodity price	EUR / MWh	-19.92	-19.92	-19.92	-19.92	-19.92	-19.92	-19.92	-19.92	-19.92	-19.92	-19.92	-19.92
			Flex Fee	EUR / MWh	-1.96	-2.17	-1.97	-2.03	-1.96	-2.03	-1.96	-1.96	-2.03	-1.96	-2.03	-1.96
			TAX	EUR / MWh	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1

The last tab contains the Mark-to-Market. The MtM is the difference between fixed contract prices versus the current forward prices. Mark-to-Market (MtM) is not calculated on indexed contracts, except for the parts of the volume which have been clicked, i.e. have a fixed price.

Positions Positions + underlying Cash flow Average price Cash flow + costs Average price + costs **Mark to market**

Mark to market



GdF Suez, TTF, Contract ID's 125, EUR

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		2015											
		Tot	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
GdF Suez	TTF Contract ID's 125	EUR 14 408	2 918	2 727	1 116	-1 317	-2 419	-4 114	-3 660	-2 930	4 835	7 567	9 687
Total		EUR 14 408	2 918	2 727	1 116	-1 317	-2 419	-4 114	-3 660	-2 930	4 835	7 567	9 687

In the following example the user has selected a MtM report of all physical & financial contracts. In the display options the user selected the trading date, the report period, the sequence (Physical/Financial, Counterparty and Commodity). The user also selected that the MtM has to be presented in the original contract currency.

				2015												
				Tot	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
BP	API2 Coal ARA	Financial	Contract ID's 92	USD	-831 327	-72 674	-79 570	-78 293	-77 745	-77 928	-76 285	-76 285	-76 285	-72 088	-72 088	-72 088
Barclays Int	NCG	Financial	Contract ID's 93	EUR	-18 182	299.52	-891.60	-1 722	-2 738	-3 226	-3 467	-3 964	-3 609	-2 786	1 895	2 027
Enovos	German Power	Physical	Contract ID's 90	EUR	50 618	31 664	1 189	-7 416	-17 856	-7 992	-3 274	-13 318	8 640	14 630	33 428	10 923
	TTF	Financial	Contract ID's 88	EUR	-389 147	-27 562	-35 018	-35 460	-39 112	-38 995	-41 910	-41 456	-39 506	-33 011	-29 009	-28 108
GdF Suez	NCG	Physical	Contract ID's 117	EUR	-102 082	11 474	11 368	9 085	7 232	5 702	5 592	4 472	4 840	7 141	17 224	17 953
	TTF	Physical	Contract ID's 125	EUR	14 408	2 918	2 727	1 116	-1 317	-2 419	-4 114	-3 660	-2 930	4 835	7 567	9 687
Shell	TTF	Physical	Contract ID's 89	EUR	1 227	2 028	2 374	2 064	1 750	1 516	-4 006	-3 812	-3 428	-190.54	1 051	1 883
			Contract ID's 95	EUR	-136 582	-9 401	-12 085	-12 474	-14 125	-14 242	-15 524	-15 297	-14 497	-11 067	-9 248	-8 623
Vitol	Naphtha	Physical	Contract ID's 129	USD	6 537 857	517 857	584 000	588 000	592 000	596 000	600 000	604 000	608 000	612 000	616 000	620 000
Total				USD	5 706 531	445 183	504 430	509 708	514 255	518 073	523 715	527 715	531 715	539 913	543 913	547 913
				EUR	-375 575	11 420	-30 337	-44 808	-66 166	-59 655	-66 702	-77 035	-50 491	-20 449	22 908	5 741

Note that contract ID 125 is now part of the total MtM overview.

15.3 Risk physical

Reports → Risk physical → Physical

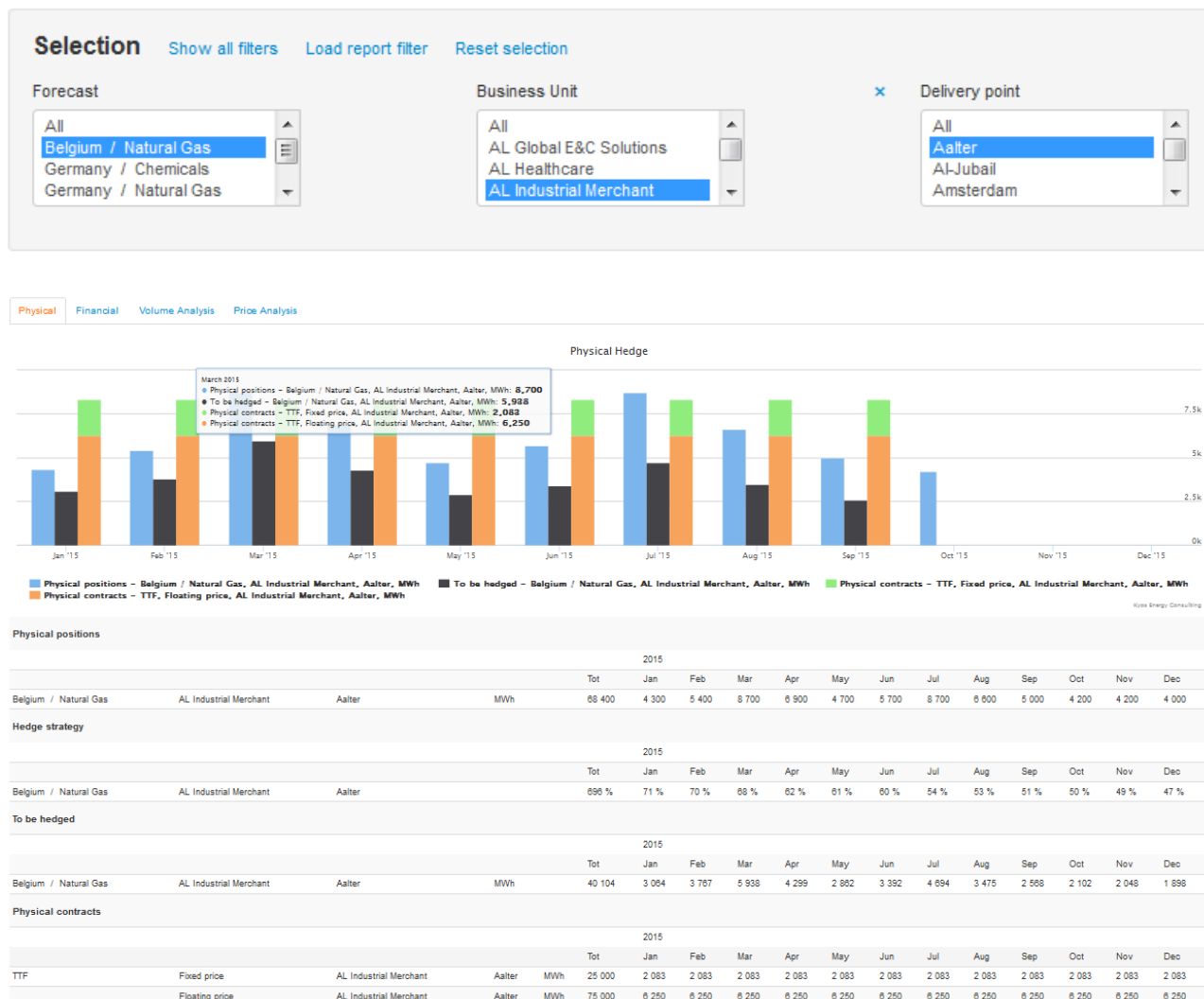
After storing relevant budgets, physical & financial contracts as well as the relevant hedging strategy the hedge report "Risk physical" can be generated. This hedge reports contains:

- Total expected commodity requirements for the future period (taken from Time series)
- What part of this total should be hedged (latest forecast from Time series * hedging strategy)

- The relevant physical contracts that have been stored

The user has to select the applicable Forecast (Country/Commodity), the Business Unit and the Delivery point to view the physical hedge report. See the following example:

Run Risk management Risk physical Invoices

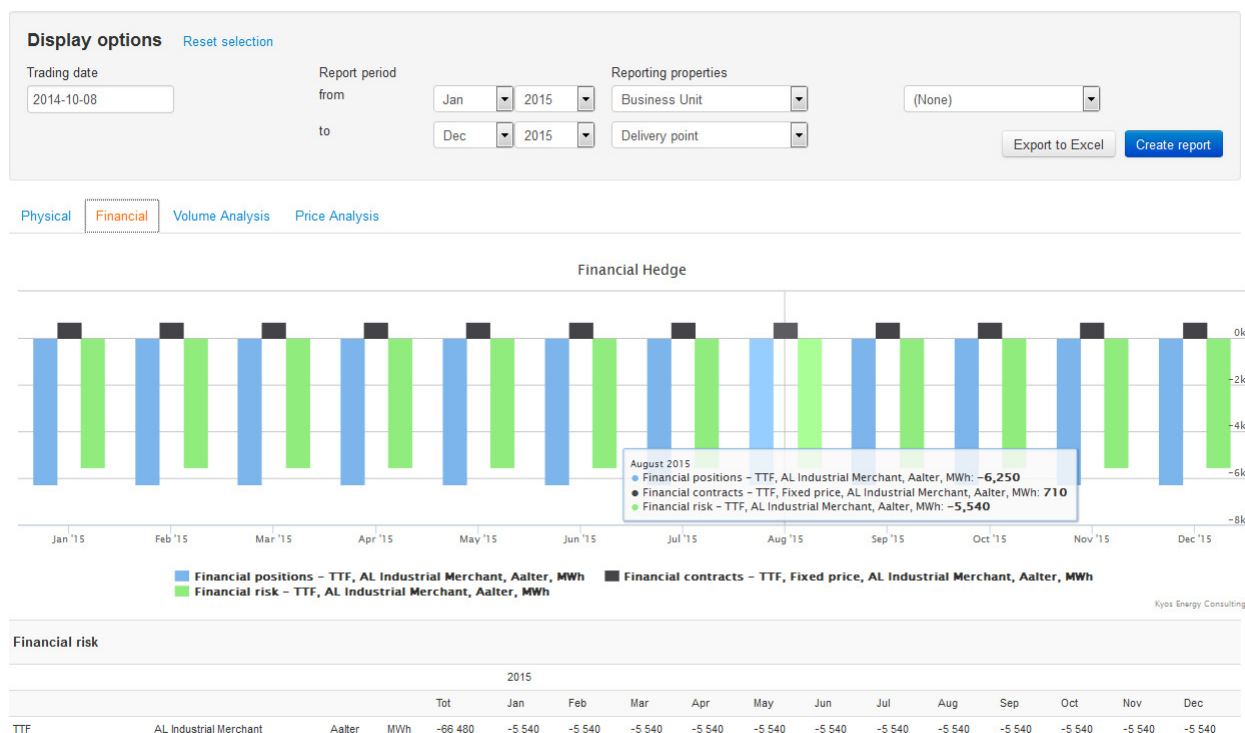


The system will present in the Physical position an overview of the latest known budget figures. See picture on the page before. The overview shows the monthly volumes of the selected Country/commodity then the selected Business Unit and finally the relevant Delivery point.

In the next row the system presents the hedging strategy for the above stated combination. The system calculates the volumes that should have been hedged by multiplying the hedging percentages with the physical positions as taken from the latest known budget.

In the last row the system presents the physical contracts that are part of this selection. The system automatically searches for physical contracts with commodities that are part of the budget figures. These commodities are defined earlier in Time series.

In **Reports → Risk physical → Financial** the user is able to view the calculated open position (financial risk). This open position is a combination of physical contracts expressed in the underlying financial positions and the financial contracts (financial hedges). The financial contracts are deducted from the financial positions to be able to calculate and present the financial risk expressed in a volume.



15.4 Invoices

Reports → Invoices

The invoice functionality of the PRM system generates expected settlement amounts for the selected legal entity and period. This enables the user to estimate the expected settlement amounts directly after the month end for example of the financial settlement of swaps. The user is also able to calculate expected invoice amount of future months.

The output of the invoice report can be exported to Excel.

Selection
[Load report filter](#)
[Reset selection](#)
[Save this filter](#)

Legal Entity

Air Malta p.l.c.
AL Benelux BV
AL Internal
Belgium Client

Counterparty

All
BASF
BP
Chemoil

Payment for delivery in

Jan 2015

[Export to Excel](#)
[Create invoice](#)

[Overview](#) [Details](#)

Invoices for Legal Entity Belgium Client, Jan 2015

Counterparty		EON Global Commodities											
Commodity		TTF											
		Physical Contract											
ID/Conf.number	Buy/Sell	Contract Period	Delivery Period	Payment Date	Delivery point	Fixed price Volume	Floating price Volume	Price	Cash flow Commodity	Entry Fee	TAX	Green Charge	Total Payment
60 /	Buy	1 Jan 2015 - 31 Dec 2015	Jan 2015	2 Feb 2015	Brussel HO	4 135 MWh	3164.7522 MWh	21.9132 EUR	-159 960 EUR	-2 000 EUR	-1 825 EUR	-12 775 EUR	176 566 EUR
Total													176 566 EUR

Invoices for Legal Entity Belgium Client, Jan 2015

Counterparty	Mitsui & Co									
Commodity	Copper									
	Physical Contract									
ID/Conf.number	Buy/Sell	Contract Period	Delivery Period	Payment Date	Delivery point	Fixed price Volume	Floating price Volume	Price	Cash flow Commodity	Total Payment
61 /	Buy	1 Jan 2015 - 31 Dec 2015	Jan 2015	2 Feb 2015	Brussel HO		100 MT	7 014 USD	-701 400 USD	701 400 USD
Total										701 400 USD

Invoices for Legal Entity Belgium Client, Jan 2015

Counterparty	Mitsui & Co									
Commodity	Copper									
	Financial Swap									
ID/Conf.number	Buy/Sell	Contract Period	Delivery Period	Payment Date	Delivery point	Volume	Fixed price	Floating price	Cash flow Commodity	Total Payment
62 /	Buy	1 Dec 2014 - 30 Nov 2015	Jan 2015	2 Feb 2015		75 MT	6 975 USD	7 000 USD	1 875 USD	-1 875 USD
Total										-1 875 USD

By clicking on “Details” the user can view the expected invoice including physical price clicks.

Invoices for Legal Entity Air France - KLM, Jan 2015

Counterparty	Total													
Commodity	Jet Fuel (USD)													
	Physical Contract													
ID/Conf.number	Buy/Sell	Contract Period	Delivery Period	Payment Date	Delivery point	Fixed price Volume	Floating price Volume	Fixed Price	Floating Price	Cash flow Commodity	Entry Fee	Crack	Green Charge	Total Payment
70 /	Buy	1 Jan 2015 - 31 Dec 2015	Jan 2015	2-Feb-2015	Schiphol	4 100 MT		1 050 USD		-4 306 025 USD				4 306 025 USD
70 /	Buy	1 Jan 2015 - 31 Dec 2015	Jan 2015	2-Feb-2015	Schiphol	4 100 MT		1 001 USD		-4 103 075 USD				4 103 075 USD
70 /	Buy	1 Jan 2015 - 31 Dec 2015	Jan 2015	2-Feb-2015	Schiphol	4 100 MT		975.2500 USD		-3 998 525 USD				3 998 525 USD
70 /	Buy	1 Jan 2015 - 31 Dec 2015	Jan 2015	2-Feb-2015	Schiphol		28 700 MT		898.4430 USD	-25 785 314 USD				25 785 314 USD
70 /	Buy	1 Jan 2015 - 31 Dec 2015	Jan 2015	2-Feb-2015	Schiphol						-10 000 USD	-102 500 USD	-82 000 USD	194 500 USD
Total														38 387 439 USD

16 KyInventory

The KyInventory model serves as an accounting tool kit that can be employed for a gas cavern storage facility. The model keeps a track of the hourly net volumes in the various tranches of the cavern and the hourly weighted average costs incurred by the production/injection/withdrawal facilities. In addition, the model facilitates the customer sales tagging process by taking into account, the daily gas volume intake of the various customers.

The display created by the KyInventory module on the platform shows:

1. WACOG: This is the Weighted Average Cost of Gas and is computed on an hourly basis for the PAD and WORKING tranches in the cavern facility. Three different methodologies (LOWEST, AVERAGE, and HIGHEST) are used to compute WACOG, thereby resulting in 6 values for each hour.
2. Net Volume: This is computed on an hourly basis for the PAD and WORKING tranches in the cavern facility.
3. Customer Sales Tagging: This is computed on a daily basis for each customer and indicates the volume intake by that customer from PAD, Back-Up and As-Available tranches. In fact, the Back-Up and As-Available are sub-tranches of the WORKING tranche. In addition, for every tranche, the sales tagging also differentiates between the so-called “Base” and “Supplemental” methodologies.

In addition to the display data, the model produces two output csv files that contain the hourly WACOG and Net Volumes for:

1. PAD/WORKING tranches, broken down by methodology (LOWEST/AVERAGE/HIGHEST)
2. Individual Plant Facilities that include all the production, injection and withdrawal units, broken down by methodology (LOWEST/AVERAGE/HIGHEST)

16.1 Create KyInventory profile – General Settings

This subsection provides a practical demonstration to create and run a KyInventory profile. The profile can be created on the KYOS Analytical Platform at:

Analytics → KyInventory □ Create profile

Refer to the Figure below for a snapshot from the KYOS platform and a description of the various entries.

Define KyInventory profile

[Back to overview](#)

General

Name

Inventory

Calculation methodology

Schedule profile ☒ Yes ☐ No

Schedule time

Automated ☐ Yes ☒ No

Inputs

Input directory

File name injection

File name withdrawal

File name vent

File name incremental costs

File name facility output

[Save](#)

Name

Enter a unique profile name.

Inventory

Choose from the list of inventory names that were previously created. An Inventory can be created at:

Assets & Contracts → **Inventory** ☐ **Create profile**

Calculation Methodology

Select the appropriate calculation methodology to be used for that day: LOWEST/AVERAGE/HIGHEST.

Schedule Profile

Select ('Yes' or 'No') whether the KyInventory profile job needs to be executed at some user-defined time during the day. If 'No' is selected then the job will not be executed at any pre-defined time on each day.

Schedule Time

Specify the user-defined time when the job is supposed to be executed. This is required if 'Yes' is selected for *Schedule Profile*.

Automated

Select whether the job execution process needs to be automated or not. If 'Yes' is selected, then the job is

run automatically whenever a new set of input files are provided or rather the job is run when all the necessary conditions for running the job are satisfied. This can happen several times during the day but it should be noted that the same job will not be run again.

Kindly note that if both, 'Schedule Profile' and 'Automated' are selected as 'Yes', then the 'Automated' option may override the 'Schedule Profile' option. Hence, it is appropriate to choose one of the two options but not both.

16.2 KyInventory profile – Inputs

The user has to mention the following input information in order to complete the profile creation process:

1. **Input Directory:** This specifies the complete path that contains the location of the input directory. e.g. C:\Users\stole.KYOS\Downloads\Test\Dec1
2. **File Name Injection:** Name of the input csv file that contains the hourly injection volume data.
3. **File Name Withdrawal:** Name of the input csv file that contains the hourly withdrawal volume data.
4. **File Name Vent:** Name of the input csv file that contains the hourly vent volume data.
5. **File Name Incremental Costs:** Name of the input csv file that contains the marginal/incremental costs for each individual (production/injection/withdrawal) plant unit for every hour.
6. **File Name Facility Output:** Name of the input csv file that contains the hourly meter readings of all production plant units and also the minimum and maximum volume capacities.

All the five input csv files need to be inside the input directory. We now describe the general format (column entries) of the respective input files. The input files containing the hourly injection, withdrawal and vent volume information have the same format which is shown in the figure below. The column entries are self-explanatory and there should be altogether 24 volume entries covering all the hours. The files are meant for a single particular date (indicated by first three columns) and there should not be multiple dates.

Year	Month	Day	Hour	Volume
2014	12	10	1	1041.75
2014	12	10	2	1041.75
2014	12	10	3	1041.75
2014	12	10	4	1041.75
2014	12	10	5	1041.75

Format for Injection/Withdrawal/Vent input files

The input file containing the meter reading data for all the production plant units should have the following format as shown in the figure below. The column entry 'Volume' refers to the realized volume of that particular production plant (identified by 'Source ID') in that particular hour. Likewise, the column entries 'Minimum' and 'Maximum' refer to the bounds on the volume capacity of that particular plant unit. The file is meant for a single particular date (specified in first three columns) and there should not be multiple dates. The file needs to necessarily contain a numerical entry for all the hours on that particular date and thereby 24 entries for each production plant.

Year	Month	Day	Hour	Source ID	Volume	Minimum	Maximum
2014	12	10	1	101	1250.1	0	1250.1
2014	12	10	2	101	1250.1	0	1250.1
2014	12	10	3	101	1250.1	0	1250.1
2014	12	10	4	101	1250.1	0	1250.1

Format for “Facility Output” file

The input file containing the incremental cost data should have the following format as shown in the figure below. The column entry ‘Process’ refers to the type of the facility, i.e. Injection or Withdrawal or Production. The column entry ‘Volume’ refers to the volume contributed by an individual unit of that (production/injection/withdrawal) plant facility and the column entries ‘NG’, ‘Elec’, ‘Water’, ‘Other’ refer to the marginal costs incurred by that individual plant unit during that particular hour. Similar to the other input files, this file is also meant for a particular date and should not contain multiple date entries. There should be numerical entries for all the hours.

Year	Month	Day	Hour	Process	Source ID	Volume	NG	Elec	Water	Other
2014	12	10	1	Production	101	41.67	1	3	2	4
2014	12	10	1	Production	101	83.34	1	3	2	4
2014	12	10	1	Production	101	125.01	1	3	2	4
2014	12	10	1	Production	101	166.68	1	3	2	4
2014	12	10	1	Production	101	208.35	1	3	2	4
2014	12	10	1	Production	101	250.02	1	3	2	4
2014	12	10	1	Production	101	291.69	1	3	2	4

Format for ‘Incremental Cost’ file

16.3 KyInventory profile – Editing

After a KyInventory profile is created, a user is provided with the following dropdown menu:

☐	ID	Name	Inventory	Methodology	Schedule	Automated	
☐	2	Test KyInventory	Haarlem Inventory	Lowest	No	No	View

20 per page

Calculate WACOG
Edit
Default sales distributions
Delete

Delete selected
Save selected as search

Edit

This allows the user to edit the various parameters in the created profile.

Delete

The user has a choice to delete the created profile.

Calculate WACOG

The user can choose this to start the WACOG calculations. In case of automatic start, this choice becomes redundant as the job is automatically run at the user defined time.

Default Sales Distribution

This allows the user to specify the default percentage breakdown of a customer volume intake into 'Base' and 'Supplemental' categories. The user can select this option and can fill in the values at:

Default sales distributions for Test KyInventory profile.

[Back to overview](#)

Counterparty	Base	Supplemental
Bayern Gas		
Kyos Energy		
Trading House		
Vattenfall		

[Save](#)

16.4 KyInventory profile – Outputs

As explained previously, a job can be manually run or can be automated. Upon pressing the 'view' button, the following outputs are displayed:

1. WACOG: The Weighted Average Cost of Gas (WACOG) represents the volume weighted average of the incremental costs involved in utilizing the production, injection and withdrawal plant units.

The WACOG is displayed each hour for all the tranches (PAD/WORKING) and for all methodologies (LOWEST/AVERAGE/HIGHEST), thus leading to overall six values for every hour.

The hourly WACOG is further broken down by individual costs incurred for Natural Gas, Power, Water and other. The figure below shows a snapshot of the WACOG display.

As can be seen, the user can select the appropriate combination of tranche and methodology to view the final display.

WACOG
Net volume
Sales tagging

Reset selection
Filter

Methodology
Tranche

All
Lowest
Average
Highest

All
Pad
Working

20 results found. Total items: 48.

« 1 2 3 » 20 per page

Date	Volume (Mscf)	Methodology	Tranche	Natural gas (USD/Mscf)	Power (USD/Mscf)	Water (USD/Mscf)	Other (USD/Mscf)	Total WACOG (USD/Mscf)	Total WACOG Adjusted (USD/Mscf)
12/02/2014 00:00:00	22221	LOWEST	WORKING	1.8567	0.317953	0.02	0	2.19465	2.19465
12/02/2014 01:00:00	23067	LOWEST	WORKING	1.85666	0.318041	0.02	0	2.1947	2.1947
12/02/2014 02:00:00	23882	LOWEST	WORKING	1.85661	0.318161	0.02	0	2.19477	2.19477
12/02/2014 03:00:00	24667	LOWEST	WORKING	1.85653	0.318305	0.02	0	2.19483	2.19483

- Net Volume:** This displays the hourly net volume for all the tranches and sub-tranches (PAD/Back-Up/As-Available) that is computed by considering the hourly injection, withdrawal and vent volumes in each tranche. All these values are displayed as shown in the figure below:

WACOG Net volume Sales tagging

Previous day values

Date and time	Injection (Mscf)	Withdrawal (Mscf)	Vent (Mscf)	Pad (Mscf)	Back-Up (Mscf)	As-Available (Mscf)	Total (Mscf)
12/01/2014 23:00:00	781.2	0	0	2	4	21341.3	21347.3
Date and time	Injection (Mscf)	Withdrawal (Mscf)	Vent (Mscf)	Pad (Mscf)	Back-Up (Mscf)	As-Available (Mscf)	Total (Mscf)
12/02/2014 00:00:00	876	0	0	2	4	22217.3	22223.3
12/02/2014 01:00:00	845.7	0	0	2	4	23063	23069
12/02/2014 02:00:00	815.4	0	0	2	4	23878.4	23884.4
12/02/2014 03:00:00	785.1	0	0	2	4	24663.5	24669.5
12/02/2014 04:00:00	754.8	0	0	2	4	25418.3	25424.3
12/02/2014 05:00:00	943	0	0	2	4	26361.3	26367.3
12/02/2014 06:00:00	876.5	0	0	2	4	27237.8	27243.8
12/02/2014 07:00:00	979.8	0	0	2	4	28217.6	28223.6
12/02/2014 08:00:00	1000	0	0	2	4	29217.6	29223.6
12/02/2014 09:00:00	756	0	0	2	4	29973.6	29979.6
12/02/2014 10:00:00	471.35	0	0	2	4	30444.9	30450.9

- Sales Tagging:** This displays the customer sales tagging information that essentially contains the total daily gas intake by each individual customer which is further broken down by: individual tranches and sub-tranches (PAD/Back-Up/As-Available) and by Base/Supplemental volume categories. The figure below provides the snapshot:

WACOG	Net volume	Sales tagging
-------	------------	---------------

Counterparty	Total (Mscf)	Base (pad) (Mscf)	Base (backup) (Mscf)	Base (as-available) (Mscf)	Supplemental (pad) (Mscf)	Supplemental (backup) (Mscf)	Supplemental (as-available) (Mscf)
Cargill International SA	14491.000	0.000	0.000	3491.000	0.000	0.000	11000.000

In addition to the display on the platform interface, two output csv files are generated upon a job completion that contains the following information:

1. WACOG and Net Volume - For all tranches and methodologies
2. Plant Tagging – WACOG and Net Volume for all individual plant units – For all methodologies and tranches.

17 KyCavern

The KyCavern model's primary purpose is to value and hedge a hydrogen cavern. The model can also calculate optimal cash flows taking the whole hydrogen system into account both with and without a hydrogen cavern. A new KyCavern profile can be created by following the steps:

Analytics → KyCavern ☐ Create profile

Edit KyCavern profile

[Back to overview](#)

General

Name	Test Cavern 2015
KyInventory	Test KyInventory ▼
Inventory	Haarlem Inventory
Tranches	☐ Pad ☐ Backup ☒ As-Available
Inventory end volume type	☒ Fixed volume ☐ Floating with start volume
Inventory end volume	0
Discount rate	1 %
Scheduled	☐ Yes ☒ No
Automate calculation directly when new inputs are available	☐ Yes ☒ No

Inputs

Input directory	a:\Desktop\Test_KyCavern_Inputs
File name Margins	Margin_Output.csv
File name Gas price simulations	NG_Scenarios.csv
File name Power price simulations	Power_Scenarios.csv
File name Net volume change	Net_Volume_Change.csv

Each KyCavern profile is directly connected to a KyInventory profile. Based on the KyInventory profile, the relevant Inventory profile is determined automatically. The KyInventory profile is necessary, since it contains the start volume of the hydrogen cavern per each trading day. The Inventory profile contains the definition of different tranches within the cavern.

Tranche

The user can choose which tranches should be used in the valuation. The tranches' volumes are determined based on the definitions within the connected Inventory profile. The available choices are 'Pad', 'Backup', and 'As-available'. Only consecutive tranches can be chosen together. For example, the user cannot choose only 'Pad' and 'As-available', without choosing the 'Backup' tranche. The model will choose the optimal actions based only on the volumes within the chosen tranches.

Inventory end volume

The 'Inventory end volume' contains the hydrogen volume which should be in the cavern at the end of the valuation period. The user can define the end hydrogen volume as a fixed number, or it could float with the start volume. If the end volume is floating it is determined by the model automatically during each run, based on the start volume.

Discount Rate

The discount rate is applied in all calculations to discount back future cash-flows to the current (valuation) date. For example, if you receive cash-flows of 1000 USD (from hydrogen withdrawals) 1 year from now, and the discount rate is 5%, this is included in the valuation as $1000 * \exp(-5\%) = 951$ USD. Note that the discount rate is interpreted as a continuous rate, which leads to the use of the exponential function.

Scheduled

The user can schedule a run of the profile every day at a certain time. The model would use the latest available files in the input directory.

Automate calculation directly when new inputs are available

KyCavern would run automatically when new files are available in the chosen input directory.

Input Directory

This is the path to the directory located on the server, where the KyCavern input 'csv' files should be located.

File Names

These fields should contain the names of the specific KyCavern 'csv' input files. The complete file name should be provided. There are in total four input 'csv' files. The content of the files is discussed in next section.

17.1 CSV Input Files

17.1.1 Total margins file

Assuming there are N simulations, the header line defines the following column names:

- Date
- Rate
- Forward
- Scenario_1, Scenario_2, Scenario_3, ..., Scenario_N

Rate defines the rate of injection (negative number) or withdrawal (positive number). The unit should be consistent with the volume unit used in the Inventory definition. For example MWh, or MMSCF (Million standard cubic feet per day) which is equivalent to MMSCFD (million metric standard cubic feet per day). A specific rate does not need to be defined for all days, except the zero rate, which should always be included. If a rate is missing for a certain date, the valuation model assumes that the specific rate is not available for the dates where it is missing. For example, this could be the case due to maintenance.

The column 'Forward' contains the margin based on forward prices (not simulated). This forms the basis for the intrinsic calculation.

The values in Scenario_1 (to Scenario_N) correspond with the margin based on prices for Scenario 1 (to Scenario N).

17.1.2 Net volume change of the start volume

This file contains the net change of volume from the last available run of the associated accounting profile. This file can be called “Net_Volume_Change.csv”. The file should contain only one number.

17.1.3 Gas and Power price simulations

The price simulations files are initially created by KySim. There will always be one power and one gas file containing spot simulations for the chosen period. The files are called “Power_Scenarios.csv” and “NG_Scenarios.csv”.

The header line describes the content of the different columns. There will always be $N + 2$ columns. The first one is the date (in Excel format), the second one is the daily forward price, and the following columns contain the daily spot simulations.

NG_Scenarios.csv												
1	Date,	Forward,	Scenario_1,	Scenario_2,	Scenario_3,	Scenario_4,	Scenario_5,					
2	41955,	22.73,	22.73,	22.73,	22.73,	22.73,	22.73,	22.73,	22.73,	22.73,	22.73,	22.73,
3	41956,	21.705,	21.618,	21.875,	21.715,	21.692,	21.714,	21.76,	21.551,	21.445,	21.48,	21.48,
4	41957,	21.709,	21.684,	22.195,	21.914,	21.589,	21.73,	21.788,	20.73,	21.405,	21.43,	21.43,

17.2 Manual Run

The model can be run manually through:

Analytics → KyCavern □ Run □ Run

Run KyCavern profile

[Back to overview](#)
[View Jobs](#)

Dates input files

Margins	2015-04-23 14:20:24
Gas price simulations	2015-04-23 14:20:24
Power price simulations	2015-04-23 14:20:24
Net volume change	2015-04-23 14:20:24

Inventory end volume

Volume	0
--------	---

[Run](#)

If all the input files are placed in designated input directory, the relevant fields are in green, and KyCavern can be run by pressing the ‘Run’ button. However, if some of the files are missing the relevant field is red, and the platform does not allow a run.

17.3 Outputs

The outputs for a specific KyCavern profile's run can be viewed by:

Analytics → KyCavern □ View □ View

17.3.1 Values

The cavern valuation model calculates the expected value of the hydrogen network. The calculation is performed twice: first including the cavern, which involves a real optimization, and excluding the cavern, which means that no volumes are injected or withdrawn. The difference shows the value contribution of the cavern.

The value is the total expected margin over the valuation horizon of e.g. 1 year. The tool calculates two different values:

- The intrinsic value, which is based on the current forward curves
- The total value, which is based on the average value across the price simulations. This is the most important value, because it takes into account the uncertainty in future market conditions. Note that the difference between total and intrinsic value is the extrinsic value.

Intrinsic Value

The forward curves for power and gas are the expected future prices. Based on these expected prices, the MTO tool calculates the expected margins per different injection and withdrawal rate into the cavern. Given the expected margins, the valuation model calculates the optimal expected action (inject/withdraw/nothing) per day. Note that for the no-cavern calculation the action is 'nothing'. The intrinsic value is calculated by adding up the resulting daily margins.

Simulation Value

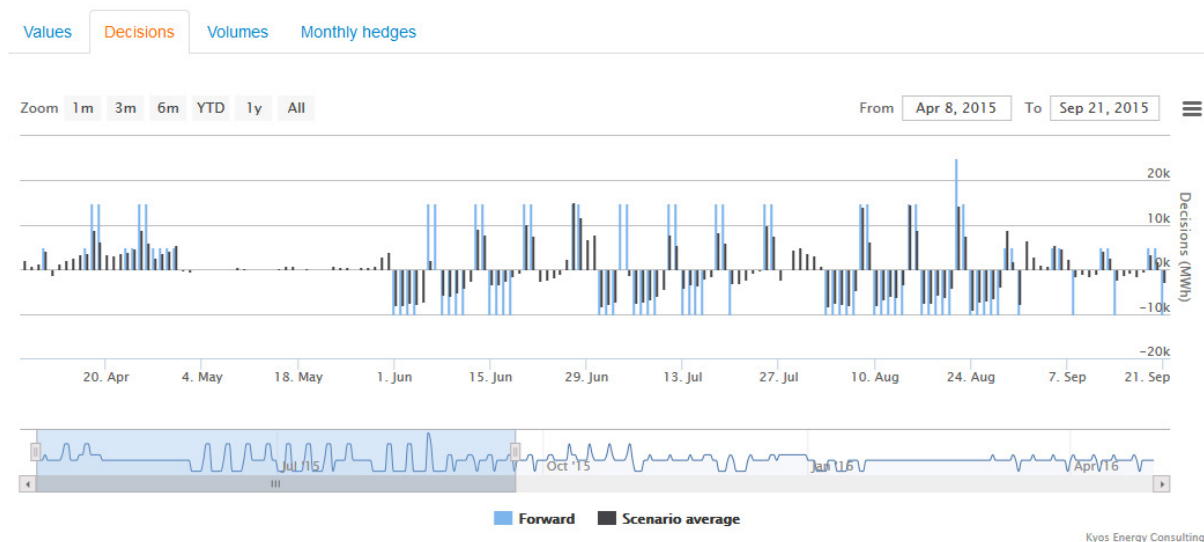
The MTO tool uses as inputs multiple price simulations around the forward gas and power curves coming from the KySim model. For each simulation path, the MTO tool calculates the relevant margins. Using these margins, the valuation model finds the optimal actions per simulation and day.

	Intrinsic value (EUR)	Simulation value (EUR)
Cavern	27 287	42 518
Network without cavern	72 328 890	72 427 325
Network with cavern	72 356 177	72 469 843

17.3.2 Decisions

The cavern valuation model calculates the optimal daily injection or withdrawal decisions. The decisions, based on intrinsic and simulation strategies, are outputted. Each simulations' optimal

decision path is calculated based on stochastic optimization. The average optimal decision over all simulations is the optimal daily strategy. For the first day, the optimal action is the same for each simulation.

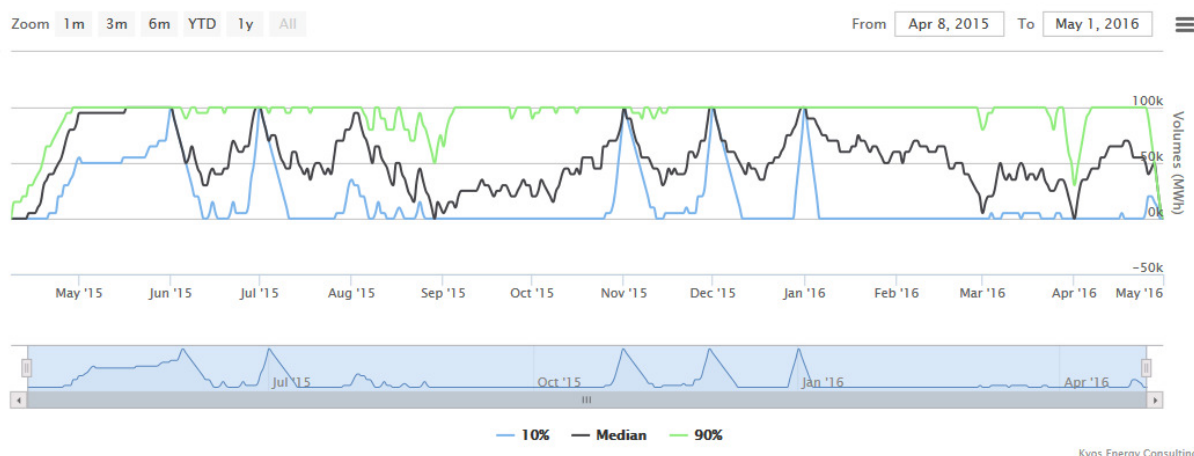


Show/hide table

Date	Forward (MWh)	Scenario average (MWh)
2015-04-08	0	2 050
2015-04-09	0	900
2015-04-10	0	1 300
2015-04-11	5 000	4 150

17.3.3 Volumes

The volumes represent the hydrogen inventory levels at the beginning of each day. If a cavern valuation is performed over 365 days, the number of data rows will be 366. The last row contains the end volume, which is the volume at the beginning of the next day following the 365 days of the valuation.



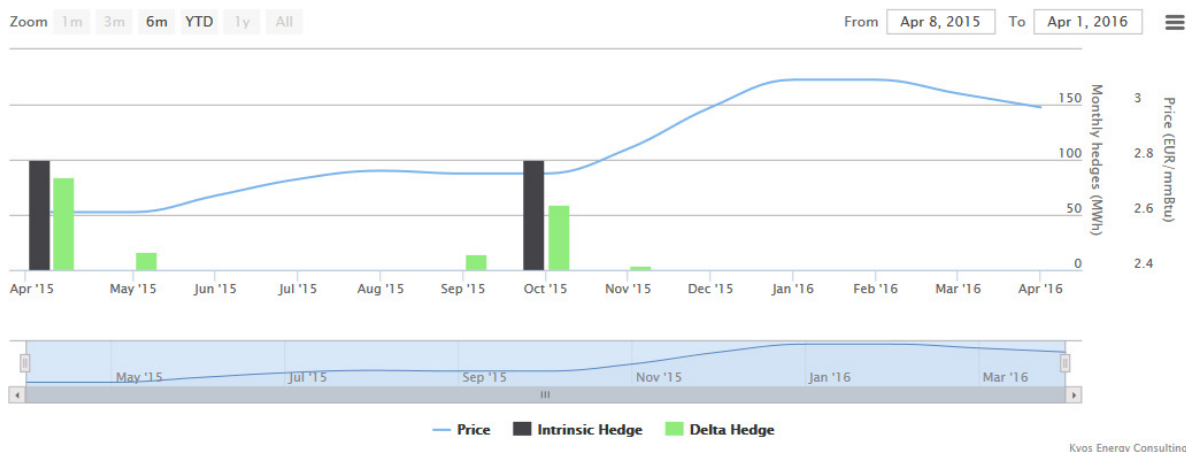
Show/hide table

Date	10% (MWh)	Median (MWh)	90% (MWh)
2015-04-08	0	0	0
2015-04-09	0	0	15 000
2015-04-10	0	0	15 000
2015-04-11	0	0	15 000
2015-04-12	0	0	20 000

17.3.4 Monthly Hedges

The model calculates monthly hedges. They are in the hydrogen's volume unit, and represent a total volume per given month. When the intrinsic hedges are executed they will result in locking in a profit, based on the intrinsic strategy. When the delta hedges are executed they would mitigate the risk associated with a spot trading strategy.

[Values](#)
[Decisions](#)
[Volumes](#)
[Monthly hedges](#)



Show/hide table

Date	Price (EUR)	Intrinsic Hedge (MWh)	Delta Hedge (MWh)
2015-04-08	2.61	100 000	84 827
2015-05-01	2.61	0	17 450
2015-06-01	2.67	0	1 962
2015-07-01	2.73	0	-22 888

18 KyVaR: Value-at-Risk model

Value-at-Risk (VaR) is a risk metric which shows the potential ‘worst-case’ loss in market value of the portfolio over a certain horizon. The worst-case scenario is expressed with a certain confidence interval. For example, a 10 day 95% Value-at-Risk of 1 mln € means that there is 95% probability that the portfolio will lose not more than 1mln €. Or alternatively, there is a 5% probability that the loss will exceed 1 mln €. Compared to Earnings-at-Risk (EaR) or Cashflow-at-Risk (CfaR), the VaR is especially useful when the market value of the portfolio is well-defined and can be measured on a daily basis. In less liquid markets or when the assets and contracts are exposed to considerable spot price or volumetric risks, the EaR or CfaR may be more appropriate.

The KYOS VaR engine in the KYOS Analytical Platform uses the variance-covariance method, also known as Delta-Normal method or parametric method. Separately, KYOS also offers a Monte Carlo simulation and historical simulation VaR model, but these are not (yet) integrated in the Platform. The variance-covariance method assumes that all the market price (log)returns have a joint Normal distribution. In practice, this means that the model needs a series of volatilities and correlations as input. In the KYOS VaR model, the first step consists of calibrating the volatilities and correlations on historical price forward curves. In the second step, the actual VaR is calculated with the calibrated numbers, with the exception of the volatilities if the user has uploaded his/her own values. Both steps automatically run one after the other.

The monthly granularity is another characteristic of the KYOS VaR model in the Platform: all exposures from contracts and assets are measured per month and all volatilities and correlations are for monthly forward prices.

18.1 Create KyVaR profile

This subsection provides a practical demonstration to create and run a VaR calculation with the KyVaR engine. The profile can be created on the KYOS Analytical Platform at:

Analytics → KyVaR □ Create profile

Name

Enter a unique profile name.

Model base currency

The VaR is always a monetary number, e.g. 1 mln USD or 2 mln EUR. All exposures which are in a different currency will be translated into this base currency, also taking into account the FX price risk.

Contract filter, KyPlant filter, KyStore filter, KySwing filter

The Value-at-Risk is calculated over the exposures arising from contracts, plus potentially the exposures from power plants, gas storage and swing contracts. Depending on the KYOS license, one or more filter types are displayed. Filters should first be created in the relevant section. For example, the user first has to create a filter (selection) of one or more contracts in the Contracts menu, before it can be selected in the VaR profile.

Confidence level and Risk horizon

The Value-at-Risk shows the potential ‘worst-case’ loss in market value over a certain horizon and with a certain confidence. The most common confidence levels (95-99%) and risk horizons (1, 5 or 10 trading days) can be selected from the drop-down. Because the model assumes a Normal distribution of the underlying returns, the following rules apply:

- The 5-day risk horizon yields a VaR which is $\sqrt{5} = 2.236$ times higher than a 1-day horizon
- The 10-day risk horizon yields a VaR which is $\sqrt{10} = 3.162$ times higher than a 1-day horizon

And:

- The 95% confidence level corresponds to a multiplier from the Normal distribution of 1.645, compared to 2.326 for the 99% confidence level. So, the 99% VaR is exactly $2.326/1.645 = 1.414$ times higher than the 95% VaR.

Volatility and correlation inputs

The VaR calculation requires a volatility for each commodity and each forward month, plus a correlation matrix for any combination of commodity and forward month. For example, if there are 4 commodities and the exposures are up to 50 months ahead, then the model needs 4 times 50 = 200 forward price volatilities and a correlation matrix with dimensions 200 x 200. The model estimates these numbers on a set of historical forward curves. So, it is very important that the user checks that there is a sufficiently complete history of forward curves for each commodity in Forward curves → Curves

In the KyVaR profile the user defines until what maturity the volatility and correlations should be estimated. Beyond that period, the numbers are assumed to be constant. For example, if the volatilities are calculated until 18 months ahead, then the volatilities for 19 months and beyond (as well as correlations) are assumed to be the same.

If the method for volatility/correlation estimation is ‘equally weighted’, then all returns from the last N trading days are used equally, where N is the user defined window. If the method is ‘exponentially weighted’, then the most recent returns have a higher impact on the estimation than returns from longer ago, using an exponential weighting scheme and a user defined decay factor.

Note that the user may overwrite the historical volatilities, but that the correlations may not. The reason is that the number of correlations easily becomes too large to manipulate by hand, and are also mutually dependent: if 1 correlation is changed, then another correlation may have to be changed as well.

An example input file for the historical volatilities can be downloaded from the Platform. The format is rather simple:

- Column 1 contains the ID’s of the forward curves
- Row 1 contains the numbers 0 to N, where N is the number of forward maturities
- Row 2 and further contain the volatilities, one row per commodity / FX rate

VaR limit

This indicates the maximum level of the VaR. It is an optional input and does not affect the calculations.

Instead, it is a number which is shown on the output screen and on the index page, also changing the background colour of the reported VaR from green (VaR is below the limit) to red (VaR is above the limit). So, it can act as a useful indicator of whether the market price risk of the portfolio is at an acceptable level for the company.

Scheduled

The VaR calculation may run automatically at predefined times and dates. It will then run with the most recent input data.

18.2 Run a KyVaR profile

The VaR calculation may start automatically on certain days and times, if the runs have been scheduled (see previous section).

It is also possible to run the VaR model on an ad hoc basis with the Run option. The calendar shows all dates, but typically the model will be run for the current trading date. In any case, the user must be careful to select a trading date which is not too far in the past, because then there may not be enough historical forward curves for the calibration. Note that for the calculation of current market value, the model selects the most recent historical trading date (if there is no curve for theselected trading date).

18.3 View results

18.3.1 VaR

There are three tabs displaying the results of a VaR calculation: VaR, Volatilities and Exposures. The VaR tab is the most important and displays the total VaR as well as various subportfolio VaRs:

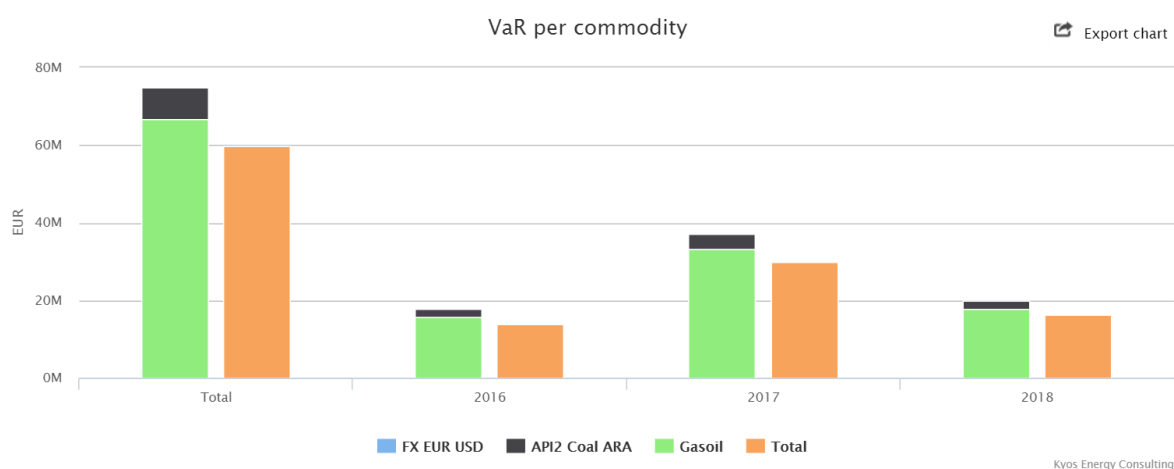
- Per year
- Per commodity (default), counterparty, book and asset type (contracts, plant, storage, swing)

The table shows all subportfolio VaRs. With first dropdown the user can compare the VaR to the VaR of a previous job, e.g. from a previous trading date. In this case the table shows both VaRs, plus the changes.

With the second dropdown the user determines what type of subportfolio VaRs are shown in the bar chart. The choice does not affect what is displayed in the table.

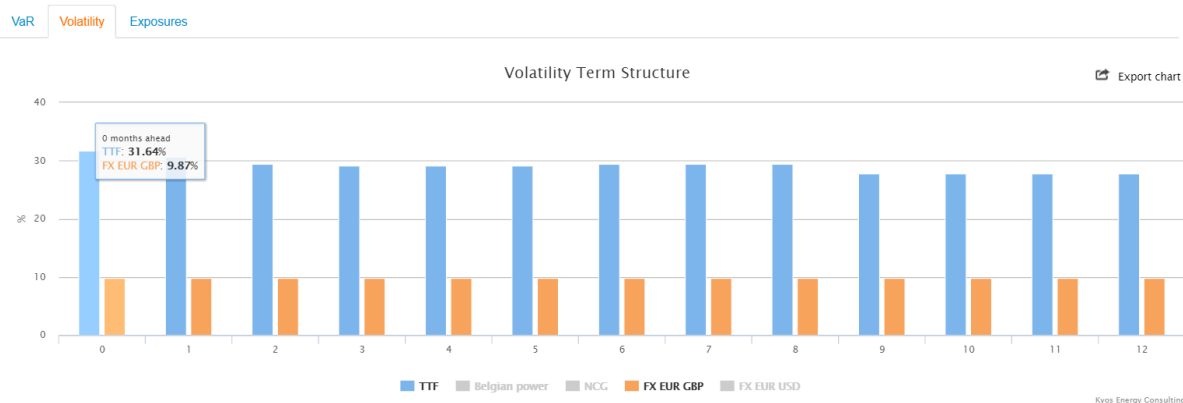
VaR (in EUR) 2016-06-06

		Total	2 016	2 017	2 018
VaR total		59758 000	13856 000	29849 000	16165 000
VaR per commodity	FX EUR USD	0	0	0	0
	API2 Coal ARA	8 067 000	2 259 100	3 896 000	1 928 000
	Gasoil	66 674 000	15 654 000	33 237 000	17 854 000
VaR per book	No book specified	59 758 000	13 856 000	29 849 000	16 165 000
VaR per counterparty	Delta NV	59 758 000	13 856 000	29 849 000	16 165 000
VaR per asset type	Contracts	59 758 000	13 856 000	29 849 000	16 165 000



18.3.2 Volatilities

The volatilities tab displays the volatility term structure which has been used in the calculation, from month 0 (= Balance-of-Month, intra-month) up to the end of the portfolio horizon. The quality of the volatilities depends directly on the quality and availability of the forward curves. Typically, you can expect either a flat curve or a (mostly) downward sloping curve. If that is not the case, please check which forward prices have been used for the calculation. A typical pattern is the following, with a downward sloping curve for Dutch TTF gas and a flat curve for an FX rate:



In order to remove small random variations in the volatility term structure, the volatilities are estimated in groups:

- Month 0
- Month 1
- Month 2
- Month 3,4,5
- Month 6,7,8
- Month 9-12
- Month 13-24

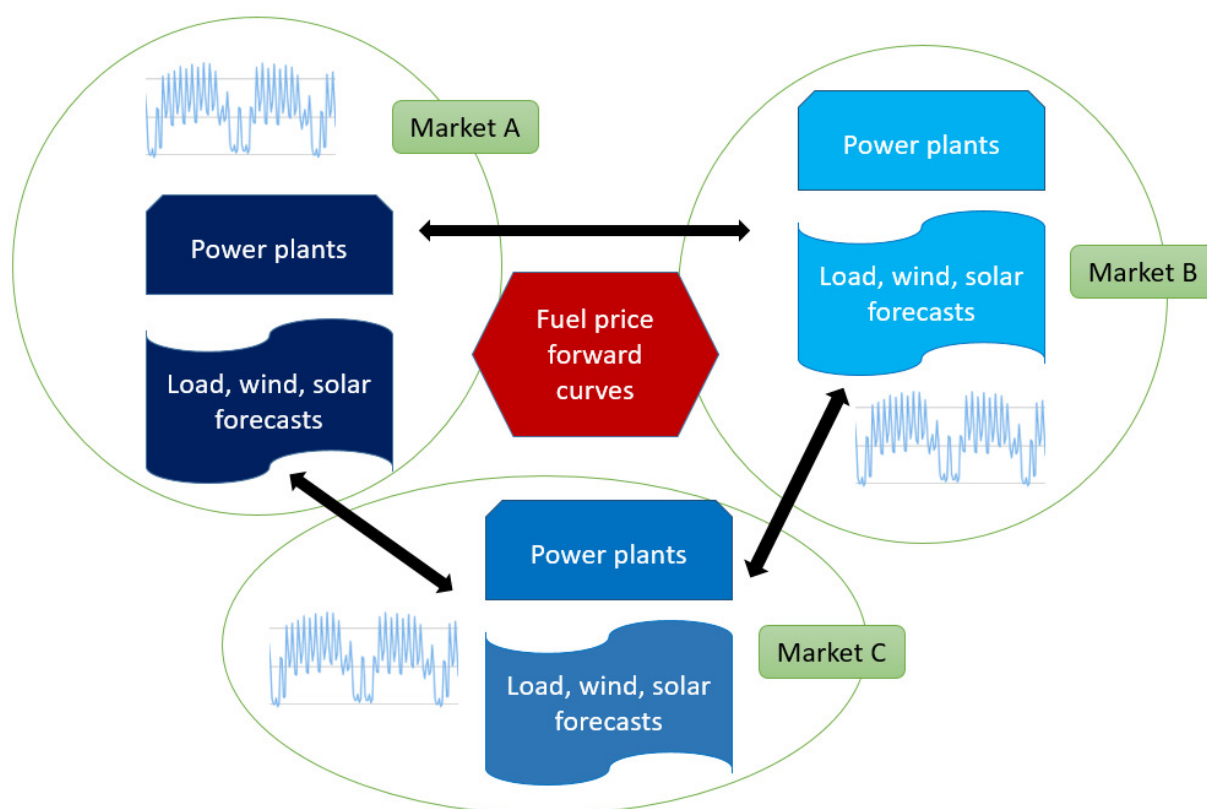
18.3.3 Exposures

In this tab the exposures are shown. The exposures are volumes (e.g. MT or MWh) and netted over the different contracts and assets. For power plant, gas storage and swing contract exposures, the VaR model uses (minus) the delta hedges.

19 KyPowerFundamentals (KyPF)

In this chapter we describe the settings, inputs and outputs of the fundamental power market model KyPowerFundamentals (short name: KyPF). Some of the inputs are defined directly here (e.g. power plant details), other inputs come from other parts within the KYOS Analytical Platform (e.g. fuel prices, load/renewable forecasts and interconnection capacities).

KyPF is a model which forecasts future power prices and generates power price simulations. It does so by simulating actual power markets, where individual plants dispatch their assets so as to earn maximum revenues. For a single power market, KyPF requires detailed inputs of all individual power stations and their fuel costs, together with a forecast of the residual (hourly) demand. This residual demand is the actual demand minus the generation from non-dispatchable sources, typically solar, wind and other renewable sources. KyPF can also jointly generate forecasts of multiple interconnected markets, and optimizes the flows on the interconnection capacities. Finally, KyPF cannot only create a single price forecast, but also a range of simulated forecasts on the basis of simulated input variables. In this case it performs the optimization for each individual input scenario.



19.1 Create KyPF profile

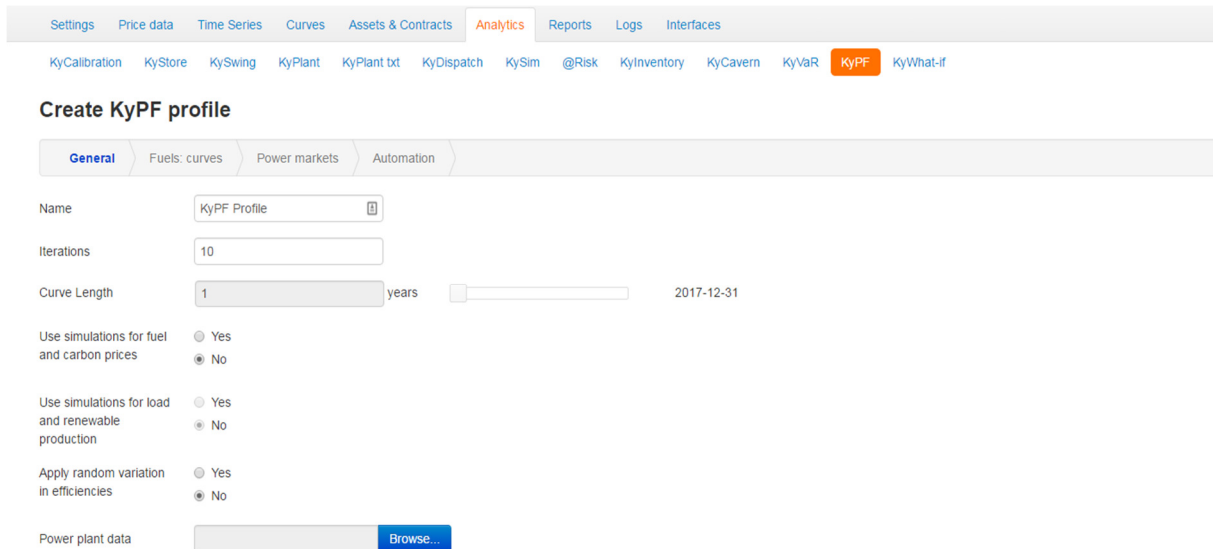
There are 4 default tabs on the screen:

- General

- Fuel: Curves
- Power markets
- Automation

Dependent on the user defined information in the General tab, the tabs may vary in subsequent steps.

Analytics → KyPF ☐ Create profile



General

Name

Enter a unique name

Iterations

Enter a number of iterations. The model uses a fixed number of repeated calculations to converge to an optimal solution. For quick results a number between 10 and 20 is usually good, but for more accuracy it is recommended to set it to 20-40. A higher number of model iterations will provide a more stable result, but will also lead to a longer calculation time (almost linearly). Note that the last 6 iterations are 'balancing', and all previous iterations are 'normal'. Both types aim to minimize the difference between production and residual demand, the imbalance. In a normal step, the model adjusts the hourly prices both upwards (if production is too low) and downwards (if production is too high). In the first 5 balancing steps the model only adjusts the prices upwards in hours with too low production. This ensures that enough plants are committed. In the final balancing step the model keeps the same commitment as in the last balancing step, but adjusts the production of some plants from their maximum output to a lower output. So, when there are e.g. 20 iterations:

- The first 14 are 'normal' steps with prices adjusted in both directions
- The next 5 are 'balancing' steps with only positive price adjustments
- The last is the 'final balancing' step

Curve Length (in years)

The end date of the calculations is set at 31 December of a selected year. For example, if today is 23 October 2015 and the curve length is 1, then the end date is 31 December 2016, i.e. the end of the next year.

Use simulations for fuel and carbon prices

Select whether simulated fuel and carbon price will be used, and is only shown when the user has a KySim license. If “No” is selected, then the forward curves of the fuel and carbon prices will be used as input. If “Yes”, the user has to define a KySim simulation profile and the model will use simulated fuel and carbon prices, and hence generate simulations of hourly power prices.

Use simulations for load and renewable production

Select whether simulated load demand and renewable production will be used (currently only possible option is “No”).

Change plant efficiencies if equal to other plants with the same fuel

This option may be useful in case there are several power plants with the same efficiency. Then it may be difficult for the model to find a stable solution. In this case, a small random adjustment to the efficiencies improves the model stability and speed of convergence. Note that the random variation is only applied to plants which have the same fuel and maximum efficiency as other plants. Also note that the adjustment is between -0.5% and +0.5%, drawn from a uniform distribution.

If “Yes” is selected, the user should decide to use either a random seed number or a user-defined number. A user-defined seed number should be used in case you want to be able to exactly replicate the results.

Power plant data

Upload the file with the format described in the section ‘Power plant data’.

Fuels: Curves

This tab is only shown when the user has NOT selected ‘Use simulations for fuel and carbon prices’ in the General tab.

When the user moves to this tab, he will see a number of commodity names, each of which is a fuel (or an emission). For each fuel a forward curve has to be selected. Note that the list of fuels displayed corresponds to the fuel id’s in power plant input file (PlantData.csv), which has been uploaded in the previous step.

Commodity	Forward curve
API2 Coal ARA	API2 Coal
TTF	TTF

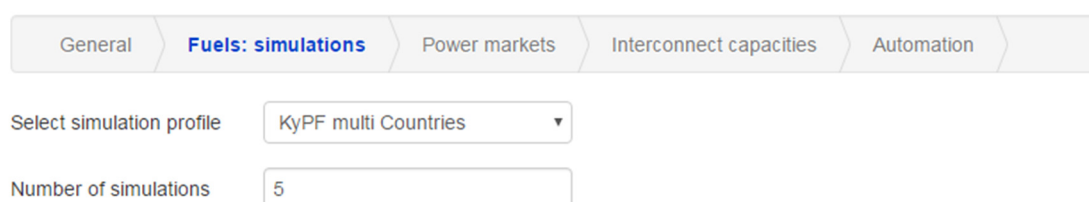
Fuels: Simulations

This tab is only shown when the user has selected 'Use simulations for fuel and carbon prices' in the General tab.

The user has to select a KySim simulation profile. The user can only select a profile with all the commodities in the power plant input file (PlantData.csv).

The second input is the number of simulations. Computational time increases linearly with the number of simulations and we advise to start with a low number (e.g. 5), check the total time, and then decide what is feasible to run within a reasonable amount of time. A large number of simulations could also lead to memory issues, depending on the specifications of the server.

Please note that it is also required to define a number of simulations while defining the KySim profile. Therefore, KyPF model will use the minimum of the number of simulations defined in KyPF and KySim.



Power markets

In this tab, the user defines a number of inputs for each power market:

- Total load: the hourly load (demand) time series. This time series is an input to KyPF. The model will find the optimal production of all power plants, such that the production of the plants, including the renewable generation as well as import minus export, equals the total load in each market.
 - The total load should be defined, and uploaded via a file, in the menu Time Series.
 - All the renewable power production (non-dispatchable), which are also hourly, should be defined in Time Series as well. Then in the file with power plant data, the applicable time-series ID should be included.
- The other dropdowns allow the user to define where the results of KyPF can be stored. In case of simulations, the average across the simulations is stored. All these inputs are optional.
 - The hourly prices may be stored in a KyCurve profile. This allows you to analyse the hourly price forward curve, zoom in, export to Excel, etc.
 - The monthly average prices (base, peak and offpeak) may be stored in a Price Source. This allows you to analyse the monthly prices over time (with multiple KyPF runs) and use them in other KYOS applications such as KyCurve.

If there are multiple markets, only one can be the primary power market for which additional statistics are created.

Dutch power

Total load	NL Power	
Store hourly electricity curve in	Dutch BasePower KyPF	KyCurve profiles for the selected power commodity, defined as direct upload.
Store baseload tradable products in	KyPF NL Base	Baseload sources for the selected power commodity.
Store peakload tradable products in	KyPF NL Peak	Peakload sources for the selected power commodity.
Store offpeak tradable products in	KyPF NL OffPeak	Off-peak sources for the selected power commodity.
Is primary	☒ Yes	

German Power

Total load	DE Power	
Store hourly electricity curve in	German BasePower KyPF	KyCurve profiles for the selected power commodity, defined as direct upload.
Store baseload tradable products in	KyPF DE Base	Baseload sources for the selected power commodity.
Store peakload tradable products in	KyPF DE Peak	Peakload sources for the selected power commodity.
Store offpeak tradable products in	KyPF DE OffPeak	Off-peak sources for the selected power commodity.
Is primary	☐ Yes	

Belgian power

Total load	BE Power	
Store hourly electricity curve in	No curves	KyCurve profiles for the selected power commodity, defined as direct upload.

Interconnections

If there are multiple markets: a new tab "Interconnections" will appear on the screen. The user must define the interconnection capacity between each power market. The interconnection capacities could be a single number, constant over time, or a time series. The time series may be daily or hourly and is uploaded via a file in the Time Series menu.

From	To	Capacity	Time series
Dutch power	German Power	1000 Mthw	☐ Select timeseries profile
Dutch power	Belgian power	1020 Mthw	☐ Select timeseries profile
German Power	Dutch power	1230 Mthw	☐ Select timeseries profile
German Power	Belgian power	1120 Mthw	☐ Select timeseries profile
Belgian power	Dutch power	1125 Mthw	☐ Select timeseries profile
Belgian power	German Power	1135 Mthw	☐ Select timeseries profile

Automation

General	Fuels: curves	Power markets	Interconnect capacities	Automation
---------	---------------	---------------	-------------------------	------------

Automatic runs

Schedule calculation daily at a specific time ☐ Yes ☒ No

Automate calculation directly when new inputs are available ☐ Yes ☒ No

Schedule calculation daily at a specific time

The user can schedule to run the profile every day at a certain time. The model uses the latest available fuel forward curves or the latest simulation job as an input. The scheduling can be done multiple times per day and on different weekdays.

Automate calculation directly when new inputs are available

KyPF runs automatically when a new forward curve or a new simulation for the selected profile is available. The default valuation date is equal to the trading date plus one day.

19.2 Power Plant data

In the General tab, the user has to upload a file containing the plant characteristics. The fields in grey are optional and the corresponding columns may be removed from the input file. You may also change the order of the columns: the model uses the headers (first row) to identify what information is in which column. The file type could be Excel or csv. In case of csv, the separator of the file is a comma (,). In case of Excel, the KYOS Platform translates the file to csv.

The input sheet expects three types of power plants:

- Fuel fired plants, including CHP's: most input columns are to describe their characteristics (for a description see below).
- Pump hydro storage plants and batteries: their characteristics are defined by the following inputs:
 - IsBattery: 1 if the plant is a pump-hydro facility or a battery or other storage type
 - InjRate: MWh of pumping (storage) capacity per hour, so the energy consumption during the pumping / storage phase
 - WdrRate: MWh of generation / production capacity per hour
 - MaxMW: maximum capacity of the battery or (for pump-hydro) the total MWh of energy which can be stored in the upper reservoir and produced when water flows to the lower reservoir
 - EffMax: overall efficiency during the pumping / storage phase. For example, if the efficiency is 0.7 (70%), then each MWh used for pumping leads to 0.7 MWh energy in the storage (upper reservoir)
 - VOMPerMWh: variable operations and maintenance costs per MWh of production
- Non-dispatchable generation, such as solar, wind and hydro.

The Column names are as follows:

- LegalEntity, string, must not have a comma among the string
- PlantName, string, must not have a comma among the string
- PowerCommID, positive integer. This should correspond with a power commodity ID in the Platform. It determines in which market the plant produces.
- TimeSeriesID, positive integer. This is only used for non-dispatchable sources. The ID should correspond with an hourly Time series ID in the Platform, containing the forecasted production.
- ReportingCommID, positive integer. This should correspond to a commodity ID in the Platform. It is used for displaying results in the Platform:
 - a. If the plant runs on a fuel, fill in the fuel commodity ID (Comm1ID)
 - b. If this is non-dispatchable production (nuclear, renewable), which is defined as a time-series, fill in the commodity ID that best captures this type of production. For example, a user may have separate forecasts for wind off-shore and wind on-shore, but report this together under 'wind' and hence define a commodity 'Wind' in the Platform.

- Comm1ID, positive integer. This is the 1st fuel used by the power plant. It should correspond to a commodity ID in the Platform
- Comm1Multiplier, floating/decimal number, any number between 0 and 2. The multiplier, the plant efficiency and the fuel's energy content determine the fuel costs. Fuel costs = fuel price * multiplier / efficiency * energy content. The reason to have the multiplier defined separately, is that a certain fuel (e.g. biomass) may not have a forward curve, but the price is derived from another forward curve via a multiplier.
- Comm2ID, positive integer. This is the 2nd fuel used by the power plant and an optional input. If the plant has no 2nd fuel, then fill in 0
- Comm2Multiplier, floating/decimal number, any number between 0 and 2, default 0
- CO2CommID, positive integer, default 0. This input is for European markets or other markets with emission trading. Then the ID should correspond to the commodity ID of the emission (CO2, carbon) market in the Platform. Otherwise, fill in 0.
- MustRun, 0 or 1, default 0. If set to 1, the plant will always produce power, except for maintenance periods. The plant production will still be optimized, between minimum and maximum load.
- IsCHP, 0 or 1, default 0. If set to 1, the plant produces both power and heat (or steam). For a CHP the optimization is different and requires extra inputs. See further on.
- ColdStart, any number larger or equal to 0. Contains the cost for a cold start, expressed in the currency of the market (e.g. €, \$, ¥, £).
- ShutDown, any number larger or equal to 0. Contains the cost for a shutdown, expressed in the currency of the market (e.g. €, \$, ¥, £).
- MinOnTime, any number larger or equal to 0 (integer), default 0. This is the minimum number of hours a plant should produce before it can shut down.
- MinOffTime, any number larger or equal to 0 (integer), default 0. This is the minimum number of hours a plant should be off, before it can start production again.
- EffMin, floating number between 0 and 1. It contains the efficiency when the plant produces at the minimum load (MinMW). It may not be larger than EffMax. Note that for natural gas plants, all efficiencies are LHV.
- EffMax, floating number, between 0 and 1. It contains the efficiency when the plant produces at the maximum load (MaxMW).
- MinMW, positive number, minimum capacity (MW) of the plant
- MaxMW, positive number, maximum capacity (MW) of the plant
- MidMW, positive number, default 0. It contains the production (in MW) in between minimum and maximum capacity.
- EffMid, floating number, between 0 and 1, default 0. It contains the efficiency at MidMW.
- HeatCurveID, positive integer, default 0. Only required for CHP's. It contains the ID of the hourly time-series of heat (in MWh) that has to be produced by this plant. The model will find the optimal dispatch and produce the heat either with the plant, the gas boiler or the electric boiler.
- MinMWAAfterHeat, positive number, default 0. Required for CHP's: minimum power output if the plant produces heat.
- MaxMWAAfterHeat, positive number, default 0. Required for CHP's: maximum power output if the plant produces heat.
- HeatAtMinMW, positive number, default 0. Required for CHP's: heat output when the plant produces at minimum capacity.

- HeatAtMaxMW, positive number, default 0. Required for CHP's: heat output when the plant produces at maximum capacity.
- EffGasBoiler, floating number, between 0 and 1, default 0. Efficiency of the gas boiler.
- EffElectricBoiler, floating number, between 0 and 1, default 0. Efficiency of the electric boiler.
- VOMPerMWh, positive number, default 0. Variable operations and maintenance costs for each MWh of power production.
- StartDate, positive number containing the start date of production in Excel numeric format, default 0
- EndDate, positive number containing the end date of production in Excel numeric format, default 1,000,000
- Main1StartDate, positive number in Excel date format, default 0
- Main1EndDate, positive number in Excel date format, default 0
(up to 5 maintenance periods can be defined)
- Jan[1] to Dec[1], floating number larger or equal to 0, default 1. Contains the availability of the plant in the first January month. Similar input for Feb, Mar, Apr, May, Jun, Jul, Aug, Sep, Oct, Nov, Dec. For example, if a plant has an unavailability of 10% on average, you may fill in 0.9. The model will (simply) shave 10% off all capacities. Also, when the capacities are seasonal, with e.g. a higher capacity in winter than summer,
- Jan[2] to Dec[2], floating number larger or equal to 0, default 1. Same as previous, but for the second year and beyond. For example, if the model is run on 15 September 2016:
 - a. Sep[1] until Dec[1] refer to the year 2016,
 - b. Jan[1] until Aug[1], and Sep[2] until Dec[2] refer to the year 2017
 - c. All other years are in Jan[2] until Dec[2]

19.3 Run KyPF

Analytics → KyPF ▾ Run

In the list of KyPF profiles click on a profile and select run.

Select from a list of available trading dates and valuation dates. The trading date determines which fuel forward curves are used or which KySim run is used.

When KyPF is run, in the background the executable Matlab model is started. The inputs are:

- KyPFSettings.xml: contains the settings
- PlantData.csv: contains the file which has been uploaded by the user, containing all the power plant characteristics
- LoadDemand.csv: contains the demand Specified further on.
- PriceData.csv: contains the daily fuel price forward curves. Specified further on.
- TimeSeries.csv: contains all time series, including the non-dispatchable production forecasts, interconnection capacities time series, heat demand time series

19.3.1 LoadData.csv

This input file is generated by the Platform.

- *Load_ID* – load profile ID
- *Year* – yyyy
- *Month* – m/mm
- *Day* – d/dd
- *Hour* – h/hh, 0 - 23
- *Load_MW* – load in MW

19.3.2 PriceData.csv

This input file is generated by the Platform.

- *ForwardCurve_id* – fuel commodity forward curve ID
- *Year* - yyyy
- *Month* – m/mm
- *Day* – d/dd
- *Price* – price

19.3.3 TimeSeries.csv

This input file contains all other time series and it is generated by the platform:

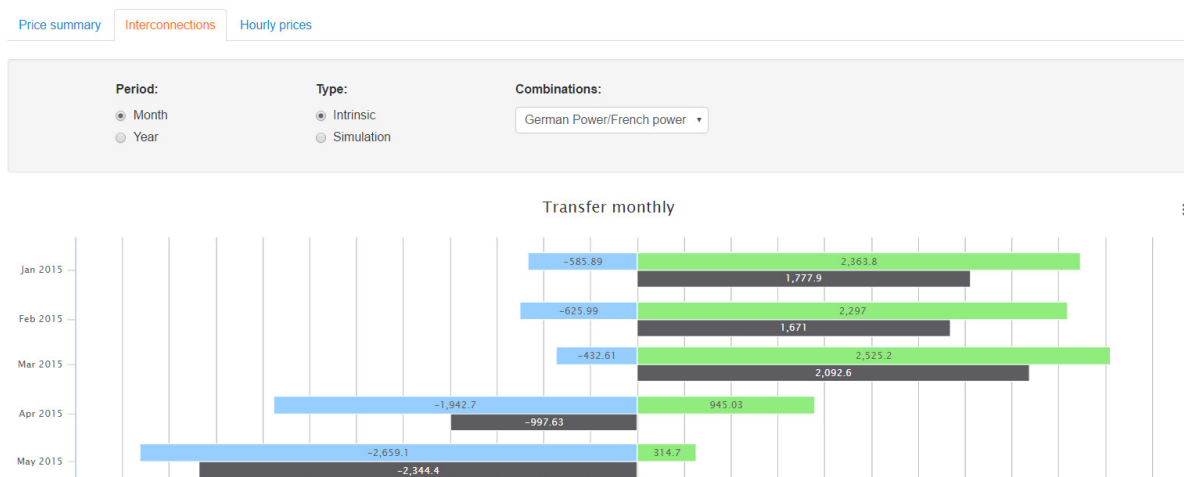
- *TimeSeriesID* – time series id on the platform
- *Year* - yyyy
- *Month* – m/mm
- *Day* – d/dd
- *Value*– time series value

19.4 Results

After a KyPF run ('job') has finished, the results are available in different places of the Platform.

First of all, three output tabs display the results in the web interface:

- Price summary: shows the average monthly and yearly prices
- Interconnections: shows the monthly and yearly interconnection flows (for multiple areas)
- Hourly prices: shows a graph of all the hourly prices



Secondly, after pressing the button 'download plant statistics' you can download a zip file with detailed hourly and yearly plant statistics. For each plant, the files contain information about the total production (base, peak, offpeak), costs, load factor, CO2 emissions and profit.

[Download plant statistics](#)

Thirdly, there are outputs available in other parts of the system:

- The hourly price forward curves are available as a forward curve in Curves → Forward curves.
- The monthly average prices are available as a price source in Price data → Price Sources.
- The detailed job outputs can be downloaded as a zip file. This is possible in Logs → Jobs. Go to the relevant job, and choose download from the drop-down menu. Here you can also find the PlantData.csv input file, and other input files, used for the calculation.

The following sections describe the outputs which are in the zip file.

19.4.1 Monthly power prices

KyPF generates 3 files per market, containing the monthly baseload, peakload and offpeak prices. These files are sent to the underlying Price Source profiles in the Platform, where they can be used as input for KyCurve. The name of the file is always:

- {commodity_id}_base.csv
- {commodity_id}_peak.csv

- {commodity_id}_offpeak.csv

The commodity_id is the ID of the power commodity as defined in the Platform. The unit of the monthly prices are defined according to the primary power commodity currency and unit, e.g. EUR/MWh or JPY/MWh. They may be adjusted according to the market convention, e.g. saving the price in JPY/kWh in Japanese market.

19.4.2 Hourly power prices

KyPF generates 1 file per market, containing the hourly forecasted prices. The name of the file is:

- {commodity_id}_hour.csv

The commodity_id is the power commodity ID as defined on the Platform; the unit of the hourly price are defined according to the primary power commodity currency and unit, e.g. EUR/MWh.

19.4.3 Monthly and yearly plant statistics

The file PlantStatisticsMonthly.csv contains detailed information for each plant and each month, including the power production, fuel consumption, CO2 emissions, costs and profit, e.g.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
1	PlantID	PlantName	PowerComm	MonthDate	Base	Peak	OffPeak	Fuel1Consum	Fuel1Consum	Fuel2Consum	Fuel2Consum	Operational	LoadFactor	CO2Emission	NrStarts	Revenues	FuelCost	CO2Cost	OMCost	StartCost	Profit
2	1	TOMATOUA	14	01-Sep-16	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
3	1	TOMATOUA	14	01-Oct-16	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
4	1	TOMATOUA	14	01-Nov-16	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
5	1	TOMATOUA	14	01-Dec-16	242544	74328	168216	607305.491	607.3055	0	0	744	1	0	0	1779691323	936932693	0	155228160	0	687530470
6	2	TOMATOUA	14	01-Sep-16	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
7	2	TOMATOUA	14	01-Oct-16	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
8	2	TOMATOUA	14	01-Nov-16	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
9	2	TOMATOUA	14	01-Dec-16	366792	112404	254388	918409.838	918.4098	0	0	744	1	0	0	2691373687	1416895146	0	234746880	0	1039731662
10	3	TOMATOUA	14	01-Sep-16	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
11	3	TOMATOUA	14	01-Oct-16	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
12	3	TOMATOUA	14	01-Nov-16	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
13	3	TOMATOUA	14	01-Dec-16	485088	148656	336432	1214610.98	1214.611	0	0	744	1	0	0	3559382645	1873865385	0	310456320	0	1375060940
14	4	DATE 1	14	01-Sep-16	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
15	4	DATE 1	14	01-Oct-16	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

This file has columns as follows:

- PlantID, the power PlantID equals to the row number of each power plant in PlantData.csv minus one, e.g. the first power plant in PlantData.csv is in the second row, and its PlantID will be assigned to be equal to 1 (= 2-1).
- PlantName, the same as the PlantName in PlantData.csv.
- PlantCommID: power commodity id of the plant, to indicate which power market the plant belongs to.
- MonthDate, positive number, the start date of each month, e.g., 42644 in excel stands for October 01, 2016.
- Base, the total production in the base hours (i.e. all the hours); the unit equals that of the primary power commodity unit, usually MWh.
- Peak, the production in the peak hours; the unit equals that of the primary power commodity unit, usually MWh.
- OffPeak, the production in the offpeak hours; the unit equals that of the primary power commodity unit, usually MWh.
- Fuel1ConsumptionMWh, the first fuel consumption in the month; the unit is in MWh

- Fuel1ConsumptionOri, the first fuel consumption in the fuel original unit, e.g. coal is usually quoted in Ton. The fuel original unit is defined by the user when the user creates a new commodity in the Settings on the Platform.
- Fuel2Consumption, second fuel consumption; the unit is in MWh
- Fuel2ConsumptionOri, second fuel consumption in its original unit
- OperationalHours, number of operational hours
- LoadFactor, number of operational hours divided by the number of total hours
- CO2Emissions, carbon emissions in the month; the unit is typically Ton
- NrStarts, the number of power plant starts
- Revenues: equal to the sum of the hourly power price multiplied by the hourly power production; the unit is in the primary power commodity currency
- FuelCost: equal to fuel consumption multiplied by the fuel price; the unit is in the primary power commodity currency
- CO2Cost, carbon emission cost; the unit is in the primary power commodity currency
- OMCost, operational and maintenance cost: equal to the sum of VOMPerMWh (defined in PlantData.csv) multiplied by power production; the unit is in the primary power commodity currency
- StartCost, start costs; the unit is in the primary power commodity currency
- Profit: equal to Revenues – FuelCost - CO2Cost – OMCost - StartCost

The file PlantStatisticsYearly.csv is the same, but per year.

19.4.4 Hourly plant statistics

KyPF creates a separate output file for each of the following statistics:

- Production
The hourly power production per plant is saved per year, e.g. HourlyProdcution2016.csv. The hourly production is in the unit of primary power commodity unit, usually MWh
Columns:
 - Col 1: year
 - Col 2: month
 - Col 3: day
 - Col 4: hour
 - Col 5 to the end: power plant PlantID, the power plant PlantID equals to the row number of each power plant in PlantData.csv minus one, e.g. the first power plant in PlantData.csv is in the second row, and its PlantID will be assigned to be equal to 1
- Fuel consumption
The hourly fuel consumption per plant is saved per year, e.g. HourlyFuel1Consumption2016.csv. The format is the same as Production (HourlyProdcution2016.csv). The unit is the fuel original unit (defined by the user on the Platform), e.g. coal is usually quoted in Ton. The hourly fuel consumption is saved for both the first (Fuel1) and the second (Fuel2) fuel.
- Production cost (e.g. in €)
The hourly production cost per plant is saved per year, e.g. HourlyProductionCost2016.csv, the format is the same as HourlyProdcution2016.csv, and the unit is in the primary power currency.
- Production by fuel type

The total hourly production per fuel type is saved per year, e.g.

Nuclear_HourlyProduction_2016.csv, the fuel type is defined by the user on the platform (Settings-Commodities), when the fuel type string has special characters, the special characters will be replaced by underscore. For example, for power plant using “Coal [JPY/MWh]”, the total production in the year 2016 from these power plants will be summed together per hour and saved into Coal__JPY_MWh__HourlyProduction_2016.csv. The columns will be as follows:

- Col 1: year
- Col 2: month
- Col 3: day
- Col 4: hour
- Col 5: Fuel Commodity name, e.g. Coal [JPY/MWh]

19.4.5 Transfer flows and market splitting

If KyPF has been run for multiple markets, with interconnections, then KyPF finds the optimal flow of power between the markets. The algorithm is based on price-coupling, i.e. power flows from a low price to a high price area until either the prices are equal or the maximum capacity is used. The resulting flows or average flows (in MW) are stored in TransferHourly, TransferMonthly and TransferYearly. The files SplittingHourly, SplittingMonthly and SplittingYearly count the frequency that an interconnection is fully used and (hence) prices in two areas are different.

- SplittingHourly
Hourly splitting information between each pair of power market, the splitting exists (equal to 1) between two markets if the price difference is bigger than 0.01% of the price level
 - Col 1: date
 - Col 2: hour
 - Col 3 to the end: power market pair information, P_{commodity_id}_P_{commodity_id}, e.g. P_15_P_16, number 15 and 16 represents the power commodity ids of two power market.
- TransferHourly
The format is the same as SplittingHourly, with the transfer amount in between each power market saved. The transfer amount is the absolute value saved in this file, with the sign signifying the direction: “positive” implies export, “negative” implies import. E.g. in column P_15_P_16, the power transfers from power market with power commodity id 15 to power market with power commodity id 16. When the number is positive, it means power market with id 15 exports to power market with id 16, when then number is negative, it means power market with id 15 imports from power market with id 16. The unit is in the primary power commodity unit, usually in MWh. The unit could also be interpreted as MW. E.g. the amount is the actual power transfer per hour, e.g. 500 MWh in a particular hour; it could also be interpreted as in this particular hour, the average transfer capacity is 500 MW (= 500MWh/1hour).
- SplittingMonthly
The hourly splitting information is summarized into monthly and saved into SplittingMonthly.csv. The column information is as follows:

- power1_commodity_id, the first power commodity id, e.g. 15
 - power2_commodity_id, the second power commodity id, e.g. 16
 - year
 - month
 - SplittingHoursInMonth, number of splitting hours in the month
 - SplittingPercentage, number of splitting hours divided by number of total hours in the month
- TransferMonthly

The format is the similar to SplittingMonthly, with column information as follows:

 - power1_commodity_id, the first power commodity id, e.g. 15
 - power2_commodity_id, the second power commodity id, e.g. 16
 - year
 - export, monthly average export. The export is calculated as the sum of the exporting amount in the month divided by the total number of hours in this month. E.g. in month 2016, Nov, there are 720 hours in total. Among the 720 hours, there are 240 hours when the first power market (assuming power commodity id: 15) is exporting to the second power market (assuming power commodity id: 16), then export amount in these 240 hours is summed together and then divided by 720 and the result is the monthly average export in this month. The unit is usually the primary power commodity unit (e.g. MWh) divided by hour, e.g. MW.
 - Import, monthly average import, the import is calculated similarly to export, it is calculated as the sum of the absolute value of import amount divided by the number of hours in the month. The unit is the same as export, usually in MW
 - net_export, monthly average transfer, this is calculated as the sum of the transfer amount divided by the total number of hours in the month. The unit is the same as export, usually in MW.
 - SplittingYearly

The SplittingYearly is the same as SplittingMonthly (with the “month” column removed), but per year.
 - TransferYearly

The TransferYearly is the same as TransferMonthly (with the “month” column removed), but per year.

19.4.6 Graphs

Several graphs are saved for the user’s reference; they are saved under the folder ~\Output\Intrinsic\Graphs, and include the following:

- Convergence_AbsoluteImbalance.png
- Convergence_Imbalance.png
- Convergence_Price.png
- Year_{YEAR_Number}_FuelProduction.png, e.g. Year_2016_FuelProduction.png

20 What-If: Scenario Analysis

The what-if scenario analysis tool allows you to assess the impact of forward price shocks on expected future cash flows. Note that the price shocks only affect floating price (=indexed) contracts, not fixed price contracts. For instance, you can add 2% to the forward curve and see how this affects the expected future cash flow of a portfolio of contracts.

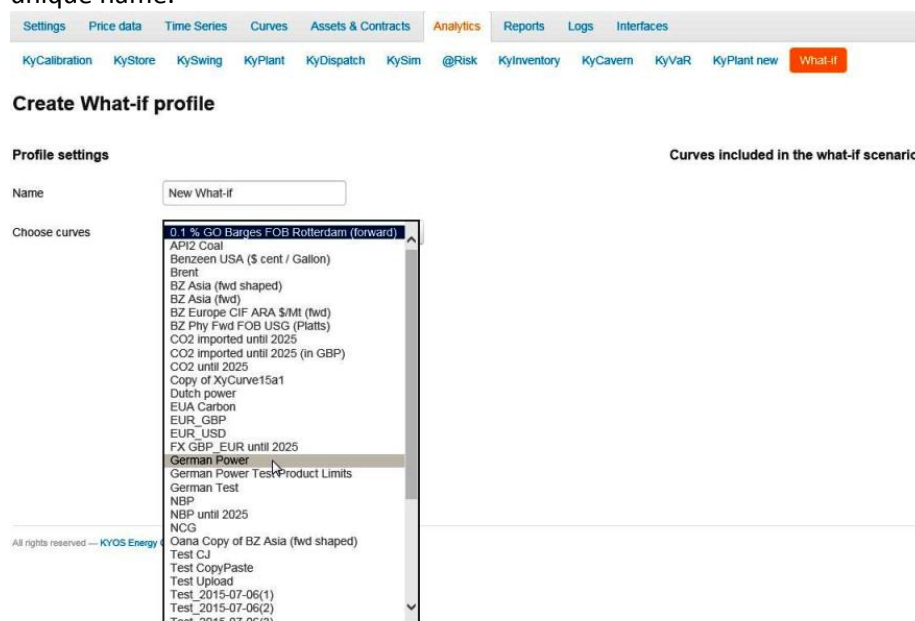
The main steps for the What-If analysis are:

1. Create a What-If profile, where you define which curves should be shocked (stressed)
2. Run the profile and choose the size of the shocks. These will be applied to all contracts.
3. View the impact of the price shock on the expected future cash-flows. In the View screen you can flexibly select which contracts to analyze.

20.1 Create a What-If profile

Analytics → What-If ☐ Create profile

A What-If profile is created by selecting one or more forward curves. The profile can be stored under a unique name.



20.2 Run a What-If Job

Analytics → What-If ▾ Run

Clicking on the arrow from a profile shows several options; choose Run.

The user can change the values of the forward curve:

- a) by a percentage
- b) by adding/subtracting a value
- c) by defining new price levels for the curve

[Settings](#) [Price data](#) [Time Series](#) [Curves](#) [Assets & Contracts](#) [Analytics](#) [Reports](#) [Logs](#) [Interfaces](#)

[KyCalibration](#) [KyStore](#) [KySwing](#) [KyPlant](#) [KyDispatch](#) [KySim](#) [@Risk](#) [KyInventory](#) [KyCavern](#) [KyVaR](#) [KyPlant new](#) [What-if](#)

Run What-If profile: German power

Please select a trading day. For disabled trading days try running KyReport first.

Trading date

Jul 2015
2015

Su	Mo	Tu	We	Th	Fr	Sa
				1	2	3
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

Selected curves

	Jul-15	Aug-15	Sep-15	Oct-15	Nov-15	Dec-15	Jan-16	Feb-16	Mar-16	Apr-16	May-16	Jun-16	Jul-16	Aug-16
German 2015-07-20	35.75	30.73	33.97	34.27	36.44	33.18	36.54	37.05	31.66	30.61	28.39	29.45	29.30	29.29
Power														
Change														

percentage

absolute

user defined

20.3 View the What-If analysis

Analytics → What-If □ View □ View

Clicking on the view button of a calculation job shows the result of the forward price shock. The results table and graph show the initial cash-flow expectation and the stressed cash-flow expectation, i.e. after the shock. In an extra tab, the initial and stressed prices are shown as well. With the filters at top, the user has complete flexibility which **contracts to include in the analysis**.

After applying the filters a graph is shown:



21 Logs

21.1 Jobs

Logs → Jobs

After running a model a job is created. A job has a status which can be one of the following:

- **queued:** the job is waiting to be executed
- **dependent:** the job needs another job to finish before it can be executed
- **preparing:** the job is prepared to be executed
- **running:** the Matlab model is currently running the job
- **processing:** the job result is processed
- **finished:** the job has finished successfully
- **obsolete:** a more recent job for this profile exists
- **error:** the job has finished with an error

21.2 Export

Logs → Export

Displays information about actions:

Date and time of the action, the user who performed the action, the analytical model involved the profile-name, the job ID, etc.

Export log

20 results found. Total items: 24.									
< 1 2 > 20 per page									
Exported	Action	User	Tool	Profile name	ID	Start and End dates	Job ID	Trading date	Job executed
2014-08-21 12:21:19	Export to Siemens	Admin istrator	Kydispatch	Corby Test	1		143688		2014-08-21 11:24:12
2017-05-21 12:48:14	Export to excel	Admin istrator	Kyplant	Diemen Intr/Scheduled	26	2017-05-21 - 2017-12-31	134606	2017-05-20	2017-05-21 11:40:02
2017-05-22 10:16:25	Export to excel	Admin istrator	Kyplant	Diemen Intr/Scheduled	26	2017-05-17 - 2017-12-31	134450	2017-05-16	2017-05-19 11:40:01
2014-08-12 11:59:13	Export to Allegro	Admin istrator	Kyswing	Profile 1	8405		143092	2014-08-11	2014-08-12 10:20:02
2014-08-12 11:59:13	Export to Allegro	Admin istrator	Kyswing	Profile 2	8406		143093	2014-08-11	2014-08-12 10:20:02

21.3 Deleting

Deleting works top down in the platform. If you delete an item all it's decendants are deleted too! If you have a storage 'my gas storage' defined and you delete it, all the items in the picture below are also deleted (white, green, blue and gray). The same is true for power plant, inventory and contracts.



Deleting 'my gas storage profile 1' results in the removal of all blue items. So all the jobs created by profile

'my gas storage profile 1' are deleted as well.

Deleting a job 'my gas storage profile 1 job no 1' will have no influence on any other data.

Deleting 'my gas storage scheduled volumes' will have no influence on any other data.

Note: deleting a user needs the user to be disabled first. If you proceed and delete the user than all the jobs he/she created are also deleted.

Deleting a 'price source' will also result in removal of all price data for that source.

22 Interfaces

22.1 Mappings

Interfaces → Mappings

There is a possibility to connect Kyos platform profile id's to Client specific id's.

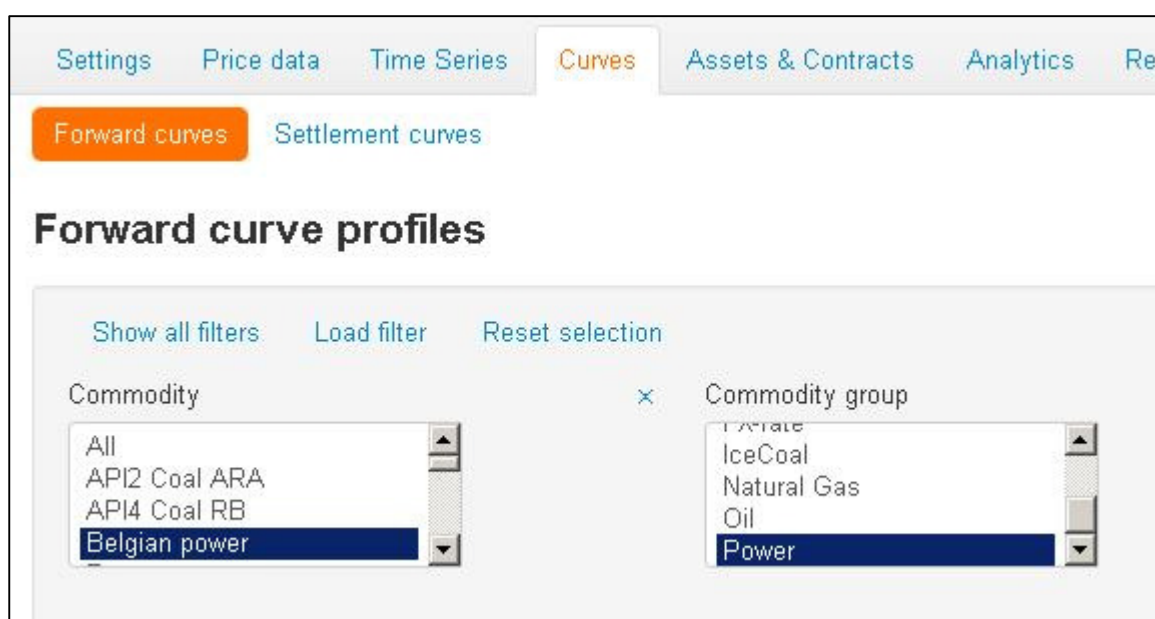
22.2 Export

Interfaces → Export

This is customer specific.

Clients can perform an action when a model is run for instance send an email to the user when KyCurve for a specific profile has run. For example we take Belgian power. To do so the go to the

Curves → Forward curves



Create a filter and click on save. A dialogue pops up.

Save current filter setting

Provide a name to remember this filter setting

BePowerForwardCurve

Or select an existing filtersetting to overwrite

TTF curves

Make this private?

☐ Yes (only you can see it)

Cancel

Save

Now go to Interfaces → Export ☐ Create Select KyCurve and email

[Mappings](#)
[Export](#)
[Short positions](#)

Create Interface profile

Select KYOS model

KyCurve

Export type

Email

Click next:

[Settings](#)
[Price data](#)
[Time Series](#)
[Curves](#)
[Assets](#)

[Mappings](#)
[Export](#)
[Short positions](#)

Create Interface profile

Profile name

Select saved search

Schedule profile
☐ Yes ☒ No

Automated
☐ Yes ☒ No

Click next:

[Settings](#)
[Price data](#)
[Time Series](#)
[Curves](#)
[Assets & Contracts](#)

[Mappings](#)
[Export](#)
[Short positions](#)

Settings for Email export type

Job status

Emails

Attachment

Job status

You can select: Finished, Error or Both.

Email

Enter your email address

Attachment

You can select: None, Output folder, Job folder

From now on every time the Belgian power KyCurve runs you receive an email.

There is also a possibility to export to a specific directory

22.3 Short positions

Interfaces → Short positions

Documentation to be developed.

23 Contact Details

For questions regarding the KYOS Analytical Platform, please contact KYOS Energy Consulting.

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